



RUNNING TRACKS



A Construction &
Maintenance Manual
Eleventh Edition

2023





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A Construction & Maintenance Manual

Eleventh Edition

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Table of Contents

Acknowledgments.....	vii
The American Sports Builders Association.....	viii
Certified Track Builders Program.....	ix
ASBA Facility Awards Program	xi
Introduction	xii
Completed Tracks Shown in this Edition	xiv
Chapter 1—Getting Started	1
1.1 Initial Planning	1
1.2 The Planning Team.....	2
1.3 Delivery Methods	2
1.4 Designers and Consultants.....	7
1.5 Planning the Facility.....	8
1.6 Budgeting	10
Chapter 2—Planning and Design Considerations	13
2.1 General Requirements.....	13
2.2 Site Considerations.....	15
2.3 Base Geometry.....	16
2.4 Track Configurations	16
2.5 Field Event Dimensions and Areas	19
2.6 Site Requirements.....	20
2.7 Transition Area	20
2.8 Impact of Region, Geography and Location	21
2.9 Permits, Codes and Authorities.....	21
2.10 Trees and Vegetation	21
2.11 Soil Conditions	22
2.12 Site Grading.....	23
2.13 Drainage	24
2.14 Curbs and Perimeter Edging.....	27
2.15 Base Construction	28
2.16 Construction Documents.....	29
Chapter 3—Choosing a Surface	55
3.1 General.....	55
3.2 Installation of Surfacing Systems.....	55
3.3 Classifying Track Surfaces.....	56
3.4 Types of Surfacing Systems.....	57
3.5 Primers, Binders and Coatings	65
3.6 Surface Evaluation, Testing and Standards	68

Chapter 4—The Contractor and Construction Processes	71
4.1 Choosing a Contractor	71
4.2 Overview of Construction	74
4.3 Construction Timetable.....	88
Chapter 5—Field Events and In-Ground Equipment	89
5.1 General.....	89
5.2 Long Jump, Triple Jump and Pole Vault	89
5.3 Throwing Events	91
5.4 High Jump	92
5.5 Steeplechase	92
5.6 In-Ground Track Equipment.....	92
5.7 Accessibility to Storage Shelters.....	93
5.8 Viewing for Spectators.....	93
5.9 Timing Systems.....	93
5.10 Sound Systems	94
5.11 Lighting.....	94
5.12 Fencing.....	96
5.13 Water	97
5.14 Other Safety Issues.....	98
Chapter 6—Marking a Track: Calculations, Layout and Painting.....	109
6.1 Purpose of Markings	109
6.2 Selecting a Marking Specialist.....	109
6.3 Phase I—Calculations.....	110
6.4 Phase II—Layout.....	111
6.5 Phase III—Painting	112
6.6 Certification	113
6.7 Common Errors	113
6.8 Suggested Symbols and Graphics.....	113
Chapter 7—Verification	115
7.1 Track Certification Levels.....	115
7.2 Class 1 and 2 (World Athletics).....	116
7.3 Class 3—Marking and Slope Verification.....	116
7.4 Class 4—Marking Verification	128
7.5 Class 5—Certificate of Accuracy	137
Chapter 8—Care and Maintenance	139
8.1 General Maintenance	139
8.2 Regular Maintenance	140
8.3 Water on Tracks	143
8.4 Winterizing	144
8.5 Repainting.....	144
8.6 Health and Hygiene	144
8.7 The Repair Continuum—From Resurfacing to Reconstruction.....	146
8.8 Safety Maintenance	148

Chapter 9—Indoor Tracks	151
9.1 Planning the Facility	151
9.2 Track Planning and Design Considerations	152
9.3 Indoor Track Construction	153
9.4 Applying a Track Surface	154
9.5 Field Events and In-Ground Equipment.....	154
9.6 Marking a Track: Calculations, Layout, and Painting	155
9.7 Verification.....	155
9.8 Care and Maintenance	155
 Chapter 10—Appendices.....	157
Appendix A—Sources of Information	158
Appendix B—Governing Bodies and Rules of Track and Field.....	160
Appendix C—Track and Field Specifications	161
Appendix D—Calculation and Slope Notations	175
Appendix E—Conversions	176
Appendix F—400 Meter Track Verification Method.....	177
Appendix G—Suggested Running Track Marking Color Coding by Governing Body....	178
Appendix H—Checklists.....	179
 Glossary of Terms.....	187
 Index.....	195

Acknowledgments

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Many thanks to the governing bodies of track and field in the U.S. (NAIA, NFHS, NCAA, NIAAA and USATF), who meet annually with ASBA representatives to discuss, clarify and suggest modifications to the rules to facilitate the design and construction of better sports facilities. Please note that currently the NAIA uses NCAA rules and the NIAAA uses NFHS rules. We'd also like to thank ASBA headquarters staff for their support of the manual's publication.

It is our goal that *Running Tracks: A Construction & Maintenance Manual, Eleventh Edition* incorporates the best and most current information from across the industry. We'd like to especially thank all the industry professionals who have taken the time to help ensure this manual indeed meets that goal, including Tim Webber and Mike McNaught of Labella Associates and Mark Baginski of Verde Design, for their work to update and enhance the figures and drawings in this eleventh edition. We would also like to give a special thanks to Kaylee Derby, ASBA Education Director, for providing much needed support and to Alison Smith for "driving the bus" and keeping us on task. We must also thank the editorial committees and many contributors of the previous editions for laying such an excellent foundation.

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The American Sports Builders Association

With nearly 500 member companies, ASBA is a national organization for builders, designers and suppliers of materials for tennis courts, running tracks, natural and synthetic turf fields, indoor and outdoor synthetic sports surfaces. It is recognized as a centralized source for technical information, including consumer-oriented information.

Founded in 1965 as the U.S. Tennis Court & Track Builders Association (USTC&TBA) by a group of contractors who recognized the need for an organization that could help them address their mutual problems, set industry standards and keep them informed on issues pertinent to their businesses, its name was changed in 2004 to reflect the broad range of work performed by its members.

Headquartered in Maryland, the ASBA works to raise professional standards and to ensure that consumers receive services of the highest quality. As part of its work to achieve this goal, the Association sponsors a certification program that permits experienced tennis court, running track and field builders to demonstrate their competency by completing an examination on all areas of construction. In addition, ASBA offers an awards program to recognize excellence in design, construction and renovation.

Another priority is education. The Association recognizes that the well-informed consumer is more likely to make appropriate decisions. Therefore, ASBA publishes online Buyer's Guides for track, tennis court and field construction, which provide information on the process of selecting a site, choosing a contractor, identifying a surface and so forth. Also available online at www.sportsbuilders.org are more detailed Construction Guidelines and a searchable database of members. For those who prefer printed material, the Association offers a directory of members at no charge; the directory allows consumers to locate design professionals, builders and material suppliers, as well as consultants, trade publications and other industry resources.

Like its publications, the ASBA is a resource not only for the professionals who design, build and supply materials for top-flight sports facilities, but for the owners, operators, players and others who enjoy them. The members of the Association recognize that the way to continue interest in sports is to provide an environment that is conducive to enjoyment and/or performance.

For more information, browse our Association's website or contact ASBA at:

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Certified Track Builders Program

The Certified Track Builder (CTB) program was developed by ASBA to help establish consistent, high-level professional standards and to improve the practice of track construction. Candidates for certification must pass a comprehensive examination on track construction and maintenance as well as fulfill a set of prescribed standards of experience that demonstrate a high level of knowledge in track construction. The purposes of certification for track builders are:

- Assisting the general public in evaluating the experience and expertise of builders;
- Recognizing professional track builders who have met a designated level of experience and demonstrated a standard of knowledge;
- Providing a means of identifying professionals who have met a standard of achievement; and
- Raising professional standards and improving the practice of track construction.

Becoming a Certified Track Builder can also prequalify a candidate for a project.

In Massachusetts, for instance, a local school district restricted bidding for a project to companies that employ ASBA Certified Track Builders, a precedent that was challenged in the courts in 2008. In defending its decision to require that the bidder employ a CTB, the school district argued that it had encountered serious problems regarding the workmanship in prior track installations, which justified the use of the certification requirement as a means of identifying qualified contractors. The school district's bidding documents also required that the "installer of track surfacing shall be a member in good standing with the American Sports Builders Association with specific qualifications in building and constructing tennis courts and running tracks."

After review, the Massachusetts Office of the Attorney General found that the district's decision was not arbitrary because these requirements "formed a rational basis" for concluding that the noncertified, nonmember low bidder was not a responsible bidder. It further found that the requirement was not overly restrictive because "certification is open to anyone who meets the ASBA's standards." Copies of the Massachusetts decision can be obtained free of charge from the ASBA headquarters. The process for becoming a Certified Track Builder involves:

1. Completing and submitting the application and verification eligibility forms, which include identifying three references.
2. Paying the applicable fees.
3. Completing the eligibility spreadsheet, which demonstrates the applicant's relevant work.
4. Passing the comprehensive exam, a four-hour written test consisting of 100 multiple-choice questions.

Certification is good for three years, at which time a CTB can recertify, either by retaking the exam or via continuing education points and the eligibility spreadsheet demonstrating

the CTB's relevant work of the last three years. Continuing education points are earned through attending technical seminars, authoring papers, participating in research and other ASBA-related activities.

For more information on the CTB program and to see a list of current Certified Track Builders, visit **www.sportsbuilders.org/certification**.

ASBA Facility Awards Program

The purpose of the Facility Awards Program is to support ASBA's goal of promoting excellence in sports facility construction and recognizes those facilities built or designed by ASBA members which have attained this goal. Awards are given to the facility with recognition of participating ASBA members.

For tracks division, the award categories include:

- Indoor Track Facility
- Outdoor Track Facility
- Track & Field Facility
- Single Field Facility
- Multi-Field Facility

Most of the tracks showcased throughout this manual are ASBA award winners. Learn more about the awards program on the [ASBA website](#).

Introduction

The purpose of this book is to promote quality planning, design, construction, maintenance and repair of track facilities by providing information to owners, administrators, coaches, builders, architects and engineers. The word “track” as used in this book refers to the running track oval and to the related field events.

Since no two track projects are exactly the same, this book cannot provide all the answers. Rather, it is intended to explain the terminology, geometry, amenities, design options and the rationale for various choices. In other words, it helps interested parties better understand the important questions to ask as they prepare to make decisions on their track facility projects.

Asking questions is not only smart, it is essential. The investment of time and energy in learning about track facilities can yield a huge return in their quality and in the enjoyment that will be derived from them. The American Sports Builders Association is committed to providing information to assist owners and users in making informed choices that promote the construction of quality track facilities.

Given this objective, this book is largely directed at the high school and college markets, where ASBA members do most of their work. Design and construction considerations for tracks at higher levels (tracks constructed for international competition, for example) are addressed by the *Track and Field Facilities Manual*, published by World Athletics. See Appendix B in Chapter 10 for contact information.

Chapter 10 contains a number of appendices meant to provide additional resources for information, calculations, specifications and other useful materials for those involved in constructing track facilities. The rules are constantly under review by the governing bodies and are subject to change. Please consult the appropriate governing body’s latest rule book. The end of the manual also features a glossary of terms commonly used in the course of designing and building a track.

Dimensional Notation

In some cases, the track calculations and measurements noted during a project will be metric and in other cases imperial. Calculations and construction may be performed either in metric or in imperial as long as appropriate tolerances are achieved. Throughout this book, measurements are expressed in either metric or imperial, depending on the most common use.

Disclaimer

The drawings and technical information in this book are intended to provide clarification of the design and construction concepts discussed in the text. They are meant to represent generic solutions, typical of those used in the design and construction of tracks. Each drawing represents only a single option to address the needs described in the text. In every

case, there are other ways to achieve the design intent. In addition, readers of this book are cautioned that site conditions can vary widely. Local climates, subsurface geotechnical conditions, construction methods, availability of materials and site engineering requirements make the development of universal construction details virtually impossible. While every effort has been made to ensure the accuracy of the drawings, ASBA is not responsible for any errors or omissions.

The recommendations in this book are suggestions for architects, engineers, contractors and owners. Except where they refer to the rules of track, these guidelines provide only options, explanations or illustrations of typical generic solutions to construction issues. Parties not experienced in track construction are advised to consult a Certified Track Builder, a qualified contractor or a design professional. Due to the differences in design intent, local regulatory requirements, climate and construction sites, each project is different. ASBA makes no representations regarding the suitability of a particular design or decision as it relates to a specific project.

Completed Tracks Shown in this Edition

Front Cover

- Franklin High School | Stockton, CA
- Danbury High School | Danbury, CT
- Universite de Sherbrooke, Quebec, Canada

Chapter 1

- Benicia High School | Benicia, CA

Chapter 2

- San Jose City College | San Jose, CA
- Nike Campus | Beaverton, OR

Chapter 9

- Clemson University | Clemson, SC
- Franklin & Marshall College | Lancaster, PA

Chapter 10

- Mt Holyoke College | South Hadley, MA
- Tennyson High School | Hayward, CA

Glossary

- Lexington High School | Lexington, MA
- Tokay High School | Lodi, CA

Index

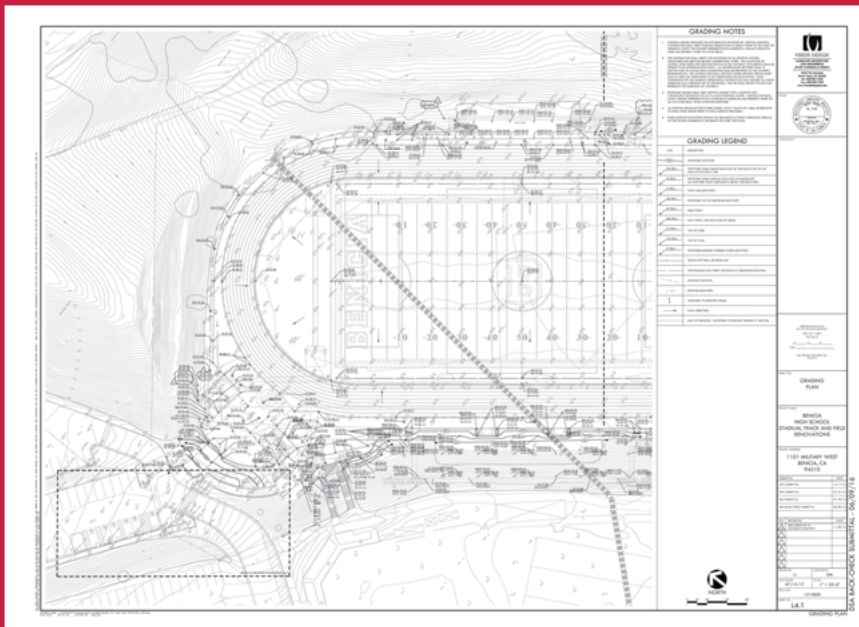
- University of South Carolina | Columbia, SC
- Mount Eden High School | Hayward, CA

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- West Point High School | Avondale, AZ
- Abessinio Stadium, Salesianum School | Wilmington, DE
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Chapter 1

GETTING STARTED



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Chapter 1—Getting Started

This chapter will cover the following planning aspects in the development of a new track facility.

- 1.1 Initial Planning
- 1.2 The Planning Team
- 1.3 Delivery Methods
- 1.4 Designers and Consultants
- 1.5 Planning the Facility
- 1.6 Budgeting

1.1 Initial Planning

The first step in any track facility project is planning. Planning involves a number of steps, including defining the project, setting a budget, choosing a suitable site, developing an overall site plan, selecting a surface, developing general specifications and so forth. Building soil testing into the early phases of the planning process is also a good idea, if feasible, as the results may influence decision making. They may require choosing a different site, for example, or increasing the overall budget.

Effective planning also involves considering the use of the facility. Who are the end users? Competitive athletes? Recreational joggers? Fitness walkers? Those using wheelchairs? Will other sports be sharing parts of the facility? The preferences and needs of these users must be considered. For example, for a facility to be used in competition, the order of events and the movement of athletes and officials from one event to another become important. Likewise, the needs of spectators must be addressed. Storage, maintenance and operations are important elements to consider. In short, it is essential to consider how the track facility will be used in order to foster and enhance a user's experience—no matter what form that experience takes.

It is important to recognize that every track design involves compromise. For example, a compact facility that is great for spectators may not allow reasonable room for staging areas, warm-ups and officials. Also, when several sports (football, soccer, etc.) share the infield area, there may be interrelationships in such aspects as geometry, drainage and slope.

Planning decisions are critical because the investment in a track facility—whether for school, community, collegiate or international competition—is substantial. If properly maintained, well-planned and well-constructed, a track facility that appeals to users and enhances programming will provide years of functional use.

1.2 The Planning Team

Depending on the size and scope of the project, an owner may involve a number of professionals on the project's planning team:

- A Certified Track Builder (CTB)
- A track facility consultant
- A design professional
- An experienced contractor
- An experienced facility manager
- Manufacturers and installers of goods and materials including surfacing material, fencing, field event equipment, bleachers and lighting

Equally important is owner representation including coaches, administrators and maintenance personnel. Their input can help ensure that, once constructed, the facility will operate smoothly, helping users to perform at their best and enhancing the enjoyment of spectators. Involving personnel who will ultimately be responsible for maintaining the facility will also help ensure value for construction dollars. Remember that the upkeep of a facility is key to its long-term use and durability.

For facilities that are being developed or adapted for regional use and higher levels of competition, it is especially important to consult with the appropriate governing body: the National Federation of State High School Associations (NFHS), the National Collegiate Athletic Association (NCAA), USA Track & Field (USATF), World Athletics or the local state organization. Some facilities will need to meet the requirements of more than one organization. These governing bodies specify requirements for most competitive events, which vary depending upon the level of competition. Typically, issues such as number and width of lanes, number and location of runways, number and location of throwing event areas and support facilities for competitors, officials and spectators will be affected.

1.3 Delivery Methods

1.3.1 Cooperative Purchasing

Cooperative purchasing—also known as group purchasing, collaborative buying or consortium procurement—may be the preferred option for engaging a vendor. With this method, members of a purchasing group can leverage the purchasing power of a large group, through a competitive bidding process or negotiation, to obtain volume pricing discounts from preferred vendors. The real advantage of this approach is it puts the owner in control and shortens the process for getting a project started. The owner is typically assured of a responsible bid as all vendors are vetted prior to eligibility. Buying cooperative members often include government entities, school districts, municipalities, colleges and universities and many other public and nonprofit entities.

A school district that is a member of a buying cooperative can utilize a cooperative contract to procure its running track project. Because a cooperative purchasing contract agreement was entered into through a competitive bidding process, it often satisfies the local procurement bid laws. It is important for an owner to become familiar with local or state bid

laws to fully understand how cooperative contracts can be used. In many cases, an owner can use a cooperative bid contract to enter into an agreement with a vendor to perform some or all of a running track project. In addition to volume pricing discounts, the benefits of a cooperative purchasing procurement might also include a streamlined bid process, improved terms and services or increased control over contractor selection.

1.3.2 Design/Bid Projects

One method for construction projects, one that is often mandated in public projects, is called the design/bid or competitive bid approach. In this format, the owner engages a design firm to devise a facilities plan; prepare construction drawings, specifications and contract documents; and put the project out to bid. Potential bidders may be prequalified on the basis of experience, financial stability and/or other factors as allowed under regulations that differ from locality to locality.

General contractors (GC) prepare bids for the work and execute contracts with subcontractors for various specialty aspects. In most cases, the contractor submitting the lowest responsive bid, one which meets all the requirements of the contract documents, is awarded the contract.

Once the bids have been submitted, they should be carefully compared. It may be helpful for the owner to develop a checklist (see Appendix H for a sample bid review checklist).

The owner, often with the help of the design professional, should note any inconsistencies between bids and should question bidders to ensure that each bid reflects the requirements noted in the specifications. An owner should be especially careful with a very low bid or a very high bid. It is appropriate to question the contractor or to invite the contractor to review the bid.

The selected general contractor then is responsible for completing the facility as designed. The design firm may or may not assist the owner in administering the contract.

While applicable to all types of projects and now a well-understood method of project delivery by all players in the construction arena, there are numerous disadvantages to the design/bid process. Among them:

- All design work must be completed prior to letting out the bids. In many cases this means that specific design details and the choice of materials and methods are made without considering their impact on construction. Since many aspects of the project are highly specialized, the designer may or may not fully understand the effect of these kinds of design decisions on the scheduling and/or cost of the project. If a participating contractor introduces new or better methods, costly redesigns may be required.
- This method can create an adversarial relationship between the design professional and the contractor. It is not uncommon, for example, for specialty contractors to note aspects of a design that are not in compliance with governing body requirements, that are unsafe in practice or both. The subcontractor, and ultimately the GC, is then faced with bidding a job knowing the design is deficient or with correcting the

design professional, who may well be involved in selecting the winning bidder and/or administering the contract. Further, the contractor may feel no obligation to—and may, in fact, be discouraged from—raising concerns about potential conflicts within the contract documents or field conditions.

- This method assumes that all contractors are equally capable of providing a high-quality product on time and on budget. The method also requires the contractor to focus on keeping costs down while meeting those goals, an almost impossible task. Such unrealistic expectations encourage the bidder to seek the least expensive methods and the cheapest subcontractors. Competitive bidding also ignores a bidder's current workload and the specific qualifications and approaches of the members of the contractor's project staff. What the owner should seek is not the lowest price, but the best value for the investment.
- Failing to involve the GC early on, and necessitating the least expensive rather than the best approach, may result in costly change orders later. Rather than keeping costs down, competitive bidding may simply delay the reality of what the project will cost.

What can the public owner do to increase the likelihood of choosing a qualified company to build the project? A 2008 decision by the Inspector General of Massachusetts upheld the right of a local school district to restrict bidding for a running track project to companies that employ an ASBA Certified Track Builder (CTB).

In defending its decision to require that the bidder employ a CTB, the school district argued that it had encountered serious problems regarding the workmanship in prior track installations, which justified the use of the certification requirement as a means of identifying qualified contractors. The school district's bidding documents also required that the "installer of track surfacing shall be a member in good standing with the American Sports Builders Association with specific qualifications in building and constructing tennis courts and running tracks." After review, the Massachusetts Office of the Attorney General found that the district's decision was not arbitrary because these requirements "formed a rational basis" for concluding that the noncertified, nonmember low bidder was not a responsible bidder. It further found that the requirement was not overly restrictive because "certification is open to anyone who meets the ASBA's standards." Several other states have also accepted prequalification based on demonstrated competence. Authorities are beginning to uphold the buyer's right to have knowledgeable professionals working for them.

Prequalification of potential bidders can, to some degree, mitigate the danger of relying on low bid alone. Bidders may be prequalified on the basis of experience, financial stability and/or other factors as allowed under regulations, which differ from locality to locality. Unfortunately, state and local laws still often permit only minimal prequalification of contractors in order to encourage the largest possible pool of potential bidders and to avoid any possible appearance of discrimination, favoritism or conflict of interest. Requiring that responsive bidders employ a Certified Track Builder is one way to help ensure that the company is capable of doing the job and providing a quality project. ASBA's certified builder program was developed in order to help raise professional standards and improve the practice of sports facility construction. In order to become a certified builder an individual must meet specific criteria set forth by ASBA; each candidate must complete an application that shows they have sufficient experience in the sports facility construction industry and then must pass a comprehensive exam on track construction. In order to retain the designation, an individual must recertify every three years by providing documentation

showing continued experience and education in the industry. Copies of the Massachusetts decision, as well as information on the certification program, can be obtained free of charge from the ASBA headquarters.

Certainly, there are other options for prequalifying bidders for both private and public owners, including reviewing past client references or professional references from architects and engineers, reviewing bonding limits and general liability insurance, performing site visits at previous projects completed by the firm, etc. These options do provide some alternative processes for getting a project.

1.3.3 Design/Build Projects

For many projects, design/build may be a cost-effective option. In a design/build project, the owner works with an expert contractor or a contractor/designer team. That contractor or team both designs and builds the project. Not only can a design/build contractor or team help an owner by formulating and implementing project plans and solving problems, they can also provide invaluable expertise about local building requirements, available materials and workable options. Ultimately, the purpose of the design/build alternative is to streamline the design and construction process, saving time and cost to the owner.

Should this option be chosen, it is essential to select a good design/build contractor or team. Extensive, current experience in track facilities is a prerequisite. ASBA's Certified Track Builder program assures such current experience by requiring recertification every three years. Also, because of the cooperative nature of the owner/builder working relationship in such a project, it is important that the parties work well together, have a similar vision of the facility being planned and demonstrate the ability to communicate effectively. Even once the ideal design/build team is identified, the input of a track consultant is worthwhile; additionally, a design professional may be needed to prepare plans and other documents for permitting.

1.3.4 Lease/Lease Back

Leasing provides a financing option for all school and collegiate clients. The terms of these lease payments typically run between two and five years with monthly, quarterly, semiannual or annual payments. Financing can be provided for up to 100% of the acquisition cost. This can provide a great deal of convenience to the purchasing entity as the lease structure is tied to their annual general fund. More importantly, it allows the owner to purchase their needed facilities at today's prices and not incur any additional increases when the project has to be put back year after year. The operating budget of the owner is impacted only by the lease payments in the current operating budget year. Typically, the interest rates are very low—lower than conventional commercial rates.

1.3.5 Negotiated Agreements/CM at Risk

An alternative to competitive bidding that is becoming increasingly accepted is the negotiated agreement. Under this method, the owner chooses one or a small number of construction firms based on an evaluation of their experience, history of performance, financial stability or other qualifications. The owner then works in partnership with the

design team and the selected construction firm(s) to finalize design documents and to negotiate an acceptable bid, which in many cases is some form of cost plus a contractor's fee. The advantages of this process are:

- The owner can prequalify contractors on any number of variables, including intangibles such as cooperative spirit or integrity, and can meet and consider the specific individuals who will supervise the project.
- The owner maintains the maximum amount of control over the project.
- The owner can benefit from input from the contractor in planning and design decisions. The contractor may be able to suggest options that will help control costs without sacrificing quality. Early aspects of construction can begin even before the full design process is completed, thereby speeding up the timeframe.
- The GC can share the details of subcontractor or specialty contractor arrangements and bring them into the design process as necessary.
- Since subcontractors need not necessarily be locked in prior to the start of negotiations, the owner, designer and GC can work together to identify and contract with the best available subcontractors and specialty contractors for the project.
- A cooperative arrangement is established early on between owner, design firm and contractor. Risk is, to some degree, shared.

No process is perfect, however, and there are some disadvantages:

- The roles of the various parties in a negotiated agreement are not as widely understood, and some or all of them may be uncomfortable with the process.
- Since a commitment is made to the GC early in the process, a premium is placed on selecting the right contractor.
- To some degree, risk must be carefully delineated in case design errors are discovered during construction.

1.3.6 Request for Proposal (RFP)

Under the Request for Proposal (RFP) model, which is becoming a popular approach to track construction, the owner issues a written notice or call for proposals to prequalified designers and contractors. These designers and contractors are invited to submit detailed proposals for the development and implementation of the project. The owner outlines general parameters including identification of program needs, a list of evaluation criteria and a target budget; invitees are required to develop and price a turn-key project. The owner then evaluates each proposal against predetermined criteria to select finalists. Usually, the finalists are invited to meet with the owner to present their proposals in complete detail. A contract is issued to the designer or contractor whose proposal best meets the needs of the owner.

Comparing proposals can be difficult. One popular method for doing so is called a “calculated formula” in which each aspect of the proposal is weighted. For example, price might be weighted at 40%, product at 30% and installer at 30%. Each aspect of each proposal is rated for each of the three components, with the company rated best in all three awarded the bid. The advantages of this process are that:

- This method, too, potentially involves the contractor early on so that their expertise can be tapped.
- This method permits factors other than bid price to be considered.
- This method provides an incentive for potential bidders to think creatively for ways to minimize price and maximize effectiveness of the finished facility.
- This method clearly establishes accountability for all aspects of the finished project.

1.4 Designers and Consultants

Depending on the scope of the project, employing the services of an expert can help control job costs by better translating the needs of the owner into proper direction for construction and by helping to avoid costly mistakes. ASBA encourages owners to seek out a Certified Track Builder (CTB) to help. A Certified Track Builder (CTB), design professional or track consultant can help the owner identify needs and refine information to the specific conformation and characteristics of the site. An expert can assist in defining the scope of work to be included in the job, planning the facility, determining a realistic budget for the project, evaluating and comparing bids, overseeing the work in progress and solving any problems that occur during construction.

While construction of a track does not always involve the use of a design professional, there are many instances in which the advice of a professional experienced in track construction will be useful. These instances are referenced throughout this book but can include such factors as:

- Determining proper dimensions and locations of the track and of field events, particularly on irregular or limited sites
- Orientation of the track and field events, site layout and planning
- Drainage, including slope and drainage of the track, site drainage and stormwater management
- Soil issues and base construction
- Evaluating surfacing options
- General information, advice and comparison of accessories and amenities
- Climatic considerations
- Fulfilling some states' statutory requirement to engage a licensed professional

If the project design professional is not experienced in track design, the use of a Certified Track Builder (CTB) or track consultant to work with the designer is highly recommended. Consultants also can assist in conducting feasibility studies, recommending features to be included in the project, suggesting the best design and utilization of space to accomplish programming objectives, evaluating surfaces, accessories and amenities to meet the owner's needs and budget, supervising construction and authorizing payments.

How can an owner find a qualified track designer or consultant? One way is to consult with ASBA. ASBA offers a directory of its members, including Certified Track Builders, design professionals and experienced consultants. A searchable database also is available on the ASBA website at www.sportsbuilders.org.

An owner may also contact such other professional associations as the American Society of Landscape Architects (ASLA), the American Institute of Architects (AIA) and the American Society of Civil Engineers (ASCE). The reference section of this book includes contact information for these organizations. Another way to find a qualified professional is by asking colleagues who have recently completed similar projects for their recommendations. See “Choosing a Contractor” below for other ideas on finding and evaluating designers and consultants.

Regardless of the delivery method chosen, it is usually advisable for an owner or owner’s representative to hold a pre-bid conference to ensure that all bidders have seen the site and understand the scope of work to be performed.

1.5 Planning the Facility

Each track facility project is unique. In planning the facility, the following factors should be considered. Many of these factors will be addressed in more detail in other chapters of this book.

1.5.1 Intended Use

Track facilities can be used for training and/or competition by anyone from the elementary school child to the world-class athlete, or they may be used for recreation by joggers and walkers. The intended uses of the facility affect such factors as the number of lanes, the layout and number of field events, the need for accessories and amenities and the appropriate surface.

1.5.2 Facility Concept

The facility should be planned for its intended purpose(s), and its construction should feature quality materials and methods. That being said, budgets, purchasing regulations, surface choices, layout, landscaping and accessories and amenities will likely vary from one project to another. The goal should be to plan the most usable track possible within the limitations of the site, budget, etc. A quality track, regardless of budget, location or intended use, should be attractive, accessible, safe and enjoyable to use.

1.5.3 Initial and Long-Term Funding

The amount of funding readily available will affect the type of track facility to be built and the accessories and amenities to be included. But the long-term costs, such as accessories and amenities to be added later, as well as short- and long-range maintenance costs, must be considered in the planning process as well. For example, it may be less expensive to include electrical conduit for future installation of lighting during initial construction than to add it later.

1.5.4 Geographic Considerations

Local climate affects both construction and use of the track facility, including the depth of base construction, footings, orientation, availability of materials, need for certain amenities

and other factors. Additionally, construction in many regions is regulated with regard to snow loads, wind loads, seismic tolerances, water consumption, energy usage and fire hazards. These regulations can significantly impact the design and construction of a track project.

1.5.5 Security

The location of a track within the overall site will affect choices for fencing, landscaping, accessories and amenities. In most cases, security is a function of lighting and of limiting access. Additionally, the track should be visible so that security personnel can monitor it when making normal rounds.

1.5.6 Scope of the Project

The definition of a project involves not only determining the location of the track, but also defining the additional facilities to be included: lighting, competitor and spectator seating, locker rooms and restrooms, parking, refreshment stands, press box, meeting rooms and maintenance facilities. Work to be completed during initial construction should be considered, as should plans for future upgrading and expansion.

Once the project has been defined, critical decisions remain, including choosing a site and choosing a surface. For more information on site considerations, see Chapter 2; for information on surfacing options, see Chapter 3.

A good track facility begins with a well-defined concept, a considered site plan, acknowledgement of safety zones and a suitable site on which a well-built base and a quality surface can be constructed. But it doesn't end there. Continued planning for the facility then shifts to selecting track and field event equipment, determining amenities and developing a maintenance plan. Accessories and amenities are addressed in Chapter 5. Maintenance plans are addressed in Chapter 8.

Finally, landscaping and aesthetics should also be considered in the design phase. The overall character of the site and the surrounding areas should be assessed, with the track facility ideally designed and constructed in such a way as to harmonize with its environment. The careful selection of building materials and architectural elements, together with the arrangement of facilities on the site, can help a track facility blend well into its surroundings. Trees, shrubs and other plantings, access paths, ground elevations, etc. should not interfere with the layout of the track, the athletes' use of the facility or the spectators' ability to view all events. See Chapter 2 for more discussion of trees and vegetation.

1.5.7 Environmental Design

The environmental impact of the construction industry is enormous. Green building practices can substantially reduce or eliminate negative environmental impacts through high-performance, market-leading design, construction and operations practices. The U.S. Green Building Council (USGBC) created the LEED (Leadership in Energy and Environmental Design) program as a way to define and measure the popular phrase "green building."

Although the LEED system is primarily focused on the construction or renovation of buildings, many of the concepts apply to site work and landscape architecture as well. In some situations, the construction of a running track may be part of a larger overall project that does include adjacent buildings and structures. In these cases, the design and construction methods for the track will impact the LEED rating for the project.

The following are examples of how the design and construction of a running track could have an impact on the overall LEED rating of a project:

- Construction Activity Pollution Prevention—Institute and maintain proper erosion and sediment control measures
- Site Selection—Avoid development of land that was previously undeveloped, habitat for endangered species or within specific distances of wetlands/water bodies
- Stormwater Design—Implement proper stormwater management systems to ensure that the post-construction stormwater quantity and quality meet or exceed preconstruction conditions
- Heat Island Effect—Utilize hardscape with a high solar reflectance index (SRI) or provide shading of low SRI materials
- Water Use Reduction—Utilize drought-resistant landscaping and eliminate the use of potable water for landscape irrigation
- Recycled Materials—Use recycled materials in the construction of the facility
- Construction Waste Management—Recycle and/or salvage construction and demolition debris, diverting this material from landfills
- Construction Waste Management—Recycle and/or salvage construction and demolition debris, diverting this material from landfills
- Regional Materials—Use materials sourced within 500 miles of the project site.

1.6 Budgeting

Developing a budget may be the most difficult step in the construction process. To make informed decisions, an owner should have a clear sense of priorities. Working within a budget involves considering various alternatives and making choices. Knowledge of what factors are most important to the planned facility and a desire to seek creative solutions can result in a quality project built at a reasonable cost.

The budget should include sufficient funding for some or all of the following:

- Feasibility study
- Site acquisition
- Planning and design
- Permits
- Site investigation
- Site preparation
- Site drainage and grading
- Site utilities
- Track construction (base and pavement, surface construction, striping)
- Fencing
- Construction of related facilities (service buildings, seating, parking, etc.)

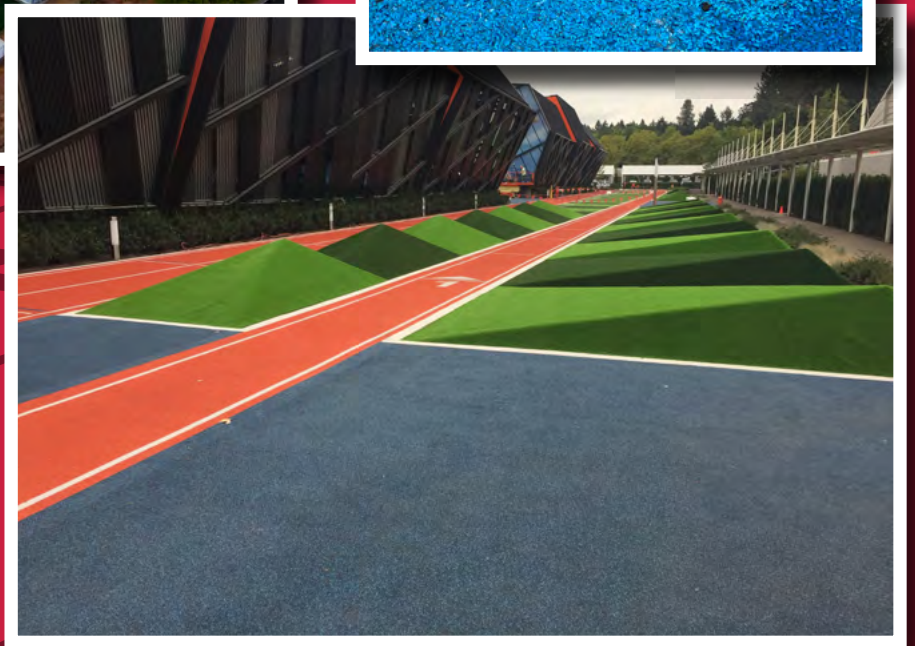
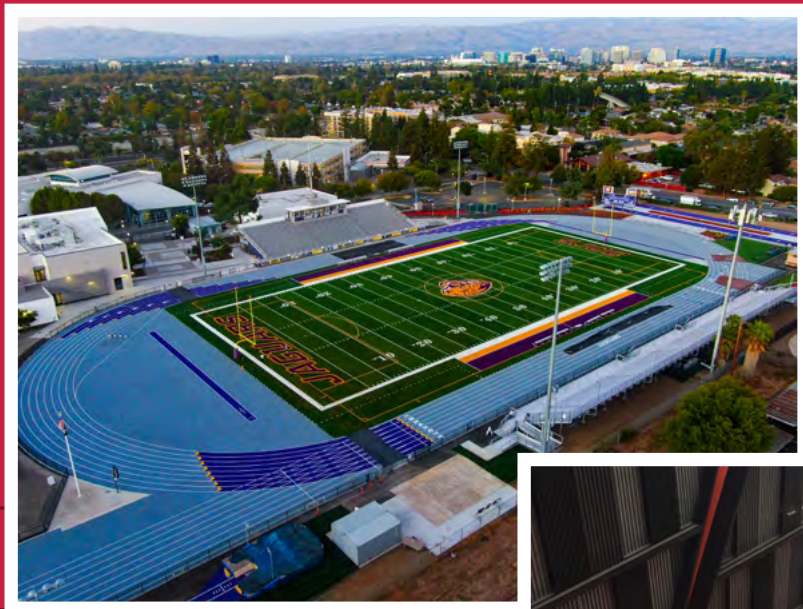
- Accessories and amenities
- Landscaping, walkways
- Operations

Other than the scope of the project and the choice of a surface, one of the most important factors in meeting a reasonable budget is the site chosen for the facility. Site factors, including permit requirements, water management considerations, excavation and fill needs and access to and availability of utilities, can be budget breakers. Site factors are addressed in more detail in Chapter 2. A Certified Track Builder, design professional, track consultant or experienced contractor can help the owner identify such factors and anticipate their impact on the facility budget.

Because planning and developing a track facility is a complex process, owners are cautioned not to purchase a site until they have at least preliminarily discussed demographics, financial feasibility and construction feasibility. Rather than delaying a project, careful advance consideration before purchasing a site can prevent expensive and time-consuming mistakes and problems further down the road.

Chapter 2

PLANNING AND DESIGN CONSIDERATIONS



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Chapter 2—Planning and Design Considerations

This chapter will further discuss planning and some of the considerations unique to a track facility.

- 2.1 General Requirements
- 2.2 Site Considerations
- 2.3 Base Geometry
- 2.4 Track Configurations
- 2.5 Field Event Dimensions and Areas
- 2.6 Site Requirements
- 2.7 Transition Area
- 2.8 Impact of Region, Geography and Location
- 2.9 Permits, Codes and Authorities
- 2.10 Trees and Other Vegetation
- 2.11 Soil Conditions
- 2.12 Site Grading
- 2.13 Drainage
- 2.14 Curbs and Perimeter Edging
- 2.15 Base Construction
- 2.16 Construction Documents

2.1 General Requirements

Tracks can be stand-alone or part of a larger, more complex athletic facility. They can require complex geotechnical engineering and civil scope or can be a straightforward resurface. With this in mind, it is important to consider when a designer is required and when one is not. With many new construction project permits, adherence to layers of codes, stormwater management, and many other potential agency reviews often require a licensed professional designer—preferably one with the understanding and experience in track and athletic facility design. Please be informed as to these requirements prior to embarking on your project.

For both the professional designer and the contractor, the design process is an opportunity to identify the goals and objectives of a project, including the development of the project program or components of a project. During this time the team can consider spatial relationships and constraints and understand the existing site and infrastructure to formulate a plan that can best achieve the stated goals. A key outcome of a design of any level is to create a plan that articulates the vision into a recipe for the contractor to build, providing the qualitative and quantitative aspects of the project.

Project delivery methods have changed dramatically in the last few years, offering greater opportunities for contractors, designers, and owners to work together to achieve a common goal. Experienced design professionals and contractors can and should learn from each other.

This chapter will introduce the reader to those factors that should be considered while the facility is still in the design phase. This is the time to evaluate every aspect of the project and to make adjustments as necessary. After all, changes to preliminary plans are relatively inexpensive, and it is far easier to tinker with the schematics for a facility than it is to make changes once the ground has been broken—or at any phase of construction that follows.

All tracks, whether public or private, for recreation or for competition, should conform to a certain set of standards. Standards exist for the safety of those using the track, as well as for consistency among facilities and, in most cases, for the purpose of allowing those facilities to be used for competition. The standards used, however, will depend to a large extent on the type of facility desired, its uses and many other factors. Consistent principles must nonetheless be followed in planning and designing a facility, regardless of those variables.

The first stage in the development of an athletic facility is to determine the desired uses for the new facility, which sets the course for its design. Types of projects include:

- International competition and training
- National and regional competition and training
- Intercollegiate competition and training
- High school and middle school competition and training
- Elementary school training
- Municipal facilities
- Private club facilities
- Community wellness facility
- Other uses or programs planned for the facility which may affect the design

Make sure you have a complete understanding of the age and interests and abilities of the users. The amenities and surface will need to be appropriate for the level of competition and training and will have to fit the parameters of anticipated use of the facility.

Think about whether the facility will be used year-round or just during the track season. Will it be open during the day or will there be evening use, perhaps necessitating lighting? Will the track be secured after hours? To what degree will the facility staff be able to provide maintenance? What equipment and expertise can they provide? The answers to all these questions will give you a jumping-off point for the track facility that is right for you.

In considering what type of track this is to be, it is of utmost importance to picture the facility as a whole. This includes having an understanding not only of what is entailed in initial construction, but also of any additional facilities and amenities that will be built later. Also take into consideration the requirements for the type of playing fields that may be part of the facility plan, either in the initial construction phase or in a later one. (For more information on these requirements, see “Base Geometry” in section 2.3.)

2.2 Site Considerations

Once a location has been selected for a track facility, it is time to begin the basic site examination that must precede design and construction. This begins with geotechnical investigation, which determines whether or not the site soils are suitable for the project or the recommended modifications to the soil are identified. Any site considered for the construction of a running track must have stable soil. Using site soils, managing the balance of cut and fill, and minimizing export and import materials are excellent tools for managing project costs. At this time, the design team can determine whether a Phase I Environmental Impact Study will be required.

The site must be of sufficient size to incorporate the requirements and desired features of the facility's layout. The dimensions and layout of the track need to conform to the governing body of the highest level of competition to be held on the track. As mentioned previously, it is of the utmost importance to know your clientele—the end users of the facility—since this will determine the governing body whose requirements must be followed.

During planning, a licensed surveyor should conduct a complete site survey. The survey will locate property lines and all existing features on the site, including buildings and structures, sidewalks, trees, shrubs, light poles, all adjacent streets and site access points and any underground utilities. Easements, right of ways, and setbacks should also be obtained and are often required for agency review. Utilities may need to be located by local utility companies—don't wait until construction is formally planned to make your calls. Contact the utility companies early in the site selection and design process. All electrical, gas, water, sewer and communication lines need to be located. Construction over these utilities should be avoided whenever possible.

While almost any site can be used for the construction of a running track, certain site conditions, such as the need for extensive cut and fill or the need to relocate utilities, can significantly affect costs. In addition, the potential for minimizing the environmental disturbances to a site should be considered (how many trees must be removed, what vegetation can remain, etc. More environmentally sensitive site considerations are discussed below). Such challenges should be identified early in a track project. Ground elevations, as well as spot elevations of all improvements identified on the site survey, will provide information that will aid in the development of a grading and drainage plan for proposed improvements. Elevations should be tied to U.S. Geological Survey (USGS) elevations. The site survey should be done in AutoCAD/Civil3D format so that it can easily be incorporated into construction drawings.

Finally, a site should be selected that will best allow for accessibility, not only for the ultimate end users but also for construction equipment and, when completed, for maintenance. Ideally, a site should encourage alternative transportation by offering access to mass transit, support facilities for bicycling and, perhaps, a preferred carpool parking area, provide bus drop-off area which isolates pedestrians from vehicles, along with accessible parking and access paths. Please review local and state codes since many codes now require improvements to elements well beyond the track, such as providing electrical vehicle charging stations (EVCS) or providing a prescribed shade coverage over all paved surfaces (athletic facilities surfaces are exempt).

With the selection of a suitable site accomplished, it is time to move to the next step: designing the actual layout of the facility and the locations of all components. Care must be taken to incorporate safety, convenience and efficiency. It is critical to the success of the finished product to consider, for example, proximity of the event areas to equipment storage sites (with careful attention being paid, for example, to the distances required to transport large landing pads). Consideration also should be given to efficient administration of events, to the location of timing devices (which may require electricity) and to staging areas for the athletes.

Having addressed the proposed facility from a competition standpoint, from the viewpoint of officials and organizers of planned events and from the perspective of those responsible for facility maintenance, planners should next focus on spectators. To be a success, any facility, particularly one built for competition, must be user-friendly on all levels.

Look at the plans and try to envision your facility. Supporting plans with the use of three-dimensional graphics is a great way to help clients envision a new facility. Is it welcoming and easily located? Are the restrooms and concession areas accessible? Where is the parking in relation to the track? Is there a clear and unobstructed view of all events that will be taking place, or will spectators need to leave their seats in order to see how well a particular team member is performing in any given event? Did we consider shade and other weather-related elements in our design? Don't let spectators' comfort and ease be an afterthought—plan for their needs as well.

2.3 Base Geometry

As mentioned above, an appropriate plan for the facility takes every aspect into consideration; this includes any playing fields that may be located within the track oval itself. The dimensions of these fields (Fig. 2-1 and Fig. 2-2) will vary according to the sport, as well as the level at which the sport is being played, and will dramatically affect the area needed.

If space will not allow for the event runways for the long jump, triple jump, pole vault, high jump and javelin to be located outside of the track oval, their locations will need to be addressed when the track configuration is determined. In addition, sufficient area along the sidelines, around field event spaces and in other areas will need to be provided for safety. Again, it is important to try to keep all events within the spectators' sight lines. Also, the layout and orientation of straights and jump events should take into consideration the prevailing winds and orientation of the sun. See Chapter 5 for more guidance on the safety considerations of a track's design.

2.4 Track Configurations

Not all 400m tracks are the same. Depending on the site available, the events to be included, the governing body of the highest level of competition to be held on the track, and the sports fields to be accommodated, the designer has the option of selecting any of several different track configurations.

The equal quadrant track consists of two straightaways of 100m each and two curves of 100m each (Fig. 2-7). In the past, many tracks were built with this design. Today, however, few tracks are built in this configuration for several reasons:

1. A wider radius, such as is found on the International Standard and Double Bend tracks, favors runners and enhances performance;
2. A wider radius lessens injury;
3. A wider radius allows greater flexibility in placing fields, especially soccer pitches, within the track oval.

The non-equal quadrant track features two equal curves and two equal straightaways that may be either shorter or longer than the curves. (See Fig. 2-4). World Athletics (WA) specifies a type of track that has straightaways of 84.39m and measure line radii of 36.80m (Fig. 2-5). This type of track is found in many existing stadiums and may be the track of choice, if space allows. It accommodates an American football and/or soccer field, and its wider radius favors runners, enhancing performance.

The double bend track (sometimes called the broken back track) has a compound radius curve (usually two small curves and a large curve formed by three different radii and, together, forming one complex curve) at each end, along with two straightaways. (See Fig. 2-6.)

Another way in which tracks differ is the location of the finish line. While most commonly located at PC#1 (Point of Curve #1, the point of transition between the home straightaway and the first curve), other options are sometimes considered. Wherever the finish line is located, additional room will need to be set aside for judges and spectators. Since the location of the finish line will influence many aspects of track layout, it must be determined early in planning. Starting line areas of a track are high-use areas. Consider adding a practice starting line further back in the chute to minimize wear and tear in the competitive area of the track. A simple line can create a value-added element to your project.

Tracks also differ with regard to the location and length of the chute and the location of the starting line for the 110-meter and 300-meter hurdles. Finish line location will directly affect the starting line for both of these events. The designer should consult the rules and recommendations of the relevant governing body in connection with chute design.

In understanding how tracks differ, it is important to know how they are measured. This, in turn, necessitates an understanding of the terms and formula commonly used in discussing track design and measurement. The two terms most commonly used to describe tracks are “measure line” and “radius point.”

The measure line is defined as a theoretical line from which the lane line locations are measured. The runner’s edge or inner lane line is located 20cm away from the measure line. When there is a raised curb for lane one, then the runner’s edge of the curb is located 30cm from the measure line.

The radius point is the theoretical point (or points) used to define the curves of the track. Because of their importance in layout and marking a track, the radius points may be permanently marked on the site by radius monuments. A standard track, whether equal quadrant or nonequal quadrant, has semicircular curves, each drawn from a single radius

point, meaning that there will be two radius points for this type of track. A double-bend track with its complex curves will have six radius points, and the steeplechase will necessitate additional radius points to define the water jump detour.

There is no one best design. An experienced track designer will assist an owner in choosing the optimum design for a specific site and usage. Differences in site, weather and soil conditions require variations in construction and repair methods and materials. Readers are advised to consult an ASBA Certified Track Builder, a design professional with experience in designing sports facilities or a qualified contractor before undertaking construction or repair of a running track facility.

While track design is complex and site specific, there are a number of commonly used formula and measurements:

Formula to calculate distance for a 400m track (excluding the double-bend facility)

$$400\text{m} = 2 \times \text{distance between the radius points} + 2 \times \text{arc length}$$

$$\text{Arc length} = \text{Pi} \times \text{radius to the measure line}$$

Formula to calculate distance for a 400m track (double-bend facility)

Important note: This equation presumes 60° sectors. For tracks with other than 60° sectors, the ratio will change. Contact a Certified Track Builder, design professional or an experienced contractor for a site-specific design.

SD—straightaway distance
 SR—short radius to the measure line
 LR—long radius to the measure line

Example for 60° sector:

$$400\text{m} = 2 \times \text{SD} + 4/3 \times \text{Pi} \times \text{SR} + 2/3 \times \text{Pi} \times \text{LR}$$

Figure 2-7 shows some examples of radius-to-straightaway length proportions. These are by no means the only acceptable configurations for tracks with radii typically between 90' and 130'. Note: Generally, larger radii are preferred.

The following calculation values are recommended for a proper balance of accuracy and workable measurements. Some common measurements used in track design and construction include:

20cm = 0.6562' or 7.87"	Distance from the measure line to the start of lane 1 (flush curb/painted line rule). See Figure 2-9
30cm = 0.9843' or 11.81"	Distance from the measure line to the start of lane 1 (raised curb rule). See Figure 2-9
0.3048m = 1'	Meter equivalent of 1 foot

1.0668m = 3.5' or 42" Width of standard lane (except WA)

1.2192m = 4.003' or 48" Width of standard lane (WA)
1.22m = 4.003'

1m = 3.28084' Feet equivalent of 1 meter

pi = 3.1416 Calculation used for curve distances

400m = 1312.336' Measure line distance in lane 1

The results of calculations should be expressed to the nearest 0.0001m (metric) or to the nearest 0.001' (imperial), with the exception of the 1m calculation. These are usually rounded up for construction and marking. Distances are rounded up to ensure that the distance is not less than 400m.

2.5 Field Event Dimensions and Areas

Figures 2-10a and 2-10b demonstrates how to calculate the D-zone, or the area of the infield made up of all or part of the semicircle inside the curve of the running track, given the radius and the centerline distance. The radius is measured from the radius point to the edge of the synthetic track surface (typically where it meets the drain or curb). The centerline distance is measured from the center of the cross curb (synthetic track surface side) to the apex of the curve.

2.5.1 High Jump

Common Dimensions	Sq. yds	Sq. m
50' x 50'–Square	278	232.4
50' x 80'–Rectangle*	444	371.2
50' x 100'–Rectangle*	556	464.8
50' Radius–Semicircle	436	364.5
70' Radius–Semicircle	855	714.8

*A rectangular high jump event area facilitates construction and allows the placement of a pit at either end. Such flexibility is useful both in practice and for event management; therefore, this configuration is recommended.

2.5.2 Pole Vault—Long Jump—Triple Jump

Common Dimensions	Sq. yds*	Sq. m*
130' x 5'	72	61
135' x 5'	75	63
150' x 5'	84	70
165' x 5'	92	77

**Rounded up to the nearest whole number*

See more discussion on recommended dimensions in Chapter 5.

2.6 Site Requirements

Once computations are concluded, it is time to get down to the actual business of laying out the track. As previously mentioned, it is important to have a view of the finished facility in your mind; this will enable you to focus on what the design should be, rather than on the diagram in front of you. Remember that certain “real world” considerations should be followed.

The site for a typical eight-lane track, not including site amenities, is approximately 70m (230') x 190m (623') to 104m (340') x 171m (560'). Figures 2.2, 2.2a, 2.2b, 2.2c and 2.3 show site dimensions for this and several other configurations. It is desirable to have the track laid out so that the straightaways and jump runways are parallel to the prevailing winds. This is important so that dashes and hurdle races can be reversed to take advantage of the wind.

Remember that you're not just planning for the track and field events. Amenities and event areas take up space, too. Additional area will need to be provided for grandstands, bleachers, site fencing, concession stands, restrooms, storage facilities and event areas. Parking areas and accessibility for emergency vehicles also should be taken into consideration.

Use the chart in Fig. 2-10c to calculate areas for a 400m running track.

2.7 Transition Area

The oval area of a track includes both straightaways and both curves. The transition area covers that portion of the straightaway extension shown in Fig. 2-12, and the transition distance is the length of the transition area. The chute area is a rectangle and can be easily calculated.

Please note that transition area and distance as shown in Fig. 2-12 are based on a radius of 105.5' to the inside edge of the synthetic surfacing and will vary slightly with other radii.

2.8 Impact of Region, Geography and Location

Consider the atmospheric forces that will affect an outdoor facility in your region over the course of the year. Northeastern and midwestern areas of the country are subject to multiple freeze-thaw cycles during the winter. The more frequent the freeze-thaw cycles and the greater the depth the ground freezes in a given region, the more carefully soil conditions, drainage and base construction must be considered. Remember that repeated freeze-thaw can be extremely detrimental to asphalt, which tends to crack under such conditions.

In southern and southwestern states and at high altitudes, sun will be the enemy of the surface. Prolonged direct sunlight and high temperatures will fade the color of the track and may have an adverse effect on its chemical stability. Make certain that your contractor has previous experience in construction within your specific geographic area in order to ensure the long-term integrity of the structure, despite the elements.

2.9 Permits, Codes and Authorities

A properly zoned site must be available for construction of any track facility. Owners are cautioned not to purchase the site, however, until zoning and permit requirements have been investigated. All too often, owners purchase a site only to learn that the facility planned cannot be built because of zoning, permit requirements or environmental limitations. Any possible variance or special exception that must be obtained from local, municipal or state governments, or from property owners' associations, should be considered. Rather than delaying the project, appropriate investigation before purchasing a site can prevent expensive and time-consuming problems later.

Once the site has been selected, the design and permitting, if required, must be completed and permits obtained prior to site preparation and excavation. Local codes should be considered including the type of zoning, lighting design, fence location and height, erosion control and drainage considerations, flood plains, access roads, parking requirements, utilities and setbacks. Conservation, wetlands and historic preservation regulations may affect some sites. An environmental impact statement also may be required. The municipal or county clerk's office usually can provide information regarding which permits and regulations affect the site. Permits often require certified surveys or stamped architectural or engineering drawings. The contract documents should specify whether the owner or the contractor is responsible for securing the proper permits. This process can be time-consuming, so be certain to allow adequate time for obtaining permits in the project schedule.

2.10 Trees and Other Vegetation

Trees, shrubs and other landscape features have a number of benefits for a track and field site, including enhancing the aesthetics of a site, providing shade, serving as windbreaks, screening parking and service areas, reducing noise pollution, providing stormwater management opportunities, improving air quality and mitigating the urban heat island effect. However, the type and location of such growth—including the future maximum size of the vegetation as well as root growth—must be considered as it relates to the functionality, maintenance, access and other conditions of the facility.

Whenever possible, avoid locating the track and events area immediately adjacent to trees. Both deciduous trees, which drop leaves, and evergreen trees, which drop pine cones and needles, can cause maintenance problems if their debris is not removed from the track surface. Such material must be removed not only for the athletes' safety but also for the staining it can cause. Shade from trees can provide a cool, damp medium for algae or fungus growth on the surface as well.

Tree roots that extend under areas to be paved and surfaced must be removed prior to construction. Where removing vegetation is not practical, root barriers are sometimes used. Keep in mind, however, that removing vegetation from a construction area can be costly. Further, it can alter the look and feel of the site. Wherever possible, aim to preserve existing natural resources and conserve energy and materials. Such environmentally responsible site design measures minimize negative environmental impacts and maximize opportunities to take advantage of the benefits of those resources, such as reducing the need for extra irrigation and chemical fertilizers.

2.11 Soil Conditions

It is advisable to retain a geotechnical engineer with experience in running track construction or to arrange for your track design professional to meet with a soils testing firm. Track construction differs considerably from the construction of roads or buildings; tracks carry minimal loads (mainly construction and maintenance vehicles), but planarity is extremely important. These requirements should be considered when evaluating the subgrade and when recommending the cross section of the base course and asphalt or concrete.

As was previously mentioned, it is essential to determine that the soil in the area is suitable for the project. This is done with a professional soil analysis, which takes into consideration the chemical properties of the soil and diagnoses any problems or conditions that may affect construction or that have the potential to negatively impact the finished product. The important soil properties to be evaluated are permeability, elasticity and plasticity, shearing strength, cohesion, compressibility, shrinkage and swell and frost susceptibility. If any of these properties tests in the unacceptable range, another site should be chosen or the construction plan should be adjusted to incorporate the recommendations of a geotechnical consultant as to the best method of mitigating the problem.

The geotechnical engineer may recommend that certain areas be stabilized by one of the following methods:

- Placement of a geotextile membrane over questionable soils
- Full-depth asphalt or concrete construction
- Uniform blending of subgrade soils
- Lowering the water table
- Use of a frost-resistant granular base
- Installation of more extensive drainage systems
- Soil stabilization and treatment
- Undercutting and replacing with suitable soils

The site with the best soil for a good, hard, nonheaving subgrade without peat, topsoil, clay, shear sand or other decomposed materials should be selected.

2.12 Site Grading

Once plans are approved and permits are in hand, the actual construction process can commence. The first steps to be taken are those that create a stable subgrade on which to build.

Initially, the site will need to be graded to conform to the rules of track construction. These rules allow for a maximum longitudinal slope of one tenth of one percent (0.1%) in the running direction over the length of any individual event.

Normally, topsoil and other unsuitable materials are removed well beyond the area to be surfaced. This should be done in a way that minimizes the disturbance of the remaining subgrade soils and facilitates placement of suitable fill materials outlined in the geotechnical investigation and recommendations or contract documents. Consult the geotechnical report for specific recommendations. Topsoil should be preserved and utilized elsewhere on the site for landscaping.

Fill is normally well-graded, granular, locally available materials. It must be free of organic or expansive material, such as clay. Fill should be placed in lifts and compacted in conformance with the site-specific recommendations of the geotechnical engineer. It may be necessary to reduce or increase the water content of the fill material in order to achieve optimum compaction. Backfill of all trenches should be granular material. Where trenching or pipe is used under an area that will be paved, that compaction must be consistent with the geotechnical engineering recommendations or contract documents to reduce the risk of settlement.

An important aspect of the geotechnical investigation is evaluating if the site soils may require or benefit from soil stabilization. Existing adjacent pavements can provide good indications of poor soil conditions. While a geotechnical investigation is always recommended, you may be able to detect signs of poor construction or poor soils from previous history with the site or by observing bulging, cracking dips in nearby asphalt surfaces that are two or more years old. Understanding these conditions is critical to the long-term success of the project.

Should weed growth be encountered, it can generally be traced back to one of three situations:

1. Excavation that cut preexisting weed growth but did not entirely eliminate it.
2. Excavation that uncovered ungerminated seeds but did not remove all of them. (These dormant seeds then germinated due to increased moisture in the soil and warmth caused by raising the subgrade and/or by heat retained by the pavement, especially asphalt pavement.)
3. Contaminated fill materials (in other words, fill materials that contained ungerminated seeds).

Soil sterilization should be undertaken by a professional service for safety and to achieve the best results. In fact, in many jurisdictions, a licensed installer is required. Herbicides are used for this process, either in liquid, pellet or powder form. Liquid herbicides are mixed with water or other substances to facilitate application. A water mixture is designed to penetrate the earth and to permit root systems to absorb the herbicide. An oil mixture works best for absorption through foliage. Herbicides in pellet form are useful where rainfall is sufficient to carry them to root systems. Depending on the type, herbicides can be applied either to the subgrade prior to installing the aggregate base or on top of the aggregate base prior to installing the asphalt or concrete. In some cases, it is possible to mix herbicide with the asphalt cutbacks that are used as primers. Your designer or contractor can determine which is the best option given your site considerations.

The timing of any herbicide application is very important. Liquid herbicide should not be sprayed during windy conditions. This can damage vegetation adjacent to the construction site. If rain or surface water washes onto the sterilized area before it is covered, some of the herbicide will leach or wash into the surrounding ground, causing harm to desirable vegetation. Leaching also may occur if herbicides are applied to water-saturated soils after heavy or persistent rainfall. Herbicides should be applied in accordance with the manufacturer's directions and in a manner that complies with all applicable environmental laws and regulations.

2.13 Drainage

Adequate drainage is one of the most important considerations in selecting a site and in designing a facility. The primary purpose of a drainage system is to move water away from the track and the event areas. To facilitate drainage, the preferred site would be a relatively level plane that is higher than the surrounding terrain to support natural drainage patterns, though new patterns typically will emerge.

Where drainage problems do occur, they are usually the result of both surface and subsurface drainage issues. Ideal surface drainage ensures that the only water that falls on the track surface is rain; all other water is directed away from the surface. Subsurface drainage prevents excessive moisture buildup in the soil under the pavement and keeps that soil as dry as possible. This is especially important in frost areas to minimize the frost susceptibility of the soil and the impact that the freeze-thaw cycle can have on the pavement.

Drainage is not an area where costs should be cut or minimized in any way that will affect the finished product. The problems that a substandard drainage system can cause will quickly become apparent and will haunt the track facility throughout its useful life or until they are corrected—often a far more expensive prospect than simply paying for a well-designed system in the first place. Consulting a design professional to determine the size, slope, and type of subsurface drainage system is recommended.

The most effective way to remove water from a track surface depends on its permeability. An impermeable surface primarily sheds water by sheet drainage in the direction of slope while any remaining water evaporates into the atmosphere. In addition to shedding water in the same two ways, a permeable surface allows some water to drain into the voids in the

surface where it migrates to the low side of the track and dissipates. Residual moisture evaporates through the voids in the surface.

One common cause of inadequate drainage is when the asphalt under a porous surface is lower than a curb or other barrier on the side where the water drains, meaning the water cannot escape. Another problem occurs when meeting governing body rules or an owner's request results in minimal slope—as they often do on high jump approaches, for example—and the slope is not adequate to move the water. This loss of planarity and slope may also be caused by unsuitable material in the subsoil, inadequate drainage design, improper slope or grade or inadequate compaction of the subgrade. Water on tracks due to paving errors is also common.

Additionally, when building an adjacent synthetic turf field on the inside of the track, ensure the collector trench for the field is far enough away and/or lower than the bottom of track subbase materials to prevent water from making its way under the track.

To identify and correct paving errors prior to surfacing, the owner or design professional should require that the track be flooded, and areas that hold water should be marked and planed or filled prior to surfacing. Unless the surfacing contractor is also responsible for the paving, it is inappropriate to hold that contractor responsible for drainage issues if the pavement has not been checked and corrected prior to surfacing. Tracks should not be surfaced until drainage issues are corrected.

If the project team determines that a site needs a drainage system to properly control water, many options exist. In general, all of them outlet to a storm sewer system, daylight or drywell. Some of these options may be limited by local building codes or environmental regulations (e.g., retention, storage and in-ground recharge requirements). Consulting with an experienced design professional is also a good option.

2.13.1 Precast Channel Drains

Precast channel drains have become the most common system for track construction (Fig. 2-19). They typically are used around the inside perimeter of the track to catch surface drainage from the track oval. They also can be used outside the track to prevent surface drainage from a hill or seating structure from flowing onto the track and to control drainage around field event runways and pads.

These drains feature a continuous opening throughout their length with a series of inline catch basins that serve as cleanouts and outlets to a subsurface drainage system. The size of the channel opening and the number of catch basins are determined by the volume of water that they are required to carry. The most common type has a neutral slope (level invert) although they are also available with positive slope.

The opening comes in two designs: a slot or a grate. The slotted design is often used when the channel drain separates a D-zone from the track oval since it results in the least interference to an athlete crossing the drain. The grated drain is removable and comes in various designs and materials. The slotted drain is cleaned by flushing with a high-pressure hose or power washer. The grated drain sections can be removed and the drain flushed with

water and/or cleaned with brooms or special scrapers. The precast channels are embedded in concrete, which allows them to also serve as a curb and provide a termination point for the track and infield. This is further discussed below under Curbs and Perimeter Edging.

2.13.2 Swales

A swale is a natural drainage channel, usually covered with grass, that carries surface water away from the outside of the track. The flow line of the swale is sloped to direct water to the desired location. It can be an effective and economical way to control surface drainage that might otherwise flow over the track surface. Swales should be avoided on the inside of the track as the slopes required along with slow drying of the flow line will interfere with athletic use of the infield.

2.13.3 Swales with Catch Basins

This design uses a swale around the inside perimeter of the track along with multiple catch basins that direct water to a subsurface drainage system (Fig. 2-18). The flow line of the swale is graded to allow water from the track and infield to reach the catch basins. Although more effective than a swale alone, the slopes and flow line of this set-up can interfere with athletic use. In addition, depending on the proximity of the field playing area to the track, the catch basins can be a safety concern.

2.13.4 French Drain

A French drain is primarily used to control subsurface moisture or as a vertical drain capturing surface and subsurface water in natural grass and landscape systems. A typical design utilizes a filter-fabric-lined trench containing a perforated pipe surrounded by pea gravel. A well-draining sandy loam in the root zone, such as USGA, in planting areas or the aggregate can continue to the surface in non-planted areas. Drain tile surrounded by drain rock and wrapped with filter fabric then outlets to a subsurface drainage system (Fig. 2-18). This type of drain can be used to intercept water moving towards a track or to reduce moisture content in the subgrade.

2.13.5 Synthetic Turf

A synthetic turf system can sometimes be used to catch surface drainage from the track, but this is not recommended by the ASBA manual, *Sports Fields: A Construction & Maintenance Manual*. When designed and built properly, these turf systems are designed to drain large volumes of water rapidly. A typical inside track perimeter around a synthetic turf field features a conventional curb bordering the track over which surface water flows onto the turf. Many states currently require impervious surfaces to be treated before being infiltrated, stored, and evacuated by downstream piping and conveyance systems.

While the descriptions of many of these systems suggest directing collected water to the storm drainage system, whenever possible it is advisable to instead harvest rainwater and stormwater for irrigation or for other uses on site or to allow it to percolate into the soil to recharge the aquifer. Stormwater may be filtered through plantings dispersed over porous surfaces such as lawns or collected and stored in ponds. Alternately, stormwater may be

released into the porous stone surrounding the perforated collection pipes under a synthetic turf system as shown in Figures 2-15, 2-16, and 2-17.

Nonporous surfaces are installed in two ways: flush to the curb or drain with the asphalt or concrete recessed to the thickness of the synthetic surface, or with the asphalt paving or concrete flush to the curb or drain and the surface above (Fig. 2-14 (edge treatments)). The advantages of the recessed asphalt/flush surface method are a cleaner-looking edge detail, easier installation and the resulting absence of a lip or edge that may present maintenance issues later. The asphalt paving for this detail must be absolutely precise to allow for the thickness of the synthetic material to be installed, and good compaction may be difficult. If moisture penetrates the area between the asphalt and concrete, potential deterioration could cause separation or edge-curling of the surface.

The asphalt flush/surface-on-top method provides a better asphalt paving project, allows for the surface to extend over the joint onto the adjacent concrete and presents a more consistent surface depth. It does, however, leave a lip of material on the surface. Typically, the surface edge is bull-nosed or tapered.

The decision on how to install nonporous pavement can be made based on the owner's wishes, the experience of the asphalt paver in the area and the manufacturer's and installer's recommendations. With either method of installation, proper maintenance is key to a long-lasting surface.

2.14 Curbs and Perimeter Edging

Curbs and perimeter edging are extremely important in the design and construction of a running track and associated event areas. These buffer zones act as controls for joining track surface and base materials with the infield areas. This zone additionally provides a vegetation barrier, allows for the placement of drainage, helps control elevation and establishes the correct length and width of the track. Concrete and concrete polymer curbs also serve as termination points for natural grass and synthetic turf playing fields. Raised curbs used to define the inside of lane 1 are discussed in later sections. Please refer to ASBA's position paper, "Raised Curb or Painted Inside Line for Outdoor Running Tracks," for more details.

2.14.1 Pavement Extension

This type of construction places the various layers in a stepladder fashion (Fig. 2-13). It allows for water to progressively leave the surface and provides a buffer zone for maintenance. Pavement extension generally is an economical method for edging the track.

In this method, the aggregate base is placed so that its inside perimeter is parallel to and 27" inside of the track measure line, as well as 17" from the outer side of the last lane line. The asphaltic concrete is constructed so that the perimeter is parallel to and 22" inside of the track measure line, 12" from the outer side of the last lane line. The synthetic surfacing is constructed so that the perimeter is parallel to and 16" inside of the track measure line, 6" from the outer side of the last lane line. This provides a 6" buffer zone between the asphalt and the surfacing as well as a 6" zone of rubber before the first stripe and after the last. On

some multilayered rubber and binder surfaces, the layers are feathered to the edge of the asphalt after achieving the specified surface width.

2.14.2 Turf Connection

Where a running track surrounds a synthetic turf field and uses the field drainage to convey the water, care must be taken to ensure water can sheet flow from the track onto the field. Furthermore, the relationship of the field infill with the edge of the track cannot be more than ½” or it will be considered non-accessible by reviewing agencies (Figs. 2-15, 2-16, and 2-17).

2.14.3 Flush Curb

A flush, solid curb can be installed on the inside and outside perimeters of the track and event areas, or on the inside only. The curb functions as a buffer between the track and surrounding materials (Figs. 2-16 and 2-17). For permeable systems, the top of the curb is installed flush with the top of the asphalt or concrete so that water cannot accumulate inside the curb.

2.14.4 Removable Raised Curb

Removable raised curbs are available in various materials (Fig. 2-9, Raised Curb Layout). The most popular materials are aluminum, polyurethane and aluminum with a firm rubber top. These curbs rest on pads that allow for the movement of water from the track surface to the drain channel or infield. They are used for setting the distance for track competition. These systems are relatively easy to install and can be stored during the off season. However, installing these curbs on the lane line on an existing 20cm measuring line track will result in an incorrect track length when the curbs are used. It is also worth noting that race distances in lane one of a track designed for use of a raised curb will be shorter than the intended distances. Lane one on a 400-meter track designed for a raised curb will be approximately 399.37 meters without the curb in place.

2.15 Base Construction

Another integral component that determines the long-term success of a track construction project is a well-built base. Once again, this is no place to cut corners or try to save a few dollars. Remember that the base is, quite literally, the foundation of your facility and can affect every aspect of it, right down to the safety of your athletes.

2.15.1 Aggregate Base

An aggregate base course of crushed aggregate, processed/ recycled asphalt or processed/ recycled concrete is installed over the properly prepared subgrade to a thickness that depends on the geotechnical investigation and recommendations for the site but is not less than 4” after compaction (see below for details on this process). The material used must meet applicable ASTM and state transportation department specifications. Your contractor will be responsible for meeting these requirements.

2.15.2 Spreading and Compacting

The material should be spread in such a way as to produce a uniform density and thickness and to provide the best possible base. The crushed aggregate is placed in layers not more than 6" in depth and then compacted. The material, once spread, should be compacted to 95% minimum Proctor Test as outlined in the geotechnical investigation and recommendations and, once compacted, should not vary more than 0.5" in 10' when measured in any direction with a 10' straight edge.

2.15.3 Specialized Fill Materials

A soils engineer may recommend additional specialized fill materials to be placed between the subgrade and the aggregate base course to further stabilize undesirable soil conditions. These materials may include lime, sand, geotextile fabric or a combination of several.

No matter whether your project is an arena for internationally sanctioned track and field events or a community track for recreational joggers and walkers, it is paramount to have an acceptable site, to use the proper drainage systems and to have a good sound base. All of this can be accomplished by becoming an educated consumer and by hiring experienced and knowledgeable professionals for the job. Remember that investing in quality materials and workmanship will pay dividends in end users' enjoyment and safety for years to come.

2.16 Construction Documents

Once all the planning and design decisions have been made and a surface and planned accessories and amenities have been chosen, a clear scope of work and a set of construction drawings with details and specifications should be issued for construction. The delivery method chosen will affect the construction documents for the project, but they will include some or all of the following:

- Detailed scope description
- Drawings
- Specifications
- Construction contract
- General conditions
- Preselected proprietary products
- Allowable substitutions
- Insurance and bonding requirements
- Responsibility for permits, taxes and utility locates
- Work done by others

FIELD SIZES

FIGURE 2-1

EVENT	LENGTH MIN TO MAX	LENGTH OPTIMUM	WIDTH MIN TO MAX	WIDTH OPTIMUM
AMERICAN FOOTBALL	360'	360'	160'	160'
HIGH SCHOOL SOCCER	300' TO 360'		165' TO 240' *	
NCAA SOCCER	**		**	
• NEW FACILITIES	345' TO 360'	360'	210' TO 240' *	225'
• EXISTING FACILITIES	330' to 360';	360'	195' TO 240' *	225'
FIFA SOCCER (METERS)	100M to 110M		64M TO 75M *	
HIGH SCHOOL BOY'S LACROSSE	330'	330'	160' TO 180'	180'
HIGH SCHOOL GIRL'S LACROSSE	330' to 420'	360'	180' TO 210'	195'
NCAA MEN'S LACROSSE	330'	330'	180'	180'
NCAA WOMEN'S LACROSSE	330' to 420'	330'	180' TO 210'	195'
UNIFIED LACROSSE	360'	360'	159.99' TO 180'	180'
IRB RUGBY (METERS)	120M to 144M	144M	70M	70M
FIELD HOCKEY	300'	300'	180'	180'

NOTES:

THIS INFORMATION IS ACCURATE AT THE TIME OF THE PUBLICATION AND IS PROVIDED AS A CONVENIENCE TO THE READER. THE RULES ARE CONSTANTLY UNDER REVIEW BY THE GOVERNING BODIES AND ARE SUBJECT TO CHANGE. PLEASE CONSULT THE LATEST RULE BOOK FOR THE APPROPRIATE GOVERNING BODY FOR THE RULE IN PLACE AT THE TIME OF YOUR PROJECT, AS WELL AS FOR THE COMPLETE WORDING OF THE RULE FOR THE RECOMMENDED BUFFERS (SPACE BEYOND THE DIMENSION LINES SHOWN ABOVE). ASBA ACCEPTS NO RESPONSIBILITY FOR THE USE OF THE INFORMATION.

*ASBA SUGGESTS THAT THE MAXIMUM WIDTH FOR FIELDS TO BE LOCATED INSIDE A RUNNING TRACK NOT EXCEED 240 FEET (INCLUDING FIELD SAFETY SETBACKS). IN MOST CASES FIELD WIDTH WILL BE LESS BASED ON SITE CONDITIONS. PLEASE REFER TO THE ASBA FIELD CONSTRUCTION AND MAINTENANCE MANUAL FOR MORE INFORMATION.

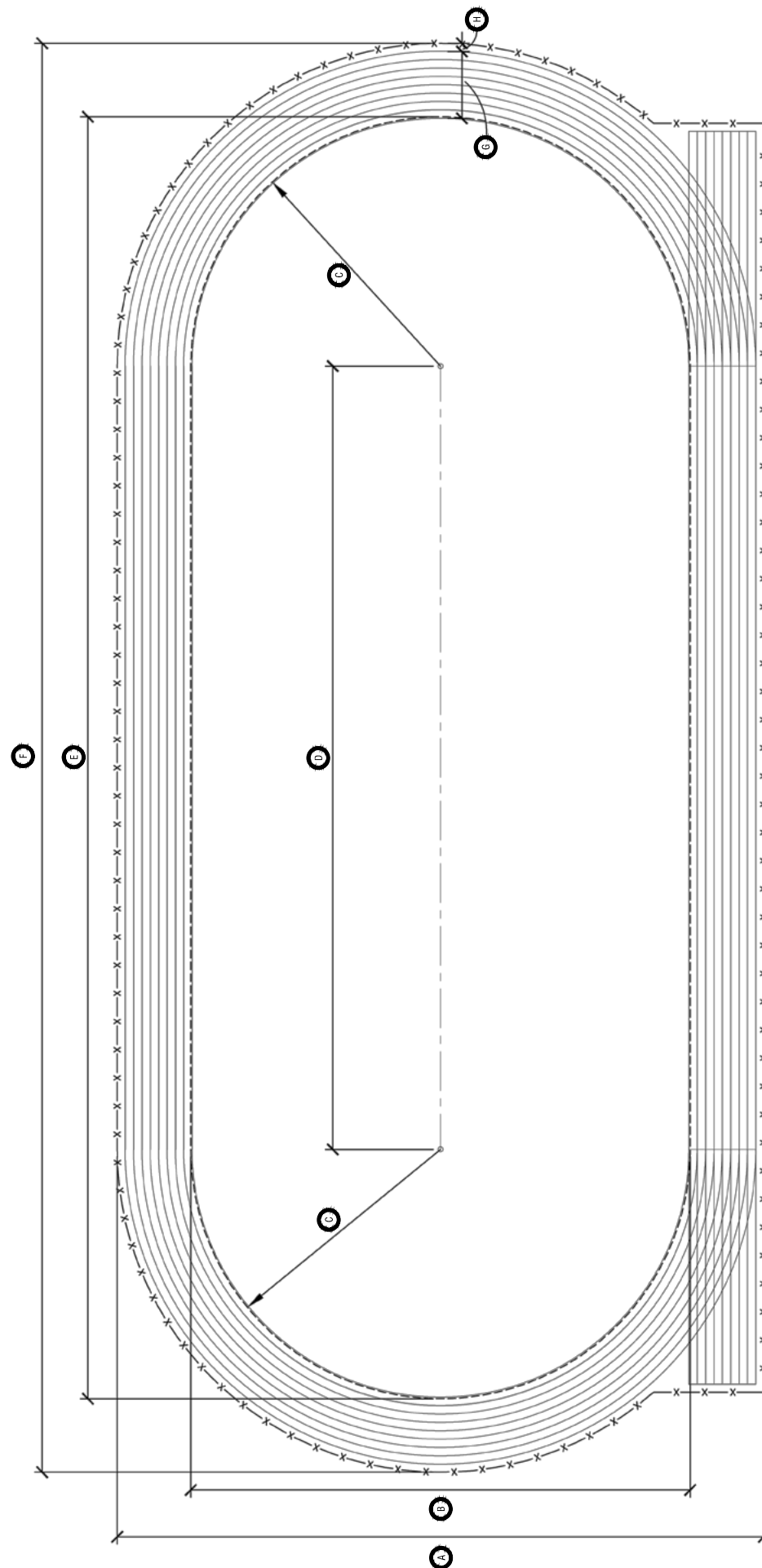
** FIELDS OF LESS THAN MINIMAL DIMENSIONS MAY BE USED BY PRIOR WRITTEN MUTUAL CONSENT OF THE COMPETING INSTITUTIONS.

SITE REQUIREMENT DIMENSIONS

FIGURE 2-2

Assumptions: Width of track: Clear zone: EML: Approximations:		29.00 Ft= number of lanes (8)* lane width (3.5 FT) + 1.0 ft (track bounded by concrete curb) •• 3.28 Ft = 1 meter 1.33 Ft= edge of asphalt and concrete curb to ML (measure line) Measurements are approximate, adjustments are necessary when you determine your exact needs.									
Input		Calculated Dimensions based on Input									
Width of Site In feet	Width of Site in meters	Measure Line in feet	Measure Line in meters	Infield Measure in feet (width)	Infield Measure in meters (width)	Radius pt to Radius pt Distance in feet	Radius pt to Radius pt Distance in meters	Length of Site in feet	Length of Site in meters		
230	70.10	84.05	25.62	165.44	50.43	392.12	119.52	622.12	189.62		
235	71.63	86.55	26.38	170.44	51.95	384.26	117.12	619.26	188.75		
240	73.15	89.05	27.14	175.44	53.47	376.41	114.73	616.41	187.88		
245	74.68	91.55	27.90	180.44	55.00	368.55	112.34	613.55	187.01		
250	76.20	94.05	28.67	185.44	56.52	360.70	109.94	610.70	186.14		
255	77.72	96.55	29.43	190.44	58.05	352.85	107.55	607.85	185.27		
260	79.25	99.05	30.19	195.44	59.57	344.99	105.15	604.99	184.40		
265	80.77	101.55	30.95	200.44	61.09	337.14	102.76	602.14	183.53		
270	82.30	104.05	31.71	205.44	62.62	329.28	100.37	599.28	182.66		
275	83.82	106.55	32.48	210.44	64.14	321.43	97.97	596.43	181.79		
280	85.34	109.05	33.24	215.44	65.67	313.58	95.58	593.58	180.92		
285	86.87	111.55	34.00	220.44	67.19	305.72	93.18	590.72	180.05		
290	88.39	114.05	34.76	225.44	68.71	297.87	90.79	587.87	179.18		
295	89.92	116.55	35.52	230.44	70.24	290.01	88.40	585.01	178.31		
300	91.44	119.05	36.29	235.44	71.76	282.16	86.00	582.16	177.44		
305	92.96	121.55	37.05	240.44	73.29	274.31	83.61	579.31	176.57		
310	94.49	124.05	37.81	245.44	74.81	266.45	81.21	576.45	175.70		
315	96.01	126.55	38.57	250.44	76.33	258.60	78.82	573.60	174.83		
320	97.54	129.05	39.33	255.44	77.86	250.74	76.43	570.74	173.96		
325	99.06	131.55	40.10	260.44	79.38	242.89	74.03	567.89	173.09		
330	100.58	134.05	40.86	265.44	80.91	235.04	71.64	565.04	172.22		
335	102.11	136.55	41.62	270.44	82.43	227.18	69.25	562.18	171.35		
340	103.63	139.05	42.38	275.44	83.95	219.33	66.85	559.33	170.48		
** If you want to construct the track using the asphalt extension method, take the additional asphalt width greater than 29 feet and add that number to the length of site and subtract that number from the infield width. This assumes that the asphalt is centered on the track.											

SITE REQUIREMENT DIMENSIONS
FIGURE 2-2A



SITE REQUIREMENT DIMENSIONS (FEET)
FIGURE 2-2b

Width of track: (G) - 28.00 Ft = number of lanes (8)* lane width (3.5 Ft) (track bounded by concrete curb) (H) - 1 M = Clear Zone (Safety) (1.0508 from internal 8th lane stripe, [1M + 2']) Assumptions: 20cm = Measure Line Offset (7.874') 3.5 Ft = Lane Width (42" or 1.0668 M) 3.28 Ft = 1 meter 8 = Total Lanes Measurements are approximate, adjustments are necessary when you determine your exact needs. Approximations: Table shown below if is for only 20cm (stripe to measureline) 1st lane and not for raised curb (30cm)	Calculated Dimensions based on Input												
	FEET						METERS						
	(A) Total Width of Track Site	(B) Width of Infield at Measure Line	(C) Measure Line Radius	(D) Radius pt to Radius pt	(E) Length of Dzones & Infield at Measure Line	(F) Total Length of Track Site	Length of Measure Line	(A) Total Width of Track Site	(B) Width of Infield at Measure Line	(C) Measure Line Radius	(D) Radius pt to Radius pt	(E) Length of Dzones & Infield at Measure Line	(F) Total Length of Track Site
	230	168.417	84.209	391.619	560.036	621.619	1312.336	70.1040	51.3336	25.6668	119.3654	170.6990	189.4694
	235	173.417	86.709	393.765	557.182	618.765	1312.336	71.628	52.8576	26.4288	116.9715	169.8291	188.5995
	240	178.417	89.209	375.911	554.328	615.911	1312.336	73.1520	54.3816	27.1908	114.5776	168.9592	187.7296
	245	183.417	91.709	368.057	551.474	613.057	1312.336	74.6760	55.9056	27.9528	112.1837	168.0893	186.8597
	250	188.417	94.209	360.203	548.620	610.203	1312.336	76.2000	57.4296	28.7148	109.7898	167.2194	185.9898
	255	193.417	96.709	352.349	545.766	607.349	1312.336	77.7240	58.9536	29.4768	107.3959	166.3495	185.1199
	260	198.417	99.209	344.495	542.912	604.495	1312.336	79.2480	60.4776	30.2388	105.0020	165.4796	184.2500
265	203.417	101.709	336.641	540.058	601.641	1312.336	80.7720	62.0016	31.0008	102.6081	164.6097	183.3801	
(1) 270.448	208.865	104.433	328.083	536.948	598.531	1312.336	82.4326	63.6622	31.8311	99.9997	163.6619	182.4323	
275	213.417	106.709	320.933	534.350	595.933	1312.336	83.8200	65.0496	32.5248	97.8203	162.8699	181.6403	
280	218.417	109.209	313.079	531.496	593.079	1312.336	85.3440	66.5736	33.2868	95.4264	162.0000	180.7704	
285	223.417	111.709	305.225	528.642	590.225	1312.336	86.8680	68.0976	34.0488	93.0325	161.1301	179.9005	
290	228.417	114.209	297.371	525.788	587.371	1312.336	88.3920	69.6216	34.8108	90.6386	160.2602	179.0306	
295	233.417	116.709	289.517	522.934	584.517	1312.336	89.9160	71.1456	35.5728	88.2448	159.3904	178.1608	
300	238.417	119.209	281.663	520.080	581.663	1312.336	91.4400	72.6696	36.3348	85.8509	158.5205	177.2909	
305	243.417	121.709	273.809	517.226	578.809	1312.336	92.9640	74.1936	37.0968	83.4570	157.6506	176.4210	
310	248.417	124.209	265.955	514.372	575.955	1312.336	94.4880	75.7176	37.8588	81.0631	156.7807	175.5511	
315	253.417	126.709	258.101	511.518	573.101	1312.336	96.0120	77.2416	38.6208	78.6692	155.9108	174.6812	
320	258.417	129.209	250.247	508.664	570.247	1312.336	97.5360	78.7656	39.3828	76.2753	155.0409	173.8113	
325	263.417	131.709	242.393	505.810	567.393	1312.336	99.0600	80.2896	40.1448	73.8814	154.1710	172.9414	
(1) Equal Quadrant Track													

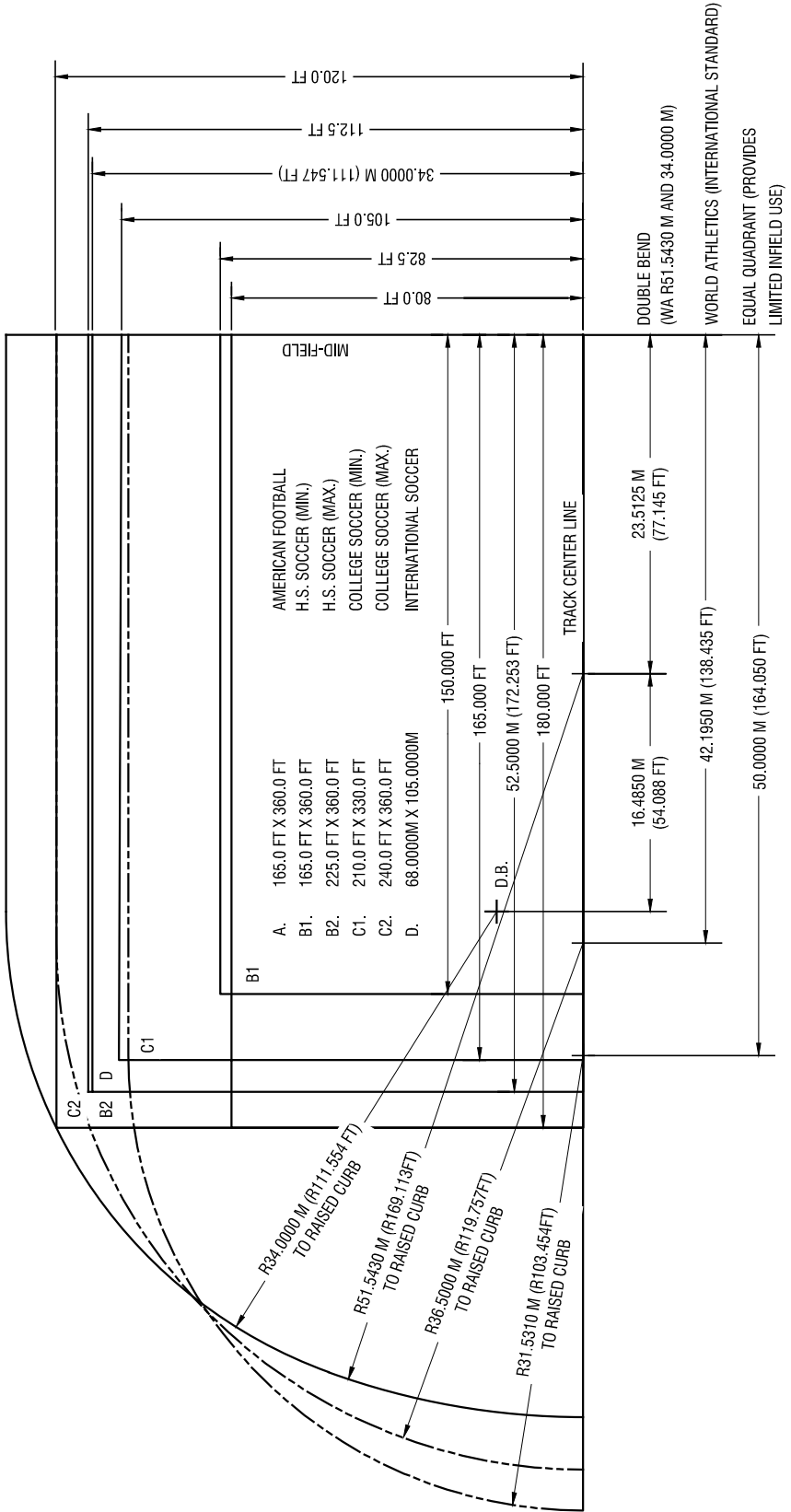
SITE REQUIREMENT DIMENSIONS (METERS)
FIGURE 2-2c

Width of track: Assumptions: Approximations: Table shown below if is for only 20cm (Stripe to measureline) 1st lane and not for raised curb (30cm)	Calculated Dimensions based on Input											
	METERS						FEET					
(A) Total Width of Track Site	(B) Width of Infield at Measure Line	(C) Measure Line Radius	(D) Radius pt to Radius pt	(E) Length of Dzones & Infield at Measure Line	(F) Total Length of Track Site	Length of Measure Line	(A) Total Width of Track Site	(B) Width of Infield at Measure Line	(C) Measure Line Radius	(D) Radius pt to Radius pt	(E) Length of Dzones & Infield at Measure Line	(F) Total Length of Track Site
75.2500	56.4796	28.2398	111.2821	167.7617	186.5321	400.0000	246.883	185.301	92.650	365.099	550.399	611.982
76.3000	57.5296	28.7648	109.6327	167.1623	185.9327	400.0000	250.328	186.745	94.373	359.687	548.433	610.015
77.3000	58.5296	29.2648	108.0619	166.5915	185.3619	400.0000	253.609	192.026	96.013	354.560	546.560	608.143
78.3000	59.5296	29.7648	106.4911	166.0207	184.7911	400.0000	256.890	195.307	97.654	349.380	544.687	606.270
79.3000	60.5296	30.2648	104.9203	165.4499	184.2203	400.0000	260.171	198.588	99.294	344.227	542.815	604.397
80.3000	61.5296	30.7648	103.3495	164.8791	183.6495	400.0000	263.451	201.869	100.934	339.073	540.942	602.525
81.3000	62.5296	31.2648	101.7787	164.3083	183.0787	400.0000	266.732	205.150	102.575	333.920	539.069	600.652
82.3000	63.5296	31.7648	100.2079	163.7375	182.5079	400.0000	270.013	208.430	104.215	328.766	537.197	598.779
(1) 82.4324	63.6620	31.8310	100.0000	163.6620	182.4324	400.0000	270.448	208.865	104.432	328.084	536.949	598.531
83.8200	65.0496	32.5248	97.8203	162.8699	181.6403	400.0000	275.000	213.417	106.709	320.933	534.350	595.933
85.3400	66.5696	33.2848	95.4327	162.0023	180.7727	400.0000	279.987	218.404	109.202	313.099	531.504	593.086
86.8700	68.0996	34.0498	93.0294	161.1290	179.8994	400.0000	285.007	223.424	111.712	305.215	528.638	590.221
88.3900	69.6196	34.8098	90.6418	160.2614	179.0318	400.0000	289.993	228.411	114.205	297.381	525.792	587.375
88.3400	69.5696	34.7848	90.7203	160.2899	179.0603	400.0000	289.829	228.247	114.123	297.639	525.886	587.468
89.3400	70.5696	35.2848	89.1495	159.7191	178.4895	400.0000	293.110	231.528	115.764	292.485	524.013	585.596
90.3400	71.5696	35.7848	87.5787	159.1483	177.9187	400.0000	296.391	234.808	117.404	287.332	522.140	583.723
91.3400	72.5696	36.2848	86.0079	158.5775	177.3479	400.0000	299.672	238.089	119.045	282.178	520.260	581.850
92.3400	73.5696	36.7848	84.4371	158.0067	176.7771	400.0000	302.953	241.370	120.685	277.025	518.395	579.978
93.3400	74.5696	37.2848	82.8663	157.4359	176.2063	400.0000	306.234	244.651	122.325	271.871	516.522	578.105
94.3400	75.5696	37.7848	81.2955	156.8651	175.6355	400.0000	309.514	247.932	123.966	266.718	514.649	576.232
(1) Equal Quadrant Track												

TRACK/PLAYING FIELD CONFIGURATION

FIGURE 2-3

NOT TO SCALE



400 METER NON-EQUAL QUADRANT TRACK

NOTES:

1. NOT FOR CONSTRUCTION
2. DUE TO ROUNDING, CALCULATION OF THESE NUMBERS MAY NOT YIELD PRECISE CONVERSIONS OR DISTANCES
3. "D" AREA SIZE VARIES DEPENDING ON REQUIREMENTS FOR FIELD SIZE AND JUMPING/THROWING ACTIVITIES IN THOSE AREAS
4. DIMENSION TO MEASURE LINE SHOWN IS AN EXAMPLE. REFER TO FIGURE 2-8 FOR LIST OF MEASUREMENTS
5. REFERENCE FIGURE 2-9 FOR LANE LINE/MEASURE LINE DIMENSIONS

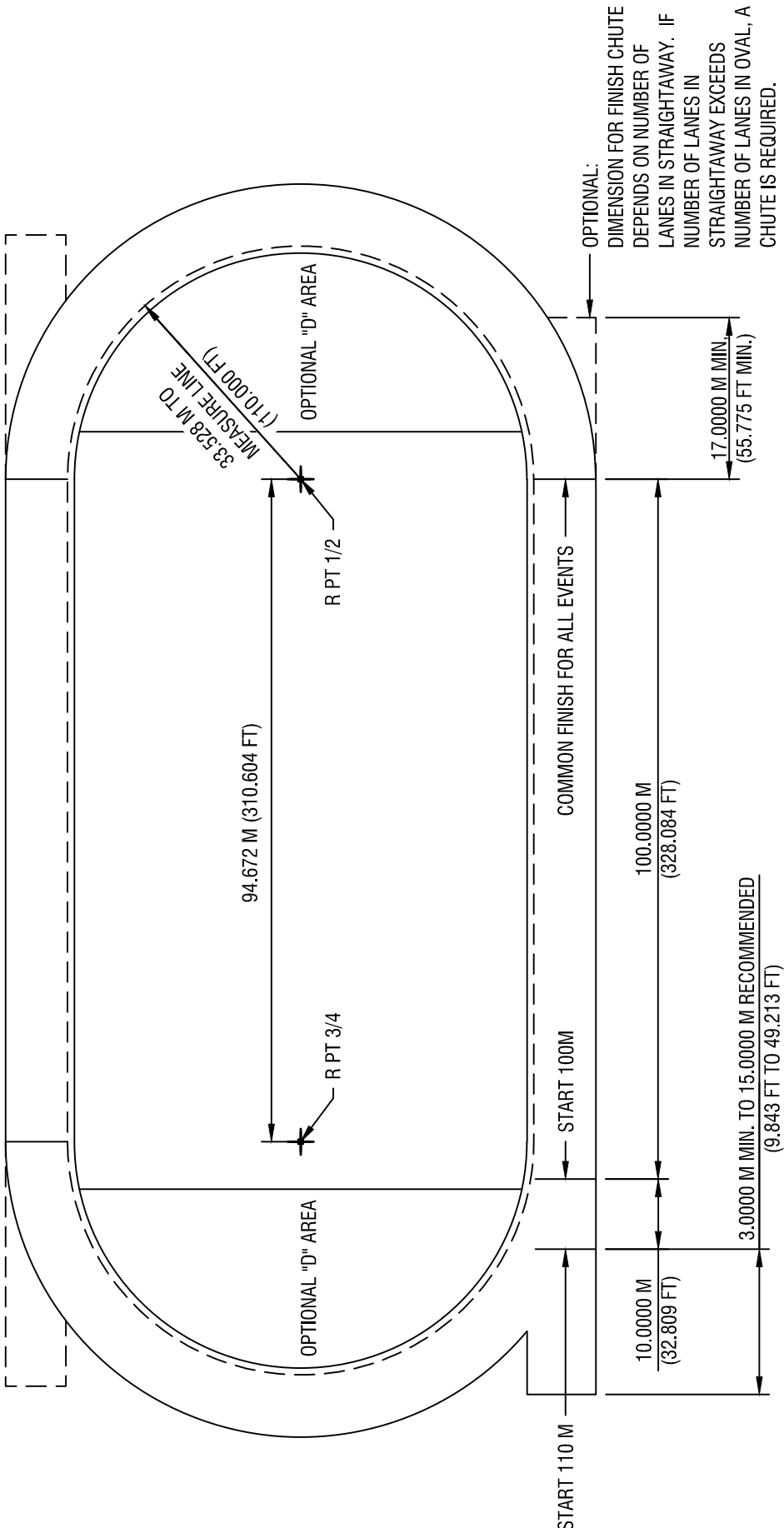
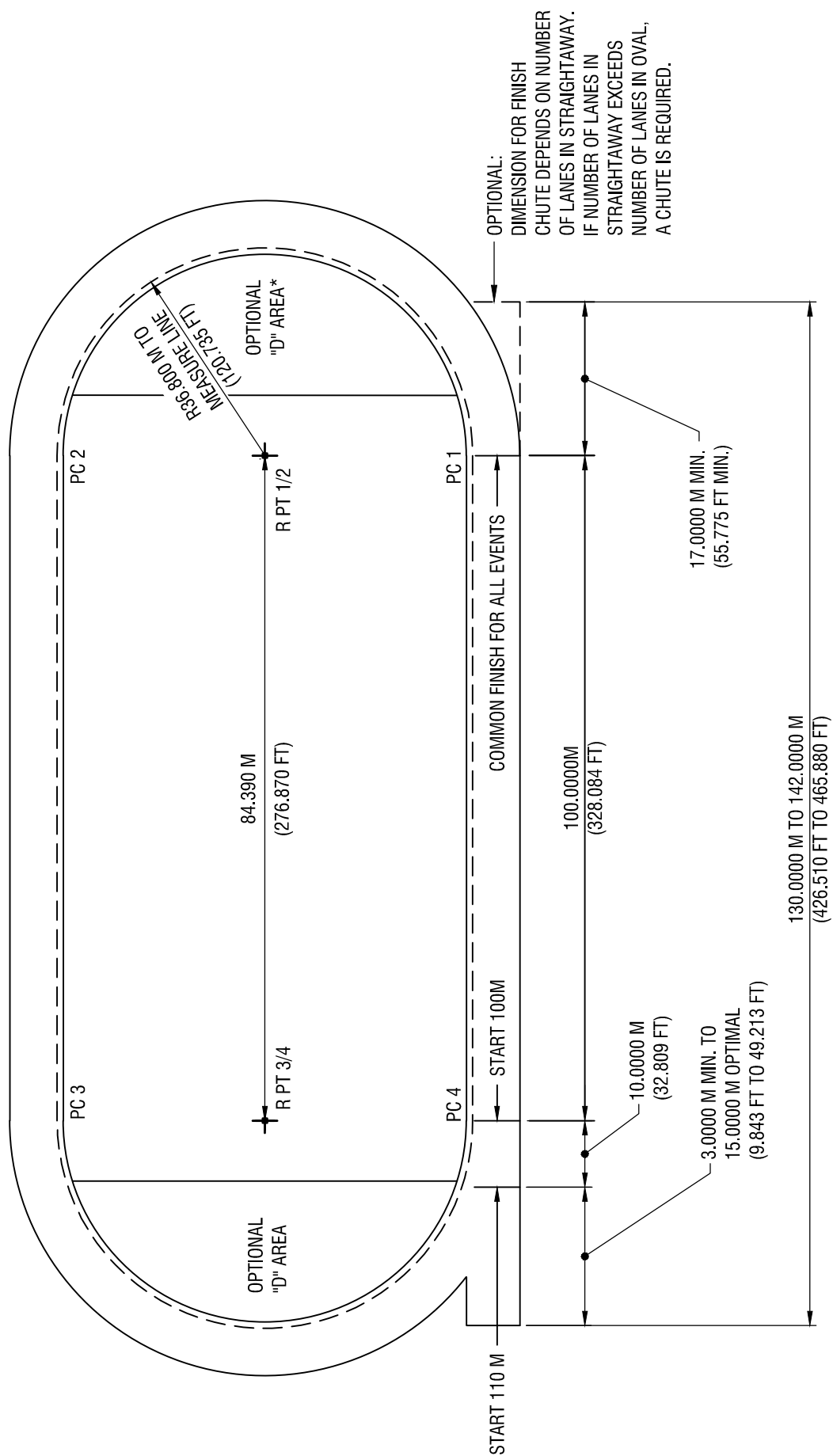


FIGURE 2-5

NOT TO SCALE

1. THIS LAYOUT IS THE SPECIFIED WORLD ATHLETICS NONEQUAL TRACK LAYOUT

2. NOT FOR CONSTRUCTION
3. DUE TO ROUNDING, CALCULATION OF THESE NUMBERS MAY NOT YIELD PRECISE CONVERSIONS OR DISTANCES
4. "D" AREA SIZE VARIES DEPENDING ON REQUIREMENTS FOR FIELD SIZE AND JUMPING/THROWING ACTIVITIES WITHIN THOSE AREAS
5. REFERENCE FIGURE 2-9 FOR LANE LINE/MEASURE LINE DIMENSIONS



400 METER DOUBLE-BEND TRACK
FIGURE 2-6
NOT TO SCALE

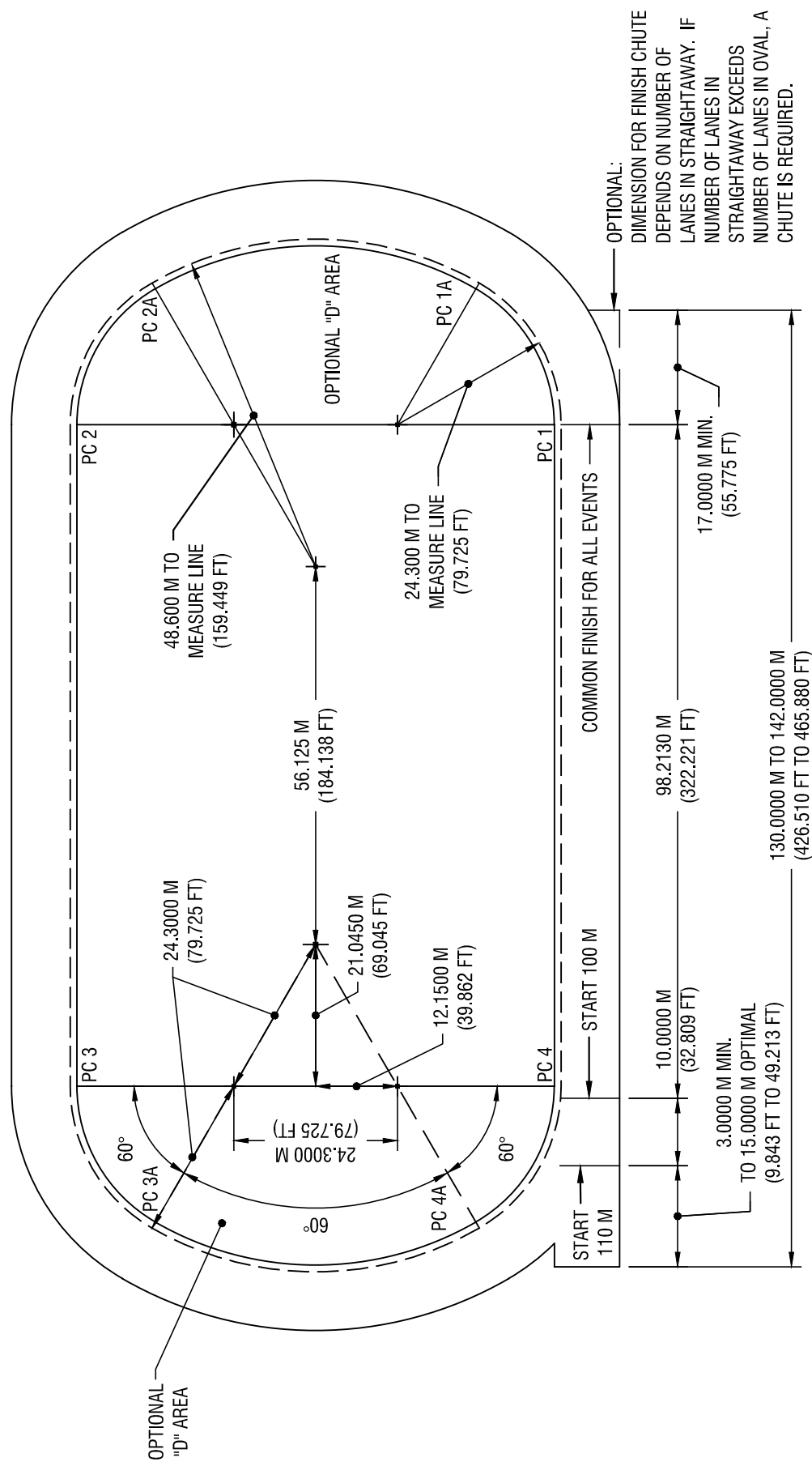
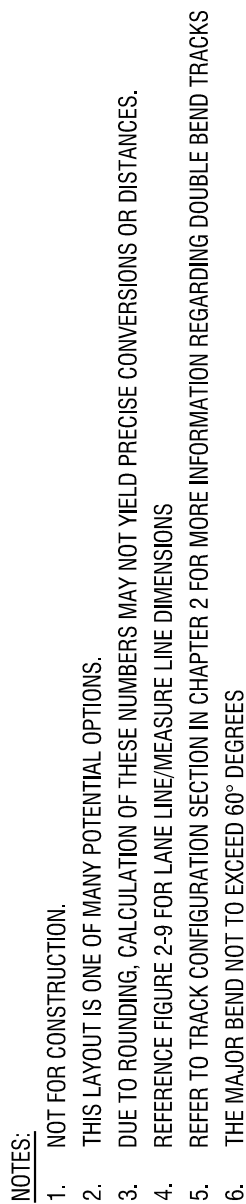
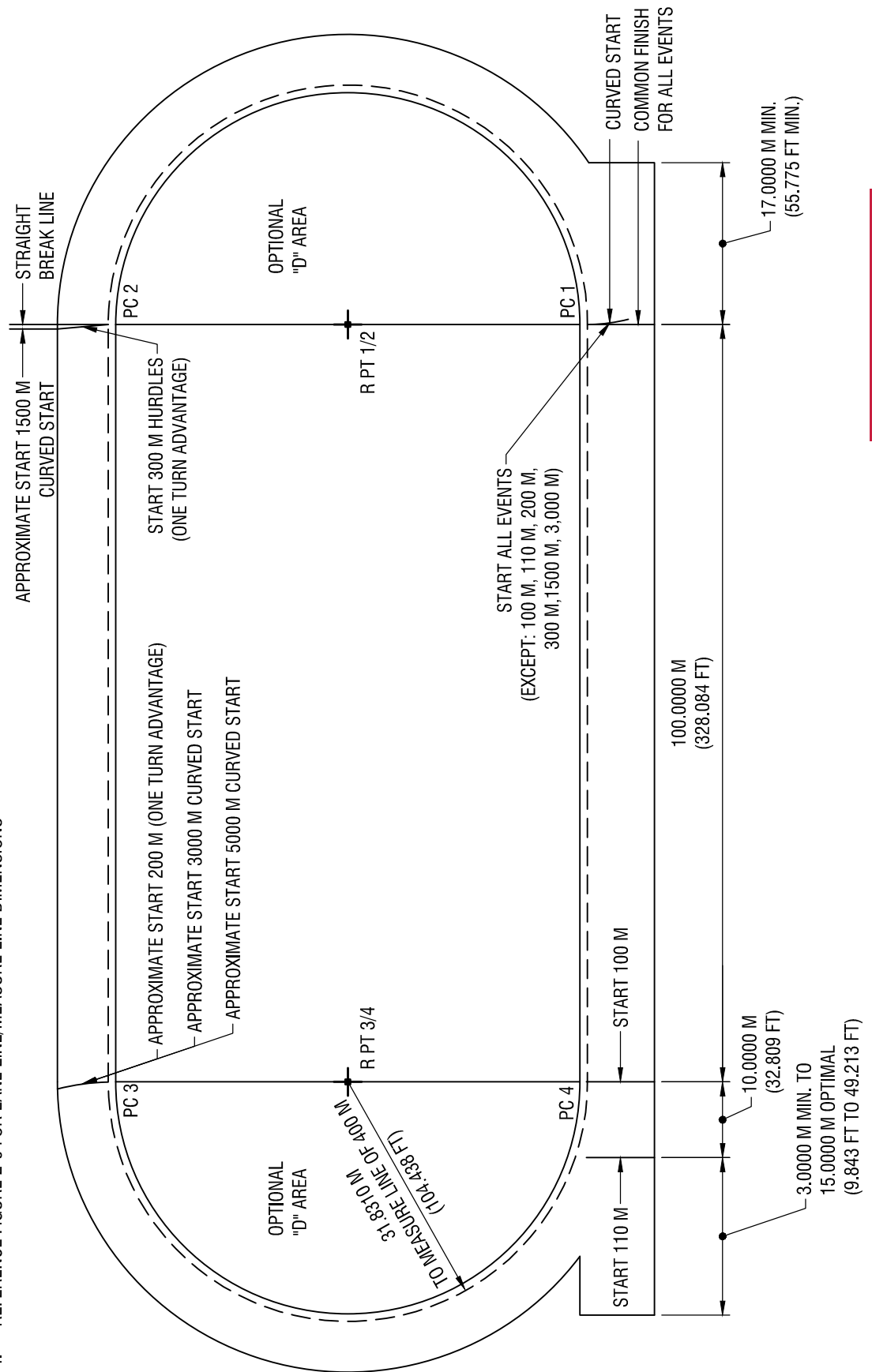


FIGURE 2-7

NOTES:

1. NOT FOR CONSTRUCTION
2. DUE TO ROUNDING, CALCULATION OF THESE NUMBERS MAY NOT YIELD PRECISE CONVERSIONS OR DISTANCES
3. FINAL LAYOUTS OF RUNNING EVENT STRIPING DEPENDS ON ACTUAL DIMENSIONS OF TRACK
4. REFERENCE FIGURE 2-9 FOR LANE LINE/MEASURE LINE DIMENSIONS



COMMON FIELD MEASUREMENTS FOR 400M TRACKS
FIGURE 2-8

Radius to Measure Line (Feet)	Distance between Radii (FT)	Radius to Measure Line (Meters)	Distance between Radii (M)	Course Length (FT)	Course Length M	Notes
90.000	373.425	27.4320	113.8198	1312.336	400	
95.000	357.717	28.9560	109.0320	1312.336	400	
100.000	342.009	30.4800	104.2443	1312.336	400	
102.000	335.726	31.0896	102.3291	1312.336	400	
104.000	329.442	31.6992	100.4140	1312.336	400	
104.433	328.082	31.8312	99.9993	1312.336	400	1
106.000	323.159	32.3088	98.4989	1312.336	400	
108.000	316.876	32.9184	96.5838	1312.336	400	
110.000	310.593	33.5280	94.6687	1312.336	400	
112.000	304.310	34.1376	92.7536	1312.336	400	
114.000	298.026	34.7472	90.8385	1312.336	400	
116.000	291.743	35.3568	88.9233	1312.336	400	
118.000	285.460	35.9664	87.0082	1312.336	400	
120.000	279.177	36.5760	85.0931	1312.336	400	
120.735	276.868	36.8000	84.3894	1312.336	400	2
122.000	272.894	37.1856	83.1780	1312.336	400	
124.000	266.611	37.7952	81.2629	1312.336	400	
126.000	260.327	38.4048	79.3478	1312.336	400	
128.000	254.044	39.0144	77.4326	1312.336	400	
130.000	247.761	39.6240	75.5175	1312.336	400	

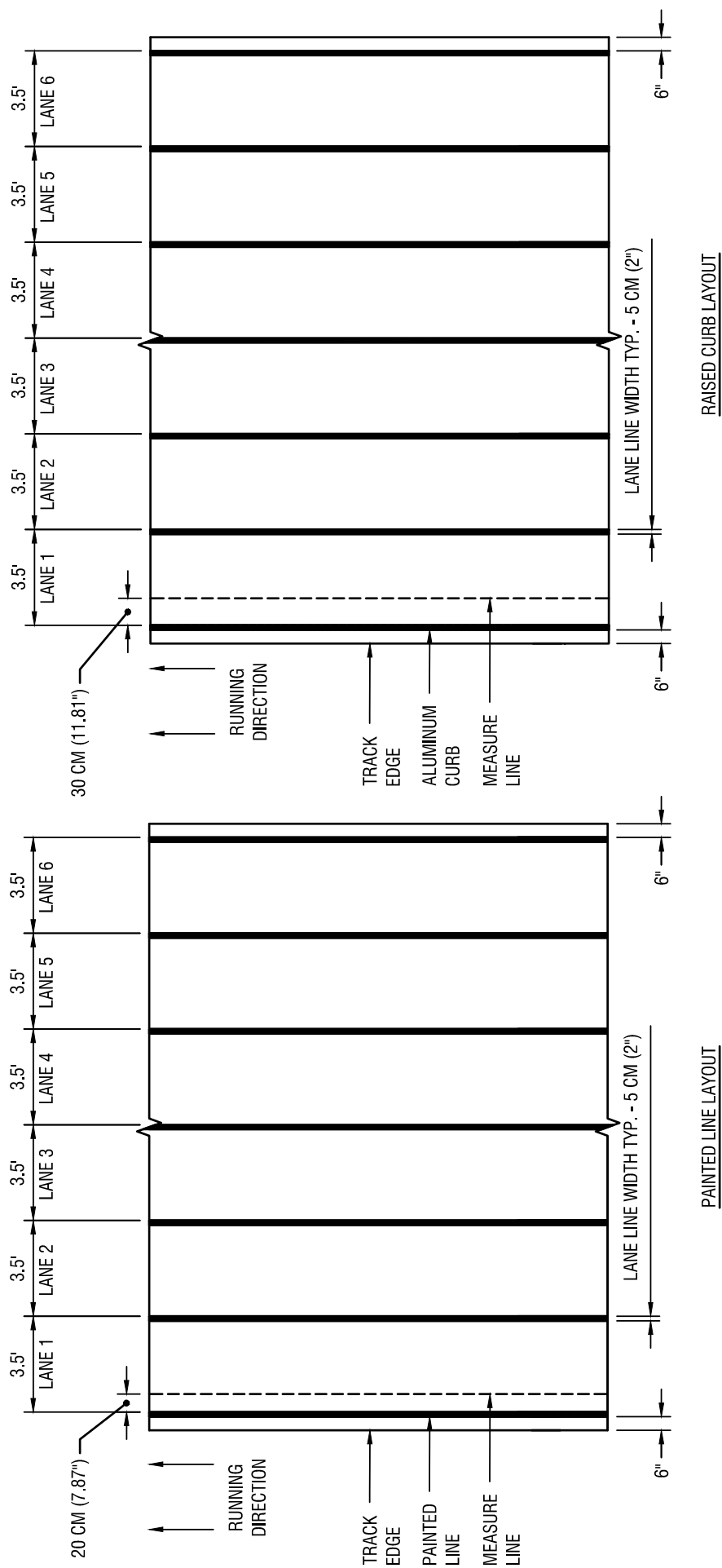
NOTES:

- 1. EQUAL QUADRANT TRACK
- 2. INTERNATIONAL STANDARD (WORLD ATHLETICS)
- 3. ALL CONVERSIONS USE 1M=3.28084FT
- 4. THE VALUE OF PL USED IN 3.1416

LANE LINE/MEASURE LINE TRACK MEASUREMENTS

FIGURE 2-9
NOT TO SCALE

- NOTES:
1. NOT FOR CONSTRUCTION
 2. PROVIDE 1 METER MIN. CL



APPROXIMATE AREA OF "D" AREA USING
RADIUS & CENTERLINE DISTANCE IN SQUARE YARDS
FIGURE 2-10a

CENTERLINE DISTANCE (FT)	60	62	64	66	68	70	72	74	76	78	80
RADIUS (FT)	SY	SY	SY	SY	SY	SY	SY	SY	SY	SY	SY
100'	881	922	963	1005	1047	1089	1131	1174	1217	1260	1304
102'	891	933	975	1017	1060	1103	1146	1189	1233	1277	1321
104'	902	944	987	1029	1073	1116	1160	1204	1249	1297	1338
106'	913	955	998	1042	1086	1130	1174	1219	1264	1309	1355
108'	923	966	1010	1054	1098	1143	1188	1233	1279	1325	1371
110'	933	977	1021	1066	1111	1156	1202	1248	1294	1341	1388
112'	943	988	1032	1078	1123	1169	1215	1262	1309	1356	1404
114'	953	998	1043	1089	1135	1182	1229	1276	1324	1372	1420
116'	963	1009	1054	1101	1147	1195	1242	1290	1338	1387	1436
118'	973	1019	1065	1112	1159	1207	1255	1304	1353	1402	1451
120'	983	1029	1076	1123	1171	1220	1268	1317	1367	1417	1467
122'	992	1039	1087	1135	1183	1232	1281	1331	1381	1431	1482
124'	1002	1049	1097	1146	1195	1244	1294	1344	1395	1446	1497
126'	1011	1059	1108	1157	1206	1256	1307	1357	1409	1460	1512
128'	1021	1069	1118	1168	1218	1268	1319	1370	1422	1474	1527
130'	1030	1079	1128	1178	1229	1280	1331	1383	1436	1488	1542
SY = SQUARE YARDS											

APPROXIMATE AREA OF "D" AREA USING
RADIUS & CENTERLINE DISTANCE IN SQUARE METERS
FIGURE 2-10b

CENTERLINE DISTANCE (FT)	60	62	64	66	68	70	72	74	76	78	80
RADIUS (FT)	SM	SM	SM	SM	SM	SM	SM	SM	SM	SM	SM
100'	736	771	805	840	875	910	946	982	1018	1054	1090
102'	745	780	815	850	886	922	958	994	1031	1068	1105
104'	754	789	825	861	897	933	970	1007	1044	1081	1119
106'	763	799	835	871	908	945	982	1019	1057	1095	1133
108'	772	808	844	881	918	956	993	1031	1070	1108	1147
110'	780	817	854	891	929	967	1005	1043	1082	1121	1160
112'	789	826	863	901	939	977	1016	1055	1095	1134	1174
114'	797	835	872	911	949	988	1027	1067	1107	1147	1187
116'	805	843	882	920	959	999	1039	1079	1119	1160	1200
118'	814	852	891	930	969	1009	1050	1090	1131	1172	1213
120'	822	860	900	939	979	1020	1060	1101	1143	1184	1226
122'	830	869	909	949	989	1030	1071	1113	1155	1197	1239
124'	838	877	917	958	999	1040	1082	1124	1166	1209	1252
126'	846	886	926	967	1009	1050	1092	1135	1178	1221	1264
128'	853	894	935	976	1018	1060	1103	1146	1189	1233	1277
130'	861	902	943	985	1028	1070	1113	1157	1200	1245	1289
SM = SQUARE METERS											

SQUARE YARDAGE (METERS) FOR 400M TRACKS
MEASUREMENTS BASED ON AN EQUAL QUADRANT TRACK
FIGURE 2-10c

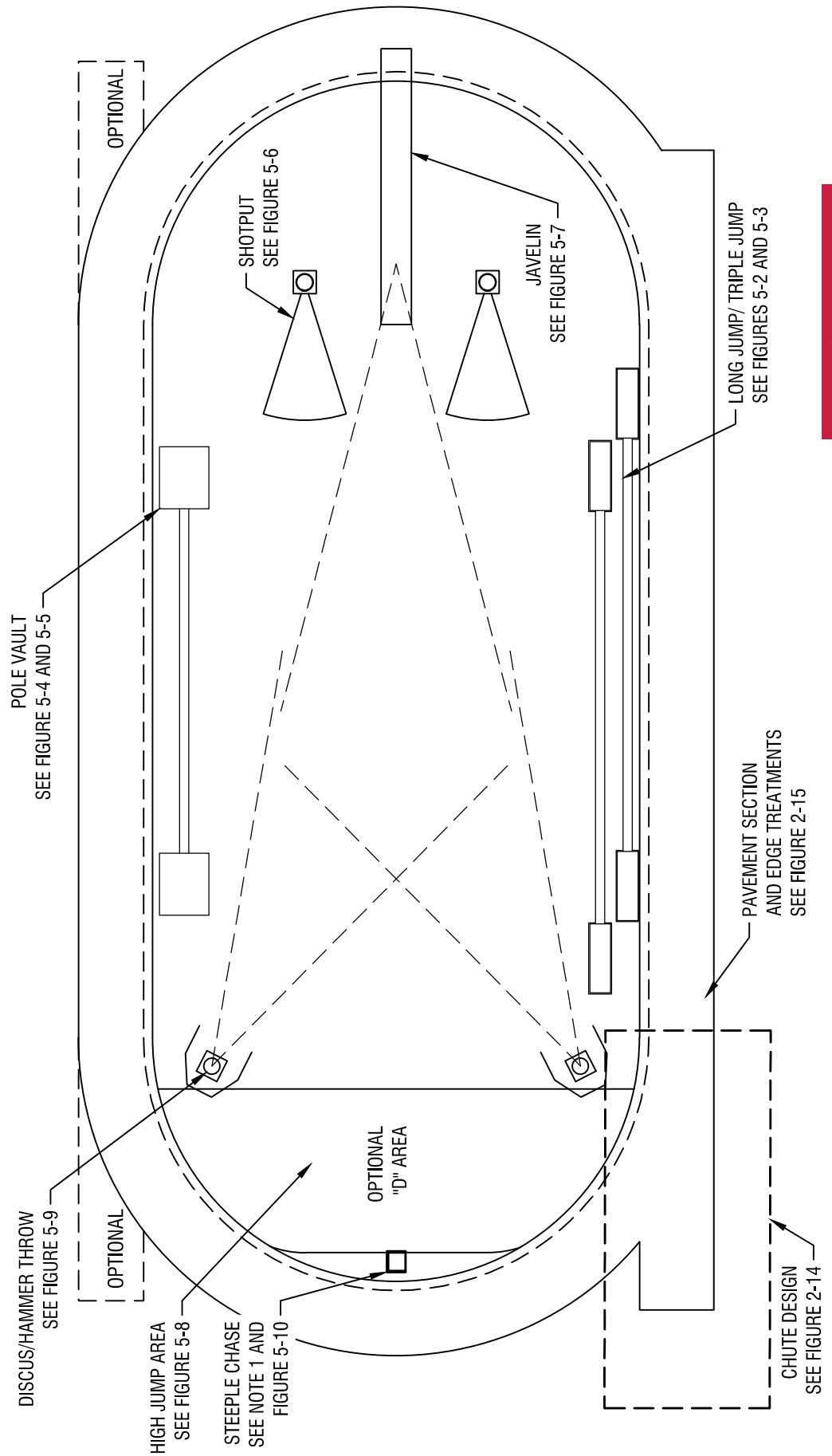
Width		Track Oval		Transition Area		Transition Distance	
Feet	Meters	Sq yds	Sq m	Sq yds	Sqm	Feet	Meters
18.0	5.49	2725	2278	41	35	63.90	19.5
19.0	5.79	2883	2410	45	37	65.80	20.1
20.0	6.10	3042	2542	48	40	67.66	20.6
21.0	6.40	3201	2676	52	44	69.48	21.2
21.5	6.55	3278	2740	54	45	70.34	21.4
22.0	6.71	3362	2810	56	47	71.27	21.7
22.5	6.86	3441	2877	58	48	72.14	22.0
23.0	7.01	3522	2944	60	50	73.03	22.3
24.0	7.32	3684	3079	64	53	74.76	22.8
25.0	7.62	3846	3215	68	57	76.46	23.3
26.0	7.92	4009	3352	72	60	78.14	23.8
27.0	8.23	4173	3488	76	64	79.80	24.3
28.0	8.53	4337	3625	81	67	81.44	24.8
28.5	8.68	4415	3691	83	69	82.21	25.1
29.0	8.84	4502	3763	85	71	83.05	25.3
29.5	8.99	4581	3830	87	73	83.82	25.5
30.0	9.14	4668	3902	89	75	84.65	25.8
31.0	9.45	4834	4041	94	79	86.23	26.3
32.0	9.75	5001	4181	99	82	87.79	26.8
32.5	9.90	5081	4247	101	84	88.53	27.0
33.0	10.06	5169	4321	103	86	89.34	27.2
34.0	10.36	5337	4462	108	90	90.87	27.7
35.0	10.67	5507	4603	113	94	92.39	28.2
36.0	10.97	5677	4745	118	99	93.89	28.6

400 METER TRACK & EVENT FIGURE REFERENCE PLAN

FIGURE 2-11

NOT TO SCALE

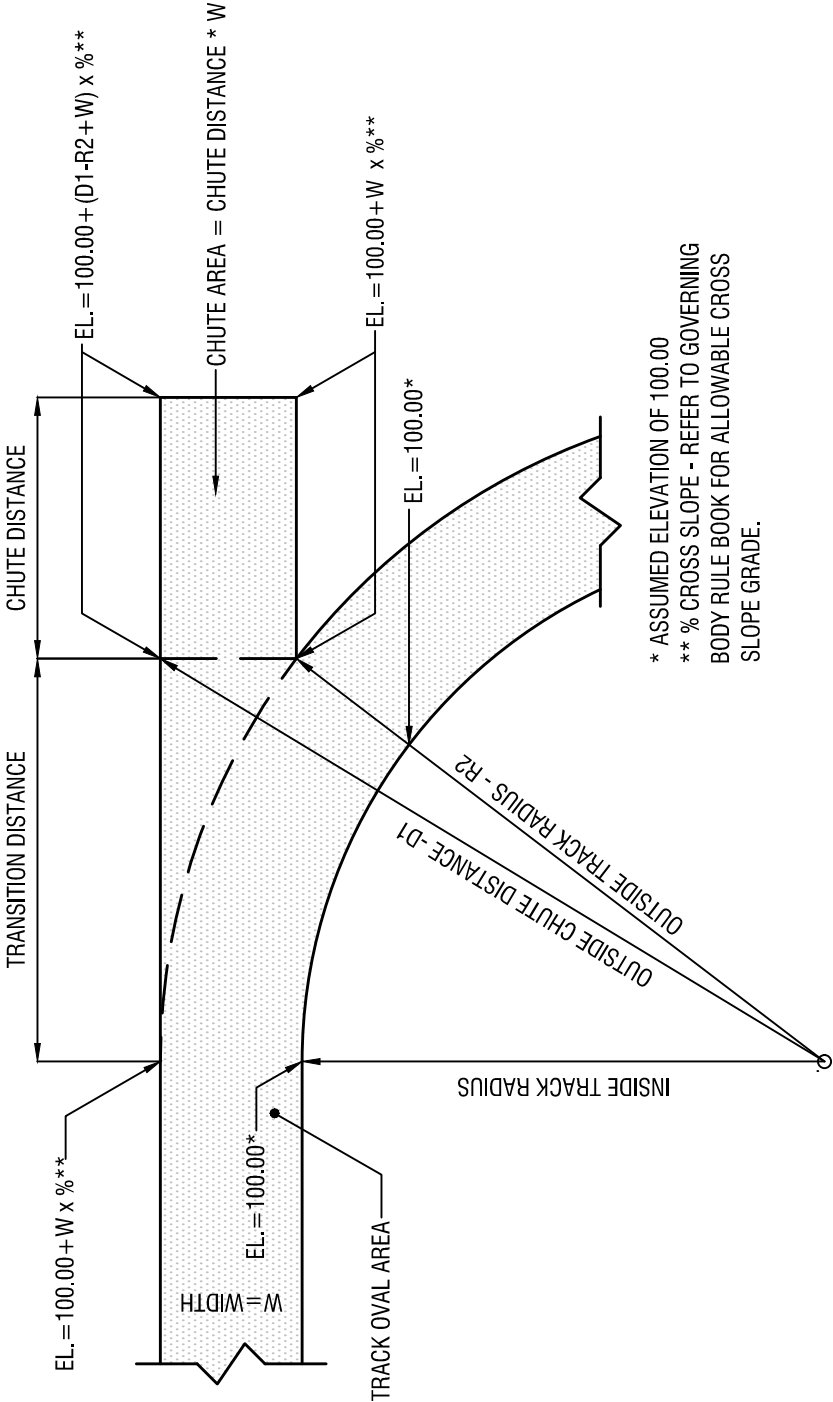
- NOTES:**
1. THE LOCATION OF THE WATER JUMP AND HURDLES VARIES AND MUST CONFORM TO THE RULES OF THE RELEVANT GOVERNING BODY.
 2. SAFETY OF ATHLETES, COACHES, SPECTATORS, ETC. MUST BE CONSIDERED WHEN LAYING OUT ALL THROWING EVENTS.



TRANSITION DETAIL SHOWING TRACK OVAL TRANSITION AREA AND
RECOMMENDED TRACK ELEVATIONS

FIGURE 2-12
NOT TO SCALE

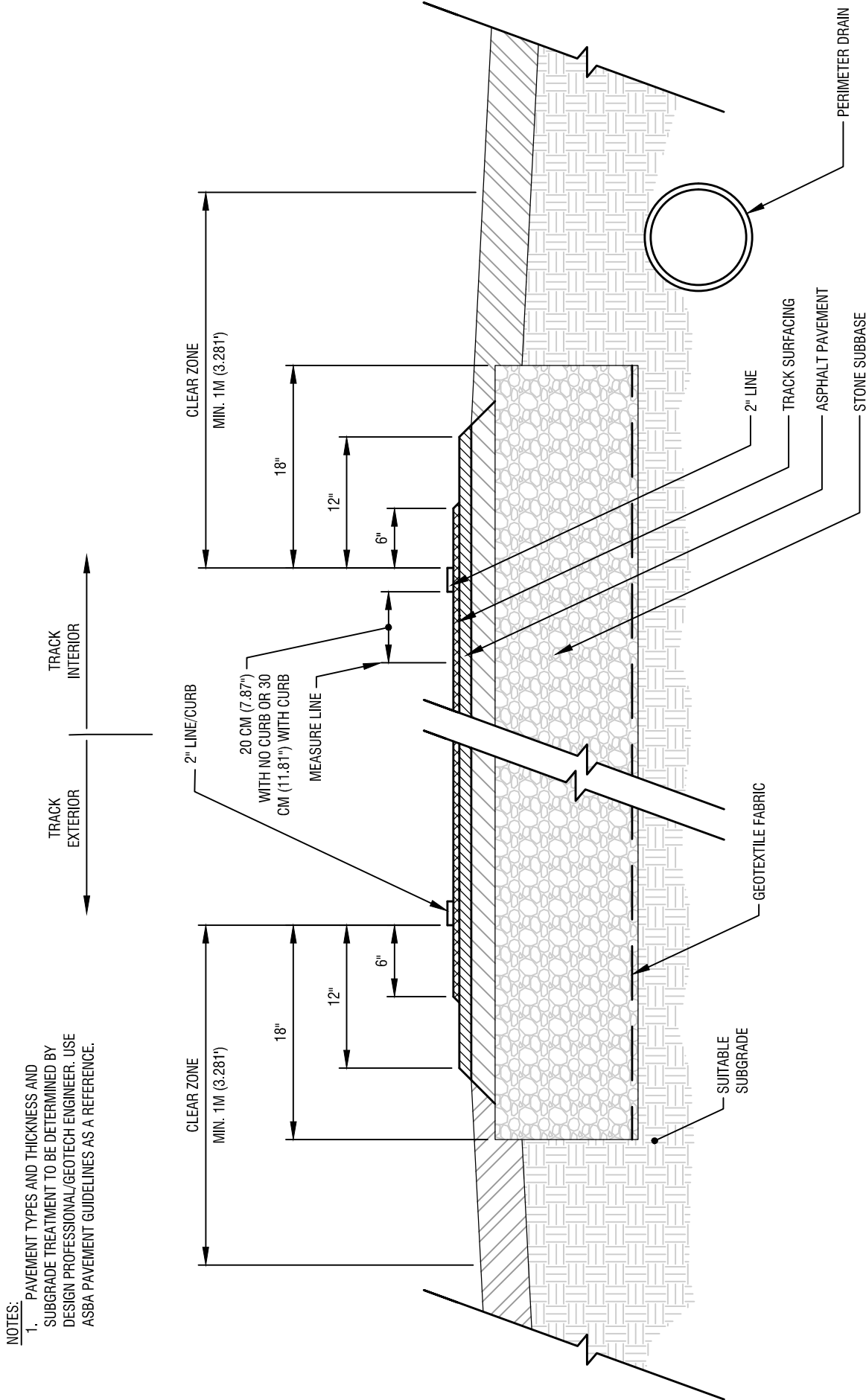
NOTE:
1. NOT FOR CONSTRUCTION



TRACK PAVEMENT SECTION WITHOUT EDGE TREATMENTS

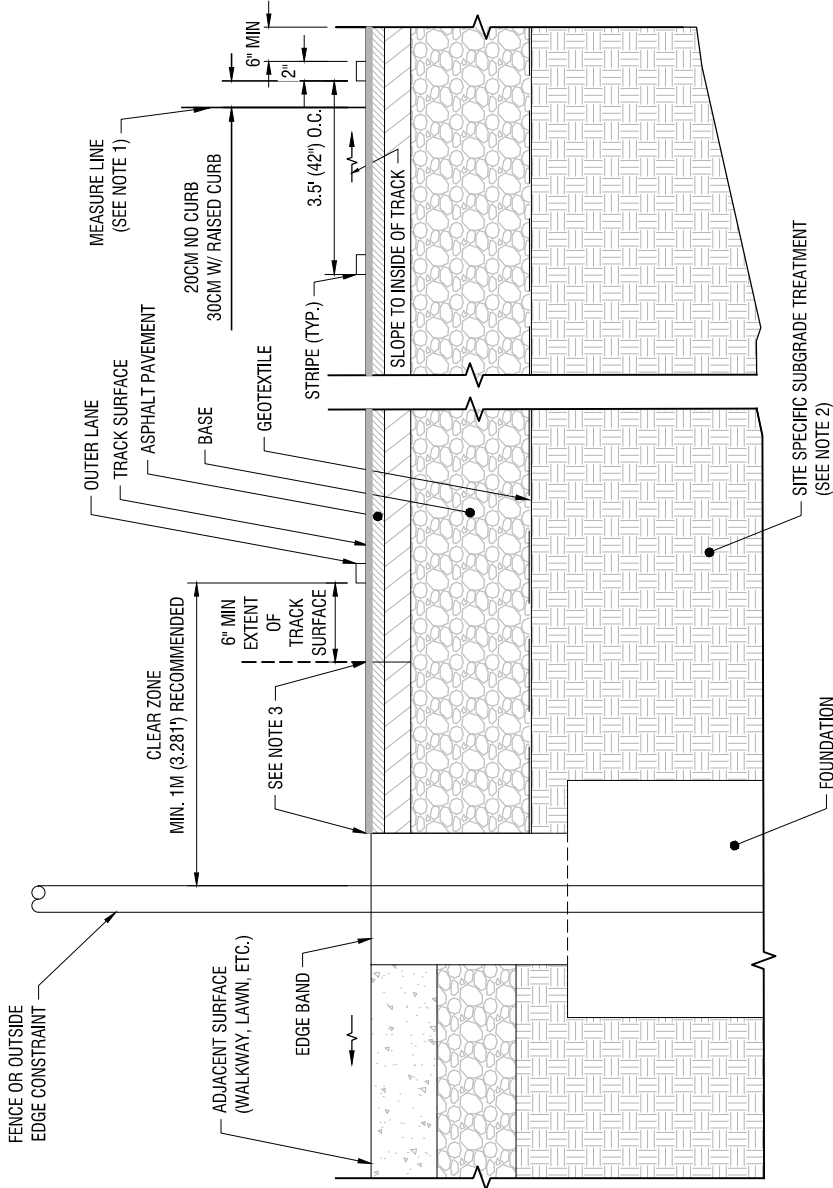
FIGURE 2-13

NOT TO SCALE



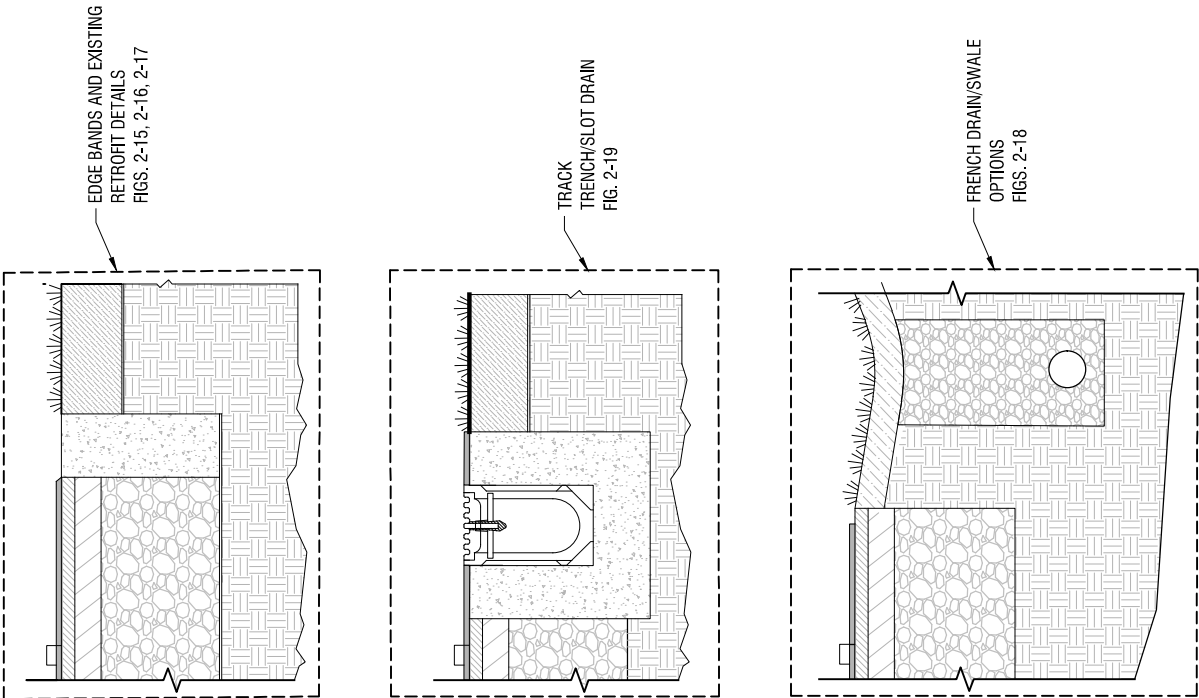
TYPICAL TRACK AND FIELD SECTION
AND EDGE TREATMENTS

FIGURE 2-14
NOT TO SCALE



- NOTES:
1. ALL MEASURE LINES TO BE 400M SEE FIG. 2-4 THROUGH FIG. 2-8 FOR MEASUREMENT DETAILS.
 2. PAVEMENT TYPES AND THICKNESS AND SUBGRADE TREATMENT TO BE DETERMINED BY DESIGN PROFESSIONAL/GEOTECH ENGINEER. USE ASBA PAVEMENT GUIDELINES AS A REFERENCE.
 3. RECOMMEND EXTENDING TRACK SURFACE MINIMALLY TO EDGE BAND. OUTSIDE EDGE OF TRACK SURFACE IS RECOMMENDED HIGH POINT TO PROMOTE POSITIVE DRAINAGE.

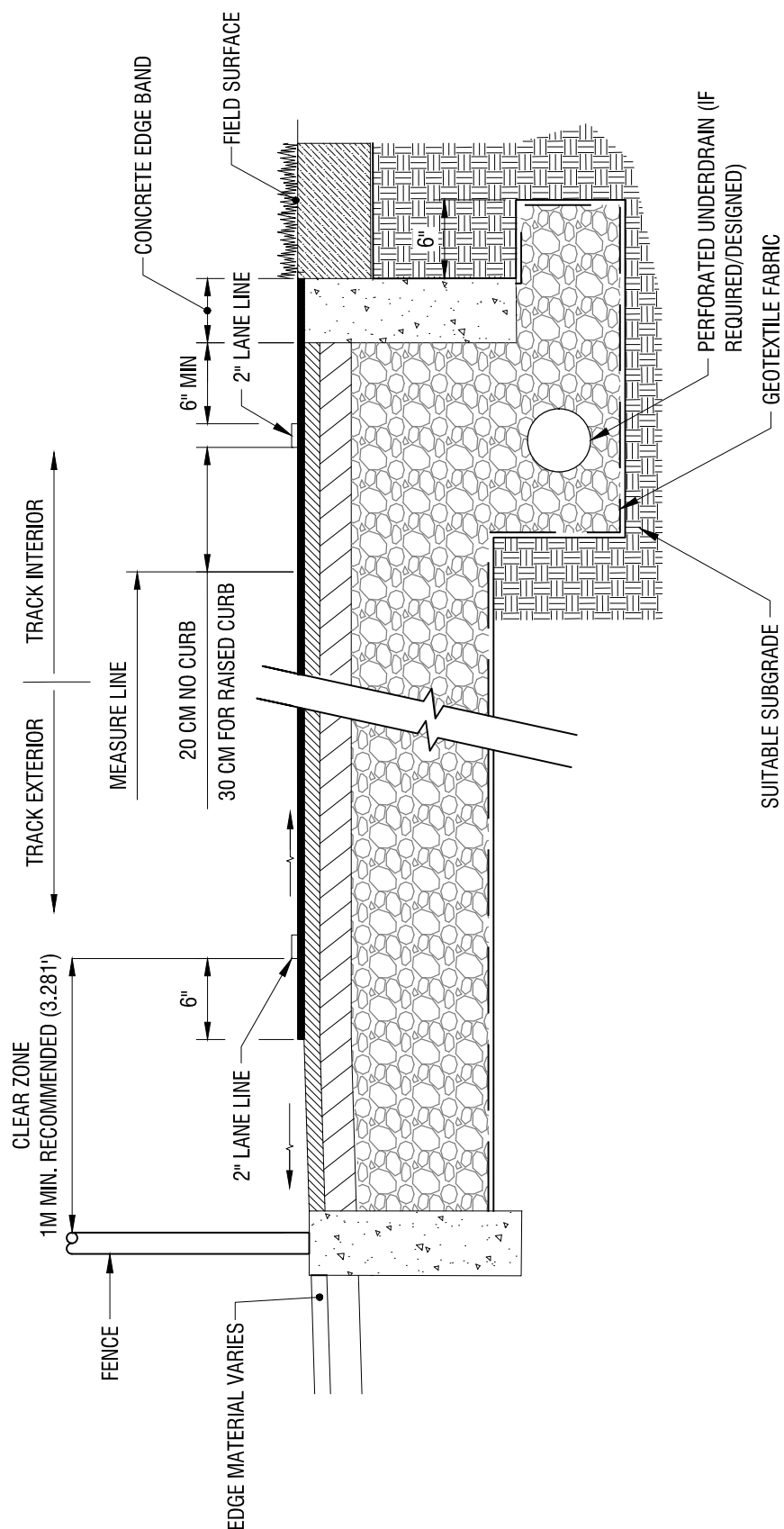
EDGE TREATMENTS



EDGE BAND DETAIL
FIGURE 2-15
NOT TO SCALE

NOTES:

1. NOT FOR CONSTRUCTION
2. CONCRETE EDGE BAND WIDTHS, DEPTHS, AND REINFORCEMENT SHALL BE DESIGNED WITH CONSIDERATION OF SITE AND SOIL CONDITIONS.
3. TRACK SURFACING MAY EXTEND UP TO EDGE OF CONCRETE OR OVER EDGE BAND. CONDITION SHALL BE REVIEWED WITH FACILITY OWNER.
4. FIELD SURFACE TO BE COORDINATED WITH FIELD BOOK DETAILS FOR CONNECTION AND/OR ADDITIONAL DRAINAGE/UTILITIES.

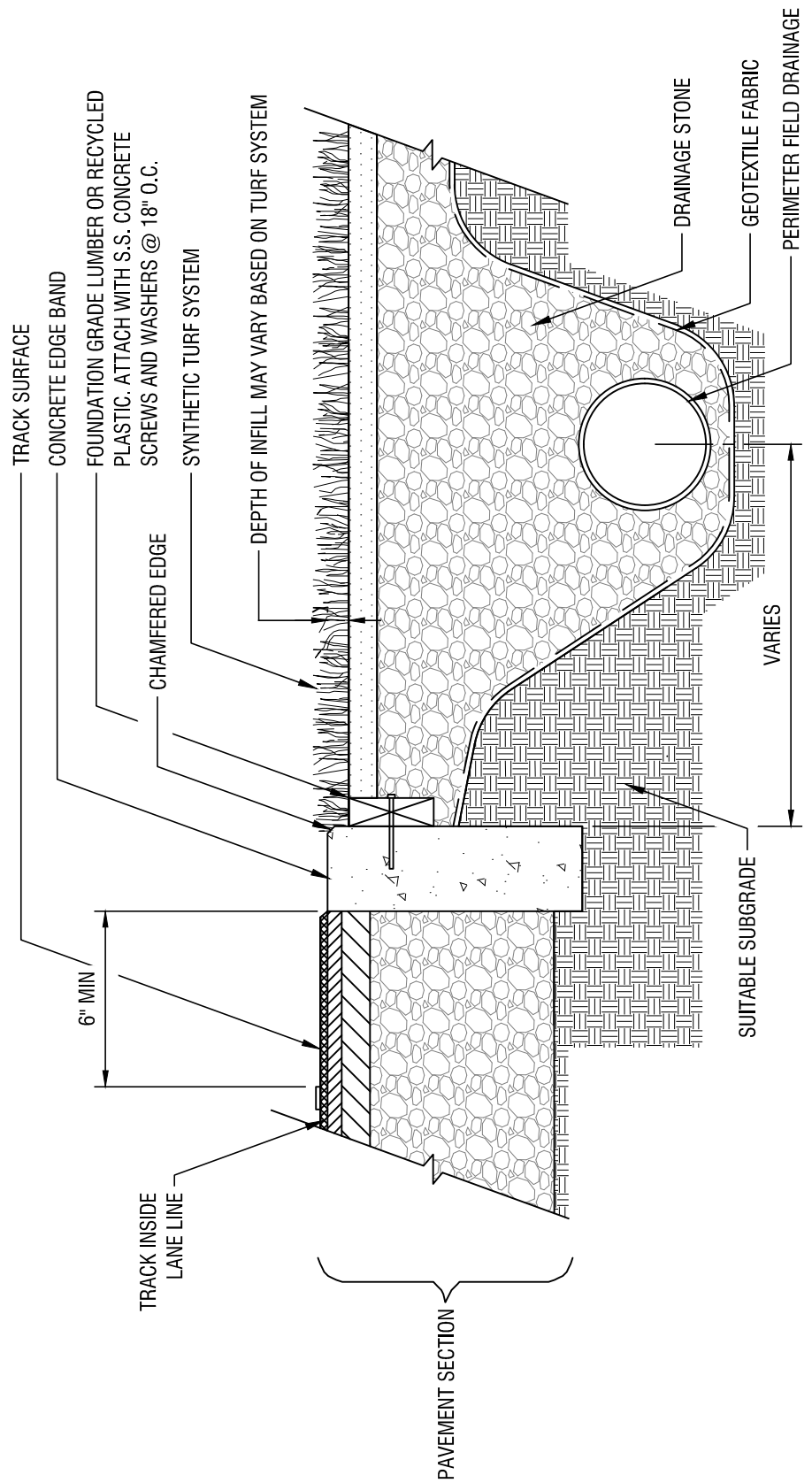


EDGE BAND WITH WOOD NAILER FOR SYNTHETIC TURF

FIGURE 2-16

NOT TO SCALE

- NOTES:
1. NOT FOR CONSTRUCTION
 2. TRACK SURFACE MAY BE EXTENDED TO COVER EDGE BAND
 3. CONCRETE EDGE BAND WIDTHS, DEPTHS, AND REINFORCEMENT SHALL BE DESIGNED WITH CONSIDERATION OF SITE AND SOIL CONDITIONS.
 4. TRACK SURFACING MAY EXTEND UP TO EDGE OF CONCRETE OR OVER EDGE BAND. CONDITION SHALL BE REVIEWED WITH FACILITY OWNER.
 5. FIELD SURFACE TO BE COORDINATED WITH FIELD BOOK DETAILS FOR CONNECTION AND/OR ADDITIONAL DRAINAGE/UTILITIES.
 6. INFILL SHALL BE WITHIN $\frac{1}{2}$ " OF ADJACENT SURFACE (TOP OF EDGE BAND OR TRACK SURFACE)



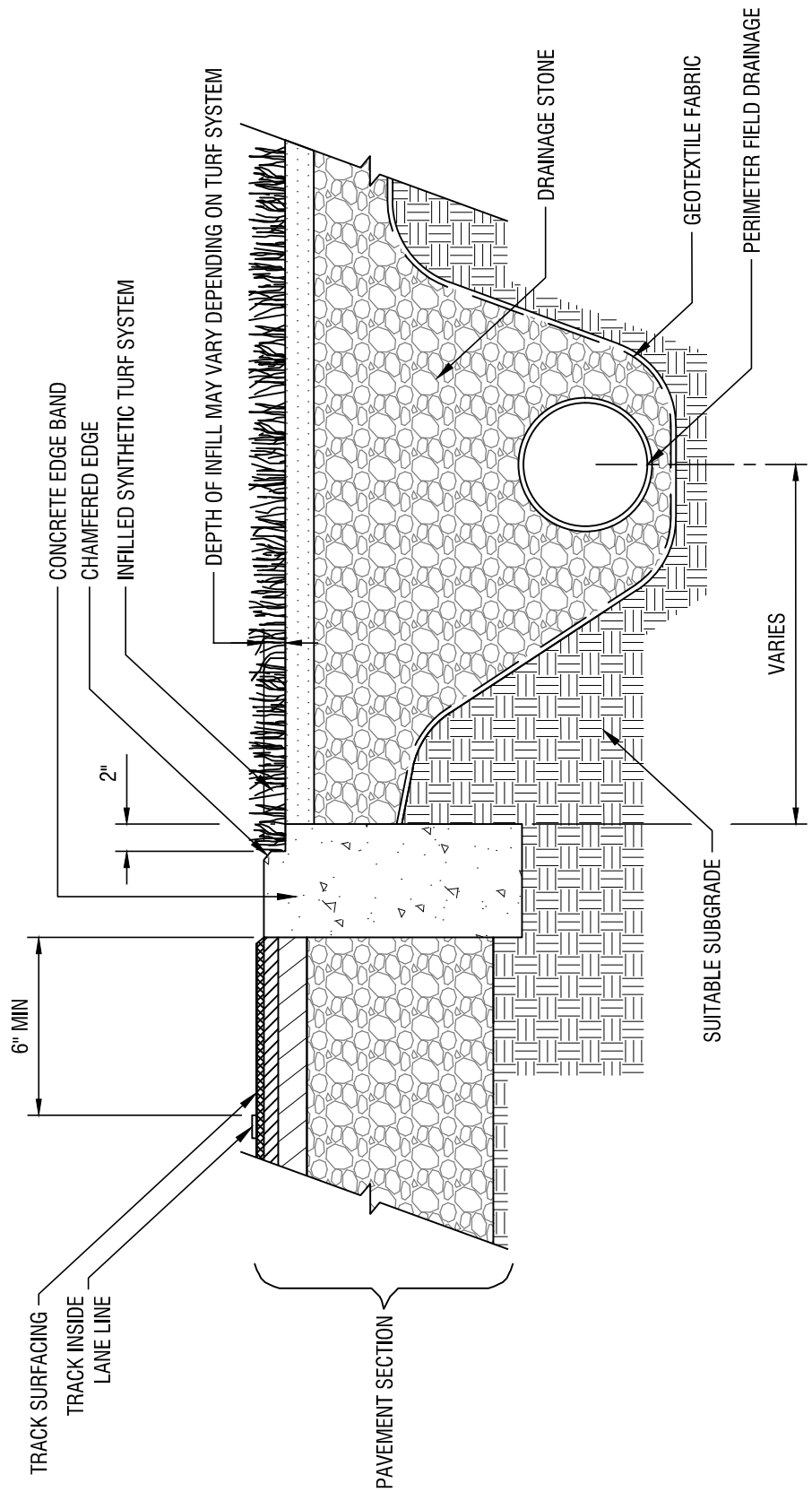
EDGE BAND WITH GLUED SYNTHETIC TURF ANCHOR

FIGURE 2-17

NOT TO SCALE

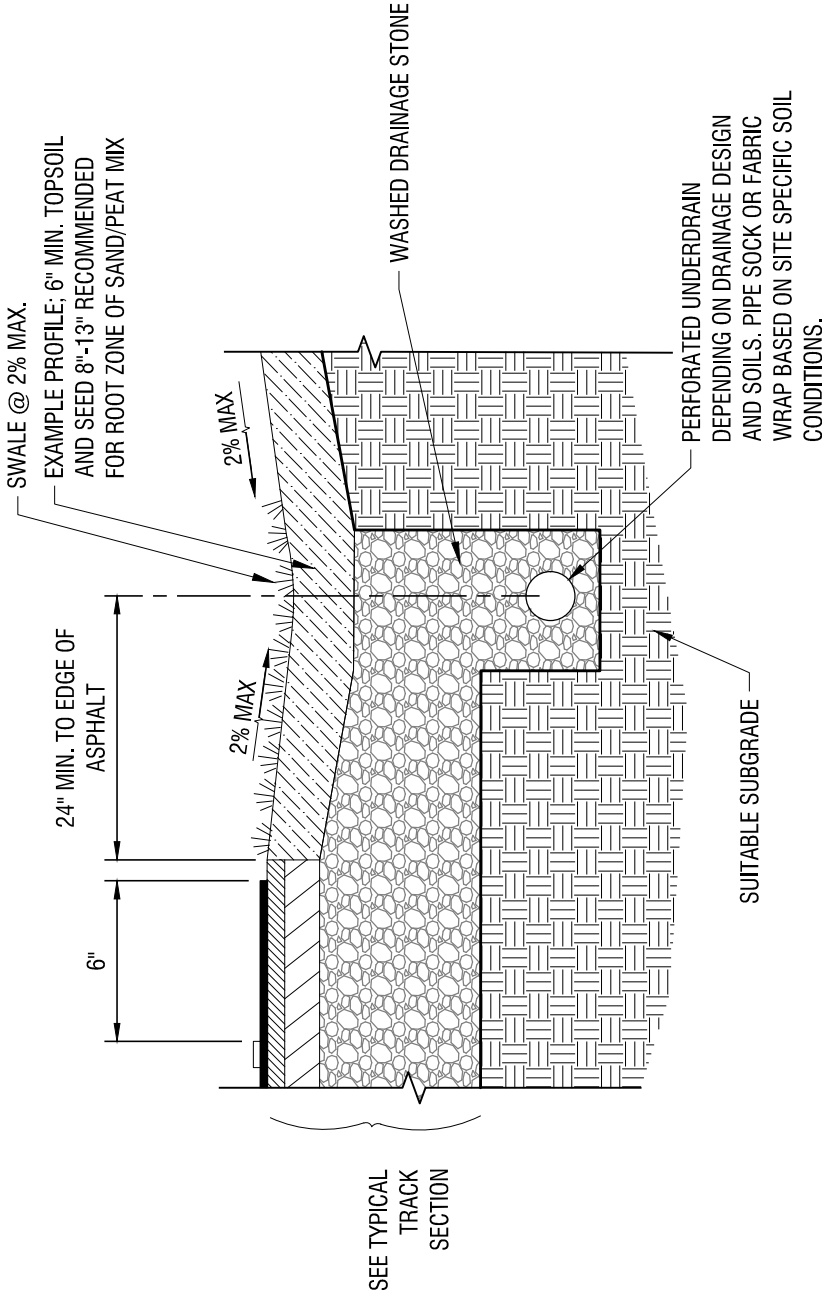
NOTES:

1. REFER TO ASBA FIELD BOOK FOR FURTHER INFORMATION ON FIELD DESIGN AND CONSTRUCTABILITY
2. CONCRETE EDGE BAND WIDTHS, DEPTHS, AND REINFORCEMENT SHALL BE DESIGNED WITH CONSIDERATION OF SITE AND SOIL CONDITIONS.
3. TRACK SURFACING MAY EXTEND UP TO EDGE OF CONCRETE OR OVER EDGE BAND. CONDITION SHALL BE REVIEWED WITH FACILITY OWNER.
4. FIELD SURFACE TO BE COORDINATED WITH FIELD BOOK DETAILS FOR CONNECTION AND/OR ADDITIONAL DRAINAGE/UTILITIES.
6. INFILL SHALL BE WITHIN $\frac{1}{2}$ " OF ADJACENT SURFACE (TOP OF EDGEBAND OR TRACK SURFACE)



FRENCH DRAIN/SWALE DETAIL
FIGURE 2-18
NOT TO SCALE

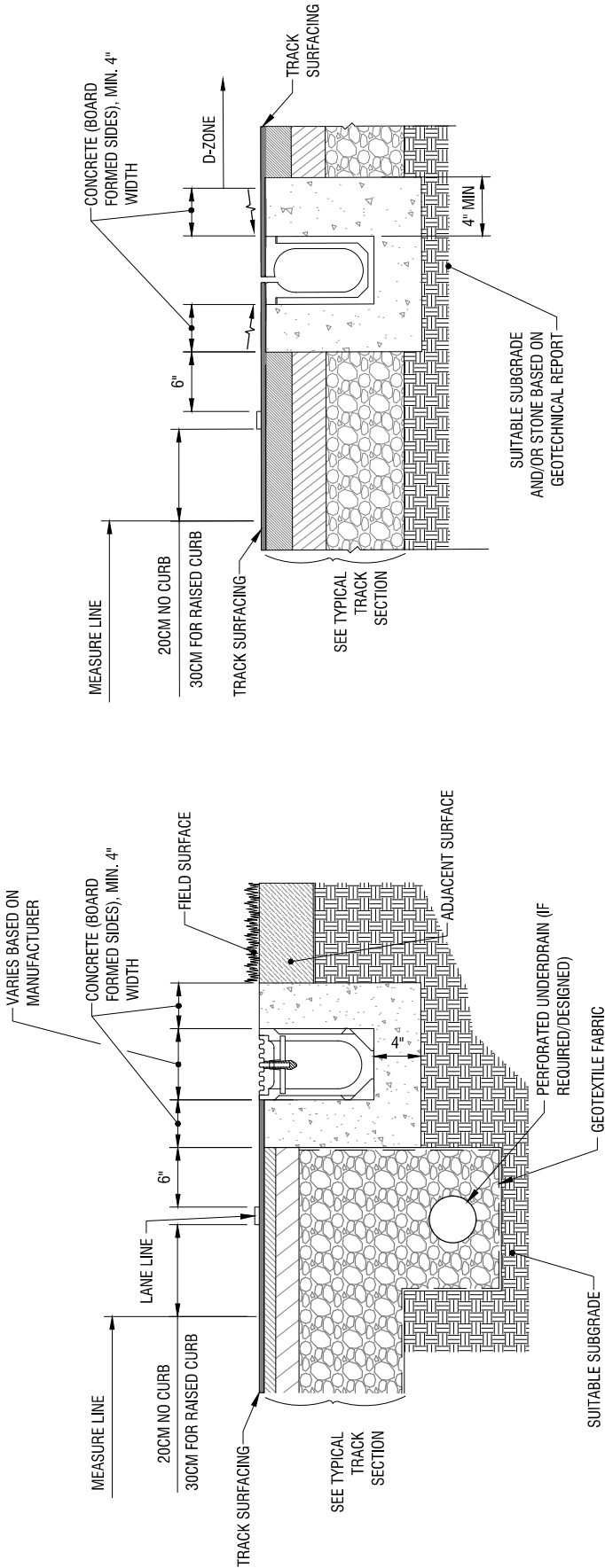
- NOTES:
1. INTERMITTENT DRAIN INLETS ARE RECOMMENDED TO PROMOTE ADDITIONAL DRAINAGE



TRACK TRENCH/SLOT DRAIN

FIGURE 2-19

NOT TO SCALE



- NOTES:
1. TRACK SURFACING MAY EXTEND UP TO DRAIN, TO EDGE OF CONCRETE OR UP TO EDGE OF DRAIN SYSTEM. CONDITION SHALL BE REVIEWED WITH FACILITY OWNER

- NOTES:
1. TRACK SURFACING MAY EXTEND UP TO DRAIN, TO EDGE OF CONCRETE OR UP TO EDGE OF DRAIN SYSTEM. OPTIONS SHOULD BE REVIEWED WITH FACILITY OWNER
 2. FIELD SURFACE TO BE COORDINATED WITH FIELD BOOK DETAILS FOR CONNECTION AND/OR ADDITIONAL DRAINAGE/UTILITIES
 3. SEE EDGE BAND DETAILS FOR SYNTHETIC TURF ANCHORING OPTIONS

TRACK TRENCH/SLOT DRAIN

TRACK SLOT DRAIN

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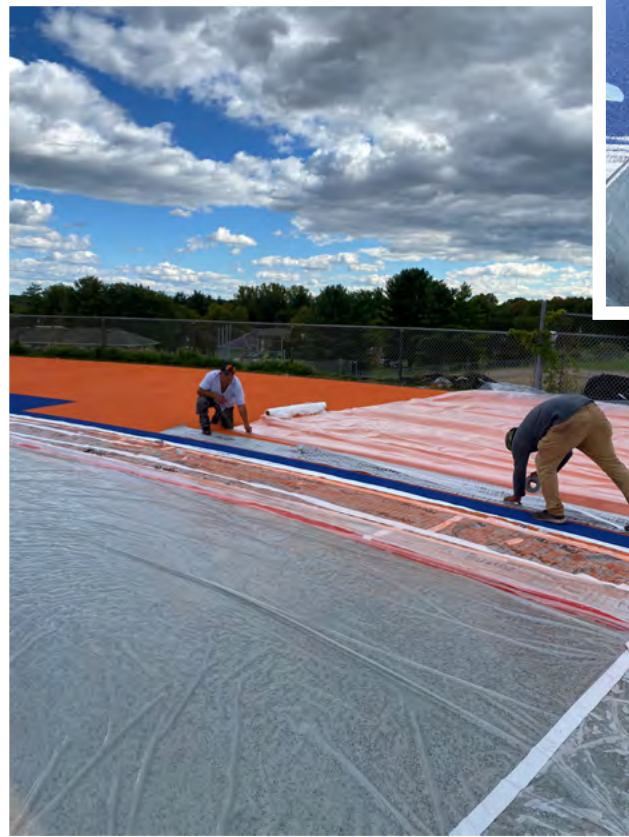
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Chapter 3

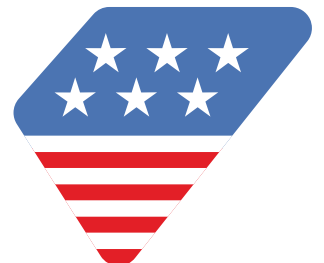
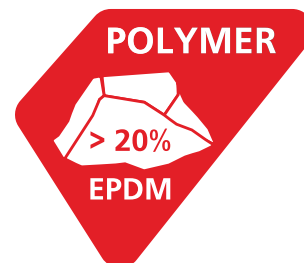
CHOOSING A SURFACE





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Chapter 3—Choosing a Surface

This chapter will cover the following aspects involved in choosing a surface.

- 3.1 General
- 3.2 Installation of Surfacing Systems
- 3.3 Classifying Track Surfaces
- 3.4 Types of Surfacing Systems
- 3.5 Primers, Binders and Coatings
- 3.6 Surface Evaluation, Testing and Standards

3.1 General

An important decision for any track owner is choosing a surface. Once the proper research has been done, the owner should consult with the design team to select the surface that is best for the intended purpose of the facility within the project budget.

Previous chapters focused on planning, design and the construction process. But critical to these phases are site selection, an accurate budget and the intended purpose of the track, including the target end user. The surface selected should take into consideration all of these factors. A given location may mandate a permeable (porous) surface over an impermeable (nonporous) one. A track intended for youth training (physical education, recreation and early competition) will not be the same as a track suitable for international-caliber competition.

3.2 Installation of Surfacing Systems

The first step in installing a surfacing system is to properly prepare and clean the substrate according to the guidelines for asphalt and concrete presented in Chapter 4. During installation, it is important that all equipment, including hand tools, be kept clean; equipment and hand tools should be cleaned thoroughly each day. It is also important that no equipment leak any gas or petroleum products and that no solvent residues be present on equipment or tools.

The next step in installing many systems is to apply a primer. The primer prepares the substrate to receive surface materials and provides an additional bond between it and the surface. Not all systems require the use of primer, however; be certain to follow the manufacturer's recommendations. It is also advisable to recheck dimensional accuracy to assure all striping requirements have been met. More information about installing specific systems is detailed below in section 3.4.

A major consideration during installation of surfacing systems is weather. Latex track surfacing is dependent on evaporation for drying and curing. For this reason, the temperature and humidity at the time of installation will affect the drying time. It is extremely important that no binders freeze. Depending on the running track surface that is chosen, existing

and new structures may require protection. Structures that require protection include fencing, drains, concrete structures, grass fields synthetic turf fields and any adjacent structures. The track surface installer should understand the product being installed and the protection required.

3.3 Classifying Track Surfaces

Although there are many different types of synthetic track surfaces available, they often fall within two distinct categories: permeable (porous) and impermeable (nonporous). Simply defined, permeable or porous pavements allow water to pass through the surface to the asphalt or concrete pavement below, while impermeable or nonporous pavements allow the water to drain and/or evaporate off the surface.

3.3.1 Permeable Surfaces

Permeable (porous) track surfaces are designed so that water will flow through the surface to the pavement below while allowing moisture vapors to escape vertically. In some cases the track will incorporate permeable asphalt pavement for the water to continue to rapidly move to the drainage system installed in the aggregate base. One of the more common permeable systems in the United States features a permeable track surface installed on typical nonpermeable highway asphalt or concrete. Such a surface can only pass water as rapidly as the water is removed by gravitational flow and slope off of the impermeable pavement. A certain amount of water will be entrenched in the track surface until it evaporates or drains. If such a surface is constantly exposed to water (as in areas with extensive rainfall or in track surfaces exposed to sprinklers), the asphalt base may deteriorate and could result in delamination of the surface.

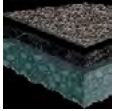
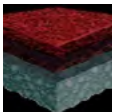
3.3.2 Impermeable Surfaces

Impermeable (nonporous) track surfaces are designed so that water does not penetrate the surface. Water is removed by slope and by gravitational flow to a surface drain system. Evaporation also will occur. Usually, an impermeable track surface will drain more quickly and freely than its permeable counterpart.

It is critical to the success of an impermeable surface that adequate drainage and base aggregate designs eliminate water from beneath the surface. Because impermeable track surfaces do not allow vapors to escape, care must be taken to ensure that moisture does not find a way beneath the surface system.

3.4 Types of Surfacing Systems

3.4.1 Latex Systems

Latex Track System		Permeability	MULTI-LAYER BASE			COLORIZATION OPTIONS			Surface depth
			Black SBR Granules	Neutral Binder	Pigmented Binder	Pigmented top coating	1/8in depth top course (color binder & EPDM)	Full depth color binder & EPDM	
Black Mat		Permeable	Y	Y	N	Y (Black)	N	N	1/2 inch
Colored Sandwich		Permeable	Y	Y	Y	Y (Colored)	Y	N	1/2 inch

Latex surfacing systems consist of rubber particles of a specified size, shape and composition, bound together by a water-based latex binder. Latex surfaces are resilient, textured, all-weather surfaces, usually of permeable design. Because water is used as a dispersing agent for the latex, most latex systems will be applied in layers. The surface thickness is normally 12.7mm (1/2"). Multilayered latex track surfaces are frequently installed using the pavement extension method of construction. After the 6" extension of the surface from the inside and outside lane lines, the layers are feathered to the edge of the asphalt. This eliminates an abrupt termination of the surface.

3.4.1a Latex Track Installation Processes

Multi-Layer Base. Dry SBR or EPDM is spread evenly utilizing a variety of techniques (drop spreader, blower, rakes or lutes). The rubber size and amount distributed are per manufacturer's recommendations. After the rubber has been properly distributed, a latex-based binder is sprayed on the dry rubber particles, saturating the rubber particles. The amount of binder required is recommended by the manufacturer. Once the binder fully dries, the next layer is installed. The number of layers required is determined by the manufacturer to achieve the proper depth, density and texture. In the case of a sandwich system, multiple layers of black SBR rubber and neutral binder are installed to a depth of approximately 3/8". Then multiple layers of colored EPDM rubber particles and pigmented binder are installed at a depth of 1/8" over the black base.



*Rubber application
(drop spreader)*

Pigmented Top Coat: Once the multi-layer base is completed and cured, the application of a UV-resistant top coating is installed by a manufacturer approved sprayer. The latex-based top coat is installed as directed by the manufacturer.

3.4.2 Polyurethane Systems

Base Mat Polyurethane Track Surfaces

Track Surface		Permeability	BASE LAYER		TEXTURE OPTIONS				Surface Depth
			Paved Mat	Pour Sealer	Structural Spray	Embedded	Embedded with Varnish	4-5mm Depth Encapsulated	
Base Mat		Permeable	Y	N	N	N	N	N	13mm
Base Mat Structural Spray		Permeable	Y	N	Y	N	N	N	13mm
Base Mat Seal and Structural Spray		Impermeable	Y	Y	Y	N	N	N	13mm
Base Mat Sandwich		Impermeable	Y	Y	N	Y	Y	Y	13-15mm

Full Pour Polyurethane Track Surfaces

Track Surface		Permeability	BASE LAYER		TEXTURE OPTIONS			Surface Depth
			Broadcast Layered	Mixed Components	Embedded	Embedded with Varnish	4-5mm Depth Encapsulated	
Multi Layer		Impermeable	Y	N	Y	Y	Y	13-15mm
Single or Double Layer		Impermeable	N	Y	Y	Y	Y	13-15mm

3.4.2a Base Layer Installation

Base Mat: A polyurethane base mat system is a homogenous blend of EPDM or SBR rubber granules and polyurethane binders that is evenly spread and screeded in a single-layer application over the prepared pavement. The system is a resilient, all-weather surface designed to allow moisture to pass through the surface. The surface has a smooth finish and may be used either indoors or outdoors. The surface thickness is normally 12.7mm (1/2") for a stand-alone base mat running surface.

In applying this type of polyurethane-bound system there are two ways to mix the polyurethane



Base mat paving

and rubber components. For batch mixing, rubber granules and polyurethane binder are first measured accurately, placed in a mixing container and mixed until all the granules are covered with binder. Once mixed, the pot life of the product is limited, as site conditions can affect it greatly. Continuous mixing requires specialized mixing equipment that is equipped with augers and pumps that produce a uniformly mixed product per manufacturers recommendations.



Base mat continuous mixing

The prepared material is then placed in front of the paving machine. Whatever procedure is used to deliver the material, the supply of material must be constant so that the paving machine does not have to stop and wait for material. The machine, which may be electric or fuel-powered, should be operated at a speed and with the screed bar at an angle to provide tight and smooth surface, free of chatter marks and voids. The screed bar itself should be constantly heated and should oscillate. Once all of these conditions are ensured, the material is placed at the specified depth.

As the material is placed, all seams should be made by the paving machine. The paver operator should only touch up the seams to ensure evenness. The edges should be straight and rounded. All cold seams, such as those from previous work, should be cut in a straight line at an inward angle and primed prior to commencing subsequent work. All seams upon completion are to be even and consistent with the track surface.

Multi-Layer Full Pour: Properly mix the two-component polyurethane coating. Mixing can be done by batch mixing or by continuous mixing equipment. This self-leveling polyurethane coating is flow-applied to the pavement in multiple layers as recommended by the manufacturer. Black SBR or colored EPDM dust and/or granules are broadcasted into the coating to attain the necessary polyurethane depth. The excess rubber particles used in the broadcast process are removed prior to any additional coating is applied. The rubber will account for between 14% and 35% of the total weight of the finished surface.

Mixed Component Full Pour: Properly mix the two-component polyurethane coating. Mix a small-graded rubber particle with the two-component polyurethane material creating a homogenous flowable coating. Mixing can be done by batch mixing or by continuous mixing equipment. The material is installed in one or two applications using a gage rake recommended by



Mixed component full pour

the manufacturer. The rubber will account for 15%-22% of the total weight of the full pour base mat.

3.4.2b Sealers and Textured Coatings



Pore sealer

Pore Sealer: This polyurethane coating can be applied by various methods. It is quite often installed by means of a flat-blade squeegee. There is also mechanical equipment that can be used to install the pore sealer. The coating is 100% two-component polyurethane or a two-component polyurethane properly mixed with rubber dust, creating a homogenous thixotropic mixture. Mixing can be done by batch mixing or by continuous mixing equipment. All coatings must be installed as per manufacturer's instruction. Once the coating is applied, sufficient material must be used to render the surface completely impermeable. It is extremely

important that the base mat and pavement are completely dry before applying the seal coat. Entrapped moisture can cause hydrostatic pressure, which may cause the surface to separate from the pavement.

Structural Spray Texture: This textured coating is a mixture of pigmented polyurethane and colored rubber granules. The coating is an all-weather surface designed to allow moisture to pass through the surface. The structural spray coating is used primarily in outdoor applications. The structural spray is prepared, using dry containers, the specified quantities of each of the components of the structural spray are weighed and mixed in accordance with the manufacturer's recommendations. Next, the specified quantity of colored EPDM rubber granules is added to the mixed material, or in the case of a single-component structural spray, to the weighed-out portion of that single-component material and mixed thoroughly. Mixing can be done by batch mixing or by continuous mixing equipment. The mixed material is then placed into a spray machine and sprayed over the base mat. Generally, two coats are applied in opposite directions. Some specifications may require a third coat. The total rate of application is also provided by the manufacturer.



Polyurethane structural spray



Embedded texture

Embedded Texture: This system is a massive flow application of a one- or two- component coating that is mixed according to the manufacturer's recommendations. Mixing can be done by batch mixing or by continuous mixing equipment. The flow application is self-leveling. The surface is typically installed with gauge rakes to ensure proper depth. The final surface must be level and free of ridges. Before the coating sets, colored EPDM rubber granules are broadcast over the surface. An excess of granules is applied to assist in the embedding process. All loose granules are then removed after initial cure of the polyurethane coating.

Embedded Texture with Varnish: Upon completion of the embedded texture as described above, a UV stable varnish coating is applied to the surface in two applications in opposite directions

Encapsulated Texture: This textured coating is a mixture of pigmented polyurethane and colored rubber granules. The coating is prepared using dry containers, and the specified quantities of each of the components of the structural spray are weighed and mixed in accordance with the manufacturer's recommendations. Next, the specified quantity and size of colored EPDM rubber granules is added to the mixed material, or in the case of a single-component structural spray, to the weighed-out portion of that single-component material and mixed thoroughly. Mixing can be done by batch mixing or by continuous mixing equipment. The mixed material is then placed into a spray machine recommended by the manufacturer and sprayed over the base mat. The coating is sprayed in multiple applications to a depth recommended by the surface manufacturer. Some manufacturers require a varnish coating to be installed over the encapsulated texture to provide UV stability.



Encapsulated texture

3.4.3 Hybrid Track Surfaces (Rolled Goods Mat with Polyurethane Track Textures)

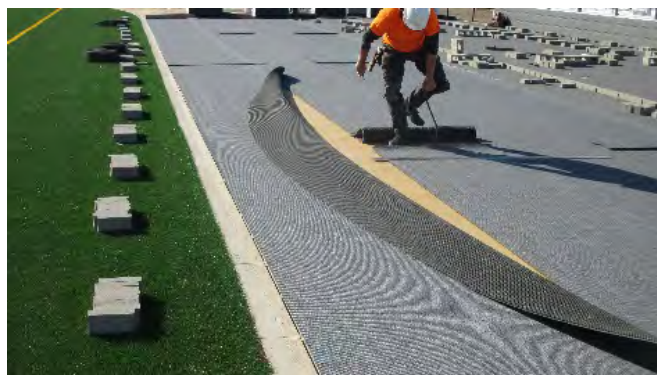
			BASE LAYER		TEXTURE OPTIONS				
Hybrid Track Surface		Permeability	Rubber Rolled Goods Mat	Pore Sealer	Structural Spray	Embedded	Embedded with Varnish	4-5mm Depth Encapsulated	Surface Depth
Hybrid Track Surface/ Seal and Structural Spray		Impermeable	Y	Y	Y	N	N	N	13mm
Hybrid Track Surface/ Sandwich		Impermeable	Y	Y	N	Y	Y	Y	13-15mm

The hybrid rolled goods track surface is often finished with one of the polyurethane textures described in the polyurethane track surface section 3.4.2. These surface textures include:

- Structural Spray Texture
- Pore Sealer Structural Spray Texture
- Embedded Texture Embedded Texture with Varnish
- Encapsulated Texture

3.4.4 Rolled Goods

Rolled Goods		Vulcanized rubber single layer surface	Single-layer surface	Manufactured texture finish
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Rolled goods

Installing Premanufactured Surfaces (hybrid/rolled goods):

Premanufactured products should be installed only when dry conditions are present and projected for time period that is acceptable to the manufacturer. Most manufacturers call for the material to be laid out and relaxed prior to installation. Adhesive is then mixed and troweled in place using a notched trowel. Application rates should be in accordance with the manufacturer's specifications. Adhesive should only be applied to the area which

will be worked on immediately. When the adhesive is tacky, the mat is placed onto the surface, making sure that each sheet is placed in a straight line or around a curve at a constant radius. Head joints should be staggered. Some systems require that the mat be rolled out so that all entrapped air is removed. Seams and joints should be level and not pinched. Placing weights on

seams and joints will help the mat to lie flat while the adhesive sets. Any excess adhesive that protrudes above the seams and joints should be removed.

3.4.5 Track Surface Manufacturer's Role

- Provide installation guidelines
- Provide specifications for the track surface selected by the owner
- Ensure that the installer is a licensed installer of the track surface system selected by the owner
- Provide samples for the track surface selected by the owner. (Samples can be misleading. It is always best to perform a site visit to get a true representation of the track surface)
- Provide materials that meet the track surface specifications.
- If required Provide test data for the track surface selected by the owner
- If required, provide a manufacturer's warranty for the track surface selected by the owner
- Provide technical assistance to the track installer.

3.4.6 The Effects of Weather on Polyurethane Sports Surface Installations

The effects of differing weathering conditions on the installation of polyurethane sports surface systems is often not understood or is misunderstood. This can lead to conflicts between the surfacing system installer, the designer, the general contractor and the owner. These weather effects on surface installation timelines and overall quality of the surface system should be of primary concern and focus to ensure a quality installation and reasonable schedule.

The majority of polyurethane surfacing products are designed and tested at a consistent 68-degree Fahrenheit temperature with 50% humidity, which are ideal conditions for the installation of high-quality, high-performance sports surfaces.

Figure 3-1 shows the effects of weather conditions on the installation of polyurethane surfaces. The effects grid:

Quadrant 1—High Temperature/Low Humidity

- Viscosity is lower than data sheet. Polyurethane will be thinner during application than design.
- Cure time is likely to be substantially longer
- Two component products are affected similarly, but not as severely
- In base mat installations, the binder can in extreme situations migrate toward the bottom third, creating changes in texture and structure.
- The effect on the surface could show as follows:
 - Base mat can show stress cracking

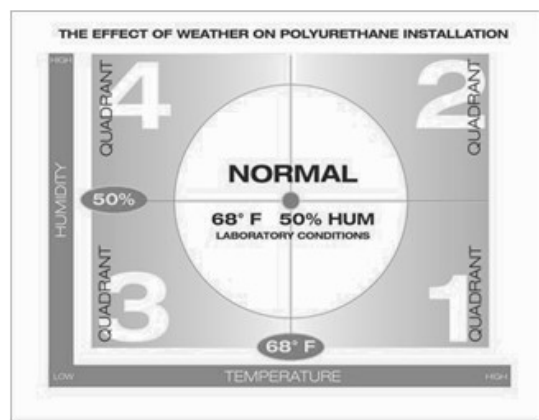


Figure 3-1

- Structural spray can wear out prematurely

Quadrant 2—High Temperature/High Humidity

- Viscosity is lower than data sheet. Polyurethane will be thinner during application than design.
- Cure time is likely to be substantially shorter.
- In base mat installations, the binder can in extreme situations migrate toward the bottom third, creating changes in texture and structure.
- The effect on the surface could show as follows:
 - Base mat applications can experience “cold seams” in which the passes are not bonded well at the seams.
 - Structural spray can wear out prematurely.
 - Can show “micro-bubbling” and “bubbling” in top portion of the application causing premature wear.

Quadrant 3—Low Temperature/Low Humidity

- Viscosity is higher than data sheet. Polyurethane will be thicker during application than design.
- Cure time is likely to be substantially longer.
- In base mat installations, the binder can be inhibited from forming a proper bond between the rubber granules.
- The effect on the surface could show as follows:
 - Base mat applications can experience “spider cracking” in which the bond between the rubber particles is not sufficient.
 - Structural spray can wear out prematurely.

Quadrant 4—Low Temperature/High Humidity

- Viscosity is higher than data sheet. Polyurethane will be thicker during the application than design.
- Cure time is likely to be substantially longer.
- In base mat installations, the binder can experience “micro foaming” changing the porosity (closed matrix) of the mat, causing premature wear.
- Two component applications can experience incomplete cure in application.
- The effects on the surface could show as:
 - Structural spray texture and appearance can be uneven and inconsistent.
 - Two-component applications can remain “gummy” and never completely cure.

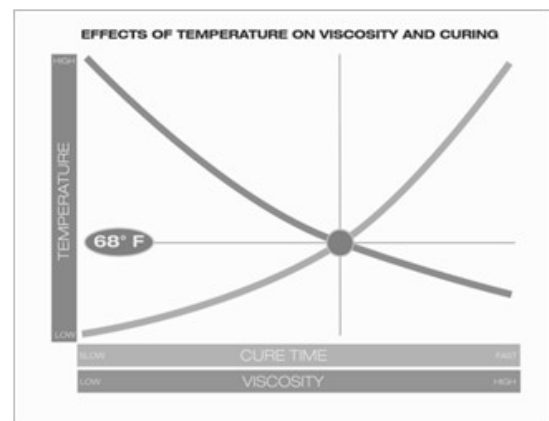


Figure 3-2

- Figure 3-2 shows the effects of temperature to viscosity and cure speed of polyurethanes.

Ideally, polyurethane sports surfaces should be installed within 50 to 90 degrees F and between 30% to 70% humidity to ensure quality and integrity in the finished surface. Outside of these conditions, special care and attention must be taken to determine the potential challenges and pitfalls of the surface installation. The polyurethane manufacturer can be a valuable source of information for these non-ideal installations. Additional time and considerations will often be necessary.

Differences in site, weather and soil conditions require variations in construction and repair methods and materials. Readers are advised to consult a qualified contractor or design professional before undertaking construction or repair of a track.

3.4.7 Asphalt Surfaces

Despite the widespread acceptance of latex and polyurethane track systems, asphalt surfaces are still being used and, occasionally, installed in certain locations. These surfaces were the predecessors to the all-weather surfaces as they are known today. These surfaces were either sand-asphalt-rubber (SAR), asphalt emulsion and rubber or roofing asphalt and rubber.

SAR systems typically consist of 9% to 14% rubber, 11% to 15% high-penetration asphalt oil and sand aggregate. They are mixed and installed in a manner similar to hot mix asphalt, usually to a compacted thickness of 1". They tend to lose resiliency as they age.

Asphalt emulsion and rubber systems are mixed on site and installed at ambient temperature to a depth of 3/8" to 1/2". They contain little or no sand aggregate, which usually makes them more resilient than SAR systems. Roofing asphalt and rubber systems are similar except that heated roofing asphalt is used as the binder. **Because these surfaces do not meet established force reduction standards, ASBA does not recommend them.**

3.4.8 Natural Surfaces

Natural material track systems, including cinder, natural clay, fired clay, expanded shale and decomposed granite, may still be installed in some regions. These systems have, however, declined in popularity due to the advantages of other systems now available. ASBA has limited information on cinder surface tracks; however, this information is somewhat dated as the Association receives relatively few requests for it. Consequently, the information has not been updated. Please note that while ASBA makes such information available to those who request it, it does so as a public service to those who want general information on this type of track surface. The available information is not an updated construction guideline nor any kind of specification.

3.5 Primers, Binders and Coatings

A variety of products may be used in the construction of the track surface.

3.5.1 Primers

- *Latex primer* is a low-viscosity latex binder that allows for a light film application.
- *Polyurethane primer* is a low-viscosity polyurethane binder that allows for a light film application.

3.5.2 Binders

- *SBR (Styrene-Butadiene-Rubber) latex binder* is the most prevalent type of water-based resin latex used to bind rubber particles. It is composed of varying proportions of styrene and butadiene monomers in water. Other types of latex binders used in track construction include vinyl, acrylics and a variety of combinations.
- *Polyurethane binder* is a single-component, moisture-cured, polyurethane product used to bind rubber granules together. It is important that polyurethane binders used in track applications not contain solvents or extenders.

3.5.3 Coatings

- *Water-based coatings*, consisting of latex binder and/or acrylics, may incorporate rubber particles and pigments and are often factory-mixed. They are most commonly used on asphalt, latex and SAR (sand-asphalt-rubber) tracks.
- *Polyurethane structural spray coatings* are available in two forms. A “single-component” coating is premixed by the manufacturer. A “two-component” coating consists of binder and pigmented paste mixed on site. Polyurethane structural sprays most often are used as a surface coating for permeable surfacing systems or for resurfacing existing systems.
- *Polyurethane coatings* are single- or two-component, self-leveling products used in the construction of impermeable systems. These coatings should not contain solvents. When polyurethane coating is used as a sealer, rubber dust is sometimes added.
- *Polyurethane aliphatic top coatings* may be of single- or two-component design and are sometimes applied to finished surfaces that include unstable colors or need to be protected from UV degradation.
- *Adhesive* is a single- or two-component polyurethane or other material specified by the manufacturer that is used to bond prefabricated materials to a prepared pavement.
- *Vapor emissions coatings* are designed to be applied to concrete surfaces prior to the installation of the track surface. These coatings prevent water vapor from the concrete to delaminate the track surface from the concrete.
- *Water-borne polyurethane structural sprays* are a single-component coating premixed by the manufacturer. The coating is mixed with a fine graded EPDM rubber particle, typically .05-1.5mm or 2.0mm, to create a textured coating. Water-borne coatings offer UV stability for colors other than red and black. The coatings are environmentally friendly and are safer to install than solvent-based coatings.

3.5.4 Rubber

Rubber track materials come in a variety of forms, as described below.

3.5.4a Black Rubber Particles

Black rubber particles are one of the most commonly used products in the construction of track surfaces. The industry describes the surface according to which binding product is used, but when taking into consideration the overall proportions of the system, black rubber particles make up the majority of the materials used in the surface.

Black rubber particles are made from post-industrial byproducts. These products may include tires, technical parts, molded goods, etc. In general, black rubber particles are made from EPDM (ethylene-propylene-diene rubber), SBR (styrene-butadiene-rubber) or NR (natural rubber). They are grouped into four classifications:

- Type 1: Black stranded rubber particles or elongated strands of black rubber. Their specific gravity should range from 1.15 to 1.40.
- Type 2: Black rubber SBR or NR granules, which are cubical, sharp-cut particles whose dimensions are specified in the particle sieve analysis. Their specific gravity should range from 1.15 to 1.40.
- Type 3: Black EPDM granules, which are cubical, sharp-cut particles whose dimensions are outlined in the particle sieve analysis. Their specific gravity should range from 1.30 to 1.50.
- Type 4: Black rubber particles for special applications produced from various feedstock other than SBR, NR or EPDM. It is also possible to provide black rubber particles in a blown or cellular form.

The quality of the black rubber granules depends on the care taken in separating the different types of feedstock before grinding them. The chemical composition of the black rubber particles may vary with feedstock and will affect their compatibility with the binder system. Rubber/binder compatibility should be verified prior to installation.

3.5.4b Colored Rubber Particles

Colored rubber particles are typically an EPDM or SBR product. They are designed for application in recreational surfaces and are categorized into two classifications:

- Type 1: Colored EPDM rubber granules made from virgin compounds are the most frequently used because of their better resistance to weathering and outdoor aging. Their environmental resistance to ozone, oxidation and water is excellent. The granules are available in different colors because nearly any color can be incorporated into the formulation of the virgin compound. The EPDM polymer content of the compound should be no less than 20%. More meaningful for quality are the tensile strength and the elongation at break of the material. The specific gravity is between 1.40 and 1.60. The hardness (Shore A) should be within a range of 55 to 70. EPDM rubber granules must be proportionally cubical, sharp-cut particles whose dimensions are specified in the particle sieve analysis.

Type 2: Colored SBR granules are not made from virgin compounds. These granules are made from colored feedstock, depending on availability. The specific gravity could range from 1.15 to 1.50.

It is important to have a blend of various granule sizes to ensure the development of a dense mat or spray coating. Surfacing systems may specify a particular sieve analysis; check with the manufacturer of the system to be applied.

3.5.3c Premanufactured Rubber Products

Two forms of premanufactured rubber products are used in track construction: rubber mats and vulcanized rubber mats. A prefabricated rubber mat is a rolled or sheet-like product that is composed of rubber particles and a polyurethane binder. This type of mat is used as a base and surfaced with a polyurethane coating. A prefabricated vulcanized rubber mat contains natural and synthetic rubber, stabilizing agents and pigments and is manufactured to a prescribed thickness and finish texture.

3.6 Surface Evaluation, Testing and Standards

When installing track surfaces, it is important to know the desired performance characteristics. Inherent to this is how much material is going to be used in the surface, the intended thickness of the completed surface and the installed weight of the surface.

3.6.1 Quantity of Material

A track specification should include detailed requirements for the amounts of various materials to be included in the construction of the track surface. The weights and quantities of materials should be listed. Track surface manufacturers should provide materials requirements needed to achieve the specified results. During construction, it may be advisable for the owner or owner's representative to verify that the material delivered to the site and installed is consistent with what is listed on the bid.

3.6.2 Thickness of Surface

The thickness of the finished track surface should be specified in the construction documents. ASBA recommends a minimum thickness of 1/2" (12.7mm). However, ASTM International's (formerly the American Society for Testing and Materials) ASTM Standard F-2157-09, *Standard Specification for Synthetic Surfaced Running Tracks*, recommends a thickness of 12mm (slightly less than 1/2") and specifies that nowhere on the track can the thickness be less than 10mm, which is slightly more than 3/8" (Class B & C). Any area with less surfacing is subject to spike penetration to the base, which is a safety concern.

ASTM Class A tracks must meet the minimum thickness stated on the selected surface's World Athletics product certificate. This is in line with the general intent behind specifying a minimum depth, which is to provide force reduction levels acceptable for the sport of track and field, in addition to preventing spikes from penetrating the pavement underlayment. Some surface systems may require more or less depth than others to achieve desired performance characteristics.

An owner may test the finished surface to verify that the track has been constructed as specified. There are instruments available through various synthetic surface machinery manufacturers that can be used to measure thickness. Additionally, laboratories can test surface performance characteristics (both in-situ and in a laboratory) using devices such as the *Berlin Artificial Athlete*.

3.6.3 Surface Weight

The weight per unit area provides a way for the owner to ensure that the relative proportions of materials (e.g., rubber, binder) in the completed track are as specified by the manufacturer of the surfacing system.

The test for weight per unit area is a destructive test, so it requires a sample. During construction of the track, an 18" x 18" independent sample is made on a strippable medium from the same materials, method and technique used in the construction of the track. The sample should be of the same thickness and should have the same texture and appearance as the completed track. The sample is then removed from its medium and weighed on a certified analytical scale or laboratory balance. The weight per unit area is calculated using standard mathematical methods. The weight per unit area of the sample should be at least 95% of the weight per unit area specified by the manufacturer.

3.6.4 Surface Standards

World Athletics (WA), the Deutsche Institute of Norms (DIN# 18035 Part 6) and ASTM International (ASTM F-2157-09) have comprehensive programs for evaluating and classifying sports surfaces. These programs set standards for track surfaces based on their physical properties and establish test methods for evaluating tracks.

Each set of standards is slightly different and may include some or all of the following properties:

- Color
- Depth
- Type of facility: governing body, type of competition
- Events to be marked

The surface properties should include:

- Evenness and imperfections
- Thickness
- Modified vertical deformation
- Force reduction
- Coefficient of friction
- Tensile strength
- Elongation
- Weathering
- Flammability

A facility is not required to meet these standards, except in the case of facilities to be used for international competition under the auspices of World Athletics, which must meet World Athletics standards. Meeting the standards may add to the cost of a facility because of the exacting requirements for construction of facilities to meet some of these standards. Each owner must evaluate the intended use of the facility and the cost/benefit of building a track within standards. A Certified Track Builder, an experienced track contractor and/or a design professional will be able to assist in answering these questions. For more information on these standards, evaluation and testing, contact the relevant organization directly (see Appendix B for contact information).



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Chapter 4

THE CONTRACTOR AND CONSTRUCTION PROCESSES





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Chapter 4—The Contractor and Construction Processes

This chapter will cover choosing a contractor, an overview of track construction processes, and the construction timeline.

- 4.1 Choosing a Contractor
- 4.2 Overview of Construction
- 4.3 Construction Timetable

4.1 Choosing a Contractor

Choosing the right contractor can determine the ultimate success of a track facility project. ASBA encourages owners to seek out a Certified Track Builder (CTB) to help. Track construction is a specialty, and a knowledgeable and experienced contractor can help the owner make the right decisions, resulting in a quality project. Track construction is a highly specialized field within the construction industry. It is vital that the contractor chosen is familiar with current products and construction techniques, as well as with the type of surface to be installed.

How can an owner find a Certified Track Builder or qualified contractor? As the trade association for sports facility builders, designers and suppliers of materials and equipment, ASBA can provide a directory of its members, including contractors who specialize in track construction. In addition, ASBA conducts a certified builder program. These experienced individuals demonstrate extensive specific industry experience and successful projects for at least three years, meeting stringent qualification requirements and passing a comprehensive examination process. Certified Track Builders must maintain current educational requirements and industry knowledge to stay certified. ASBA also conducts an inquiry program, requesting information on behalf of owners from contractors and suppliers.

Only a small number of contractors consider themselves specialists in athletic and recreational facilities. Trade industry magazines may be helpful in locating these specialists. A listing of such periodicals is included as an appendix in Chapter 10. Finally, an owner can contact municipal facilities and/or schools that have recently completed track projects, asking whether or not they would recommend their contractor, their findings during the bid process and the reason for choosing one contractor over another. Other questions to ask include:

- What was the final construction cost or project cost?
- Were there any hidden costs?
- Was the job completed on time?
- Did the job meet the owner's expectations?
- Was the contractor able to solve problems that arose during construction?
- How is the track performing?
- Have there been any post-construction problems and, if so, was the builder responsive in taking care of them?
- How does the track look?

Both experience and reliability of the prospective builder are important. An owner should consider the following, as well:

- How many years has the company been in business?
- How financially stable is the company?
- If it is a relatively new company, what is the work experience of its principals?
- How many tracks has the company built? And at what level of competition?
- How many tracks has the assigned project manager or superintendent built, and at what level of competition?
- Was the company responsible for the complete project, just for surfacing, just for site work?
- What percentage of the track project will be self-performed by the contractor?

An owner should look for individuals or for a company with specific knowledge and experience in track construction. Consider the following:

- Does the company have experience in the type of project contemplated? An owner should look for a company with experience in projects similar in size and scope to the project planned.
- Ask for references and for a complete list of recent projects. If a significant project is omitted from the list of references, there may be a reason for that omission. Call references and ask questions.
- Determine as much information as possible about a prospective contractor's knowledge, experience, workmanship, ability to meet schedules, financial responsibility and accountability. If possible, visit completed projects and talk to owners.
- Get references from design professionals, subcontractors, bankers and bonding companies.
- Ask about a contractor's insurance; have there been any major accidents or claims against the builder?
- Ask about awards and recognition. Has the contractor won any awards for their work? Is the company certified or accredited by any trade organization?
- Check on lawsuits. If the contractor has been or is currently involved in litigation, find out the details. Check with the local Better Business Bureau, or with any local licensing agency, for consumer complaints.
- Ask to meet the individuals who will be involved with the project, particularly the job superintendent. Do they seem knowledgeable about size requirements and orientation of tracks? Do they understand grading, drainage, site preparation and base materials? Are they familiar with different track surfaces? Can they make recommendations regarding specific track surfaces to meet the owner's needs? Are they familiar with amenities and accessories including fencing, lighting and field event equipment? What is their current workload? Can they realistically handle the project within a reasonable time frame?
- Consider communication. Owners want a contractor who listens and responds to their needs. The contractor should be someone with whom the owner feels comfortable, someone with whom the owner can establish rapport. The owner will want a contractor who will build the facility the owner wants. Don't underestimate the value of a good working relationship.
- Ask for proposals in writing and compare them carefully. Ensure that products to be used and methods of construction are identical to your specifications. Even if

construction materials and methods are identical and items included in the contract are consistent, look beyond price when comparing proposals.

- What is included and what is not included in the contract price?
- Who—contractor, designer or owner—is responsible for such items as geotechnical survey, testing, permits, site preparation, electric power, taxes, insurance, removal and replacement of trees or shrubbery or restoration of the construction site access route? Such items, while essential to the project, may or may not be included in the bid; whether or not they are included can significantly affect the contract price and the overall project cost.
- Compare proposed construction schedules, progress payments and guarantees and warranties. Understand what is included in any guarantee—materials, workmanship or both—and for how long.
- Rank the proposals and then attempt to negotiate a contract with the first-choice builder. While price is not the only consideration, if the proposal of the preferred contractor seems high, ask the contractor to review the bid or try to negotiate additional services into the package to make the higher bid more attractive. Remember, the lowest bid doesn't necessarily mean the best value. Best value takes into consideration the contractor's reputation, thoroughness, experience, personnel, quality, reference, and similar factors.
- When offering a public bid for the project, it is extremely important to ensure that products to be used and methods of construction are identical to your specifications. Everything that is to be included in the contract price, such as responsibilities for permits, site preparation, electric power, taxes, insurance, removal and replacement of landscaping and restoration of disturbed areas, must be identical in order to choose a contractor based on lowest bid only.
- Make certain that all documents provide a time frame for construction, how payments are to be made and guarantee and warranty requirements. If something is left out of a public bid, it can adversely affect the project and selection of the lowest qualified bidder. Generally, there can only be further negotiation with the lowest qualified bidder. In most cases, public bid offerings cannot be negotiated like proposals for private work.

Once a contractor has been chosen, confirm the agreement in writing. The contract documents, signed by both the owner and contractor, should be as specific as possible and should include, where appropriate, an agreement, conditions of the contract, drawings and specifications defining the scope of work including labor, materials, equipment and transportation to produce the project. Remember to retain all contract documents.

Once the contract has been awarded to a well-qualified contractor, a preconstruction meeting should take place as the first step in the physical construction process. A contractor should be prepared to deliver project submittals, safety data sheets (SDS) and a detailed construction schedule, while an owner should be prepared to answer questions about billing, lines of communication and site access. It is recommended that a formal meeting take place to set the tone for the project and make sure the project begins on the right foot.

Another option is to hold a partnering meeting. This meeting includes all the elements of a preconstruction meeting but adds a section on building a partnership for the project in which all parties are satisfied with the outcome. A partnering meeting is often conducted by a professional facilitator.

4.2 Overview of Construction

4.2.1 Site Preparation, Grading, Compaction and Drainage

Site preparation includes stripping the site of vegetation and topsoil as indicated on construction documents. Topsoil may be stockpiled and reused for landscaping at the completion of the pavement construction and prior to track surfacing.

Next, depending upon the existing grade, high areas may need to be excavated or low areas filled to establish an appropriate grade. Excavation and/or fill will add to the cost of construction and to the time required for construction. However, it is important to provide a proper grade for the track and surrounding area. The appropriate grade carries the water away from the track and other critical areas while blending the track into its surroundings. Grading is critical to avoiding erosion and the need for excessive ongoing maintenance. In establishing the grade, the acceptable subgrade material must be compacted in accordance with the geotechnical survey. Care must be taken to ensure that the subgrade is compacted uniformly over the entire site.

Proper drainage is critical to ensuring a facility's longevity, including both the subsurface and surface drainage. It is important not to cut costs or take shortcuts when it comes to drainage. Subsurface drainage, if required, usually involves the construction of an interceptor drain within the planned facility. Surface drainage collects and redirects surface water. Water collected in both subsurface and surface drains may be retained on site or moved away from the facility to an approved drainage system. Check local statutory requirements.

For more information on site preparation, excavation and fill, drainage and subgrade construction, see Chapter 2.

4.2.2 Constructing Concrete Curbs and Perimeter Edging

Concrete curbs provide several benefits, help maintain the integrity of the pavement edge and put forth a better finished project. Concrete curbs:

- Strengthen the pavement surface
- Minimize maintenance
- Create a visual border for the track and/or a mow or maintenance strip
- Confine flexible pavement
- Improve subbase compaction
- Control dimensions and elevation grades

For more information on curbs and perimeter edging, see Chapter 2.

4.2.3 Asphalt Guidelines

This guideline is intended to assist owners, contractors, design professionals, and hot mix asphalt suppliers in the design and installation of hot mix asphalt concrete to meet exacting sports construction standards. Because of the wide variation in climate, construction methods, site conditions and materials availability, the services of a qualified and experienced design professional should be obtained before use to ensure suitability for a specific project.

4.2.3a Qualified Suppliers

Hot Mix Asphalt (HMA) shall be manufactured from a state approved/certified HMA manufacturing facility. Work consists of one or more courses of HMA constructed on a properly prepared foundation. The low-volume (aka athletic use) asphalt concrete consists of a mixture of dense (aka well) graded aggregate and specified type and grade of asphalt binder. The manufacturing facility shall be capable of producing HMA in accordance with the following requirements and all applicable local agency specifications on an ongoing and consistent basis.

Ensuring uniform material is produced and selecting the vendor for these asphalt projects will require timely submittal of documents and qualifications to the satisfaction of the owner. Contractor/material supplier shall demonstrate the existence of the following documents:

1. Approved vendor certificate for the locality (state/county/city, et. al.) where work is being done,
2. Quality control manual for material production oversight and testing measures being performed both at the asphalt plant as well as on the job site, and
3. List/organizational chart showing personnel responsible for use of equipment and actions of the crew on the grade while paving and compacting asphalt.

Calibrated equipment and qualified personnel must be always accessible during the construction of the HMA. The contractor installing the HMA shall provide the necessary equipment, materials, and labor to complete the job acceptable to the owner and in accordance with applicable contract documents. Variations in the size and amount of equipment will depend on the size of the area being paved.

It is imperative that all documents list a “Person-in-Charge” who is responsible for the oversight of the previously listed activities. This individual will be the point of contact for the owner and shall work with the owner to ensure timely project completion and specification compliance. This individual shall be knowledgeable in all aspects of asphalt design, production, and installation and shall be an employee of the contractor installing the asphalt, even if the HMA is being produced and supplied by a separate vendor.

This individual shall be knowledgeable in all aspects of asphalt mix design, production, and installation and shall be an employee of the company holding the contract with the owner, even if the HMA is being produced and supplied by a separate vendor. This person shall have authority to take corrective actions needed to make the athletic asphalt fit for use.

For the asphalt installation as well as aggregate base, it is best practice for the owner and or owners’ representatives to review and approve all data and reports prior to the application of athletic surface materials. This includes design submittal, construction observation, and quality control and assurance sampling and testing.

4.2.3b Definitions

1. Surface Course—The low volume surface/wearing course shall be installed uniformly, to all finished lines and grades, smooth, durable, impervious thus protecting lower layers, and stable. Workmanship of the finished surface course shall be of the highest industry standards (NAPA, AI, ASBA, and NHI references) and applicable to sports

surfaces prior to acceptance by the owner. The surface course shall be built with a fine-graded, ½” maximum aggregate particle size (Nominal Maximum Aggregate Size of ¾”) or smaller. Surface course shall be installed to a minimum average 1-½” compacted thickness; ensure that no thickness of less than 1- ¼” compacted thickness is placed.

2. **Leveling Course**—The course and location of the recreational area that requires placement of a variable thickness of HMA to ‘true up’ the area prior to placement of the surface course. This course has a Maximum Aggregate Size (MAS) no greater than that of the surface course.
3. **Base Course**—The lower courses of the pavement structure below the surface and leveling course with a MAS of between ¾” and 1”. Base courses shall not be allowed to remain without the surface course placed over an extended period and as approved by the engineer. The base shall be kept clean and must be completely dry before proceeding. If the minimum thicknesses shown above cannot be met, then install surface mixture as base course.
4. **Tacking/Priming**—The process of applying one coat of emulsified asphalt to all horizontal and vertical surfaces of either an existing pavement for an overlay or between lifts while building an improved or new structure (tacking), or upon the aggregate base (priming).

Designer Note: For other low-volume, fine-graded asphalt mixtures, if permitted by the Engineer, follow the National Center for Asphalt Technology recommended guidelines shown below for minimum lift thicknesses. [Commentary: Absolute minimum is typically achieved by specifying ¼” greater than listed below.]

HMA Mixture NMAS	NCAT Fine Graded HMA Compacted Thickness, minimum
⅝” (9.5 mm)	(3) x ⅝” = 1- ⅛” (1-½” preferred)
½” (12.5 mm)	(3) x ½” = 1- ½” (2” preferred)
¾” (19.0 mm)	(3) x ¾” = 2- ¼” (3” preferred)

4.2.3c Materials

- Tack Coat and Prime Coat: AASHTO M140 or M208 (Reference the Asphalt Institute MS- 19 for Handling, Storage and Application criteria: typically, 0.05 gal / sy to 0.15 gal / sy (0.02 gal / sy to 0.05 gal / sy residual AC) depending on the existing surface condition; see Surface Preparation section and current AI and NAPA publications).
 - Tack Coat: SS-1, SS-1h, CSS-1 or CSS-1h diluted with an equal amount of water, or agency acceptable product.
 - Prime Coat: Prime Coat materials of MS-2, CMS-2, or HFMS-2s.
- Hot Mix Asphalt (Low Volume):
 - Aggregates, mineral filler, and asphalt binder shall meet or exceed the requirements of local specifications for asphalt pavements placed under this

- contract for qualities and types. The coarse aggregate shall be sound, angular crushed stone, crushed gravel, or crushed air-cooled blast furnace slag (not steel). The fine aggregate shall be well graded, moderately sharp to sharp (angular) sands. No aggregates known to cause rust spots or pop-outs (steel slag, iron pyrite, and / or dust balls) are allowed in the asphalt. No recycled concrete is allowed in any of the asphalt mixtures.
- All HMA mix designs shall be performed in accordance with the Asphalt Institute Manual Series #2 (MS-2), current edition. The HMA mix designs developed shall meet the requirements of one of the following for compactive effort in the laboratory:
 1. Marshall, 50-Blow,
 2. Superpave, 50-Gyratation, or
 3. Hveem, Low Volume Mix.
 4. Alternate **Low Volume** Asphalt Mix Designs may be allowed with the Engineers approval prior to time of bidding.
 - HMA mix designs shall be performed by qualified personnel with proven past experience and successes in the mix design and quality control of asphalt production. Resumes of the signing “individual-in-charge” may be required by the owner and shall be supplied if requested. The design shall meet the following requirements and be less than 24 months old. However, the mix design method used shall be the Contractors option, as stated previously, based on various methods which currently exist around the nation. A completed design shall be signed by a professional engineer and require submittal of documentation as detailed within this specification. This is required by the owner for the producer to demonstrate knowledge of asphalt mix design and production criterion needed to supply athletic asphalt.
 - Bidding documents shall include the Contractors proposed Asphalt Mixture Design sheets. (Reference Mix Design Submittal Checklist in Appendix H.) Designs will be for HMA to be placed for each of the uses anticipated on each project; patching, base, leveling, and/or surface course. Different asphalt suppliers shall require different design submittals. In addition to mix designs, submit Asphalt Placement Work Plan, indicating paving pass widths, paving directions, site access, and timing/coordination of athletic equipment installation (tennis net posts, vault boxes, fencing, etc.). Bid document except the asphalt mix design shall be submitted within 10 days of project award; asphalt mix design cover sheet which summarizes all materials and optimum asphalt content is required with project bidding.
 - All submitted HMA mix designs shall have a completed Mix Design Submittal Checklist (reference Appendix H) and contain at a minimum the following information:
 - All aggregate gradations and quality measurements
 - Plot (0.45 power graph) of final aggregate blend
 - Bulk (dry) specific gravity of all aggregates and final blend (Gsb) including worksheets for natural (virgin) as well as reclaimed asphalt pavement (RAP)
 - Statement of asphalt binder (PG) being used in asphalt mixture
 - Optimum % asphalt binder (Pb)
 - Mix air voids at optimum (Va)
 - Bulk specific gravity of mix at optimum (Gmb)
 - Theoretical maximum specific gravity at optimum (Gmm)

- Voids in the mineral aggregate (VMA) and voids filled with asphalt (VFA)
- Dust to total AC ratio
- All design data and associated design curves

Mix Design Method Requirements

Measures	Superpave	Marshall	Hveem
Stability, lbs.	n/a	1,200 min.	30 min.
Flow, 0.01 in.	n/a	8 to 16	n/a
Swell, in.	n/a	n/a	0.030 max.
Air voids @ optimum AC with 2-hr. aging (cure-time)	3.5%	3.5%	3.5%
VMA based on NMAS			
#4 (4.75 mm)	16.0 min.	16.0 min.	16.0 min.
3/8" (9.5 mm)	15.0 min.	15.0 min.	15.0 min.
1/2" (12.5 mm)	14.0 min.	14.0 min.	14.0 min.
3/4" (19.0 mm)	13.0 min.	13.0 min.	13.0 min.
VFA	73 to 80	73 to 80	73 to 80
Dust to effective AC (design & production)	0.7 to 1.3	0.7 to 1.3	0.7 to 1.3
Tensile strength ratio	80% min.	80% min.	80% min.

In addition, all design methods and measures listed above shall meet the following criteria for design and throughout production.

- Base mixes shall have a minimum of 45% passing the #4 sieve.
- Surface and leveling mixes shall have a minimum of 45% passing the #8 sieve.
- Coarse aggregate fraction shall have a minimum of 85% / 75% crushed faces.
- Fine aggregate angularity (AASHTO T-304, Method A), shall be $\geq 40\%$ with no more than 20% natural sand allowed.
- Mix designs shall include a breakdown factor, increase to minus #200, introduced during the design stage to mimic production values.
- VMA is based on the aggregate bulk (dry) specific gravity, G_{sb} , as determined by AASHTO T-84 and T-85.
- Performance Graded (PG) binder shall meet typical agency specification for new construction low volume roadways (reference LTPPBind, current edition; 98% reliability).
- Reclaimed Asphalt Pavement (RAP):
 - a. May be used up to 20% in the HMA binder/base and leveling courses.
 - b. Typically, 0% for HMA surface course due to potential blemishes forming from unknown or varying aggregate qualities; however, may be used up to 15% in the

- HMA surface course only if approved by the architect/engineer of record.
 - c. Requires a signed and notarized letter stating that no pyrite, steel slag, or aggregates known to rust or deteriorate are within the RAP being utilized is required.
 - d. Requires the Gsb of the RAP to be determined and used in VMA calculations, not the Gse. The RAP Gsb shall be determined after running the RAP Gmm and then from calculating the RAP Gse minus 0.1 for high absorptive aggregates and RAP Gse minus 0.05 for low absorptive aggregates.
 - e. Requires that all RAP shall be crushed and screened over a 5/8" screen deck or smaller for binder/base and leveling courses and 1/2" screen deck for surface mixtures—no exceptions.
- Reclaimed Asphalt Shingles (RAS) are not allowed in athletic facilities asphalt.
 - Current quality control testing of the mixture, aggregates, and RAP proposed to be used on the project shall be submitted to the owner prior to acceptance of the proposed mix design.

4.2.3d Testing

- Testing required to validate or control the mix supplied is the Paving Contractor's responsibility and will be included in the bid cost for providing these HMA items. Daily maximum theoretical specific gravity (Gmm) values must be made available to the Contractor's density technician for verifying in-place density within four hours of start of production. Asphalt content, gradation, and bulk specific gravity (Gmb) testing shall be performed on the first day of installation for each product used, then done a minimum of once every 400 tons of HMA supplied or every third day for low tonnages that when added together successively do not equal 400 tons. Acceptable average measures are made by use of a correlated nuclear density gauge, a correlated Pavement Quality Indicator or PaveTracker (non-nuclear) or by cutting (4) cores per lift, per day and testing per AASHTO T-166, Method C. Additional testing shall be performed on any given day once 400 tons of asphalt is placed on that day.
- The average sub-lot (daily or 400 tons; whichever is less) in-place density measure for surface course mixtures shall be 94.0% of Gmm with no value less than 92.5% of Gmm. Base and leveling installation of asphalt shall meet local DOT specifications for in-place density measures or average of 92.0% of Gmm, whichever is greater. Surface course longitudinal joints shall be measured directly upon the joint, centered upon by core or density gauge, and shall meet the mat density requirements. Base and leveling course longitudinal joint density measures shall achieve between 95% - 102% of maximum achievable individually, with an average of 98% on any given day.
- Process control testing shall be in accordance with state standards for frequency and methods where the work being performed is done with a minimum of testing meeting the above QC requirements.
- Process Control Voids and minus #200 gradation shall target mix design with no test outside plus/minus 1.0% and VMA shall target the asphalt mix design value or greater, with no test value less than minimum allowed minus 0.3%.
- Print outs of ingredients used shall be supplied for each run of asphalt; data logger or computer screen shot. Print outs shall be supplied daily with the final load of asphalt ticket.

4.2.3e Recommended Means & Methods

Equipment

- Tack/prime coat distributor truck must have an insulated tank, heating system, and a calibrated distributor capable of maintaining a uniform application of emulsified asphalt under pressure throughout the area to be paved. This requires a pump in good working order, full circulating spray bars, and free flowing nozzles. Small, isolated areas may be tacked with a wand.
- Trucks shall have smooth, clean, and tight metal beds that do not have mixture sticking to the truck bed and from which the entire quantity of HMA can be discharged smoothly into the spreading equipment. Trucks shall have a tarp and insulation as needed to protect the asphalt mixture from wind, rain, and cold temperatures. Trucks for hauling asphalt mixture shall be in good, safe working condition.
- Paving equipment must be capable of placing, spreading and finishing courses of HMA to the specified thicknesses. HMA shall be free of marks, segregation and be placed to the required uniform elevation with a smooth texture not showing tearing, shoving, or gouging. Auger extensions are required while pavers are extended beyond the basic screed width. Paving equipment shall be self-propelled and capable of maintaining the line and grade shown on the plans with suitable electronic equipment. The screed shall be straight and true with no bow and utilizing a vibratory screed. Hand work shall be minimized to ensure the best possible finished surface. It is recommended that paving equipment be equipped with sonar pods or no contact skis for sports asphalt construction. Additionally, it should be equipped with automatic slope control to maintain required tolerances. Finally, paving equipment should have fully functional screed heaters and joint preheaters.
- Rollers shall conform to the manufacturer's specifications for all ballasting. At least one vibratory roller shall be required for each project with two rollers required as a minimum. (Three rollers shall be required when tonnage is greater than 300 tons / day.) Rollers shall be of good condition and capable of compacting the HMA to the minimum in-place density required by this specification.

Surface Preparation

1. Repair pavement failures and perform crack repair according to specification requirements prior to HMA installation.
2. Cold-milling and/or grinding may be necessary to ensure that the asphalt edges at concrete abutments such as approaches, sidewalks, curbing, and drainage basins have smooth transitions.
3. After site review, detail whether wedge milling is necessary to assure positive drainage and transition. Install leveling course, if required, on the project per the site details and quantities shown on the plan sheets.
4. Existing surfaces to receive HMA must be clean prior to the installation of any portion of the work. Clean the surface on which the asphalt concrete is to be placed, and keep it free of accumulations of materials that would contaminate the mixture, prevent bonding, or interfere with spreading operations. Methods used may include but not be limited to the use of a sweeper that can wet and vacuum the area free of dirt and debris, clay, and dust, or any other foreign material.

5. Any oil or grease spots shall be scraped and treated to prevent bleeding through the tack coat. Bad oil spills may require removal with a wire brush or other suitable tool. Maintain clean pavements prior to applying emulsified tack coat. When approved sub-grade or pavement courses previously constructed under the Contract become loosened, rutted, or otherwise defective, the Contractor must correct the deficiency according to the contract item or items involved before the spreading of a subsequent pavement course.
6. If subsequent lifts are laid beyond 24 or 48 hours, apply tack coat at the diluted rate of 0.05 gal / sy (0.02 gal / sy residual AC) over newly constructed asphalt leveling or base mixes, 0.10 gal / sy (0.04 gal / sy residual AC) over existing asphalt pavements and 0.15 gal / sy (0.05 gal / sy residual AC) over milled surfaces. The higher rate shall be used on dry and brittle surfaces. All vertical edges abutting proposed asphalt surfaces shall receive a tack coat. Excessive asphalt applications, drooling, or pooling shall be swept with a broom to ensure proper bonding of the HMA. Install the HMA after the asphalt emulsion has 'broken'; i.e. turned from a brown to a black color, indicating water has evaporated. If pick up occurs, wait until emulsion cures.
7. If shown on the plans, apply prime coat at the diluted rate of 0.30 gal / sy over newly placed aggregate base course prior to the installation of the base asphalt.
8. Install tack / prime coat during appropriate weather conditions and protect the tack / prime coat from traffic so as not to wear and track. Allow the tack / prime coat to 'break', i.e. turn from brown to black prior to installation of the HMA.
9. Perform work in appropriate weather conditions that are dry with no rain, snow, or other forms of precipitation falling or imminent (anticipated during installation of the HMA).

Pavement Placement

1. Install HMA which shall generally arrive on the project between 270–300° Fahrenheit (see producer recommendation) asphalt in accordance with above weather conditions and with a temperature of 50° F. and rising for all asphalt lifts.
2. Establish an acceptable rolling pattern with the assistance of a density technician on the first day of construction. Record temperatures, equipment, rolling pattern, and in-place density results throughout the project.
3. Surface course longitudinal joints shall be smooth and true; no deviation from level and true as required of the mat will be allowed. Detail and submit to the owner a paving plan on the site plan sheet prior to placement of asphalt.
4. The entire athletic surface course shall be paved on the same day. The timing and process should be discussed with and approved by the owner before proceeding with the work. If a cold seam will occur, it must be agreed to with the owner in advance, such as near or at a planned saw and seal joint or under the fence line.
5. Rolling shall start as soon as the HMA can be compacted without displacement. Rolling shall continue until the HMA is thoroughly compacted and all roller marks have disappeared. Compact the HMA to a minimum in-place density of 94.0% of the Theoretical Maximum Specific Gravity, Gmm. (Reference density pay factor table.)
6. Smoothness shall meet the requirements of no greater than ¼" in 10 ft. for base and leveling courses and ¼" in 10 ft. for surface course.
7. Thickness of the overall mat shall be within ¼" (surface course, no minus) of the specified plan thickness at all locations. However, the yield for the day and for the entire site shall meet calculated theoretical based on 94% of Gmm supplied from the contractor's mix design and daily test values.

8. Paving joints (install/build control joints per the architectural drawings)

- Minimize construction, longitudinal, and transverse joints left open for an extended period.
- Construct longitudinal joints by paving in a hot fashion with a temperature of not less than 220°F to ensure maximum performance.
- Compact all joints to provide for a neat, uniform and tightly bonded joint that will meet both surface tolerances and density requirements.
- Cut straight and true (vertical) construction or transverse joints if the material has cooled to less than 220°F prior to the placement of the next pass to ensure the best performing joint possible.
- Off-set joints a minimum of 6" between lifts of asphalt.

9. Allow positive drainage off of the athletic facility and towards drainage outlets. Any ponding of water is not acceptable and shall require correction or replacement at the Contractor's expense and as directed by the Engineer. Flood pavement areas as directed by the owner and in the presence of the architect/engineer and surfacing contractor to determine positive drainage acceptability. Review ASBA's "Water on Tracks" position paper for more details.

10. Protect the HMA until such time that coating can be placed upon the properly compacted asphalt, particularly during other construction activities between asphalt installation and athletic surface installation.

11. If excessive segregation is occurring during placement operations, the Contractor will investigate the cause(s) and make appropriate changes to the satisfaction of the owner. (Reference AI MS-22.)

12. Excessive leveling and smoothness correction required to be performed by the surfacing contractor shall be the responsibility of the paving contractor.

Work Timeliness

HMA Full-depth pavements/overlays (a/k/a Resurfacing): The repairs, HMA overlay, and coating shall be accomplished in such a manner as not to unduly delay the progress of the project. Every attempt should be made to complete the surface course placement process in one continuous placement with no cold joints. The timing and process should be discussed with the owner before proceeding with the work.

Site Specific Identification, contractor shall:

- Remove all waste materials from the site and dispose of according to local ordinances.
- Complete all work in compliance with the American Sports Builders Association (ASBA) requirements or as modified here, whichever is more restrictive.
- Notify owner when work is complete.
- Supply owner with Notarized Certificate of Compliance for all products used on the project.
- Supply owner with yield calculations for all products used on the project. (For example, placement of 1,300 sq. yds. of Hot Mix Asphalt, 1-3/4" compacted thickness will require 128 tons when the unit weight = 150 pcf.)

References:

- *Asphalt Institute, Lexington, KY [MS-2, Asphalt Mix Design and MS-22, Asphalt Construction.] National Asphalt Pavement Association, Lantham, MD. [Athletic Asphalt.]*
- *National Center for Asphalt Technology [HMA Materials, Mixture Design, and Construction.]*

4.2.3f Asphalt Acceptance

It is the responsibility of the asphalt contractor or the general contractor to supply a base that meets the specifications of the designer and/or the rules of the sport. It is far more difficult and costly to wait until the surface has been installed to identify issues with slope, planarity, elevation or drainage. Additionally, unless the surfacing contractor is also responsible for the paving, it is inappropriate to hold that contractor responsible for such issues if the pavement has not been tested and corrected prior to surfacing. It is highly recommended that track projects include testing to demonstrate adherence to specifications and, if necessary, corrective work to achieve compliance at the cost of the asphalt contractor or the general contractor. These tests should be completed prior to the installation of track surfacing. An independent testing firm should check the dimensions, slopes and tolerances of the asphalt pavement. Results confirming compliance should be submitted to the surfacing contractor and owner prior to mobilization. If the initial assessment shows noncompliance, the necessary corrective work should be executed and the pavement retested prior to the commencement of surfacing. A few more points to consider:

- The asphalt mix design should be approved by the design professional, the surfacing contractor and the asphalt contractor. Tests that verify that the mix, as installed, meets the approved design should be provided.
- The surface course of asphalt shall be installed so that on a localized level, there shall be no bumps or depressions beneath a 10' straight edge exceeding 1/4 of an inch. Particular attention should be paid to seams and joints in the asphalt surface. The intent is to ensure the safety of the athlete and provide an even surface. Since the resilient running surface to be installed over the asphalt has tolerances with respect to thickness, it is important that the asphalt surface meets the tolerance described above. It is important to remember that the use of extra track surfacing materials to correct planarity deficiencies is not recommended.
- The installed pavement must be at the correct elevation at all curbs, channel drains and other adjoining surfaces. Incorrect areas should be repaired or reinstalled to correct elevations.
- All bond breakers, such as petroleum-based products like oil, hydraulic fluid and gasoline, must be removed.
- The asphalt should be flooded to check for low areas and surface drainage. Out-of-tolerance areas should be repaired or reinstalled.
- Prior to surfacing, the asphalt should be cured in accordance with the recommendations of the specification writer and the manufacturer of the surfacing system. This is typically 28 days based on depth of asphalt and oil content.
- If the track requires a certification that includes a topographic survey, the survey should be performed prior to surfacing.

Corrective work should be accomplished using appropriate methods, such as:

- For high spots:
 - grinding
 - heating and rolling (only if approved by the surfacing contractor)
- For low spots:
 - removing and replacing
 - keying into existing asphalt
 - hiring the surfacing contractor to apply leveling materials approved by the surfacing manufacturer

Because of the relatively rigid specifications and tolerances for asphalt pavements intended for track installations, it is recommended that only asphalt paving contractors with prior track building experience be considered.

4.2.4 Concrete Guidelines

Concrete is the product of mixing crushed stone or gravel and sand with Portland cement, making it an extremely rigid pavement. As such, it is also exceptionally durable, meaning low maintenance costs after initial construction, though initial cost may be somewhat higher than that for asphalt due to both installation and additional preparation involved with applying surface materials to concrete. If problems occur, concrete is more difficult to repair than asphalt.

Because of its durability, concrete may be useful on a site with poor soil or in a geographic region with extreme freeze-thaw cycles. Concrete pavement will be resistant to settling, heaving and vegetation encroachment; however, some cracking can be expected.

Depending on site conditions, various kinds of concrete pavement may be suitable for running tracks: nonreinforced concrete, reinforced concrete and post-tensioned concrete.

4.2.4a Materials

Concrete should comply with the standards of industry sources such as ASTM International or the American Concrete Institute. These standards address blending, mix percentages and similar issues. Depending on the size and the complexity of the facility under construction, the construction documents may detail the characteristics of the concrete to be used. The aggregate used in the concrete mix should be clean, hard, durable and graded within specified limits.

4.2.4b Reinforcement

Reinforcement materials are commonly incorporated in concrete construction to provide additional stabilization and crack resistance. Proper reinforcement will extend the life

of concrete pavements and structures. Reinforcement materials may include steel mesh, reinforcing steel (rebar) or specially developed reinforcing fiber.

4.2.4c Post-Tensioned Concrete

The use of post-tensioned concrete for tracks is continuing to evolve. It's highly recommended that an ASBA Certified Track Builder be involved throughout the design phase of a project using this construction method to ensure that ASBA recommendations and tolerances are adhered to.

This method of construction serves the owner with the longevity of a concrete foundation without the concrete slab maintenance issues inherent with standard reinforced concrete construction. The surfacing system installed over the concrete slab shall be maintained as usual. The track oval and chute or chutes are installed as one monolithic unit. This eliminates any expansion joints and therefore there are no un-tensioned joints.

Any cracking that may occur is minor as the tensioning cables hold them together and prevent cracks from attaining any significant size. This option is becoming more popular in areas that have a lack of any asphalt paving contractors and/or are located long distances from any available asphalt batch plant.

The post-tensioned concrete pavement is reinforced by encapsulating the sheathed cables in the concrete. These cables are then stressed and locked in compression.

A post-tensioned concrete pavement is constantly tensioned and therefore remains a rigid slab. There is an increased resistance to settling and/or heaving, making this an ideal alternative for unstable soils. Concrete can be installed using a laser-controlled screed or an A-frame screed using laser technology to set such. This provides a precise cross-slope.

If installed correctly, post-tensioned concrete provides the necessary inside curb detail inherent in the construction process for a synthetic turf installation. The strength of the concrete mitigates any edge curl. The concrete construction will support any future interior field construction.

Due to the nature of this process and the tremendous pressure generated, this type of installation should only be attempted by a contractor experienced in this type of construction and supervised by a Certified Track Builder.

4.2.4d Expansion Joints

An expansion joint is an area of separation between cast-in-place concrete segments and is filled with a material that allows for expansion and contraction of the concrete. Expansion joint materials may include sealants, fiber, wood and elastomers. New materials are constantly under development. Expansion joints are used in curb and sidewalk construction. However, because expansion joints may reopen, they are not recommended for use within the track or field events where surfacing material will be applied.

4.2.4e Control Joints

A control joint is a saw-cut or troweled joint partially through the pavement. Because concrete is subject to expansion and contraction due to temperature changes and curing, it will crack. A control joint provides a place for the concrete to crack and, at least to some degree, minimizes cracking in the remainder of the slab. A control joint may reflect through the track surfacing. Caulking has been found to push up in the areas of cold/hard winters due to moisture pulling through cracks from below.

4.2.4f Mix Design

A concrete mix design for track construction as specified by the design professional should result in a mix that is uniform and workable and should produce concrete with adequate compression strength. Concrete mixes should be tested to ensure compliance with the mix design requirements.

4.2.4g Placement of Concrete

Concrete should only be placed on the specified subgrade and base (see Chapter 2). It is important that before the concrete is placed, the base is free from contamination and has no free water on the surface. It is also highly recommended that the base be proof-rolled or tested prior to placement of concrete.

Ideally, concrete should be installed only when the air temperature is above 35 degrees F, or below 85 degrees F. When temperatures are outside the optimal range, installation techniques may be modified to enhance the curing process.

Typically, concrete should be at least 4" thick; however, the thickness of specific concrete components of the track project may vary depending upon the subgrade materials, the freeze-thaw activity in the location of the project and other factors. Curing compounds are not typically allowed by the track surfacing contractor. A design professional, geotechnical engineer, experienced contractor or consultant can advise as to specific requirements for the site.

Concrete is formed, spread, consolidated, screeded, bull-floated and finished. Those areas not receiving track surfacing should be finished according to the owner's recommendations. The final finish texture in the areas where surfacing material is to be applied should be in accordance with the surface manufacturer's recommendations. Typically, areas to be surfaced should have a light to medium broom finish to improve the mechanical bond for the track surfacing. Concrete pavements should be finished so that the surface does not vary by more than 1/4" in a 10' straight edge when measured in any direction.

4.2.4h Curing

Concrete slabs must be cured before they are surfaced with a coating system. To cure, concrete slabs must be kept continuously moist for approximately seven days (time varies according to the surfacing system to be applied; consult the manufacturer) by a covering of polyethylene film, burlap or curing paper, or by sprinkling or ponding, followed by additional

curing. Curing compounds should not be used as they may affect the bonding of the surfacing system to the concrete.

Depending on the concrete formula, thickness of the slab, temperature, humidity and other factors, it may take anywhere from 21 to 60 days for the slab to reach sufficient compressive strength to be referred to as “cured.”

“Cured,” however, is not the same as dry. Long after the concrete is cured, it will continue to emit water vapor as it dries. Installation of a surfacing system should not begin until the concrete has been tested and determined to be dry in accordance with specifications provided by the manufacturer of the surfacing material; otherwise, the surface may not bond or adhere properly.

All tests should be conducted by an independent third-party testing agency. Results indicating acceptable levels of vapor emissions should be supplied to the surfacing contractor in writing prior to the commencement of surfacing. Testing may be accomplished in accordance with any of the following ASTM standards:

- ASTM F-710-11–*Standard Practice for Preparing Concrete Floors to Receive Resilient Flooring*
- ASTM F-1869-10–*Standard Test Method for Measuring Moisture Vapor Emission Rate of Concrete Subfloor Using Anhydrous Calcium Chloride*
- ASTM F-2170-09–*Standard Test Method for Determining Relative Humidity in Concrete Floor Slabs Using in situ Probes*

It is important to note that each athletic surfacing manufacturer will have its own requirements and acceptable test methods, and these should be strictly adhered to.

Once curing is completed, concrete may be prepared for the application of a surfacing system. This is typically after 28 days. Concrete surfaces may require surface preparation prior to the installation of the track surface. Such preparation may include surface grinding, shot blasting and the installation of a vapor emissions coating. Always install as per vapor emissions manufacturer and track surfacing manufacturer recommendations. Naturally occurring salts in concrete may prevent surface systems from adequately bonding to the pavement. Generally, the concrete pavement is acid-etched or shot-blasted to clean it and to neutralize the salts in the concrete. Next, the pavement is primed with a primer approved by the manufacturer of the surfacing system.

4.2.5 Installing Fencing, Lighting and Other Amenities

Typically, fencing, lighting and amenities in the immediate vicinity of the track should be installed prior to application of the track surfacing. It is important to protect the track and restrict access with fencing.

Fencing and other amenities, however, should be installed outside of a recommended safety zone from the track. World Athletics and NCAA recommend a 1m safety zone both inside and outside of the running track. NFHS rules do not address a clear zone. In many cases in practice, tracks are designed and constructed such that fences, bleachers, light poles

and other obstructions are much closer than 1m from the edge of a running lane, runway or landing pit.

ASBA recommends that, at a minimum, a 4' fence be placed around the outside perimeter of the track *at least 1m from the outside edge of the running lane*. A 1m clear zone inside and outside the running track is recommended.

4.3 Construction Timetable

The time required to construct a track facility depends upon the permits required, site conditions—especially the need for excavation and/or fill—the size and complexity of the facility under construction, the type of surface being installed, the resources and current workload of the contractor and the weather during the period of construction.

Planning, design and permits may take as little as a month or two or as much as several years, depending upon the size, scope and location of the project. As a general rule, permitting may take 30 to 90 days, while the bid process may take an additional 30 to 90 days. Bidding for a project should not begin before necessary environmental and other types of approvals are obtained because revisions required by local authorities may lead to costly change orders.

For a track and field event area, an owner can count on an actual construction period of up to five months after permitting and bidding is completed. Stadium-type construction can take upwards of two to three years.

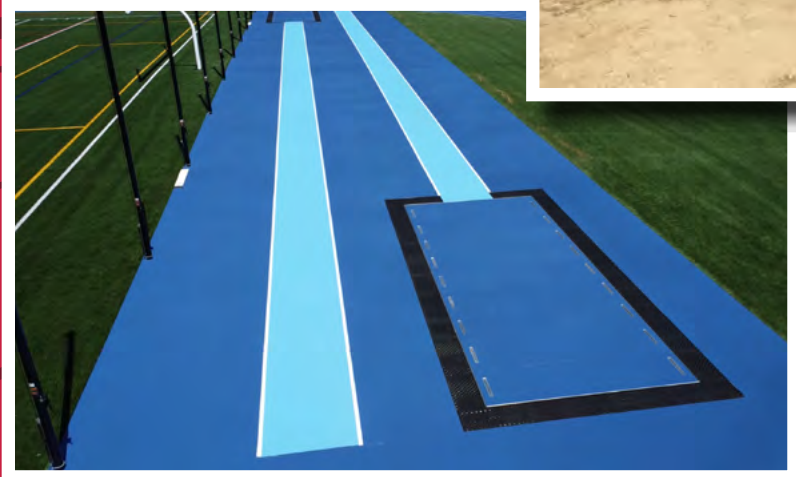
Construction will be delayed by bad weather, especially rain. Overcast skies and cooler temperatures, even without rain, can delay the construction schedule. At a minimum, any track construction schedule should anticipate lost time due to poor weather or other unforeseeable construction difficulties. Consider the cure time required for the specific base composition prior to installation of synthetic surface materials. This may add up to 30 days to duration of the project.

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Chapter 5

FIELD EVENTS AND IN-GROUND EQUIPMENT



Chapter 5—Field Events and In-Ground Equipment

This chapter will cover the following specific issues for field events.

- 5.1 General
- 5.2 Long Jump, Triple Jump and Pole Vault
- 5.3 Throwing Events
- 5.4 High Jump
- 5.5 Steeplechase
- 5.6 In-Ground Track Equipment
- 5.7 Accessibility to Storage Shelters
- 5.8 Viewing for Spectators
- 5.9 Timing Systems
- 5.10 Sound Systems
- 5.11 Lighting
- 5.12 Fencing
- 5.13 Water
- 5.14 Other Safety Issues

5.1 General

Field event facilities, such as those designed for the long jump, triple jump, pole vault, discus throw, etc., are an integral part of any track and field facility designed for competition. The same care and consideration should be afforded to the planning and construction of these venues as is given to the planning and construction of the track itself. Taken as a group, field events are often more difficult than the track oval to design and construct.

Drawings are provided as a direct reference for the events discussed in this chapter. Aside from the discus and shot put areas, construction and surfacing of field events is very similar to the construction and surfacing of the track. This chapter will briefly review construction of specific events and will focus on placement of events for meet management and on safety information.

5.2 Long Jump, Triple Jump and Pole Vault

5.2.1 Design Considerations

The runways for the long jump, triple jump and pole vault should be constructed so as to provide at least the minimum distance recommended, which may vary from one governing body to another (see Appendix C). Any additional length that can be provided usually enhances the events. The rules for laying out the runways are of utmost importance for reasons of accuracy, for certification of records and, most importantly, for the safety of the athletes. Consider the following guidelines when constructing these types of facilities:

- The length of a runway is not determined from the edge of the landing areas but from the back of the vault box or the foul lines. It is imperative that, in laying runways, correct measurements are followed per governing body recommendations.
- For the long jump/triple jump, the construction of a longer pit with the takeoff board placed closer to the pit is highly recommended. This design is appropriate for athletes at all ages and skill levels. The long jump takeoff board must be close enough so the youngest athlete can reach the pit and elite athletes cannot land too close to the back end. A single long jump takeoff board should suffice for all athletes, provided the landing pit has been constructed at an adequate length.
- The width of stand-alone runways is frequently under-designed. Runways are often constructed at a too narrow 42" wide and the takeoff boards extend out an additional 3" on each side. This is a safety concern as well as a potential area for erosion. Make the runways wide enough to allow for correct placement of the takeoff boards, ideally 5' of concrete or asphalt, the center 52" of which is surfaced (Fig. 5-1).
- The landing pits for the long jump and triple jump may need to have some form of drainage system in order to ensure that the sand does not become a sponge (Fig. 5-2). This will depend on soil percolation rate and expected rainfall.
- The single wide landing pit is more easily manageable as a competition facility (Fig. 5-3) because of the width constraints defined in the rules of competition. The sand of these pits is also easier to level. For training purposes, larger-width landing pits (with multiple runways) accommodate more athletes in a shorter period of time.
- A javelin runway should be placed in areas where spectators can view the event but not be in harm's way. Placement should also consider how the facility is used for other sports or activities to ensure safety. (Fig. 5-7)
- The location of these events is very important. Most facilities will not have a budget to incorporate separate long jump and triple jump runways. However, it is a good idea to incorporate at least two runways for these events in the facility. They should be placed in close proximity to each other.
- Landing pits can be located at each end to make better use of the facility. It is extremely important, however, that the landing areas not interfere with the running of the track events. A landing area placed near start or finish lines can only add to confusion in the staging of the running events and will present difficulty in managing the finish line.
- This same criterion would hold true for the pole vault. A single pole vault runway and landing area is shown in Fig. 2-11. It is beneficial to have two pole vault runways if the facility hosts large meets. Two runways can also allow for two runway directions to account for wind and the orientation of the sun. Again, no landing area should interfere with the running events. Ensure that the vault box is constructed and placed to specification (Fig. 5-5).
- As an aid to coaching, the runways may be marked with bordering lines and with distance marks. Chapter 6 further discusses required and optional markings. Also refer to Figures 5-3, 5-4 and 5-7 for painted runway lines.
- Also note the presence of firm and unyielding surfaces and any obstacles (e.g., light poles, fences) and ensure that they are a safe distance from the landing areas of all jumping events.
- Preferred wind directions for each field event need to be discussed with the facility owner.

- Recent rule changes allow for runways to be built in D-zones, following grade changes. Be sure to check with the governing body rules to make sure the runways are in compliance.

Note: Since soccer has become an integral part of most school sports programs, a facility designed with these three events being conducted either in the D-Zones or outside the oval is preferable.

5.2.2 Type of Sand

A frequently asked question concerns the type of sand that should be used in landing pits for long jump and triple jump events. This is an appropriate question, yet it is difficult to answer. The only association to actually issue a specification for sand is World Athletics. According to the *World Athletics Facilities Construction Manual*, “the sand must (to avoid hardening as a result of moisture) consist of washed river sand or pure quartz sand, without organic components, maximum 2mm granules, of which not more than 5% in weight is less than 0.2mm.”

What can be taken from this specification and used in the general construction of the landing areas for the long jump and triple jump is that for “safety purposes,” the sand must not harden when moist, it must be clean, it must have no organic components, it must be similar to washed river sand, it must not be abrasive and it must be relatively fine (granules not larger than 2mm in diameter). This type of product can usually be found in the various state construction codes for aggregates and is normally similar to masonry sand. USGA bunker sand, where available, meets the criteria.

At a minimum, the sand should be 12” deep. Drainage may be incorporated in the bottom of the pit where natural drainage is not effective. A flush curb, commonly constructed of reinforced concrete or landscape timbers, holds the sand in place.

5.3 Throwing Events

The shot put, discus and hammer areas should be located in close proximity to each other, as many athletes double up in these events, which also tend to share one coach. The throwing pads should be constructed with concrete in accordance with the description in previous chapters (Fig. 5-6). The center of each circle should be clearly marked. Recessed circles should have provision for drainage, either weep holes or saw-cut grooves at the depth of the recess. The shot put landing sector should be composed of compacted aggregate or similar materials. The landing areas for all other throwing events should be composed of grass. All elevations should conform to the applicable rules and regulations for that specific event.

The javelin area should be set in a part of the facility that is level and has grass. The javelin runway should be extended 1m beyond the painted foul line when synthetic surfacing is used (Fig. 5-7).

The tremendous growth in the use of synthetic turf fields at all levels, including high school and at small colleges, raises questions about where to place throwing events when the track surrounds a synthetic turf field. Although practices are changing and increasingly

use landing areas of synthetic turf, before conducting any of these events on a synthetic turf surface, be sure to consult the governing body and the manufacturer. Please note that such activities may void the manufacturer's warranty.

5.4 High Jump

The high jump is usually located inside a fully paved D-Zone, semicircle or rectangle. Consideration should be given to placing the takeoff in such a way that the athlete does not need to look into the sun, which may lead to an improper take off and cause injury. Construction of a fully paved D-Zone (Fig. 5-8) allows for the placement of the landing pad to take into consideration sun and wind direction and is likely to result in better quality underlying pavement. If large enough, it also allows for moving the takeoff around to distribute wear. The semicircle somewhat limits the options for the direction of competition.

5.5 Steeplechase

The location of the start and finish lines, the water jump and the hurdles for the steeplechase must conform to the rules of the relevant governing body. Refer to Appendix C - Track & Field Specifications for detailed rules and recommendations regarding the steeplechase course and the water jump. The steeplechase water jump pit can be constructed either inside or outside the track oval (inside is recommended). There is flexibility as to its placement; however, it must be placed on the curve prior to the finish line.

The water jump pit can be hand-formed for the concrete pours of the sloped bottom and sides. Also, stay-in-place pit forming systems are readily available from many track industry equipment manufacturers. These prefabricated aluminum systems typically are set up for bolt-together construction and included an integrated drain valve.

Water pit modular cover sets should be utilized to ensure a safe condition when the jump pit is not in use. Normally the water jump pit systems include an adjustable steeplechase hurdle with an aluminum barrier seal attached to the hurdle supports.

5.6 In-Ground Track Equipment

In-ground equipment includes such support features as vault boxes, toe boards, rings, throwing circles, raised curbs, protective cages, hurdles, water jumps, steeplechase hurdles and barriers, track curbing and landing mats. The primary criterion for selecting this equipment should be whether the product meets the needs of the owner. There are many excellent equipment manufacturers who can offer various options depending on the needs of the facility, the type of competition and track usage. They will provide the owner with working drawings for each piece of equipment, assist in ordering the correct equipment, provide material samples, etc. This type of equipment is a major investment for the owner and should not be taken lightly or considered an afterthought. The athletes' performance and safety should be foremost in making purchasing decisions. The goal should be to acquire the highest-quality, safest, best-performing equipment possible so that it will last through many seasons of use.

5.7 Accessibility to Storage Shelters

Locking storage buildings should be constructed in an inconspicuous place near the field event areas so that equipment such as pads, hurdles, starting blocks, etc. can be stored in a safe, centralized location after use. Keep in mind that equipment will need to be moved back and forth, set up and taken down. A location near the track facility will minimize physical labor on the part of staff or students and also will minimize wear and tear to the equipment. Access between the track and storage building should be over track surface, asphalt, or concrete—not dirt, mud or grass.

5.8 Viewing for Spectators

Every effort should be made to locate each event for maximum visibility while taking into account spectator safety. Ideally, each event should be viewable from the grandstands, although this will depend on the budget and on-site conditions. At a minimum, each event should be accessible to spectators watching from within the event area. Keep in mind, however, that when spectators must leave their seats in order to watch field events, over time one of two things happens: either they do not bother to watch the field events at all, or they do not take advantage of seating and instead tend to congregate in areas where they may interfere with event management or where they may be at risk. Try to think ahead to an actual meet when planning the facility and make notes of where all events will be in relation to the spectators. As always, remember the safety impacts of your decisions.

5.9 Timing Systems

Fairness in competition is of utmost importance. There is no place where the challenge of fairness is more evident than at the finish line. It is important to note that the primary function of any “timing” system is to determine the order of finish as that determines scoring. A secondary function is to record times. Up until the latter part of the 20th century, the finish of the race was determined by judges using hand-held stopwatches to record times and to pick the order of finish. Although this type of timing is still used, modern technology has progressed from hand-held timing to digital photo timing systems. The new timing systems also have the capacity to connect to the scoreboard to provide spectators instantly with an approximate elapsed time for the event.

With the advent of many types of automatic timing systems, owners must make an informed decision regarding the proper system for their venue. Some states also now require a timing system for competition as it applies to entry to the regional and state meets. Simply put, the timing system must meet the needs of the field of competition. A common example of differing needs is a high school that is only conducting local dual meets as opposed to a high school that intends to host regional and state competitions. Obviously, the timing equipment for the latter venue will be necessarily more specific and high-tech than the former. Likewise, the owner who desires to host large national or international competitions must have systems that are fully approved by the national and international governing bodies of track and field competition.

5.10 Sound Systems

A high-quality audio system is important not only for the pleasure of the spectators, but, more importantly, for the smooth conduct of the competition. Athletes, coaches and officials must be able to clearly hear information provided to them by the meet announcer, in warm-up as well as competition areas. Of course, the spectators also want to be kept informed as to the latest results, competitions, etc.

As with timing systems, sound systems will vary depending upon the level of competition for which the facility is designed. A system for a high school facility will be different from a system installed in a larger stadium. In either case, clear sound must reach all areas of the facility where spectators, athletes, coaches and other personnel congregate. Speakers need to be placed at locations and at heights to achieve this objective. Establishing locations for speakers is important because people do not want to encounter sound feedback. But more sound is not necessarily the answer. In fact, noise “pollution” is being regulated in many communities. The sounds of a meet must not carry into local residential areas. Depending upon the complexity and purpose of the sound system, the services of a design professional may be helpful in determining what best fits the owner’s objective.

5.11 Lighting

Lighting extends the hours of use of a track facility both in winter, when days are short, and in summer, when high daytime temperatures increase the desirability of evening competition. Unless lighting has been designed specifically for a track and field venue, however, the owner is liable to encounter some problems in obtaining consistent, high-quality illumination. Therefore, it is essential to ascertain that the selected lighting system has been designed for this type of facility. Prospective facility owners and managers are advised to proceed carefully and to contact their colleagues to learn as much as possible about various systems, including their advantages and disadvantages, the costs and any problems associated with them. As always, choose equipment from a reputable company with strong references and a history of successful installations in sports facilities.

Many governing bodies specify the exact quantity and type of light necessary for the conduct of meets. For national and international competitions, World Athletics has a detailed format with respect to illumination; this information can be found in the *World Athletics Track and Field Facilities Manual*. These specifications cover illumination for all areas of the venue including the finish line, spectator egress, competition areas, etc. NCAA also has “Best Lighting Practices” guidelines, which can be found on NCAA.com

Common practice for lighting high school track and field facilities is based upon lighting generally used for soccer and football. Unfortunately, the problem with this arrangement is that while there is an excellent amount of light on the athletic field itself, there is only minimal illumination for the curves of the running track and for the D-Zones. Field events outside the track oval may have little or no lighting. For the safety of athletes, officials and spectators, and for visibility, night meets demand total illumination of the entire facility.

Lighting for throwing events must ensure that the implement will not be obscured by glare. Dark corners of a facility are an accident waiting to happen—or, at the very least, leave

open the possibility for accusations of illegal or unfair behavior that an official cannot see well enough to confirm or deny.

Even if you don't intend to host night events, consider facility lighting, which increases the sense of security among facility users. Further, a well-lit area is less likely to be a magnet for those who want to loiter or cause trouble after hours. Lights can be set to operate on a timer, or they can be operated by a push-button system that an employee can manipulate. Generally speaking, it is also good to have some motion-activated lights in and around the facility to allow for safe entry and exit. These lights are smaller than the big lamps that illuminate the track itself, yet still allow for increased visibility. They are also an excellent way to tell if someone is in or near the facility.

Light poles should be designed and used exclusively for that purpose. Poles should not be used as fence posts nor should electrical conduit be attached anywhere on the fence. Either practice may create a safety hazard. Additionally, where light poles are incorporated into the fence design, local codes may require extensive grounding.

Certain light poles are designed for direct burial. These poles are manufactured with a nipple just above the grade where a junction box is installed to connect feeder wiring to the fixture wiring. It is not unusual for maintenance equipment—mowers, carts, etc.—to strike this junction box and break it from the pole, leaving the serious safety hazard of exposed wiring. Owners with light poles featuring such junction boxes should caution maintenance personnel to exercise care around light poles and should regularly inspect junction boxes to ensure that they are intact. A better option for installing wiring is to insert a conduit into the pole from below grade to feed the power source to the pole or to use poles with anchor-bolt base plates.

When working with a design professional, make sure that the professional understands your intention to have night track and field events and the need to have all applicable areas of the facility illuminated. The professional should then design the system to add extra lamps to specifically illuminate the curves of the track, the D-Zone and field events that would otherwise be left in shadows. These extra lamps can be wired separately so that they need not be used during football and soccer competition.

In choosing lighting, it is important to consider all of the variables and select the type of lighting that most closely matches the needs of the particular facility. Variables include light output, glare, lamp life, color, degree of control, initial cost and maintenance requirements. The choice of a lighting system also will be affected by frequency of use, climate, relative importance of light pollution and other factors. At a minimum, the athletes and the officials must be able to clearly see the competition and their surroundings for safety and in order to make good decisions. Spectators must be able to safely enter, move around within and leave the venue and should be provided with sufficient light to enjoy the competition in all events. Lighting adequate for television is substantially greater than that for spectators. For infrequent televising, portable lighting is available.

Design criteria should provide the initial illumination for the lighting plan as well as the average maintained illumination. When a lamp (or light source) is new, it produces a certain level of illumination. Over time, the amount of light produced by the lamp is reduced. This is called the light loss factor (LLF). Most manufacturers count on 20% to 40% depreciation. Climatic conditions, dust and dirt, voltage variations, luminaire design and maintenance will affect the amount of depreciation.

5.12 Fencing

Running tracks and their accompanying facilities normally represent a substantial investment. The primary purpose of fencing is to protect that investment against unauthorized use and against vandalism, theft and any activity that might create a liability for the owner. Fencing may be necessary or useful for spectator control, for controlling the locations where athletes and officials enter and leave the venue and for ensuring the safety and well-being of athletes, officials and spectators.

Chain-link fencing is the most common type of material used for these facilities, although various forms of wrought iron are sometimes employed. Chain-link is readily available, durable, reasonably economical and effective in providing the function typically desired by the owner. A vinyl- or polyester-coated frame with vinyl-clad fabric is preferred, although galvanized frame and fabric are sometimes used when cost is a consideration. Vinyl- or polyester-clad fencing is more expensive but provides benefits in terms of durability and aesthetics. Particularly in highly corrosive environments (such as inner-city areas where there is a high level of pollution, or in seaside areas where exposure to salt water is hard on fencing), vinyl-clad fencing is highly recommended. Fence coating comes in standard colors of green, black or brown. Sometimes vinyl-clad fabric is combined with a galvanized framework, which may be painted to match the vinyl-clad fabric.

Galvanized fencing and posts, if used, should be free from barbs or other projections which sometimes occur in the galvanizing process. These sharp projections may constitute a hazard. As an alternative to galvanized fencing, the new aluminized fencing and posts can be used. Fencing, whether galvanized or aluminized, should have selvages or edges, both top and bottom, that are bent-double (knuckled) to prevent injuries and to minimize debris from collecting against the fence.

Areas requiring fencing are the site perimeter, the running track and field event areas. Perimeter (site) fencing is normally 6' to 8' high and is designed to provide overall protection for the facility. The fence may enclose just a portion of the facility, such as the running track and football/soccer field, or it may enclose a complete athletic facility. Its purpose is to restrict the use of the facility to the persons and to the equipment for which it was designed.

A 4' fence with a top rail is usually placed around the outside perimeter of the track. *This fence should be placed at least 1m from the outside edge of the running lane. The chain link fabric shall be installed on the inside (track side) of the fence.* In fact, the NCAA has recently amended its rule 1-1.2 by adding, "Wherever possible, it is recommended that there be an obstacle-free zone on the inside and on the outside of the track at least one meter in width." The purpose of the fence is to keep spectators back from the competition areas, to confine athletes to areas they are using and to protect the track surface from excessive foot traffic, wheeled vehicles and other implements.

Fencing bordering the throwing events, when they are conducted in areas where spectators have open access, is a good idea. Normally, it is not desirable to have fencing inside the running track itself.

Gates are placed for strategic flow of athletes, officials and spectators. The position of perimeter gates is an important consideration for handling the volume of pedestrian traffic that is expected and to direct this traffic in an orderly fashion to the appropriate areas of the facility. Maze gates or turnstiles are an effective way to allow continual, consistent pedestrian access to the facility while restricting bicycles and motorcycles. It is recommended that all gates be equipped with lockable hardware.

Gates in the fence surrounding the track should be located to allow easy team access to the infield. Additionally, gates should be placed for competitors' ingress and egress to and from event areas where they will not interfere with the flow of the track meet. Typically, the arrangement will be for a gate near the head of the straightaway (for athlete ingress to check into the event) and a gate at the far end of the straightaway (for egress from the event). These gates are not to interfere with the start and finish lines. Any gate that is placed between the start and finish lines should be locked during track and field competition.

Pedestrian gates are usually 3' to 4' wide (36" to 48"). The Americans with Disabilities Act (ADA) requires that accessible gates be at least 36" wide. The ADA, as well as various state and local codes, require that all public facilities be accessible to persons in wheelchairs. Be certain to check requirements in all relevant codes when planning a facility.

In addition to wheelchair-accessible gates, gates should be wide enough to allow for maintenance and emergency vehicles to enter and leave the facility. These gates should be located in the perimeter and track fences. For ease of maintenance, it is recommended that the fencing design avoid small and narrow strips of sod or other high-maintenance areas. A primary example is the area between the outside edge of the running track and the fence, which may only be 1m wide. Do not attempt to place grass or other vegetation in this area or to use landscape materials such as chips or rocks that can end up on the track. Instead, surface the area with asphalt or concrete.

Fences are there to assist and protect. They should be as inconspicuous as possible, but still attractive and appropriate for the site. Fences can be selected to blend in with natural turf or with the background or possibly to reflect a school or facility color scheme.

5.13 Water

A source of drinking water should be located near competition areas, but outside the track oval. The start and finish areas on the main straightaway are especially important. As an added feature, a faucet for rinsing hands and faces will be appreciated by athletes. Any pipe supplying a drinking fountain or faucet should be buried at least 48" deep to prevent winter freeze or to prevent the pavement from heating the water in the summer. In addition, the ADA requires that all drinking fountains be fully accessible.

Where permanent water connection is not available, portable water cooler stands may be used. Cooler stands may include a cup-holder and trash can. Water also needs to be available for the steeplechase and sand pits.

5.14 Other Safety Issues

5.14.1 Track Oval Clearances

World Athletics has stated that there should be a 1m safety zone both inside and outside of the running track. NCAA has recently amended its rule 1-1.2 by adding, “Wherever possible, it is recommended that there be an obstacle-free zone on the inside and outside of the track at least one meter in width.” NFHS now states in Rule 5, Section 1, Article 5, that it is “recommended that there will be an obstacle-free zone on the inside and on the outside of the track at least 1 meter in width.

ASBA recommends that, at a minimum, a 4’ fence be placed around the outside perimeter of the track *at least 1m from the outside edge of the running lane*. A 1m clear zone inside and outside the running track is recommended. A wider clear zone outside the main straightaway can be useful for temporary hurdle storage, judges and timing systems.

Jumping Events

Safety considerations for these events are not merely important, they are essential.

Wind direction should be discussed with the facility owner. Sun and light reflection also are concerns for pole vaulters. Sun or lighting can impair or blur an athlete’s vision, leading to improper takeoff, which can result in injury.

The pole vault landing area should be just wide enough to accommodate the landing pit. (In this case, bigger is not necessarily better.) In fact, the pole vault event area may be constructed without a hard surface landing area. The National Federation of State High School Associations’ rules mandate that any exposed areas be covered with cushioned material. ASBA recommends that mats should overlap any hard surface by at least 3”.

It is important to consult the latest rules published by all of the relevant governing bodies for which the facility may be used to ensure that the pole vault facility meets those guidelines.

Also, note the presence of obstacles such as light poles or fences and ensure that they are a safe distance from the landing areas of all jumping events.

Throwing Events

Again, wind direction should be discussed with the facility owner. It is mandatory that the hammer and discus throwing area be protected with a cage designed to minimize the risk to spectators, officials and athletes in the event of an inadvertent release (Fig. 5-9). The cage should be constructed of netting (not chain-link) that will absorb energy and prevent bounce-back. Similarly, perimeter fencing should be constructed so as to minimize the risk of errant throws reaching the spectators, officials or athletes.

All throwing events should be conducted so as not to interfere with other activities held during the meet. Temporary or permanent fencing should be used to provide spectator safety.

Chapter 5

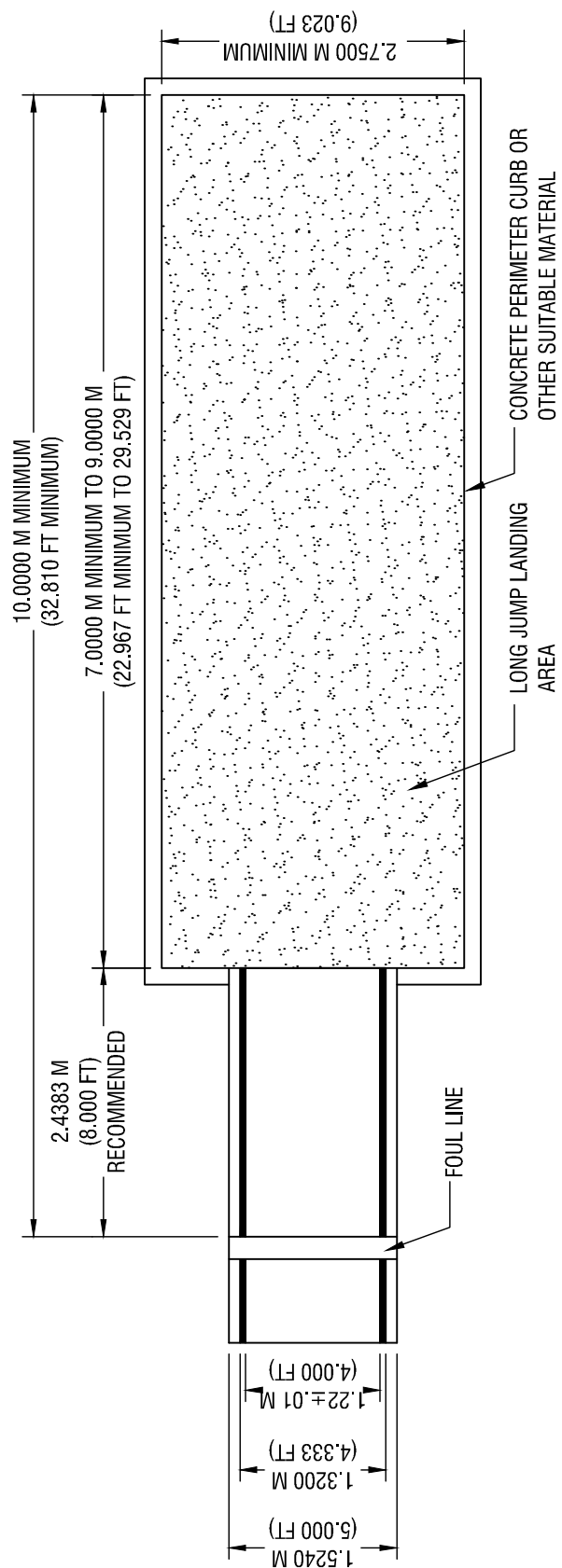
Field Events and In-Ground Equipment



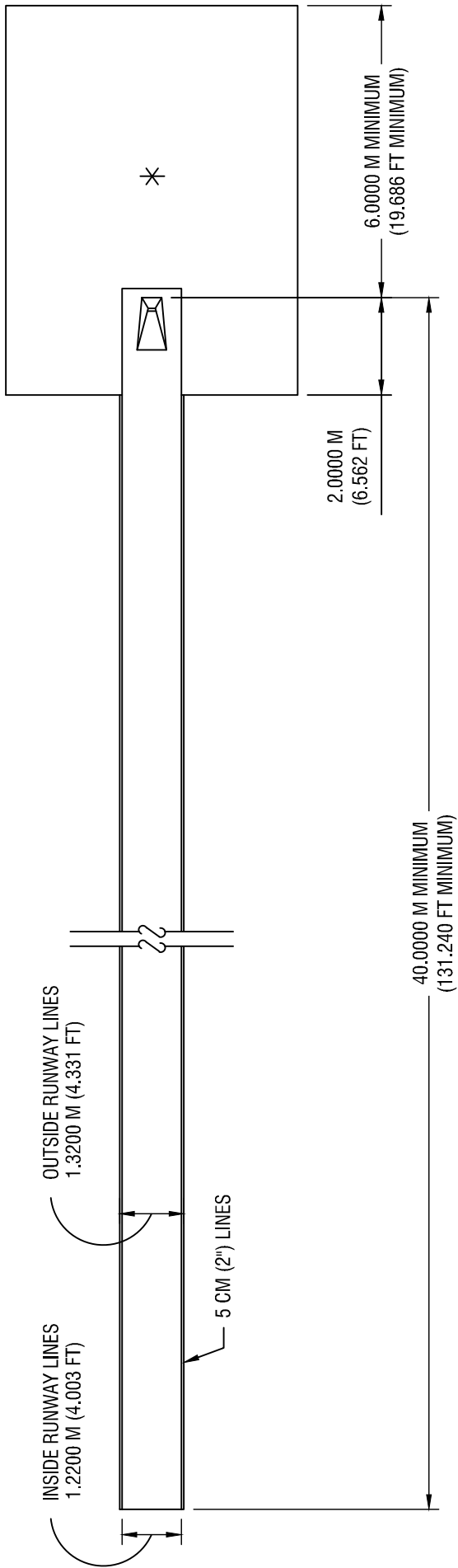


1. CONCRETE CURB TO BE DESIGNED IN ACCORDANCE WITH THE GEOTECHNICAL REPORT AND REGIONAL FROST CONDITIONS.
2. CONSULT WITH A CERTIFIED TRACK BUILDER FOR MODIFIED DESIGN IN HEAVY RAIN AREAS

LONG JUMP/TRIPLE JUMP PIT PLAN
FIGURE 5-3

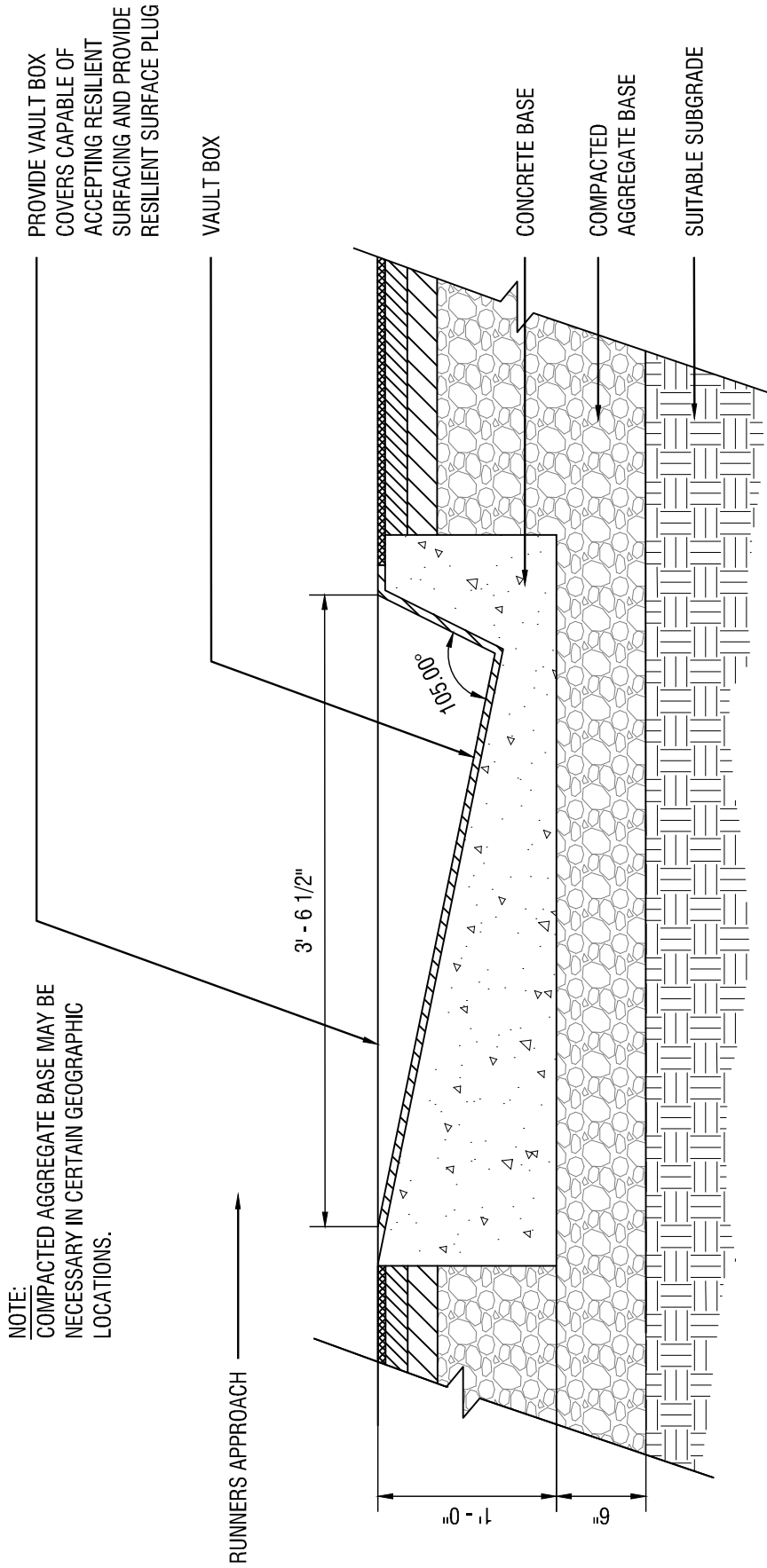


POLE VAULT EVENT AREA
FIGURE 5-4

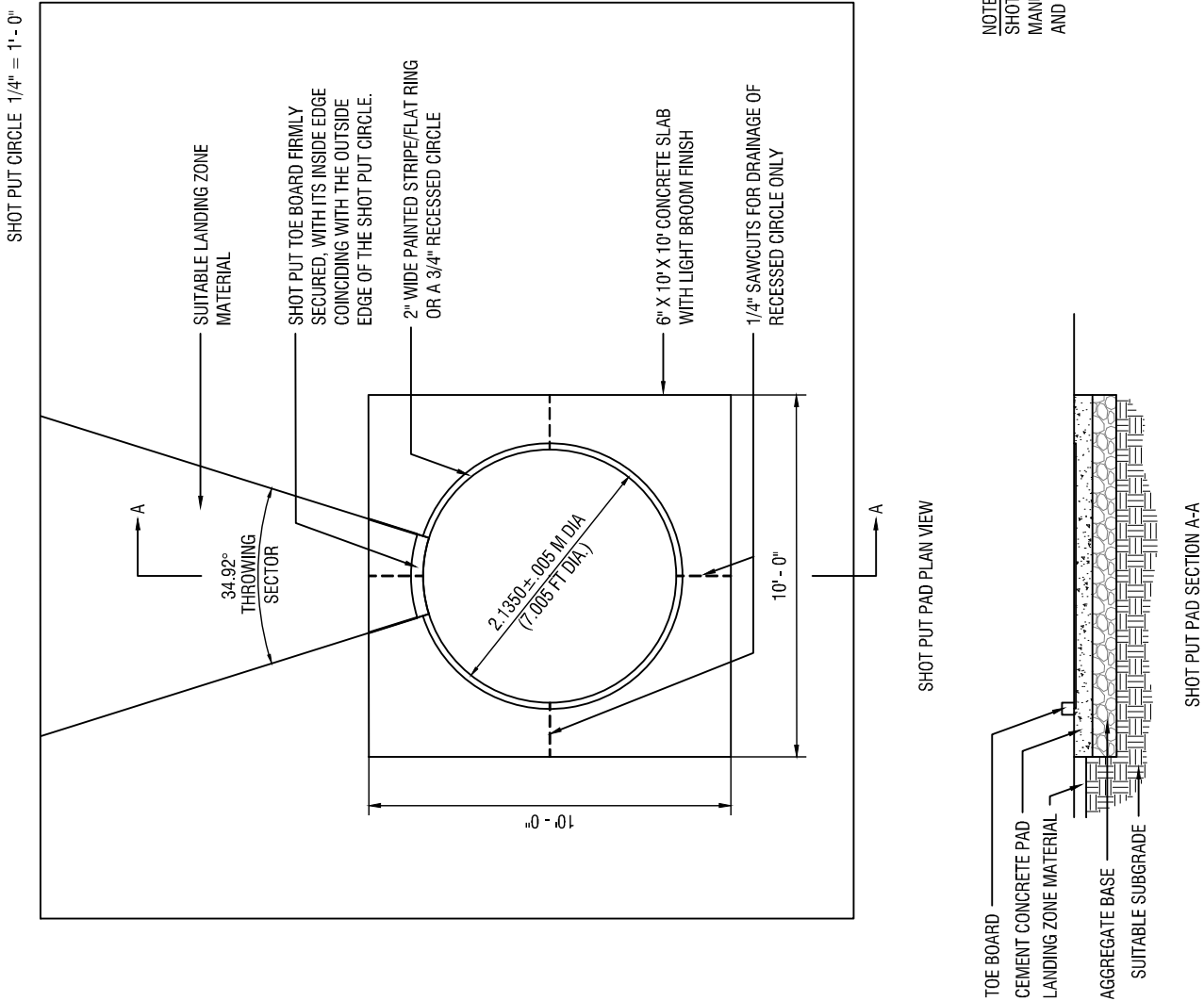


* LANDING AREA IS OPTIONAL BUT, IF USED, SIZE SHOULD BE AT LEAST 6" SMALLER THAN LANDING PADS.

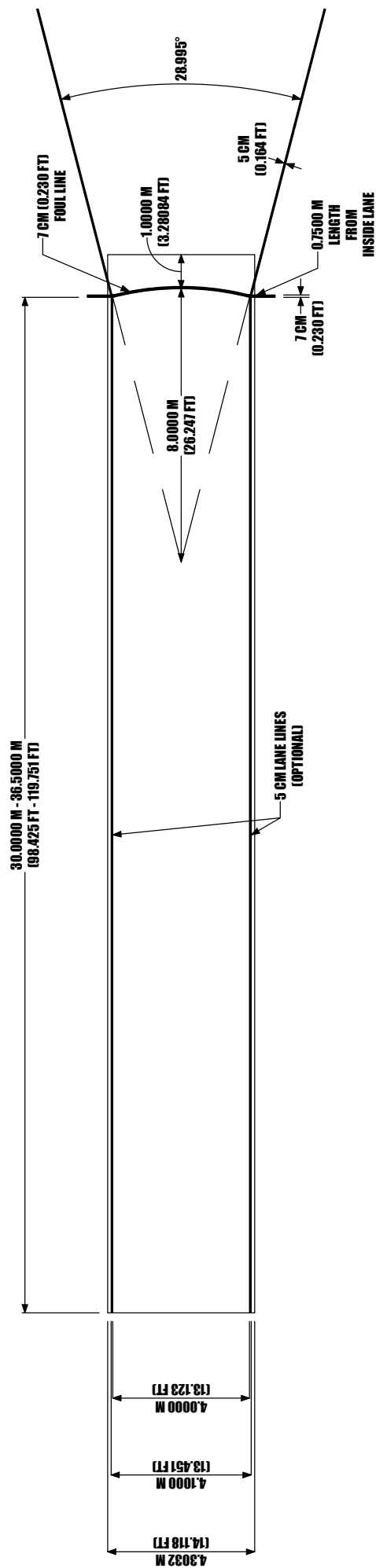
POLE VAULT SECTION
 FIGURE 5-5
 NOT TO SCALE



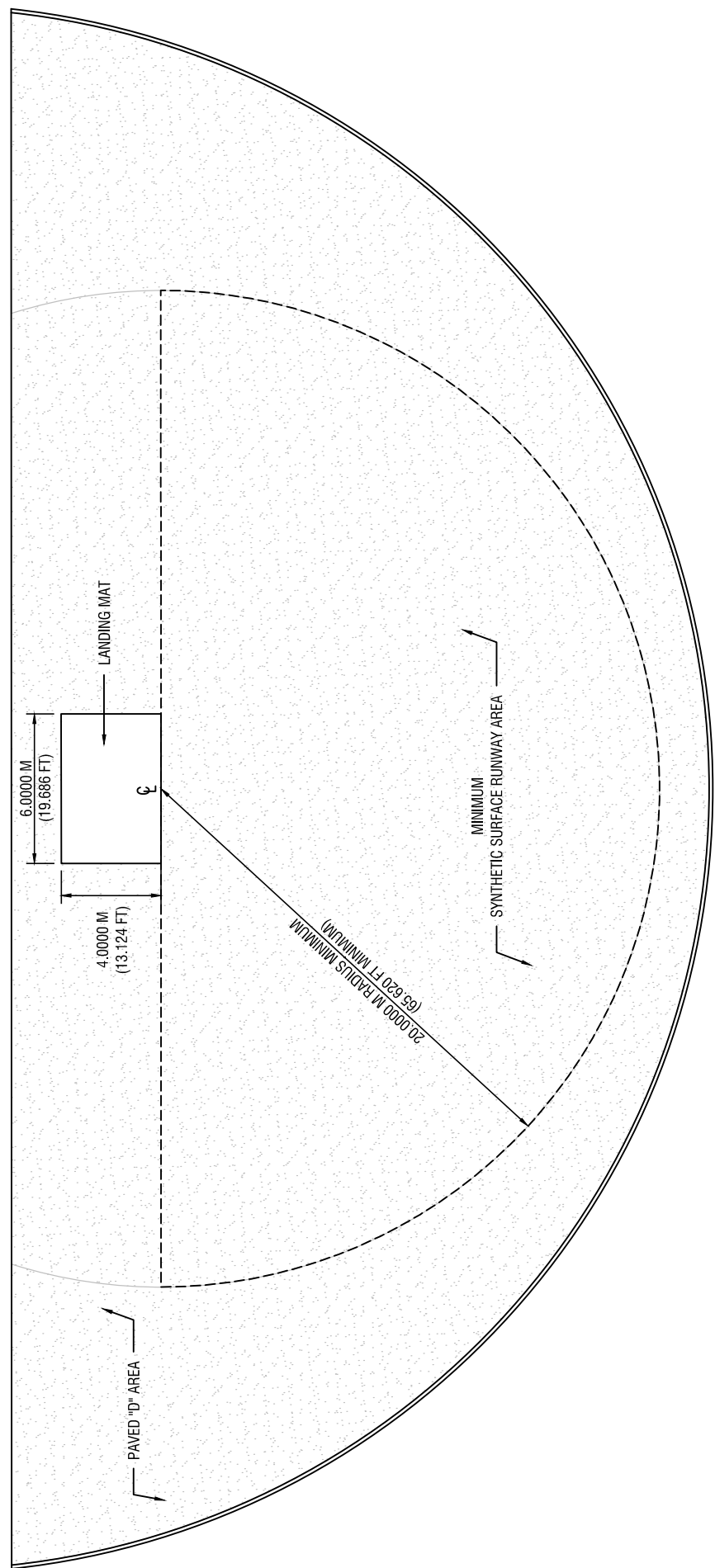
SHOT PUT PAD
FIGURE 5-6



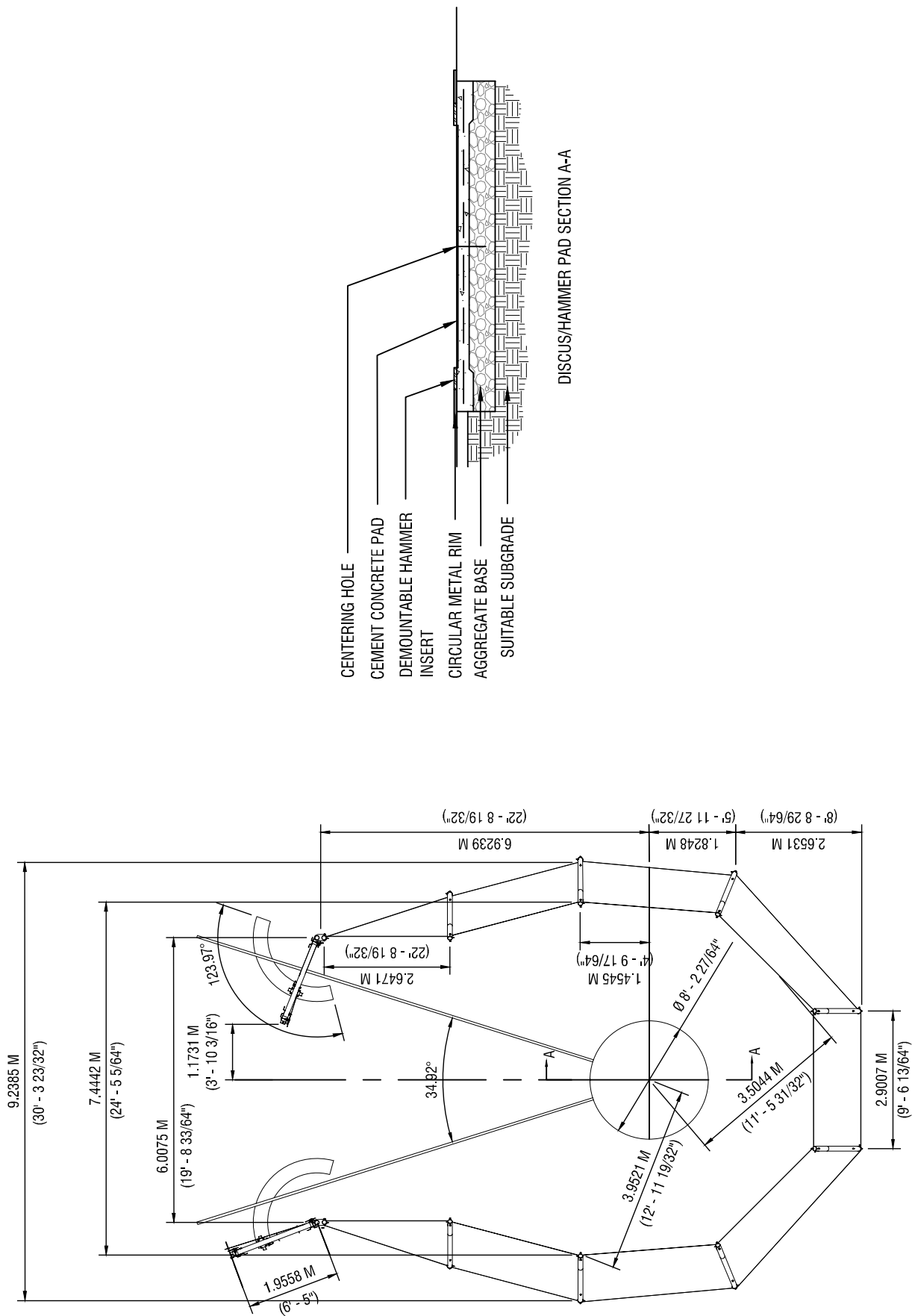
JAVELIN RUNWAY
FIGURE 5-7



HIGH JUMP LOCATION IN D-AREA
LOCATED OFF THE EDGE OF TRACK
FIGURE 5-8



DISCUS/HAMMER TOSS PAD PLAN
FIGURE 5-9
NOT TO SCALE



Chapter 6

MARKING A TRACK: CALCULATIONS, LAYOUT AND PAINTING





TRACK STRIPING IS OUR SPECIALTY

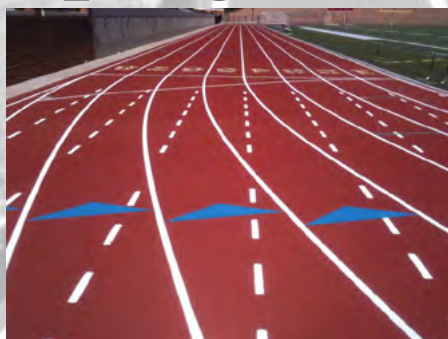
Whether you need lines and markings on your *NEW* surface or your existing track *RESTRIPED*, you can count on us for the highest quality track striping job in the industry.

We have over 60 years combined experience in striping running tracks. We stripe 250 to 300+ tracks nationwide per year and offer helpful service and competitive pricing.



Certification, computer-generated calculations, and full color large scaled drawings of your completed facility are also available.

www.trackstriping.com



Call us today for a free estimate or visit our website for more details.

John Wettstein,

770-356-2446



<http://www.facebook.com/trackstriping>

Chapter 6—Marking a Track: Calculations, Layout and Painting

This chapter will cover the process for calculating, laying out and painting a track.

- 6.1 Purpose of Markings
- 6.2 Selecting a Marking Specialist
- 6.3 Phase I—Calculations
- 6.4 Phase II—Layout
- 6.5 Phase III—Painting
- 6.6 Certification
- 6.7 Common Errors
- 6.8 Suggested Symbols and Graphics

6.1 Purpose of Markings

Markings on a track are placed there for the benefit of the users—those who compete on the track. The individuals responsible for those markings, be they coaches, owners, designers or contractors, should keep this purpose in mind. This chapter will provide guidance on track marking; it is intended to be used in conjunction with official publications and drawings available from the various governing bodies.

The markings on any track are the single most important feature when the track is used for competition. They represent distances that are very important, both to competitors and to the governing bodies who certify records. Many marking errors have been made because of misinterpretation of rules or drawings or because of poor workmanship. Often, owners do not realize the importance of accuracy in track marking and do not place enough emphasis on securing appropriate guidance. But accuracy is extremely important. No governing body permits any minus tolerance, no matter how small, and some have a maximum plus tolerance. An error of any kind may result in costly corrective work, in unfair competition or in uncertifiable records. In many cases, inaccurate markings prevent athletes from winning the recognition they have earned and may even affect their opportunities for scholarships or future competition. In short, a small, seemingly inconsequential error in track marking can have a huge significance.

6.2 Selecting a Marking Specialist

In planning and constructing a track to be used for competition, it is absolutely essential to employ the services of a knowledgeable specialist for track markings. The selection of the specialist should be based upon credentials, past performance and proof of ability. References, years of experience and knowledge of the rules are several basic considerations.

The specialist must have a complete knowledge of basic track geometry, as well as a thorough understanding of the rules of the specific governing body or bodies under which

competition will be conducted. This body of knowledge will include prescribed paths of runners, break-line compensation requirements and number of laps run in lanes for any given race.

There may be more than one specialist involved. It is not uncommon to have separate specialists for calculations, layout and painting.

Before any calculations are made, the specialist should meet with the owner's designated representative to discuss options and unique requirements. These may include the color code, design of the markings and any special events to be marked. When planning markings, it is important to recognize their location. It is recommended that, whenever possible, the design includes a common finish line located at the end of the main straightaway, also known as PC#1: point of first curve.

6.3 Phase I—Calculations

Calculations may be done manually or via computer. Computer calculations are generally more accurate, provided that the computer program has been properly written and tested. Computer calculations also are faster, allow for new computations to be easily run if assumptions or other input data change and eliminate the inherent human error of manual calculation.

A track should have two equal and parallel straightaway sections connected to two equal curves, which together add up to a specified distance when measured in accordance with the governing rules (see Chapter 2). It is not necessary—or even desirable—that a track be “equal quadrant,” i.e., each straightaway and each curve at 100m in length.

In the past, many tracks were built with equal quadrant design. Today, however, fewer tracks are built in this configuration for several reasons:

1. A wider radius allows greater flexibility in placing fields, especially soccer pitches, within the track oval.
2. A wider radius, such as is found on the World Athletics Standard (Fig. 2-5), favors runners and enhances performance.

A clear understanding of the rules for measure-line location and its relationship to lane-line location is essential. The runner's side of the inner lane line is located 20cm from the measure line for each lane. Where the track features an official raised curb, the runner's side of the raised curb is 30cm from the measure line for lane one only. then keep the last sentence. All other lanes use the 20cm rule. (See Chapter 2 for more information on track geometry and measurement.)

Lane lines are located by calculating a radius dimension from the radius point. Each individual lane must be evaluated to verify that it has the correct running distance for one lap of the oval. The start line for each event and any intermediate markings for each lane must be calculated to provide the correct distance to the finish line.

Note that no governing body allows a minus tolerance of any kind in any track measurement. As a practical matter, this makes it necessary to build in a plus tolerance. It is generally accepted that a planned plus tolerance of 6 to 8cm for a 400m lap will ensure that there will be no minus tolerance and will keep the race length as close as possible to the correct length. If World Athletics (WA) and NCAA rules apply, the plus tolerance cannot exceed 4cm per 400m lap. For NFHS, the plus tolerance is 8cm per 400m lap. As a practical matter, it may be necessary to build in a plus tolerance that meets the governing body rules.

The following values are recommended for calculations for a proper balance of accuracy and workable measurements. Some common measurements used in track design and construction include:

20cm = 0.656' or 7.874"	Distance from the measure line to the runner's edge of lane 1 (painted line rule)
30cm = 0.984' or 11.811"	Distance from the measure line to the runner's edge of lane 1 curb (raised curb rule)
0.3048m = 1'	Meter equivalent of 1 foot
1.0668m = 3.5' or 42"	Width of standard lane(except WA)
1.2192 m = 4.003'	Width of WA standard lane
1m = 3.28084'	Feet equivalent of 1 meter
pi = 3.1416	Calculation used for curve distances
400m = 1312.336'	Measure line distance in lane 1

The results of calculations should be expressed to the nearest 0.0001m (metric) or to the nearest 0.001' (imperial). These may be rounded up to the nearest 0.001m or 0.01' for construction and marking. Distances are rounded up to ensure that the distance is not less than 400m. It is recommended that all calculations be provided to the striping specialist with the exact accurate location of each marking, without any plus tolerances built in. Plus tolerance should be added in the field by the marking specialist based on site conditions and level of certification needed (if any).

6.4 Phase II—Layout

The key pieces of equipment needed for track layout may include a total station or a transit/theodolite that will read directly to a minimum accuracy of 20 seconds of a degree, along with a steel engineer's tape that will read directly to 0.001m or 0.01'. Temperature and correct tension should also be applied as they will affect this level of precision. Cloth tapes, fiberglass tapes and measuring wheels should never be used for track layout. However, numerous methods can be used to locate markings for a track, including:

1. Location of the markings in the straightaways using a steel tape, scribing the arcs

for the placement of the curved lane lines and then using a transit/theodolite for markings located in the curves. Using this method, measurements are taken from the radius points and the four tangent points of the track (PCs 1, 2, 3 and 4). The location of each mark along the curve is converted to an angle, then located via the transit or theodolite.

2. Location of all track markings using only a steel tape. This is typically referred to as the “chord method.” This method is similar to the first except that the markings around the curve are converted to a chord length rather than an angle. Care needs to be taken to prevent inaccuracy with this method.
3. Location of all markings using a total station. Using this method, all markings on the track are converted to coordinate points (northings and eastings), then transferred to a data collector and located on the track via the total station.
4. Combination of some or all of the above methods.

Simply draping a tape measure around a curve is not considered to be an accurate method of layout, nor is pacing off or guessing a mark’s location relative to another’s.

Normally, layout and painting of the lane lines is performed first. Then layout of such markings as start/finish lines, hurdle placements and exchange zones is completed. It is also acceptable (particularly when using method 3 above) to complete all layout before any paint is applied.

6.5 Phase III—Painting

ASBA recommends applying one solid coat for lane lines. Multiple coats can lead to shadowing and increased width of lane lines. Please refer to the ASBA position paper, “Line Striping: One Coat vs. Two Coats,” for more guidance.

Tracks are usually painted with a striping machine equipped with an auxiliary hand-operated spray gun. There are basically two types of striping machines: conventional and airless. Conventional machines mix air and paint at a specific ratio of pressure (usually 30 to 90psi combined pressure). Airless technology is newer, applying only paint (no air), typically between 800 and 3000psi.

The type of paint used for marking the surface should be that recommended by the surfacing system manufacturer. Acrylic latex or polyurethane paint is commonly used.

In most cases, traffic paints are unsuitable due to their brittle nature. While insufficient quantities of applied paint can cause premature wear, an excessive thickness of paint may cause cracking and curling. Typically 9mls is specified for an application rate, although this rate may vary depending on the material.

Lane lines on the straightaways and chutes are applied using a string line pulled from end to end as a guide. Lane lines on the curves are applied using scribed interval markings as a guide. A high degree of skill is required to achieve the necessary accuracy.

Most rules specify a width of 5cm (approximately 2”) for lane lines and many other markings; the rules also specify which side of the line or marking must match the layout mark to provide the correct distances.

Each governing body has a recommended color plan for the various markings (see Appendix G). School colors may be incorporated into the track via lane numbers, lines or other markings where governing bodies allow. The owner may request a drawing or color code document specific to their track to share with other schools prior to competitions.

Repainting of existing tracks is discussed in Chapter 8.

6.6 Certification

Certification of a track gives the owner and end users assurance that the track was built and marked correctly and precertifies the accuracy of all events so that a record achieved at that facility can be submitted to a governing body without additional verification. Track verification and certification is discussed in Chapter 7.

6.7 Common Errors

The following are errors commonly committed when marking tracks:

1. Using painters and/or surveyors who are not familiar with the measurements or do not use acceptable techniques and equipment required to accurately mark a track.
2. Using a construction drawing without verifying that the track geometry is correct.
3. Assuming the offset from the measure line radius is the same for a raised curb and a painted line.
4. Assuming 8” = 20cm and 12” = 30cm when calculating track measurements.
5. Expecting sample drawings to give details for calibrating and marking tracks.
6. Not knowing the end users’ desires and/or the prevailing rules of the local governing body.
7. Not using the minimum recommended conversion factors for calculations.
8. Not being familiar with which part or side of the marking the actual measurement represents.
9. Not utilizing available sources of information.
10. Assuming all staggers are identical, whether on a straightaway or curve.
11. Not applying break-line compensation to start lines and exchange zones.
12. Not understanding the concept of the measure line.

6.8 Suggested Symbols and Graphics

World Athletics is very specific with regard to the size and shape of all ancillary markings. Other governing bodies do not specify. Below are suggestions:

1. Exchange Zones
 - Large (12” to 24” high) triangles (common for NCAA and NFHS)

- Checkmark symbol (World Athletics). The checkmark symbol may also be used for NCAA and NFHS, but it is required by World Athletics

2. Hurdle Settings

- 1” to 4” diamond, triangle, square, or rectangular-shaped marks on or next to the lane line

3. Photo Timing

- Black marks at the lane line/finish line intersections. World Athletics and NCAA have specific requirements (see the rules books)

4. Graphics (for runner orientation and aesthetic appeal)

- Large 36” to 42” lane numbers (can be shadowed)
- Event identification; 3” to 4” stencils in lane indicating purpose of adjacent marking
- School logo or name stenciled on track or event areas

5. Other Markings to Consider

- Sprint races in both directions on straightaway
- Runway borders (outline)
- Distances for pole vault run-up
- Long jump or triple jump “boards”
- Throwing circle

Chapter 7

VERIFICATION



APPLICATION FORM

WORLD ATHLETICS

ATHLETICS FACILITY CERTIFICATION

This form must be sent to: technicalofficer@worldathletics.org
Conditions of application & fees are in the Certification System Procedures: www.worldathletics.org

We, _____ hereby apply for:

☐ **CLASS 1 Certificate:**
Measurement Report in accordance with the requirements of World Athletics and a full in-situ testing of the synthetic surface by an Accredited Laboratory in accordance with Rule 2 of the Technical Rules are required.

☐ **CLASS 2 Certificate:**
Measurement Report and current valid Product Certificate for the facility synthetic surfacing material are required.

☐ **INDOOR Certificate:**
Measurement Report and current valid Product Certificate for the facility synthetic surfacing material are required.

☐ **CONFIRMATION OF COMPLIANCE**
Measurement Report and the reasons why the full certification cannot be applied for are required.

FACILITY / STADIUM / VENUE

Name of the facility / stadium*	
City	Country
Address	
GPS coordinates (finish line)**	
Usage (e.g. athletics, football)	
Roof: open/closed/retractable	Facility built (year)

*as it should appear on the Certificate
**Latitude and longitude in decimal degrees (DD) or in degrees, minutes, seconds (DMS); elevation, if available.

APPLICANT

Name of applicant	
Address	
City	Postal (Zip) Code
Country	State / Province
Email	Telephone

Chapter 7—Verification

This chapter will discuss the five levels of verification for a track facility, summarized in the chart below and further described in the following sections. Verification for Class 1 and 2 are specific, rigid certification procedures that were developed by World Athletics (WA), primarily for international competition. Class 3, 4 and 5 verifications were developed by the ASBA primarily for varying levels of collegiate and high school competition.

- 7.1 Class 1 and 2 (World Athletics)
- 7.2 Class 3: Marking and Slope Verification
- 7.3 Class 4: Marking Verification
- 7.4 Class 5: Certificate of Accuracy

7.1 Track Certification Levels

	Class 1	Class 2	Class 3	Class 4	Class 5
Facility Use	Required for international championship or cup meets	Required for other international meets or meets authorized for foreign athlete participation	Recommended for any track needing assurance of accuracy, on which national collegiate or NCAA records may be set	Recommended for any track needing assurance of accuracy, on which state or national high school records may be set	Recommended for any track needing assurance of general accuracy
Document Required	WA Report of Measurement	WA Report of Measurement	ASBA Class 3 Certification Form	ASBA Class 4 Certification Form	Letter from Certified Track Builder attesting measurements conform to rules
Stamp or Signature Required	Signature	Signature	Signature	Signature	Signature
Product Certificate Required	Yes	Yes	No	No	No
<i>In Situ</i> Surface Testing Required	Yes	No	No	No	No
Certification of Track Slopes	Yes	Yes	Yes	No	No
Certification of Event Area Slopes	Yes	Yes	Yes	No	No
Verified, Computer-Generated Calculations	Yes	Yes	Yes	Yes	Yes
Minimum Equipment Required	EDM over 20 meters, theodolite for curves (certified equipment)	EDM over 20 meters, theodolite for curves (certified equipment)	EDM over 20 meters, theodolite for curves (certified equipment)	Temp.-corrected steel chain for straights, 20 sec. theodolite for curves	Temp.-corrected steel chain for straights, 20 sec. theodolite for curves

Course Length Measurement Technique	Radius measured for each lane in 26 designated locations in curves, 2 in each straight	Radius measured for each lane in 26 designated locations in curves, 2 in each straight	Radius measured in lane 1 in 26 designated locations in curves, 1 in each straight	Radius measured for lane 1 in 26 designated locations in curves, 1 in each straight	Radius measured for lane 1 in 12 random locations in curves, 1 in each straight
Course Length (CL) Tolerance (Plus)	.0001 x CL	.0001 x CL	.0001 x CL	.0002 x CL	.0002 x CL
Course Length Tolerance (Minus)	None	None	None	None	None
Lane Width Tolerance	1.22m +/- .01m	1.22m +/- .01m	Design lane width +/- .01m	Design lane width +/- .02m	Design lane width +/- .03m
Expiration of Certification	4 years	4 years	Upon resurfacing	When changes are made such as restriping or resurfacing	When changes are made such as restriping or resurfacing

7.2 Class 1 and 2 (World Athletics)

The forms and procedures for these levels are available on the World Athletics website at worldathletics.org. There is also a general introduction to the World Athletics Certification System in the *World Athletics Track and Field Facilities Manual*.

Please note that this system is intended for high-level, international competition and is not appropriate for use at the high school level.

7.3 Class 3—Marking and Slope Verification

Class 3 is generally used to verify tracks that will be used for collegiate events. Procedures include verifying the distance run for each event and that slopes are within the rules. This level can be used where national collegiate or NCAA records may be set. Procedures for Class 3 verification are explained on the following pages.

Class 3 Markings and Slopes Certification for Running Tracks

Instructions to Surveyor

1. Purpose of this document:

1.1 Intended use.

This document is intended to serve as verification to the owners of the facility that their track meets the requirements of the NCAA in the categories described below. This process can also be used for NFHS tracks by adjusting the slopes and distances to conform to NFHS rules. Class 3 Certification is intended to be completed as an integral part of the striping, and not as an after-the-fact undertaking. Although the process can be performed after the fact, corrective work to either slopes or markings can be difficult, time-consuming and costly. This form has been published by the American Sports Builders Association to assist owners of track facilities in verifying that their track facility meets the official requirements of the relevant governing body of the sport. The American Sports Builders Association makes no representation with respect to the requirements of any governing body, nor does it warrant or guarantee the accuracy or correctness of the measurements or conclusion of the certifying party.

1.2 Completion of the form.

This form may be completed by a competent measurer, such as a Registered Land Surveyor, Professional Engineer, Certified Track Builder (CTB) or experienced track striper. Note that the Report of Slopes (see item 3.6 Track and Event Area Slopes) should be completed before the surface is installed to allow for slope corrections, if necessary, prior to installation of said surface.

2. Relevant parameters of the sport:

2.1 Slopes of track and event areas.

The inclination should be checked to ensure that the track does not exceed current allowable maximums for slope. Keep in mind that the maximum inclination permitted in the running direction for the track and field events applies only in the downward direction. For renovation of existing facilities that may not completely meet these requirements, one may consider using Class 4 Certification in lieu of Class 3.

2.2 The track oval.

The 400m oval typically consists of two semicircles of equal radius, which are joined by two straights of equal length. The inside of the oval track shall be bordered with a white line 5cm wide and may include a curb of suitable material, approximately 5cm in height and width, installed on top of the white line. The theoretical running length of the oval shall be measured on the surface of the track along the theoretical path of the runner (known as the measuring line). The painted line is to be located 20cm inside the 400m

measuring line (toward the inner field) if no curbing is to be installed over the line. If designed and surveyed for curbing to be installed over the line, the painted line is to be located 30cm inside the 400m measuring line. All other lane lines are to be located 20cm inside the corresponding measuring line for that lane. The actual length of the 400m measuring line should not be less than 400m, or more than 400.06m.

2.3 Individual races.

The horizontal variance allowed for the individual races is between 1 and 1.00015 times the specified length along the measuring line for each lane. The location of incidental markings, such as hurdles, steeplechase barrier placements and baton exchange zones should be noted.

3. Report of Slope and Report of Measurement:

3.1 Surveyors identification data and survey conditions.

Complete the document for Report of Slope and/or Report of Measurement (whichever is appropriate for your work). Attach valid certificates of accuracy of instruments.

3.2 Oval lines layout—before painting the lane lines.

- Locate the centers (radius points) of the two semicircles.
- Measure the distance between the centers of the two semicircles. The deviation from the desired value should not exceed +0.007 meters.
- Place marks on the synthetic surface at the specified radius for the outer edge of the inside line at points 1-13, which are 15 degrees between points. These are the radius control points. At the same angles, place marks in the same manner for all other lane lines.
- Repeat this procedure for points 14-26. At the option of the contractor and surveyor (if the two are separate), the surveyor may place controls for all other lanes in these locations.
- Measure the distance between points 13 and 14 (line PC #2/3). The deviation from the desired value should not exceed 0.007m. Record the measured distance on Table 1.
- Measure the distance between points 1 and 26 (line PC #4/1). The deviation from the desired value should not exceed 0.007m. Record the measured distance on Table 1.

3.3 Oval lines verification—after painting the lane lines, before installing the curb (if applicable).

- At points 1-13 and 14-26, measure the distance between the radius control points and the outer edge of the painted line. Record the measured deviations on Table 1.
- Calculate the average deviation from the desired value of radius of curve PC #1/2 and PC #3/4. Record the average deviations on Table 1.
- Calculate the length of curves PC #1/2 and PC #3/4, the length of the inside border, the theoretical running distance of the track and the deviation of the theoretical running distance from 400m following the instructions on Table 2. If such deviation

does not exceed 0.06m (and is not short of 400m), the 400m oval can be considered dimensionally accurate.

- d. Although a table is not provided, the surveyor should verify that radii for all lanes meet the same criteria as 3.3.c, and that the average line radii do not vary more than .01m from the design radius when averaged at points 1 through 26, with no individual point varying more than 0.025m.

3.4 Track event layout—before painting.

- a. Using certified, computer-generated calculations, or data file derived from such, place a mark at the locations for the starting and finishing lines of all events. These are the start/finish control points. At the option of the contractor and surveyor (if the two are separate), the surveyor may also layout the locations of all other track markings, such as hurdles, steeplechase barrier placements and baton exchange zones.

3.5 Track event verification—after painting.

- a. Measure the distance between all start/finish control points and the leading edge of each painted start and finish line. Record the deviations on Table 3. If the combined total deviation of each start line and the finish line for that event do not exceed .0002 x design course length (and is not short of design length), the event can be considered accurate.
- b. The surveyor should verify the location of the steeplechase barriers (if required), and randomly, at least 9 incidental markings (hurdle placements, exchange zones, etc.) and record the deviations on Table 4.

3.6 Track and event area slopes—before applying the synthetic surface

Measure the relative elevations at the inner and outer edge of the track at the common finish line, then clockwise from the finish as follows: 100 meters and 110 meters in a straight line back from the finish (not into the curve), 200 meters (around the curve), and 300 meters. Record the slopes of track and event areas on tables 5 through 8 of the Report of Slopes.

Class 3 Report of Measurement

1. *Identification of the facility*

Name of Track/Stadium: _____

Address: _____

2. *Surveyor Information*

Company: _____

Address: _____

Tel: _____

Fax: _____

Surveyor Name: _____

Qualifications: _____

Theodolite: _____

Distance Meter: _____

Instrument Certificates: Attached _____

3. *Survey Conditions*

Date of Survey: _____

Weather: _____

Temperature: _____

4. *Oval Track Desired Layout Data*

Desired radius at outside edge of inner curb or line: _____ m

Desired distance between radius points: _____ m

Theoretical length of oval at measuring line: _____ m

Number of lanes: _____ lanes

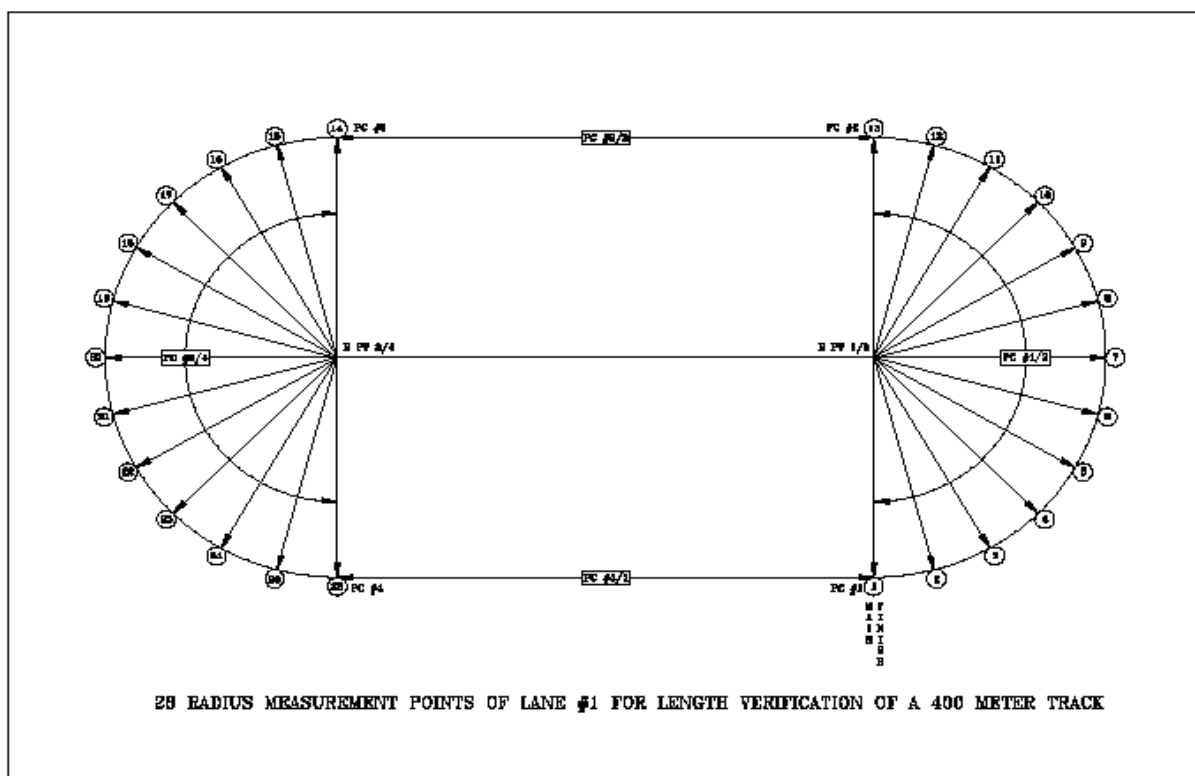
Width of lanes: _____ m

Type of inner edge: Painted line only _____ Raised curb _____

5. *Dimensional Accuracy of the 400m Oval Track:*

5.1 28 radius measurement points of lane #1

The 28 radius measurement points of lane #1 that are described in the diagram below have been carried out and the readings recorded in Table 1.



Straight PC #2/3				Straight PC #4/1		
Measured Length (m)				Measured Length (m)		
Curve PC #1/2				Curve PC #3/4		
Point	Angle	Measured Deviation (m)		Point	Angle	Measured Deviation (m)
1	0			14	0	
2	15			15	15	
3	30			16	30	
4	45			17	45	
5	60			18	60	
6	75			19	75	
7	90			20	90	
8	105			21	105	
9	120			22	120	
10	135			23	135	
11	150			24	150	
12	165			25	165	
13	180			26	180	
Average Deviation (m) Sum of deviations / 13				Average Deviation (m) Sum of deviations / 13		

Table 1—Calculated average deviation from desired radius of curves PC #1/2 and PC #3/4—Measured length of straights PC #2/3 and PC #4/1

5.2 Calculation of the length

The theoretical running distance of the oval track and its deviation from 400m are calculated as described in Table 2.

(1)	Desired radius of curves PC #1/2 and PC #3/4 in meters:		
(2)	Average deviation from desired radius of curve PC #1/2 in meters:		
(3)	Average deviation from desired radius of curve PC #3/4 in meters:		
(4)	Length of curve PC #1/2 in meters:	$= [(1) + (2)] \times 3.1416$	
(5)	Length of curve PC #3/4 in meters:	$= [(1) + (3)] \times 3.1416$	
(6)	Measured length of straight PC #2/3 in meters:		

(7)	Measured length of straight PC #4/1 in meters:		
(8)	Length of inside border in meters:	$= (4) + (5) + (6) + (7)$	
(9)	Theoretical measuring line if curb installed (0.30m):	$= 0.30 \times 3.1416 \times 2$	
(10)	Or theoretical measuring line if no curb installed (0.20m):	$= 0.20 \times 3.1416 \times 2$	
(11)	Total length of track if curb installed:	$= (8) + (9)$	
(12)	Or total length of track if no curb installed:	$= (8) + (10)$	
(13)	Deviation from 400 meters in meters:	$= 400 - (11)$ or $400 - (12)$	

Table 2—Theoretical running distance and deviation from 400m

5.3 Record start line deviations

The deviations from design locations of start and finish lines are recorded in Table 3.

Start or Finish	Deviation (in meters)									
100m	Lane 1									Outer Lane
110m	Lane 1									Outer Lane
200m	Lane 1	Lane 2	Lane 3	Lane 4	Lane 5	Lane 6	Lane 7	Lane 8	Lane 9	Lane 10
300m	Lane 1	Lane 2	Lane 3	Lane 4	Lane 5	Lane 6	Lane 7	Lane 8	Lane 9	Lane 10
400m	Lane 1	Lane 2	Lane 3	Lane 4	Lane 5	Lane 6	Lane 7	Lane 8	Lane 9	Lane 10
800m	Lane 1	Lane 2	Lane 3	Lane 4	Lane 5	Lane 6	Lane 7	Lane 8	Lane 9	Lane 10
4x200m	Lane 1	Lane 2	Lane 3	Lane 4	Lane 5	Lane 6	Lane 7	Lane 8	Lane 9	Lane 10
4x400m	Lane 1	Lane 2	Lane 3	Lane 4	Lane 5	Lane 6	Lane 7	Lane 8	Lane 9	Lane 10
1500m	Lane 1	Lane 2	Lane 3	Lane 4	Lane 5	Lane 6	Lane 7	Lane 8	Lane 9	Lane 10
10,000m	Lane 1	Lane 2	Lane 3	Lane 4	Lane 5	Lane 6	Lane 7	Lane 8	Lane 9	Lane 10
3000/5000m	Lane 1	Lane 2	Lane 3	Lane 4	Lane 5	Lane 6	Lane 7	Lane 8	Lane 9	Lane 10
Steeplechase	Lane 1	Lane 2	Lane 3	Lane 4	Lane 5	Lane 6	Lane 7	Lane 8	Lane 9	Lane 10
Finish	Lane 1									Outer Lane

Table 3—Record of deviation from design location of event start/finish

5.4. Deviation of incidental markings

Record deviations of randomly chosen marks. A positive number indicates deviation toward the finish line in the running direction. A negative number indicates deviation toward the start line, in the running direction.

Name of mark									
Hurdle or zone #									
Lane #									
Deviation from design location									
Notes									

Table 4—Deviation of incidental markings

6. Certification of length

- We hereby certify that all measurements and information shown on this report are accurate and are the result of a well-conducted survey.
- The measurement for lane one was 0.30 meters ____ 0.20 meters ____ (check one) outward from the inside border.
- The control of the inside length of the track gives a length greater than 400 meters.
- The calculated distance of ____m is within the acceptable plus tolerance of 0.06 meters.

Date: _____

Surveyor Name: _____

Signature: _____

Class 3 Report of Slopes

1. *Identification of the facility*

Name of Track/Stadium: _____

Address: _____

2. *Surveyor Information*

Company: _____

Address: _____

Tel: _____

Fax: _____

Surveyor Name: _____

Qualifications: _____

Theodolite: _____

Distance Meter: _____

Instrument Certificates: Attached _____

3. *Survey Conditions*

Date of Survey: _____

Weather: _____

Temperature: _____

4. *Desired Slope Data:*

Maximum lateral inclination of 1.00% to the Inside ____ Outside ____ of the track
 Maximum downward inclination toward the finish line of 0.10% in the running direction
 Maximum lateral inclination of 1.00% for all runways

Maximum inclination of 0.10% in the running direction for all runways

Maximum high jump inclination of 0.40% in the running direction, toward the center of the crossbar

5. *Record of Slopes on Track and Event Runways*

Percent Lateral Slope Between Locations on Running Track				
Finish in to out	100m in to out	110m in to out	200m in to out	300m in to out

Table 5–Lateral slope of track

Percent Slope From Finish to Various Start Locations			
Finish in to 100m in	Finish in to 110m in	Finish in to 200m in	Finish in to 300m in
Finish out to 100m out	Finish out to 110m out	Finish out to 200m out	Finish out to 300m out

Table 6–Longitudinal slope of track

Percent Slope on Runways								
	Runway 1		Runway 2		Runway 3		Runway 4	
Pole Vault	End to End		End to End		End to End		End to End	
	Lateral		Lateral		Lateral		Lateral	
Long/Triple Jump	End to End		End to End		End to End		End to End	
	Lateral		Lateral		Lateral		Lateral	
Javelin	End to End		End to End		End to End		End to End	
	Lateral		Lateral		Lateral		Lateral	

Table 7–Slope of runways

Percent Slope on High Jump Area in the Running Direction Toward the Center of the Crossbar	
High Jump Area #1:	
High Jump Area #2:	

Table 8–High jump slope

6. *Certification of Slopes*

- a. We hereby certify that all measurements and information shown on this report are accurate and are the result of a well-conducted survey.
- b. The slope of the track oval is no greater than 0.10% in the running direction.
- c. The slope across the track is no greater than 1.00%.
- d. The slope of the runways is no greater than 0.10% in the running direction.
- e. The slope across the runways is no greater than 1.00%.
- f. The slope of the high jump in the running direction toward the center of the crossbar is no more than 0.60%.

Date: _____

Surveyor Name: _____

Signature: _____

7.4 Class 4—Marking Verification

Class 4 is generally used to verify tracks that will be used for major high school events. Procedures include verifying the distances run but not any slope verification. Work should be done by an independent party. This level can be used where state or national high school records may be set. Procedures for Class 4 verification are explained below.

Class 4 Markings Certification for Running Tracks

Instructions to Surveyor

1. Purpose of this document:

1.1 Intended use.

This document is intended to serve as verification to the owners of the facility that their track meets the requirements of the NFHS in the categories described below. Class 4 Certification is intended to be completed as an integral part of the striping, and not as an after-the-fact undertaking. Although the process can be performed after the fact, corrective work to the markings can be difficult, time-consuming, and expensive. This form has been published by the American Sports Builders Association to assist owners of track facilities in verifying that their track facility meets the official requirements of the relevant governing body of the sport. The American Sports Builders Association makes no representation with respect to the requirements of any governing body, nor does it warrant or guarantee the accuracy or correctness of the measurements or conclusion of the certifying party.

1.2 Completion of the form.

This form may be completed by a competent measurer, such as a Registered Land Surveyor, Professional Engineer, Certified Track Builder (CTB) or experienced track striper.

2. Relevant parameters of the sport:

2.1 The track oval.

The 400m oval typically consists of two semicircles of equal radius, which are joined by two straights of equal length. The inside of the oval track shall be bordered with a white line 5cm wide and may include a curb of suitable material, approximately 5cm in height and width, installed on top of the white line. The theoretical running length of the oval shall be measured on the surface of the track along the theoretical path of the runner (known as the measuring line). The painted line is to be located 20cm inside the 400m measuring line (toward the inner field) if no curbing is to be installed over the line. If designed and surveyed for curbing to be installed over the line, the painted line is to be located 30cm inside the 400m measuring line. All other lane lines are to be located 20cm inside the corresponding measuring line for that lane. The actual length of the 400m measuring line should not be less than 400m, or more than 400.08m.

2.2 Individual races.

The horizontal variance allowed for the individual races is between 1 and 1.0002 times the specified length along the measuring line for each lane. The location of incidental

markings, such as hurdles, steeplechase barrier placements and baton exchange zones should be noted.

3. Report of Measurement:

3.1 Surveyors Identification Data and Survey Conditions:

Complete the document for Report of Measurement. Attach valid certificates of accuracy of instruments.

3.2 Oval lines layout–before painting the lane lines

- Locate the centers (radius points) of the two semicircles.
- Measure the distance between the centers of the two semicircles. The deviation from the desired value should not exceed 0.007 meters.
- Place a mark on the synthetic surface at the specified radius for the outer edge of the inside line at points 1-13, which are 15 degrees between points. These are the radius control points. At the same angles, place marks in the same manner for all other lane lines.
- Repeat this procedure for points 14-26.
- Measure the distance between points 13 and 14 (line AB). The deviation from the desired value should not exceed 0.007m. Record the measured distance on Table 1.
- Measure the distance between points 1 and 26 (line CD). The deviation from the desired value should not exceed 0.007m. Record the measured distance on Table 1.

3.3 Oval lines verification–after painting the lane lines, before installing the curb (if applicable)

- At points 1-13 and 14-26, measure the distance between the radius control points and the outer edge of the painted line. Record the measured deviations on Table 1.
- Calculate the average deviation from the desired value of radius of curve DA and BC. Record the average deviations on Table 1.
- Calculate the length of curves DA and BC, the length of the inside border, the theoretical running distance of the track and the deviation of the theoretical running distance from 400m following the instructions on Table 2. If such deviation does not exceed 0.08m (and is not short of 400m), the 400m oval can be considered dimensionally accurate.
- Although a table is not provided, the surveyor should verify that radii for all lanes meet the same criteria as 3.3.c, and that the average line radii do not vary more than .015m from the design radius when averaged at points 1 through 26, with no individual point varying more than .03m.

3.4 Track event layout–before painting

- Using certified, computer-generated calculations, or data file derived from such, place a mark at the locations for the starting and finishing lines of all events. These are the start/finish control points. At the option of the contractor and surveyor (if the two are

separate), the surveyor may also layout the locations of all other track markings, such as hurdles, steeplechase barrier placements and baton exchange zones.

3.5 Track event verification—after painting

- a. Measure the distance between all start/finish control points and the leading edge of each painted start and finish line. Record the deviations on Table 3. If the combined total deviation of each start line and the finish line for that event do not exceed $.0002 \times$ design course length (and is not short of design length), the event can be considered accurate.
- b. The surveyor should verify the location of the steeplechase barriers (if required), and randomly, at least 9 incidental markings (hurdle placements, exchange zones, etc.) and record the deviations on Table 4.

Class 4 Report of Measurement

1. *Identification of the facility*

Name of Track/Stadium: _____

Address: _____

2. *Surveyor Information*

Company: _____

Address: _____

Tel: _____

Fax: _____

Surveyor Name: _____

Qualifications: _____

Theodolite: _____

Distance Meter: _____

Instrument Certificates: Attached _____

3. *Survey Conditions*

Date of Survey: _____

Weather: _____

Temperature: _____

4. *Oval Track Desired Layout Data*

Desired radius at outside edge of inner curb or line: _____ m

Desired distance between radius points: _____ m

Theoretical length of oval at measuring line: _____ m

Number of lanes: _____ lanes

Width of lanes: _____ m

Type of inner edge: Painted line only _____ Raised curb _____

5. *Dimensional Accuracy of the 400m Oval Track:*

5.1 28 point control measurements

The 28 point control measurements described in the diagram below have been carried out and the readings recorded in Table 1.

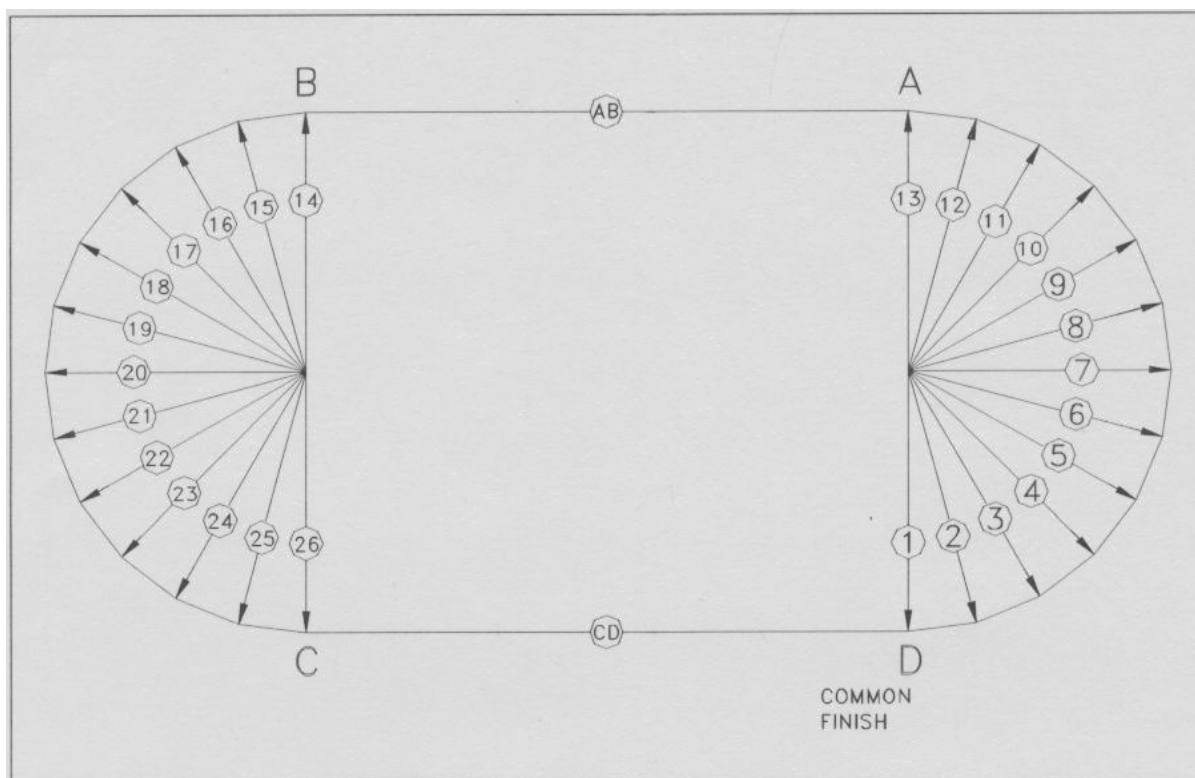


Figure 5.1 – 28 point control measurements of 400m oval track.

Straight AB				Straight CD		
MeasuredLength (m)				MeasuredLength (m)		
Curve DA				Curve BC		
Point	Angle	Measured Deviation (m)		Point	Angle	Measured Deviation (m)
1	0			14	0	
2	15			15	15	
3	30			16	30	
4	45			17	45	
5	60			18	60	
6	75			19	75	
7	90			20	90	
8	105			21	105	
9	120			22	120	
10	135			23	135	
11	150			24	150	
12	165			25	165	
13	180			26	180	
Average Deviation (m) Sum of deviations / 13				Average Deviation (m) Sum of deviations / 13		

Table 1–Calculated average deviation from desired radius of curves DA and BC - Measured length of straights AB and CD

5.2 Calculation of the length

The theoretical running distance of the oval track and its deviation from 400m are calculated as described in Table 2.

(1)	Desired radius of curves DA and BC in meters (to line or curb):		
(2)	Average deviation from desired radius of curve DA in meters:		
(3)	Average deviation from desired radius of curve BC in meters:		
(4)	Length of curve DA in meters:	$=[(1)+(2)] \times 3.1416$	
(5)	Length of curve BC in meters:	$=[(1)+(3)] \times 3.1416$	
(6)	Measured length of straight AB in meters:		

(7)	Measured length of straight CD in meters:		
(8)	Length of inside border in meters:	$= (4) + (5) + (6) + (7)$	
(9)	Theoretical measuring line if curb installed (0.30m):	$= 0.30 \times 3.1416 \times 2$	
(10)	Or theoretical measuring line if no curb installed (0.20m):	$= 0.20 \times 3.1416 \times 2$	
(11)	Total length of track if curb installed:	$= (8) + (9)$	
(12)	Or total length of track if no curb installed:	$= (8) + (10)$	
(13)	Deviation from 400 meters in meters:	$= 400 - (11)$ or $400 - (12)$	

Table 2–Theoretical running distance and deviation from 400m

5.3 Record start line deviations

The deviations from design locations of start and finish lines are recorded in Table 3.

Start or Finish	Deviation (in meters)									
100m	Lane 1									Outer Lane
110m	Lane 1									Outer Lane
200m	Lane 1	Lane 2	Lane 3	Lane 4	Lane 5	Lane 6	Lane 7	Lane 8	Lane 9	Lane 10
300m	Lane 1	Lane 2	Lane 3	Lane 4	Lane 5	Lane 6	Lane 7	Lane 8	Lane 9	Lane 10
400m	Lane 1	Lane 2	Lane 3	Lane 4	Lane 5	Lane 6	Lane 7	Lane 8	Lane 9	Lane 10
800m	Lane 1	Lane 2	Lane 3	Lane 4	Lane 5	Lane 6	Lane 7	Lane 8	Lane 9	Lane 10
4x200m	Lane 1	Lane 2	Lane 3	Lane 4	Lane 5	Lane 6	Lane 7	Lane 8	Lane 9	Lane 10
4x400m	Lane 1	Lane 2	Lane 3	Lane 4	Lane 5	Lane 6	Lane 7	Lane 8	Lane 9	Lane 10
1500m	Lane 1	Lane 2	Lane 3	Lane 4	Lane 5	Lane 6	Lane 7	Lane 8	Lane 9	Lane 10
10,000m	Lane 1	Lane 2	Lane 3	Lane 4	Lane 5	Lane 6	Lane 7	Lane 8	Lane 9	Lane 10
3000/5000m	Lane 1	Lane 2	Lane 3	Lane 4	Lane 5	Lane 6	Lane 7	Lane 8	Lane 9	Lane 10
Steeplechase	Lane 1	Lane 2	Lane 3	Lane 4	Lane 5	Lane 6	Lane 7	Lane 8	Lane 9	Lane 10
Finish	Lane 1									Outer Lane

Table 3–Record of deviation from design location of event starts/finish

5.4. Deviation of incidental markings

Record deviations of randomly chosen marks. A positive number indicates deviation toward the finish line in the running direction. A negative number indicates deviation toward the start line, in the running direction.

Name of mark									
Hurdle or zone #									
Lane #									

Deviation from design location									
Notes									

Table 4—Deviation of incidental markings

6. Certification of length

- We hereby certify that all measurements and information shown on this report are accurate and are the result of a well-conducted survey.
- The measurement for lane one was 0.30 meters ____ 0.20 meters ____ (check one) outward from the inside border.
- The control of the inside length of the track gives a length greater than 400 meters.
- The calculated distance of ____m is within the acceptable plus tolerance of 0.08 meters.

Date: _____

Surveyor Name: _____

Signature: _____

7.5 Class 5—Certificate of Accuracy

Class 5 is usually reserved for tracks that need some assurance of general accuracy. The certificate of accuracy consists of a letter from a Certified Track Builder. This level can be used for small high schools or other facilities that are not intended to host a state or national meet. A sample letter is shown below.

Class 5—Verify measurements meet striping design and write the following letter:

February 3, 2022

Mr. Howard Watkins
Harvey High School
123 Spruce Ave.
Williams, TN 77062

Subject: Certification of the Harvey High School Running Track

Dear Mr. Watkins:

This letter is to serve as certification of accuracy for the running track at Harvey High School, striped by Magic Stripes, Inc. on January 27, 2022.

I, John Johnson [CTB, PE or any other credentials, if applicable], hereby certify that the subject running track has been striped to all rules, requirements, standards and suggestions of the National Federation of State High School Association (NFHS) and the American Sports Builders Association (ASBA).

The measured length of one lap in lane one at the theoretical measuring line is [state actual measured distance].

Please do not hesitate to call me with any questions.

Sincerely,

John Johnson, CTB Magic Stripes, Inc.

Chapter 8

CARE AND MAINTENANCE





PIONEER AND LEADER IN TRACK & TURF EQUIPMENT

WashMatic WM800



SMG's WashMatic WM800 is a ride-on machine for easy cleaning of your synthetic track surfaces



EASY HANDLING



PRECISE



CONTAINED

SMG Equipment, LLC
2002 West Valley Hwy N. Suite 200
Auburn, WA 98001
253-249-7050 office
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Fax +49 (0) 73 06 / 96 65 - 50
info@smg-gmbh.de
www.smg-gmbh.de

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Chapter 8—Care and Maintenance

This chapter covers the issues, challenges and considerations for maintaining a track and its surface.

- 8.1 General Maintenance
- 8.2 Regular Maintenance
- 8.3 Water on Tracks
- 8.4 Winterizing
- 8.5 Repainting
- 8.6 Health and Hygiene
- 8.7 The Repair Continuum—From Resurfacing to Reconstruction
- 8.8 Safety Maintenance

8.1 General Maintenance

A well-designed, well-constructed and well-maintained track will offer years of service. To maximize the useful life of a track regardless of the type of surface, the owner should develop and implement a regular schedule of maintenance. Regular inspection of the track and repair of minor irregularities is more cost-effective than allowing the track to deteriorate to the point where it requires major repair or reconstruction, or to the point where it has become dangerous or even unusable.

Note, however, that even with regular maintenance, over time, every track—even the best ones—will need some repair and, eventually, every track will need to be replaced.

The most important step in maintaining all types of track surfaces is regular cleaning. Keeping the surface clean involves removing debris immediately and spot cleaning spills as soon as they occur. Practice preventive maintenance by prohibiting food and beverages (except water) on the track surface, prohibiting chewing gum and tobacco and prohibiting smoking or providing limited smoking areas at the facility. Provide waste containers with lids to encourage athletes and spectators to keep the area clean.

Require proper footwear. Remember that track shoe spikes of excessive length can accelerate surface wear. Spikes should never exceed 1/4" (6mm) in length and should be conical or pyramidal in design. Needle spikes or Christmas tree spikes will accelerate wear on the track and should be discouraged.

Remove dirt, mud or debris of any kind tracked onto the surface. Abrasive materials such as sand or dirt can cause premature wear to the surface.

The amount of maintenance required by a particular track facility will vary depending upon its geographic location, the amount and type of use and alternative uses, if any. Maintenance is particularly important after a period of inactivity. In any case, the owner should develop an appropriate maintenance plan, ensure that maintenance is performed at timely intervals and keep clear, written records of maintenance procedures, conditions and

problems noted. The need for excessive maintenance may be an indicator of more serious problems. A qualified Certified Track Builder, design professional or experienced contractor will be able to assist an owner in evaluating the need for and type of repair and maintenance.

8.2 Regular Maintenance

8.2.1 Cleaning Equipment

A large-size push broom with soft bristles or a power blower may be used to remove leaves and other debris. Wet/dry vacuums, jet spray cleaners (known as water brooms) and power washers also may be used to clean and remove stains. Power equipment will require a 110V outlet, and jet sprayers and pressure washers will require a hose connection with adequate pressure and volume. Always exercise caution when using a power washer. If the pressure is too high, the surface of the track may be damaged.

Any equipment used to assist in cleaning or maintenance should be inspected for fluid leaks. Leaking oil, hydraulic fluid, gasoline, etc. will damage the track surface. Care should be taken to inspect the equipment before it's used on the surface. Fuel and oil should never be added while the equipment is on the track surface to prevent spills or overfilling.

8.2.2 Crossing Protection

A protected crossing area should be defined for heavy foot traffic, maintenance personnel and equipment that need to cross over the track to the infield. Protect the track surface in this area by laying down rubber matting, artificial turf, carpeting, etc. There are also new products on the market made from woven fabric specifically designed for crossing area track protection. These protective coverings should be removed when not in use. The best products are easily removable and minimize the trapping of moisture.

To help protect the track from vehicles and heavy equipment that must access the track area, it is recommended that plywood or steel plating be used over the protective product to help distribute the point load of tires and or outrigger footings. This material should be removed when not in use and extreme care should be taken during use to protect the track surface beneath. Strong winds and gusts can pick up and “tumble” plywood sheeting, causing damage to the track surface.

8.2.3 Stains

If leaves, twigs, grass or other debris are allowed to stand on the surface of the track, stains may develop. To remove stains, start with the gentlest treatment. A soft brush and a mild cold water detergent solution may remove the stain. Try a small amount of the cleaning solution in an inconspicuous spot off the competition area as a test for compatibility with the track surface. If a simple soap and water solution is unsuccessful in removing the stain, contact the track surfacing installer or surface manufacturer for a recommendation. Do not use petroleum or solvent-based cleaning solutions.

Tree sap, fruit, dead insects and bird droppings also can stain the surface. They should be removed promptly. Once such debris has hardened, stain treatments can be tried but are

likely to be unsuccessful. Prevention is the best option. Likewise, trees should not be allowed to overhang the track and field event areas. Regular trimming should be made a part of the maintenance schedule.

If mold, mildew or algae have appeared in shaded areas, use a power washer to remove them. Mold and mildew often grow where surfaces are contaminated by food spills, soft drinks or decaying matter. Unattended mold or mildew can promote accelerated growth of contaminants to a larger area. Clean surfaces do not support fungus growth.

If any kind of stain has been allowed to remain on the surface for a period of time, natural bleaching of the color coating may have occurred around the stain. Unfortunately, there is no way to correct such bleaching. Do not try to touch up a stain or bleached area by dabbing on leftover surfacing material. Since all tracks fade from exposure to the sun, new material applied to touch up a surface blemish will have a different appearance than the existing track surface and may leave the track with a patchy, freckled look. Some fading of colored surfacing may be expected; after all, the sun is a powerful force. Take photos and document this wear from year to year to help establish an appropriate maintenance schedule.

8.2.4 Landing Areas

Particular care must be taken to assure that landing areas for field events are prepared annually for service. Low areas that may have developed in the landing sector of the shot put need to have additional product placed in order to bring them back to level. It is a good idea to lightly scarify the sector to mix in new material, followed by leveling and compact rolling. All edges should be neatly trimmed, and vegetation should be removed. Check all safety cages and barriers used in the throwing events.

At the long jump/triple jump landing areas, sand should be added to bring the level of sand to the top of the border surrounding the pit. The new and existing sand should be mixed and leveled. Any sand or vegetation build-up around the perimeter should be removed. For those pits that have perimeter sand traps, the traps should be cleaned.

8.2.5 Equipment

Depending on the schedule of use for the track, identify a time (at least once a year) to inspect all equipment. The end of the season may be a good time to shop for bargains and to order replacement parts so that equipment can be repaired during the off season.

8.2.6 Fencing Maintenance

Regularly inspect all fencing including the frame, fabric, hardware and footings. A fence that is heaving or buckling should be repaired. Fence footings should be inspected for any movement. Gates should be tested; they should move freely and the latches should function properly.

Straightening the posts, rails and gates on an existing fence can dramatically improve the look of any track facility. The structural integrity of bent posts may be compromised,

however, in which case it will be necessary to remove and replace them. If fence fabric is in good condition, it is possible to remove snags and tears and tighten the fabric on the frame.

Consider painting a galvanized or aluminized fence dark green or black to provide a better background and to blend into the environment. If fence fabric is rusted, bent or torn, replace it.

8.2.7 Landscape Maintenance

Perform a seasonal walk-through of the facility. Prune back trees and shrubs in the fall. Pay special attention to overhanging branches, which pose a potential hazard, create shadows and drop leaves and sap on the track or field areas.

If lawns around the facility or fields in the center of the track are fertilized, care should be taken to avoid seeding or fertilizing close to the track surface or allowing fertilizer to wash onto the surface. Fertilizers may burn, discolor or otherwise damage track surfaces. Special care should also be taken in using string trimmers. They, too, may damage track surfaces, light posts and fence posts. See also Chapters 2, 4 and 5 for additional information relative to maintenance and repairs.

8.2.8 General Wear and Tear

All fixed equipment, such as takeoff boards, vault boxes, stop boards, etc. should be checked to ensure that they are anchored securely in place. Rotted or dilapidated boards and boxes should be replaced.

Some wear is normal. Track surfaces will tend to wear in lanes 1 and 2 due to high use, as well as at the takeoff areas in the field events. Encourage recreational users and physical education classes to use outside lanes.

After a heavy rain, inspect the track and field areas. Look for any depressions where standing water collects. Depending on severity, these areas may need to be repaired.

Take a few minutes each day before activities begin to check the field event areas to ensure that all equipment is in good working condition and that it is secured in its proper place.

Keep a record of cracks in asphalt and irregularities in the track surface over time. Some cracks may merely be signs of normal wear and tear, while others may signify more serious problems. Weeds growing through cracks will accelerate their expansion. Use an approved herbicide and soil sterilant to prevent regrowth. Experienced professional track surfacing contractors repair cracks and are experienced in determining the types of cracks and the appropriate methods of repair. Only crack repair materials specifically designed for use on track surfaces should be used. In most cases, it is more cost effective to ask for professional help in repairing a crack than to use an unsuitable product or method, which ultimately worsens the problem.

Keep an eye out for bubbles, surface wear, peeling and flaking. Deficient areas of the track surface should be repaired by patching, but note that those spots may take on a different texture or appearance than the original track surface. Look for any irregularities, which could potentially cause an athlete to trip, slip or fall.

Regularly trim the grass and landscaping around the track to permit proper drainage of rainwater from the track surface. Weeds left growing around the perimeter of the track will invade the surface at its edges. A border around the perimeter of the track should be kept free of vegetation to prevent surface damage. Use a commercial herbicide and soil sterilant, available from any lawn and garden center, or hand-pull weeds to keep them under control.

When using lawn care equipment, take care not to damage the edges of the track surface. Blades from mowers, edgers and string trimmers can easily damage the track. Ideally, create a mow strip a minimum of 6" wide to help keep equipment away from the track edges. Make sure all cuttings are blown away from the track surface. Also, when fertilizing or lawn seeding, make sure that no fertilizer or overseeding goes onto the track surface. Grass has been known to take root in track surfaces and fertilizers may stain the surface.

8.3 Water on Tracks

Water on tracks generally does not delay use of the track, but it may cause staining and ultimately, lead to delamination or degradation of the surface.

Even if the track is properly designed and constructed, some water may still accumulate on the track surface. One common cause of inadequate drainage is that the asphalt under a porous surface is lower than the curb or other barrier on the side where the water drains and the water cannot escape. Another problem occurs where owners or the governing body rules require minimal slope, as they often do on high jump approaches, for example, and the slope is not adequate to move the water. However, among the most common causes is improper placement of sprinklers.

Grass fields, particularly newly seeded or sodded fields, need regular deep watering. In areas where rainfall is limited, sprinkler systems may be installed. No track is designed to handle constant watering. Even when weather conditions are ideal—hot and dry—a track simply cannot move that much water, and a porous surface will become saturated. Therefore, irrigation systems should minimize overspray onto the track.

Weather also plays a role in causing excess water on tracks. Tracks can absorb and move more water when the weather is sunny, hot and dry, less when the weather is cloudy or cool. Maintenance practices also can cause problems. For example, improper maintenance may allow grass or dirt to build up on the edge of the track, thereby creating a dam, which prevents water from draining.

Some water in or on the track, at times, is normal. Care should be taken in design, construction and maintenance to minimize the causes of standing water, thus protecting the track surface and extending its useful life. If the track has a slot or trench drain on the interior or exterior of the track surface, be sure to remove debris in the clean outs and in

the drains. Steeple chase water jumps should also be drained and cleaned after the season. Please refer to the ASBA position paper "Water on Tracks" for more information.

8.4 Winterizing

Be sure to do a walk-through of your facility prior to winter. Note any cracks and, if possible, repair them before winter sets in. If water settles in a crack and freezes, it will enlarge the crack. Periodically remove any debris over the course of the winter to minimize staining and bleaching.

In colder climates, if possible, resist the temptation to shovel snow from a track since shoveling may damage the surface. If the snow must be removed, consult with the track surface installer or manufacturer for recommended snow removal methods. Do not apply salt or other ice melt products or attempt to chip ice off.

Check drainage systems and clear drainage structures and pipes. If necessary, repair any damaged structures or pipes that are not functioning correctly. Inspect the entire facility for any evidence of drainage problems or erosion and repair them as necessary.

8.5 Repainting

Repainting the lane lines and event markings every four to seven years, depending on the type of surface and facility use, is essential for providing sharp and clear indicators for the athlete. Certain track surfaces may require initial repainting as early as two years after construction. The owner must be certain of the following:

1. The new lines and markings should be the same as those currently in place on the surface.
2. The existing lines and markings should be verified for accuracy.
3. Should there be lines and markings that are not accurate, the owner must decide on the corrective measures.
4. After the owner makes the decisions on the above issues, the owner and line-marking specialist will sign off on the procedures to be followed in the repainting process.

Prior to repainting, the track surface should be inspected for surface bubbles, cracks, wear, peeling and flaking. Any necessary repairs should precede repainting. In addition, the existing lines and markings should be brushed and cleaned so that all loose particles and debris are removed. This will provide for a better bond between the existing line paint, the track surfacing and the new line paint. The paint used must be compatible with the existing line paint and track surfacing.

8.6 Health and Hygiene

Those responsible for athletic facilities and programs should be sensitive to the general societal concern regarding transmission of infections, some of which are resistant to antibiotics and very difficult to treat. Such infections are being seen more frequently in a wide variety of settings outside of hospitals and nursing homes, where they were first

identified, including in athletic programs. Experts say that good hygiene throughout the athletic program is the key to preventing the spread of infections. For reference, the National Athletic Trainers' Association has addressed the issue via this statement on its website (www.nata.org):

Official Statement from the National Athletic Trainers' Association on Community-Acquired MRSA Infections (CA-MRSA)

In an effort to educate the public about the potential risks of the emergence of community-acquired methicillin-resistant staphylococcus infection (CA-MRSA), the National Athletic Trainers' Association (NATA) recommends that health care personnel and physically active participants take appropriate precautions with suspicious lesions and talk with a physician.

According to the Centers for Disease Control and Prevention (CDC), approximately 25% to 30% of the population is colonized in the nose with *Staphylococcus aureus*, often referred to as “staph” and approximately 1% of the population is colonized with MRSA¹.

Cases have developed from person-to-person contact, shared towels, soaps, improperly treated whirlpools, and equipment (mats, pads, surfaces, etc). Staph or CA-MRSA infections usually manifest as skin infections, such as pimples, pustules and boils, which present as red, swollen, painful, or have pus or other drainage. Without proper referral and care, more serious infections may cause pneumonia, bloodstream infections, or surgical wound infections.

Maintaining good hygiene and avoiding contact with drainage from skin lesions are the best methods for prevention. Proper prevention and management recommendations may include, but are not limited to:

1. Keep hands clean by washing thoroughly with soap and warm water or using an alcohol-based hand sanitizer routinely.
2. Encourage immediate showering following activity.
3. Avoid whirlpools or common tubs with open wounds, scrapes or scratches.
4. Avoid sharing towels, razors, and daily athletic gear.
5. Properly wash athletic gear and towels after each use.
6. Maintain clean facilities and equipment.
7. Inform or refer to appropriate health care personnel for all active skin lesions and lesions that do not respond to initial therapy.
8. Administer or seek proper first aid.
9. Encourage health care personnel to seek bacterial cultures to establish a diagnosis.
10. Care and cover skin lesions appropriately before participation.

National Athletic Trainers' Association March 2005

1 Methicillin-resistant *Staphylococcus Aureus* (MRSA) Infections. Centers for Disease Control and Prevention. Available online at www.cdc.gov/mrsa/

8.7 The Repair Continuum— From Resurfacing to Reconstruction

At times during its useful life, a track will require repair. There are five general levels of repair, but in actuality, a repair continuum exists from the simplest to the most complex. An initial inspection can only evaluate what is visible. If problems appear to extend beyond the surface, then additional investigation may be necessary.

Surfacing contractors seldom, if ever, warranty what they cannot see or inspect, therefore, the risk level for each of the options must be assumed by the owner.

Spot Repairs

Spot repairs may be required, with or without resurfacing, from time to time due to wear in high stress areas, accidental damage or vandalism. Patch kits are available for very minor repairs; however, most repairs should be made by a qualified surfacing contractor.

Cuts made for repairing the surface should always be symmetrical. The cuts should be made large enough and angled sufficiently so that all damaged surfacing can be safely removed and so that new materials can be placed with a good joint to the existing surface. When larger cuts are required, they should be made from lane line to lane line.

New material should be placed so that the finish elevation is the same as that of the existing surface. The surface texture of the new and existing materials also should be as similar as possible, but some dissimilarities in appearance and texture will be apparent.

Resurfacing

Resurfacing implies spot repair of the existing synthetic surface followed by covering that surface with new material. If the symptoms of surface wear and/or cracks do not go beyond the synthetic surface, resurfacing is a viable option. Resurfacing has the lowest cost but the highest risk since it will not identify or correct structural problems that may exist.

Resurfacing normally will be required during the life of the facility; it is defined as the placement of additional material to cover the entire existing surface at a specified application rate (sometimes called “consumption rate”), providing a finish that appears new.

Prior to resurfacing, the existing surface should be inspected for bubbles, cracks, wear, peeling and flaking. Any necessary repairs should be made prior to the application of new material.

Next, the track surface should be cleaned, removing all dust, debris, stains and surface degradation. Cleaning may be performed with an air blower and/or with a pressure washer. However, care should be taken since overly vigorous cleaning may damage the surface.

Next, old markings should be blacked out (or covered with another color to match the track surface) to minimize ghosting, a faint reminder of the old markings on a newly

resurfaced track. Remember that in most cases, the surfacing material will be porous. Old paint can easily show through the pores in a new surface. In addition, if the old lines have been repainted several times, there may be a build-up of height so that the new surfacing actually will be thinner there and more subject to wear.

Finally, the resurfacing product should be applied in accordance with the manufacturer's specifications. Various products and procedures are available, including but not limited to the following:

- Applying one or more coats of latex and/or layers of rubber to a latex track.
- Applying an aliphatic polyurethane spray (varnish) to a polyurethane track.
- Applying a solvent-based or waterborne polyurethane structural spray over an existing polyurethane base mat track.
- Applying a polyurethane structural spray over an existing polyurethane track surface.
- Applying a 3mm–5mm self-leveling polyurethane coating with colored EPDM rubber granules to an existing polyurethane full-pour or sandwich track.
- Applying a seal coating to an existing polyurethane base mat or a base mat/ structural spray track, rendering the surface impermeable. Then applying polyurethane structural spray or a 3mm–5mm self-leveling polyurethane coating with colored EPDM rubber granules over the sealed surface to provide a new impermeable surface.

Surface Replacement

If, in addition to wear and minor cracks, the synthetic surface is starting to delaminate, surface replacement should be considered. Surface replacement involves removing the existing synthetic surface, making minor repairs to the asphalt and installing a new synthetic surface. Surface replacement incurs moderate risk and moderate cost. Removing the old synthetic surface allows the asphalt to be inspected, which may expose structural problems; however, it still may be difficult to determine whether problems exist beneath the pavement.

The finish elevation of the new asphalt repairs must be level with the existing pavement. Additionally, the pavement repair must be even and free from imperfections. As with initial surfacing and resurfacing, there are various materials and methods for the application of new material.

Reconstruction

Reconstruction involves removing the synthetic surface and pavement, reworking or reinstalling the road base and building the track back up. If the symptoms in the existing track include cracks that extend into the asphalt (structural cracks) or loss of planarity (wavy or bumpy surface), reconstruction is necessary. A reconstructed track is similar to a new track except that it may be possible to reuse some of the road base as well as the existing drainage and curbing. In order to take advantage of existing materials and to limit waste, it is popular to recycle the pavement by pulverizing it and mixing it with the existing aggregate, compacting and leveling it to form a new base. However, it is recommended that a layer of aggregate base course be installed on top of the pulverized pavement. Since all or nearly all of the key components are exposed and/or replaced, reconstruction involves the lowest risk but the highest cost.

Whether a surface replacement project will require only replacement of the running surface (surface replacement) or whether it will require complete reconstruction, including replacement of the pavement, may not be evident until the existing surface is removed. An excessive number of cracks in the asphalt may be found, deep birdbaths may be located, undulating asphalt becomes apparent and, in some cases, the existing asphalt may be found to be no longer serviceable. It is better to have a plan in place to address potential problems before they are encountered. A topographical or dimensional survey, and geotechnical study and report will help determine what problems may exist.

Allowance for base repair and leveling can be a part of the quotation and contract. Additionally, contract wording should provide a plan of action for additional needs such as asphalt overlay, asphalt replacement or post-tension concrete, should those become necessary. Providing for these potential problems at the bidding and contract stage offers a clear path of action and minimizes any delay in completing the project.

8.8 Safety Maintenance

A running track should maintain the recommended 1-meter safety zone between runners' lanes and all fixed objects. This safety zone is designed to promote the safety of athletes in case of missteps, falls, etc. A facility must first be designed for safety, but operating and maintaining a facility to promote the safety of staff, competitors and spectators is equally important. Do not assume that athletes will accept the blame for any injuries they might incur. While participants in athletic activities are generally expected to understand and accept the risks inherent in the sport, the facility owner or operator may be held responsible for risks that are not inherent in the sport and are not generally foreseeable, such as those resulting from improper maintenance, improper design or hidden hazards. For additional protection, an owner should consider posting signs requiring appropriate footwear, advising that surfaces are slippery when wet and stating that participants use facilities at their own risk. Having participants sign a release in advance of use may be helpful but may not be practical. Measures such as warnings and releases are not guaranteed to be effective, but they can be a useful part of the risk management effort.

8.8.1 Facility Hazards

To check for internal hazards, an owner or manager should make a regularly scheduled walk-through of the facility with a notepad, pen and schematic. Any problems should be noted so that they can be addressed.

Lighting

Light poles may corrode over time. Eventually, corrosion will reach a point at which the pole becomes structurally unstable; the possibility then exists that the lighting assembly might collapse. Corrosion issues are most common at facilities in coastal areas where the salt air is especially harsh. Corrosion also commonly occurs at track facilities that include a sports field with above-ground irrigation. Poles that show minor corrosion should be sanded and recoated to prevent further damage. Poles that show severe corrosion should be replaced immediately.

Certain light poles are designed for direct burial. These poles are manufactured with a nipple just above the grade where a junction box is installed to connect feeder wiring to the fixture wiring. It is not unusual for maintenance equipment—mowers, carts, etc.—to strike this junction box and break it from the pole, leaving exposed wiring, a serious safety hazard. Owners with light poles with such junction boxes should caution maintenance personnel to exercise care around light poles and should regularly inspect junction boxes to ensure that they are intact. A better option for installing wiring is to insert a conduit into the pole from below grade to feed the power source to the pole or to use poles with anchor bolt base plates.

Fencing

As mentioned above, regular inspection of fencing is recommended. For safety purposes, fencing should be free from snags, rusted areas or sharp edges. If neither a bottom rail nor a tension wire is used, the bottom edge of the fence should be bent double (knuckled) to prevent injuries and to minimize debris from collecting against the fence. A buildup of leaves and debris slows or stops water from running off, and wet leaves can cause slippery footing.

Surface

Maintenance requirements for specific track surfaces should be discussed with a Certified Track Builder, experienced surfacing contractor or surfacing manufacturer. During a walk-through, any surface issues, such as cracks, tears or blisters, low or high spots, delamination or other irregularities should be noted. A small spot may worsen, becoming a potential tripping hazard. If the owner is reluctant to conduct a surface examination, an industry professional should be consulted. A professional may be more adept at spotting potential trouble.

Debris or litter should be removed before it rots and causes a slippery spot or surface irregularity. Conditions that may trap surface water, including debris trapped along the fence or low areas around the facility, should be eliminated so the drainage system works properly.

Equipment

Track equipment such as hurdles, pads and throwing cages should be inspected at the beginning of the season and regularly throughout. Any damaged equipment should be repaired or replaced. Maintenance equipment also should be inspected.

8.8.2 Other Factors to Consider

Other factors to consider in managing your facility include the following:

Facility Hours

Whether the facility is open 24/7, dawn to dusk or only during specific periods of the day is a matter of preference or policy. When the facility is closed, gates should be locked to deter the possibility of unsupervised, and possibly unsafe, use.

Lighting

Do a night walk-through every week or so. Look for lights that might be out, for dark, shadowy areas where users might feel unsafe—anything that needs repair—and have it addressed immediately.

Security Force

If the facility has a security force, an officer should be in the area of or near the track in the evening or early in the morning when athletes or recreational joggers might be present. Particularly if the facility will be used at night, the presence of a security officer can be reassuring. Some schools and parks are installing emergency alert systems that allow individuals to call for an escort or for help, should assistance be needed.

Rest Rooms

If a facility has rest rooms, consider locking them at night. In an effort to promote safety, it is recommended that facilities are only open during hours of high traffic volume. In many cases, the “safety in numbers” rule should be followed.

Equipment

If the facility will be used after hours, lock away all equipment when the track is unsupervised.



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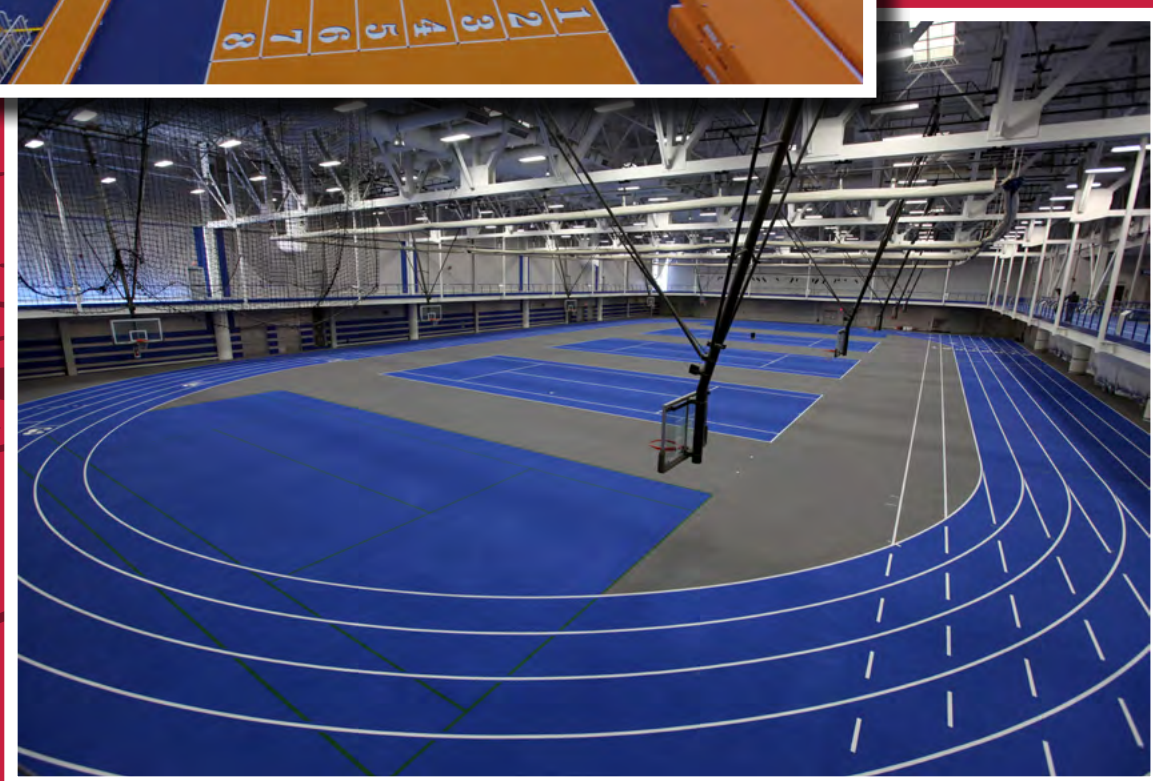
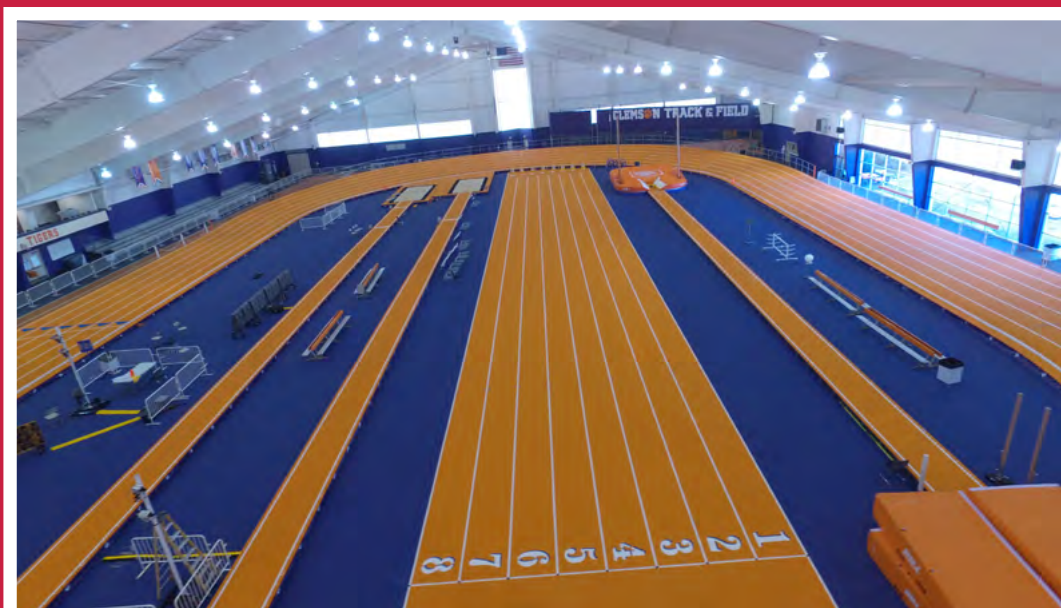
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Chapter 9

INDOOR TRACKS



Chapter 9—Indoor Tracks

This chapter provides information on the following important elements in building indoor tracks. These sections highlight areas specific to indoor tracks, but more details on many of these topics can be found throughout this manual.

- 9.1 Planning the Facility
- 9.2 Track Planning and Design Considerations
- 9.3 Indoor Track Construction
- 9.4 Applying a Track Surface
- 9.5 Field Events and In-Ground Equipment
- 9.6 Marking a Track: Calculations, Layout and Painting
- 9.7 Verification
- 9.8 Care and Maintenance

9.1 Planning the Facility

In general, indoor tracks follow outdoor rules except for specific differences noted for indoor competition. World Athletics, USATF, and NCAA have sections in their rules dealing with indoor tracks. The NFHS does not, although a few state associations may have them.

Accommodating all end users often means sharing the same indoor floor surface, despite different uses. That means the synthetic surface may need to be one that is *suitable* for all users, but not perfect for any, in terms of texture and other physical properties. Surfaces that normally “shed” rubber particles as part of normal wear are often encapsulated to reduce or eliminate this characteristic. And in many cases, only a portion of the floor may be synthetic turf.

The level of track competition programmed for the facility will also have a major effect on the building, not only for the needs of its track events, but also the amenities required to host certain competitions. Guidance on the specific needs for various types of competition is usually available from the governing bodies as noted above and from the conference or other organization sanctioning the event.

Buildings housing indoor track facilities make it difficult or impossible to follow all rules “to the letter.” Reasonable variations are often made by the governing/sanctioning bodies to such components as runway distances, lane width, ceiling heights, and runouts. If variations are required, they should be reviewed and approved as early in the planning process as possible.

Building design is beyond the scope of this document. Therefore, such features as lighting levels and HVAC are not included.

9.2 Track Planning and Design Considerations

9.2.1 Types of Indoor Tracks

Generally speaking, there are three types of indoor tracks. Which type of track to install for a particular facility depends largely on the events are expected to be held there. In other words, a multipurpose facility will require a different type of design than one meant only for typical running events.

1. Permanent (banked or flat). These tracks have a fixed design and, thus, are well suited to a single-use facility. Because they are permanent, start line measurements and other details do not vary.
2. Permanent design with height-adjustable bends. This type of track employs a fixed design supplemented with hydraulic or portable banking to accommodate a wider variety of events.
3. Portable/prefabricated components. For this type of track, design and construction typically involves simply pouring the foundational floor and assembling the track pieces on top of it. This approach allows for total flexibility for the facility, including accommodating sprints in the infield. The challenge, however, is the track specifications can vary slightly each time the components are taken apart and reassembled.

9.2.2 Basic Geometry

Nearly all new construction is for 200-meter tracks, although some of the governing bodies allow variations. Indoor tracks may be completely flat or banked on the curves. General rules of each governing body for the track oval are as follows:

WORLD ATHLETICS

- Minimum of 4 lanes, with 6 preferred
- Lanes 0.9 to 1.1 meters wide, with a raised curb preferred
- Banked turns are preferred with 0.2 meters of safety zone
- Banking angle should be 10 to 15 degrees and the same in all lanes
- The radius should be 15 to 19 meters, with 17.2 meters preferred
- A common finish is preferred towards or at the end of the straight
- Maximum cross slope is 1%; maximum longitudinal slope is 0.1%

USATF

- Same as the World Athletics

NCAA

- 200 meters is standard, but it may be shorter or longer
- Those built after January 1, 2004 cannot exceed 300 meters
- Minimum of 6 lanes

- Lanes should be 0.9 to 1.2 meters (1-1.2 meters preferred) with a raised curb on the curves
- Curves may be banked
- Banking angle should not exceed 18 degrees and be the same in all lanes
- A radius of 18 to 21 meters is preferred, as measured to the track side of the 5 cm line
- Maximum cross slope is 1%; maximum longitudinal slope is 0.1%
- Tracks should not employ a double-bend design
- As of December 1, 2018, all newly constructed or resurfaced indoor tracks shall be surveyed for a curb and shall have a regulation curb in place for competition

NFHS

- No national rules; states determine the rules
- An infield straight is often desired or, in some cases, required, with specific rules:
- 60 meters long
- An additional 3 meters before the start and 10 meters beyond the finish

9.2.3 Field Event Dimensions & Areas

Field events for everything except the weight throws are the same as for outdoor tracks. Throwing events indoors are the shotput for both WA and NCAA, and the weight throw for the NCAA.

- Circle size and landing sector angle are the same as outdoor
- Cage is almost always portable
- End portion of the landing sector is rectangular, at least 9 meters wide, and has stop barriers
- Landing sector should use suitable material since it will leave imprint

9.3 Indoor Track Construction

Most indoor track facilities include a standard, flat building slab built to the building's specifications. Installing a track surface typically does not add enough weight to change the loading of that slab.

When constructing an indoor track, slope does not have quite the same considerations as with an outdoor track since drainage and weather conditions are not an issue. Running direction slope still applies, however.

As noted by World Athletics, the banked bends on indoor tracks should include a safety zone minimum of 0.2 meters wide as measured from the outer edge of the outer lane outer marking. The outer edges should include a protective railing from the beginning of the transition curve, throughout the bend, up to the beginning of the next straight.

If the inside edge of the track has a vertical drop in excess of 0.1 meters, a safety zone should be added that is at least 0.30m wide, flush with the inner edge of the track.

When the substrate for the floor is asphalt, concrete curbs or other suitable materials should be used to border sand pits and synthetic turf areas. These areas should be designed to be completely covered with an extension of the synthetic surface.

Concrete is the most common material used in substrates for indoor tracks. It is relatively easy to accurately control both dimensions and elevations, and it typically has a longer life than asphalt. A properly designed concrete slab can also eliminate the need for curbing and the related joint and elevation issues.

Asphalt is occasionally used for the substrate, usually to reduce initial cost. The advantages noted for concrete are obviously disadvantages for asphalt. In addition, repairs are not as durable or accurate in terms of elevation as concrete.

9.4 Applying a Track Surface

If your project involves an existing building with a slab already constructed, the same types of synthetic surfaces used outdoors can be applied to indoor tracks. A variety of surface types are available depending on the use (multipurpose, track-specific, pad-and-pour, etc.). Whichever surface you choose, several factors unique to an indoor application should be considered.

Spray applications are difficult to use indoors. Initial installation may be relatively easy during construction when masking is easy and there is no HVAC to deal with. However, it can be anywhere from difficult to impossible to control overspray when the building is complete, such as when resurfacing at a later date.

Resurfacing with an embedded finish and a sprayed encapsulation coat will have the same overspray problem.

An embedded finish that is not encapsulated will have some “shedding” for the first couple of years. Use of smaller rubber particles in the embedded finish will help control this problem.

Full pour and premanufactured surfaces are the most commonly used indoors.

9.5 Field Events and In-Ground Equipment

Ceiling height requirements will vary depending on the event. Higher ceiling heights are required above the pole vault, high jump, long/triple jump, and weight throw. Running events usually do not require special considerations.

9.5.1 Long Jump, Triple Jump, and Pole Vault

The sand pit should be covered when not in use. If it is only used for the long/triple jump and will have no other traffic, a fabric cover with weighted edges can be used as it does not require removing and replacing sand. If there will be other traffic, a rigid aluminum cover recessed for synthetic surface should be used. This will require that sand be removed to

allow for cover thickness, then added when prepared for track use. The vault box should have either a rigid cover with track surface or a polyurethane plug in place when not used for track.

9.5.2 Throwing events

Portable cages are common for the indoor throwing events as they can be easily removed to allow other uses of this space. Synthetic turf is becoming more common for the throwing events landing sector.

9.5.3 Accessibility to Storage

A significant amount of storage is needed for the track equipment such as hurdles, starting blocks, and throwing implements. This storage area should be in reasonable proximity to the track.

9.6 Marking a Track: Calculations, Layout and Painting

Marking an indoor track is essentially the same as marking an outdoor track, except when the curves are banked. This makes the calculations significantly more complex. When a hydraulic banked track is used, calculations are usually done with the curves banked. When the banks are not used and the track remains flat, times are adjusted using conversion factors. Refer back to the appropriate governing body on specific conversions and calculations.

9.7 Verification

World Athletics does not have a Class 1 and Class 2 verification for indoor tracks. There is only one class indoors which requires verification of both distance and slope, similar to Class 2 outdoors. As of December 31, 2015, only two tracks in the US are World Athletics-certified, although many others are designed and constructed to be certifiable. As of December 31, 2019, the ASBA has not developed procedures for indoor tracks.

9.8 Care and Maintenance

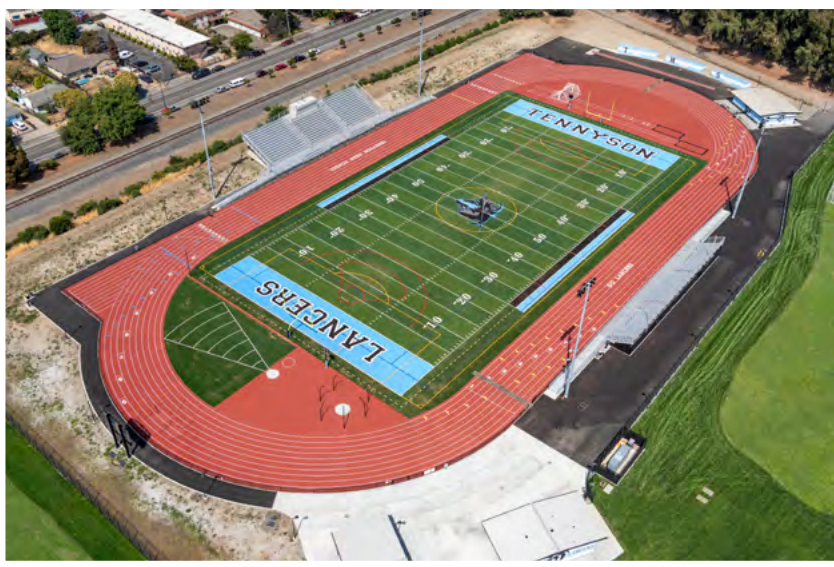
The timing and need for care and maintenance is not as difficult indoors since temperature and facility use are easier to control. However, the frequency and the procedures required are more difficult.

- There is no wind and rain to help keep it clean.
- Food and beverage spills are more common.
- Hoses and brooms typically can't be used; special machinery to confine cleaning residue is needed.
- Resurfacing and any type of other construction is more difficult (see Chapter 3).

Regular cleaning is typically performed with a powered sweeper capable of applying cleaning fluids and chemicals, as well as recovering the resulting debris. The synthetic surfacing manufacturer or installer should be consulted for appropriate cleaning products.

Chapter 10

APPENDICES



Chapter 10—Appendices

Appendix A—Sources of Information

Appendix B—Governing Bodies and Rules of Track and Field

Appendix C—Track and Field Specifications

Appendix D—Calculation and Slope Notations

Appendix E—Conversions

Appendix F—400 Meter Track Verification Method

Appendix G—Suggested Running Track Marking, Color Coding by Governing Body

Appendix H—Checklists

Appendix A—Sources of Information

There are few references available on the subject of track and field construction, and some of those are outdated. This list is provided as a service. Readers are cautioned to evaluate each listed reference on its own merits.

Reference Books

The following publications provide information concerning various aspects of track and field-related topics.

Membership Directory, American Sports Builders Association, Forest Hill, MD. www.sportsbuilders.org

Court & Field Diagram Guide, National Federation of State High School Associations, Indianapolis, IN. www.nfhs.org

Track and Field Facilities Manual, International Association of Athletics Federations, Monaco. www.worldathletics.org

Track & Field Rules Book, National Federation of State High School Associations, Indianapolis, IN. www.nfhs.org

Cross Country/Track & Field Rules, National Collegiate Athletic Association, Indianapolis, IN www.ncaa.org

Competition Rules, USA Track & Field, Indianapolis, IN. or www.usatf.org

Trade Publications

The following magazines and other publications serve various trade groups and professions associated with track and field. Contact individual publications for back issues or for reprints of articles on appropriate topics.

Athletic Business
22 E. Mifflin Street, Suite 910
Madison, WI 53703
608-249-0186
www.athleticbusiness.com

Track & Field News
2570 West El Camino Real, Suite 220
Mountain View, CA 94040
650-948-8188
www.trackandfieldnews.com

Trade Associations

The following organizations are trade associations serving the industries involved in various aspects of track and field design, construction, supply, competition and more.

American Concrete Institute (ACI)
38800 Country Club Drive
Farmington Hills, MI 48331
248-848-3700
www.concrete.org

American Sports Builders Association
(ASBA)
2331 Rock Spring Road, Suite 100
Forest Hill, MD 21050
866-501-ASBA (2722)
www.sportsbuilders.org

National Asphalt Pavement Association
(NAPA)
6406 Ivy Lane, Ste. 350
Greenbelt, MD 20770-1441
888-468-6499
www.asphaltpavement.org

US Track & Field and Cross Country
Coaches Association
1100 Poydras Street, Suite 1750
New Orleans, LA 70163
504-599-8900
www.ustfccca.org

Associations of Design Professionals

American Institute of Architects (AIA)
1735 New York Avenue, NW
Washington, DC 20006-5292
800-AIA-3837
www.aia.org

American Society of Landscape Architects
(ASLA)
636 Eye Street, NW
Washington, DC 20001-3736
202-898-2444
www.asla.org

American Society of Civil Engineers (ASCE)
1801 Alexander Bell Drive
Reston, VA 20191
800-548-2723
www.asce.org

Construction Specifications Institute (CSI)
123 North Pitt St, Ste 450
Alexandria, VA 22314
800-689-2900
www.csinet.org

Appendix B— Governing Bodies and Rules of Track and Field

The following is a compilation of track and field rules from the four organizations that are responsible for determining the rules and standards by which competition is conducted in the United States of America. Complete information is available in the rules books published by each organization. Some of the organizations' rules are available online at the respective organization's website.

USATF	USA Track and Field 132 East Washington Street, Suite 800 Indianapolis, IN 46204 317-261-0500 www.usatf.org Governing body for open competition in the USA
NCAA	National Collegiate Athletic Association PO Box 6222 Indianapolis, IN 46206-6222 317-917-6222 www.ncaa.org A governing body for collegiate competition
NAIA	National Association of Intercollegiate Athletics 1200 Grand Boulevard Kansas City, MO 64106 816-595-8000 www.naia.org A governing body for collegiate competition; currently follows NCAA rules
NFHS	National Federation of State High School Associations PO Box 690 Indianapolis, IN 46206 317-972-6900 www.nfhs.org Governing body for high school competition
NIAAA	National Interscholastic Athletic Administrators Association 9100 Keystone Crossing, Suite 650 Indianapolis, IN 46240 317-587-1450 www.niaaa.org A governing body for high school competition; currently follows NFHS rules
World Athletics	6-8, Quai Antoine 1er, BP 359 MC 98007, Monaco Cedex 377 93 10 88 88 www.worldathletics.org Governing body for international competition <i>*formerly IAAF</i>

Appendix C—Track and Field Specifications

These specifications were correct at the time of publication and are provided in a summary format as a convenience to the reader. The rules are constantly under review by the governing bodies and are subject to change. Please consult the appropriate governing body's latest rule book for the rule in place at the time of your project as well as complete wording for the rule.

The construction of a 400m track with a common finish line for all events at the juncture of the straight and the curve is preferred. Direction of the straight races should be with the prevailing wind. Direction of running around the 400m race course is so that the inside lane boundary is to the runner's left.

Length of Track

WA	The length of a standard running track shall be at least 400m and not more than 400.04m. It shall consist of two parallel straights and two bends whose radii shall be equal.
USATF	The nominal length of an outdoor track shall preferably be 400m. It shall consist of two straights and two turns.
NCAA	Same as World Athletics
NFHS	400m track standard, and not more than 400.08m for tracks built starting in 2024. It shall consist of two straights and two curves. The radius of tracks may differ depending on the configuration for activities inside the oval and available space.

Width of Track

WA	Oval:	At least 4 lanes
	Straight:	At least 6 lanes
	International Competition:	At least 8 lanes (Rule 1 a, b and c)
	Safety:	At least 1 meter on inside and 1 meter preferred to the outside of track
USATF	8 lanes standard.	
NCAA	Minimum width of 6.4m; 9 lanes of 1.067m desirable for larger meets.	
NFHS	No minimum width.	

Lane Width

WA	1.22m +/- 0.03.
USATF	Same as World Athletics; under special conditions lane width can be decreased to as little as 91.4cm (36").

NCAA Lanes shall be the same width.
Recommended minimum width is 1.067 (+/- 0.03) meters (42").
Maximum width to be 1.22 (+/- 0.01) meters (48").

NFHS Standard lane width 42" (1.07m).
Each lane is the same width.

Note: The 5cm lane line on the right is included in the lane measurement.

Radius Requirements

WA Standard 400m track radius is 36.50m to the runner's edge of the first lane line (36.8m to measure line).

USATF Inside radius between 35m and 38m, except double-bend minimum of 24m.

NCAA Maximum of 50m for records or championship qualifying.

NFHS No recommendation.

Inclination

WA Maximum lateral inclination shall not exceed 1:100. Lateral inclination shall be towards the inside lane. Inclination in the running direction shall not exceed 1:1000 downwards.

USATF Same as World Athletics.

NCAA Same as World Athletics except inclination is preferably toward the inside. The maximum overall downward inclination permitted in the running direction for the track, the running direction for all runways, and the throwing direction for all landing sectors, measured over the full length of any straight track distance, the last 20 meters of the javelin runway, the last 40 meters of other runways and the full length of landing sectors, shall not exceed 1:1000, one-tenth of one percent (0.1%). Note: This rule is presently not being enforced for "the last 40 meters of other runways."

NFHS Maximum lateral inclination is 2:100. Inclination in the running direction is the same as World Athletics.

Curb Dimensions

WA Height: 5cm
Width: Minimum of 5cm, recommended maximum width 25cm

USATF Height: 5cm, except may be raised to 6.5cm to allow for drainage 5cm
Width:

NCAA	Height:	Approximately 5cm
	Width:	5cm minimum
NFHS	Height:	2in (5cm)
	Width:	No recommendation

Note: Rules were recently revised so that a raised curb is only needed on the curves.

Curb Material

WA	Suitable material.
USATF	Same as World Athletics.
NCAA	Suitable material. Edges shall be rounded.
NFHS	Solid curb. Edges shall be rounded.

Note: For World, American and NCAA championship qualifying and championship meet records, a regulation curb must be in place.

If No Curb

WA	Curb is mandatory, unless track is grass.
USATF	Same as World Athletics.
NCAA	Curb is mandatory.
NFHS	When a curb is not used, then a 2" (5cm) or wider line denotes the inside lane marking.

Measurements

WA	Lane 1 is measured 30cm outward from the runner's edge of the curb. Remaining lanes are measured 20cm outward from the runner's edge of the left-hand lane line.
USATF	Same as World Athletics.
NCAA	Same as World Athletics.
NFHS	Same as World Athletics if using a curb for lane line 1. If using a painted line, measure 20cm outward from the runner's edge of the first lane line. Remaining lanes are measured 20cm outward from the runner's edge of the left-hand lane line.

Finish Line Location

WA	General practice is to place the finish at the juncture of the straight and curve.
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- USATF** Same as World Athletics.
- NCAA** A common finish line is recommended for all races.
- NFHS** When a common finish line is used, it is recommended that the finish line be located at the point of the curve or within 20 meters of that point.

Start Line Locations

- WA** In all races not run in lanes, the start line shall be curved so that all the runners start the same distance from the finish.
- USATF** Same as World Athletics. (Races start from alleys; the lines shall be curved.)
- NCAA** Same as World Athletics.
- NFHS** Same as World Athletics.

Color of Lane Lines

- WA** All lane lines as well as start and finish lines are to be white.
- USATF** Same as World Athletics.
- NCAA** All lane lines shall be white. Finish lines white. Start lines see NCAA Rule 1.1.4.
- NFHS** Lane lines do not have a specified color. Finish lines are white. Start lines see NFHS Rule 5.2.4.

Steeplechase Course and Water Jump

- WA** The standard distances shall be 2000m and 3000m.
- There shall be 28 hurdle jumps and 7 water jumps in the 3000m event. There shall be 18 hurdle jumps and 5 water jumps in the 2000m event.
- There shall be five jumps in each complete lap with the water jump as the fourth.
- The jumps shall be evenly distributed so that the distance between the jumps shall be approximately one fifth of the nominal length of the lap.
- In the 3000m event, the distance from the start to the beginning of the first lap shall not include any jumps, the hurdles being removed until the competitors have entered the first lap.
- In the 2000m event, the first jump is at the third barrier of a normal lap. The previous hurdles are removed until the competitors have passed them for the first time.

The hurdles shall be 0.914m high for the men's events and 0.762m high for the women's events (+/- 3mm for both).

Width shall be at least 3.94m.

RECOMMENDATION: First hurdle in the race should be at least 5m in width.

Section of the top bar of the hurdles including the water jump shall be 12.7cm square. The top bars should be painted with white and black stripes, or with other distinctive contrasting colors, such that the lighter stripes, which should be at least 22.5cm wide, are on the outside.

The weight of each hurdle shall be between 80kg and 100kg.

Each hurdle shall have on either side a base between 1.2m and 1.4m. The hurdle shall be placed on the track so that 30cm of the top bar will extend inside the inner edge of the track.

The water jump, including the hurdle, shall be 3.66m (+/- 2cm) in length. The width shall be 3.66m (+/- 2cm).

The water shall be level with the track surface.

At the hurdle end, the water depth shall be 50cm for approximately 30cm. From this point, the bottom shall have a regular slope upwards to the level of the track at the other end of the water jump.

The hurdle shall be firmly fixed in front of the water and be of the same height as the others in the competition.

To ensure safe landing of the competitors, the bottom of the water jump shall be covered at the further end with a suitable material, at least 3.66m wide and 2.5m long, the thickness of which should not exceed 2.4cm.

USATF Same as World Athletics.

NCAA Same as World Athletics except standard distance is 3000m.

NFHS Event not governed by NFHS.

Example of Steeplechase Hurdle Placements

	Lap of 390m	Lap of 410m
Race distance	3000m	3000m
Distance from start to beginning of first lap	270m	130m
Seven laps at distance	2730m	2870m

Hurdle Space:

Beginning of 1 st lap to 1 st hurdle	10m	10m
From 1 st to 2 nd hurdle	78m	82m
From 2 nd to 3 rd hurdle	78m	82m
From 3 rd to 4 th hurdle (water jump)	78m	82m
From 4 th to 5 th hurdle	78m	82m
From 5 th to finish line	68m	72m

High Jump

- WA** The minimum length of the runway shall be 15m except in Rule 1a, b or c competitions, where it shall be 20m. If possible, the runway should be 25m. The runway and takeoff areas are usually covered with the same surface as the track. The maximum overall inclination of the runway and takeoff area shall not exceed 1:167 in the direction of the center of the crossbar. The takeoff area shall be level.
- USATF** Same as World Athletics except the high jump apron is a 15m semicircle surrounding the standards. For championship competitions, the minimum suggested length for the high jump apron is 20m.
- NCAA** It is recommended that the approach be an octagon or square with a surface of at least 21m. The minimum length provided shall be 15m. In the high jump, the maximum inclination of the last 15 meters of the approach and takeoff area shall not exceed 1:167, six-tenths of one percent (0.6%), in the running direction toward the center of the crossbar. The takeoff area is the semicircle enclosed by a 3m radius whose center point is directly under the center of the cross-bar.
- NFHS** The approach shall consist of a semicircle or rectangle of level and unvarying surface. The center of the semicircle or rectangle is to be the midpoint between the standards. The depth of the approach should be a minimum of 50'. The inclination in the approach shall not exceed 1:100. Hard and unyielding surfaces, such as, but not limited to, concrete, asphalt or wood, that extend out from beneath the sides and back of the high jump landing pad shall be padded with a minimum of 2" dense foam or other suitable material. It is recommended that hard and unyielding materials that extend out from beneath the landing pad be removed.

Pole Vault—Runway

- WA** The minimum length of the runway shall be 40m or, where conditions permit, 45m. The width shall be 1.22m +/- 0.01 and shall be marked by white lines 5cm in width. Maximum allowance for lateral inclination of the runway is 1:100. Overall inclination of the runway in the running direction shall not exceed 1:1000. At the end of the runway, the box must be mounted flush with the runway and installed such that the top inside edge of its end board lines on the O-line and at the same height. The O-line shall be marked by a white line 0.01m wide that extends beyond the outside edges of the uprights.
- USATF** Same as World Athletics.

NCAA The minimum length of the runway shall be 40m. Recommended width of the runway is 1.22m. Maximum allowance for lateral inclination of the runway is 1:100. The maximum overall downward inclination permitted in the running direction, measured over the last 40 meters of the runway, shall not exceed 1:1000, one-tenth of one percent (0.1%). The center of the runway shall be marked by a maximum of seven permanent lines at 30cm increments from the plant box back toward the start of the runway. These lines shall be a maximum of 90cm in length and 5cm in width. See NCAA Rule 1.5.3.

NFHS The minimum length of the runway shall be 40m; 45m is recommended. The minimum recommended width of the runway is 1.07m. Maximum allowance for lateral inclination is 2:100. Overall downward inclination of the runway in the running direction is not to exceed 1:1000. Calculate the slope by comparing the elevation at the start of the runway to the elevation at the zero line. Intermediate measurements are not considered.

Pole Vault—Vaulting Box

WA The vault box shall be constructed from metal, wood or another suitable rigid material. If constructed of wood, the bottom shall be lined with sheet metal for a distance of at least 0.80m from the front of the box. It shall be sunk level with the ground and shall be 1.00m in length measured along the bottom of the box, 0.60m in width at the front end and tapered to 0.15m wide at the bottom of the stop board. The angle between the bottom of the box and the stop board shall be 105 degrees, and the stop board shall be 0.224m long. The side walls of the box shall slope outwards to form an angle of approximately 120 degrees to the base.

USATF Same as World Athletics.

NCAA The vault box shall be constructed of suitable rigid materials. Dimensions are the same as World Athletics. The box shall be painted a contrasting color to the runway.

NFHS The minimum length of the runway shall be 130ft (40m); 147ft 6in (45m) is recommended.

Pole Vault—Landing Area

WA The landing area shall measure no less than 5m by 5m with 6m by 6m recommended for Rule 1 a, b and c meets. The sides of the landing area nearest the box shall be placed 10cm to 15cm from the box and shall slope away from the box at an angle of approximately 45 degrees. The distance from the back of the landing area to the front of the side pads shall be at least 7m, with 8m recommended for Rule 1 a, b and c meets.

USATF Same as World Athletics except minimum landing area shall measure no less than 6m by 6.15m.

- NCAA** The pole vault pad measurement beyond the vertical plane of the back of the vaulting box shall be a minimum of 6m wide by 5m deep. For more information, see NCAA Rule 2.6.1.
- NFHS** The overall size of the pole vault landing system shall be a minimum of 19', 8" (6m) wide by 20' and 2" deep. The landing surface measured beyond the back of the standard bases shall be a minimum of 19', 8" (6m) wide. The dimension of the landing surface in back of the vaulting box to the back of the landing system shall be 16', 5" (5m) deep. Hard or unyielding surfaces such as concrete, metal, asphalt or wood around the landing pit, or between the planting box and the landing system, shall be padded or cushioned with a minimum of 2" (50mm) of dense foam or other suitable material(s). It is recommended that any excess material such as asphalt or concrete that extends out from beneath the landing pad be removed. For more information, see NFHS Rule 7.5.7-9.

Long Jump—Runway

- WA** The minimum length of the runway shall be 40m. The width shall be 1.22m +/- 0.01. The maximum allowance for lateral inclination of the runway shall not exceed 1:100. Overall inclination of the runway in the running direction shall not exceed 1:1000. The runway should be marked by white lines 5cm in width or broken lines 0.05m wide, 0.10m long and 0.50m apart.
- USATF** Same as World Athletics.
- NCAA** For new construction, the minimum length of the runway shall be 40m from the edge of the takeoff board nearest the pit. Runway width is the same as World Athletics. When the runway is not distinguishable from the adjacent surface, it may be bordered with lines 5cm wide. The maximum overall downward inclination permitted in the running direction, measured over the last 40 meters of the runway, shall not exceed 1:1000, one-tenth of one percent (0.1%).
- NFHS** Minimum runway length is 130' (40m) (147', 6" [45m] recommended). Width of the runway should be between 42" and 48". Lateral inclination shall be a maximum of 2:100. Overall inclination in the running direction shall not exceed 1:1000 downward. Calculate the slope by comparing the elevation at the start of the runway to the elevation at the long jump take-off board. Intermediate measurements are not considered.

Note: All runway lengths are measured to the respective foul line. When possible, the runway lengths should be at least 45m.

Long Jump—Takeoff Board

- WA** The takeoff shall be marked by a board sunk level with the runway and the surface of the landing area. The edge of the board that is nearer to the landing area shall be called the foul line.

The takeoff board shall be rectangular, made of wood or another suitable rigid material, and shall measure 1.22m +/- 0.01 long, 20cm (+/- 2mm) wide and 10cm +/- 0.01 deep. It shall be painted white.

The distance between the takeoff board and the far end of the landing area shall be at least 10m. The takeoff board shall be placed between 1m and 3m from the nearer end of the landing area. The plasticine indicator board shall consist of a rigid board, 10cm wide (+/- 2mm) and 1.22m +/- 0.01 long, made of wood or another suitable rigid material. The board shall be mounted in a recess or shelf in the runway, on the side of the takeoff board nearer the landing area. The surface shall rise from the level of the takeoff board to a height of 7mm (+/- 1mm).

The edges shall either slant at an angle of 45 degrees with the edge nearer to the runway covered with a 1mm-thick plasticine layer along its length or shall be cut away such that the recess, when filled with plasticine, shall slant at an angle of 45 degrees. When mounted in this recess, the whole assembly shall be sufficiently rigid to accept the full force of the athlete's foot. The surface of the board beneath the plasticine shall be of a material in which spikes of an athlete's shoe will grip and not skid.

USATF Same as World Athletics.

NCAA Same as World Athletics with the following exceptions:

Length:	Takeoff board shall be at least 1.22m long.
Width:	Takeoff board shall be approximately 20cm wide.
Thickness:	Not more than 10cm thick.
Other:	Distance between the takeoff board and the nearer edge of the landing area shall be no less than 1m or greater than 3m. The plasticine indicator board (not mandatory) shall be prepared so that the top of the plasticine is level with the takeoff board. Color of plasticine shall be a contrasting color.

NFHS The takeoff area shall be marked by a rectangular-shaped takeoff board, manufactured of wood or synthetic material. The takeoff board must be set firmly in the ground at least level with the runway and the surface of the landing area. A firm base must be provided. The width shall be between 8 inches (minimum) and 24 inches (maximum). The length shall be 42 to 48 inches long.

If the takeoff board is 8 inches wide, an additional 8in of firm, resilient material may be placed so that it abuts the edge of the takeoff board farthest from the scratch line.

On hard-surfaced runways, a painted scratch line of contrasting color and with the same size specifications may be used in lieu of a takeoff board. The boys' scratch line shall be located 12ft from the nearer edge of the landing area. The girls' scratch line shall be located 8ft from the nearer edge of the landing area. These distances may be adjusted to accommodate different levels of competition.

Long Jump—Landing Area

- WA** The landing area shall have a minimum width of 2.75m and a maximum width of 3m. The landing area shall be placed, if possible, so that the middle of the runway, if extended, would coincide with the middle of the landing area. The landing area should be filled with soft, damp sand, the surface of which shall be level with the takeoff board.
- USATF** Same as World Athletics.
- NCAA** For new construction, the landing area shall be no less than 2.75m or more than 3m in width. The damp sand shall be identical in elevation with the takeoff board.
- NFHS** The landing area shall have a minimum width of 9ft and a minimum length of 15ft. Sand shall be level with the takeoff board.

Triple Jump—Landing Area

- WA** Same as for the World Athletics Long Jump with the following exceptions:
- Distance between the takeoff board and the far end of the landing area shall be at least 21m.
- Recommended takeoff board locations for international competitions:
- Men: No less than 13m from the nearer end of the landing area.
 - Women: No less than 11m from the nearer end of the landing area.

Note: For other competition, the distance shall be appropriate for the competition.

- USATF** Same as World Athletics Triple Jump except that the women's takeoff board location shall be at least 10m from the nearer end of the landing area.
- NCAA** Same as for NCAA Long Jump with the following exceptions:
- Men: The distance from the foul line to the nearer edge of the landing area shall be at least 11m (recommended is 12.5m).
 - Women: The distance from the foul line to the nearer edge of the landing area shall be at least 8.5m (recommended 11m).
- NFHS** Same as for NFHS Long Jump with the following exceptions:

The boys' scratch line shall be located 32ft from the nearer edge of the landing area. The girls' scratch line shall be located 24ft from the nearer edge of the landing area.

Note: The length of the triple jump runway is measured from the long jump scratch line.

Shot Put, Discus and Hammer Throw—Circle Delineation

WA Circles shall be made of band iron, steel or other suitable material.

The top of the circle shall be flush with the ground outside.

The interior of the circle may be constructed of concrete, asphalt or some other firm but not slippery material.

The surface of this interior shall be level and 14–26mm lower than the upper edge of the rim of the circle.

A portable circle meeting these specifications is permissible.

The rim of the circle shall be at least 6mm thick and painted white.

A white line 5cm wide shall be drawn from the top of the metal rim extending for at least 75cm on either side of the circle. It may be painted or made of wood or other suitable material.

The rear edge of the white line shall form a prolongation of a theoretical line through the center of the circle at right angles to the center line of the landing sector.

USATF Same as World Athletics with the addition that the metal rim shall have a height of 76mm.

NCAA Same materials as World Athletics (rubber is not acceptable). The surface of this interior shall be 19mm +/- 6mm lower than the surface outside the circle.

NFHS The throwing circles shall be marked with a metal, wood or plastic band which shall not rise more than 3/4in (19mm) above the level of the circle. If the circle has a surface of asphalt, concrete, wood or other hard material, a painted line 2in (5cm) may be substituted for the band. A concrete surface with a 1/64 (1mm) roughness is recommended for shotput. Projecting lines 2in (5cm) wide and 8in (20cm) long, lying on the diameter extended and outside the circumference, shall be used to designate the back half of the throwing circle. The inside edge of the line or band is the limit of the throwing circle.

Note: Lines indicating the front and back halves of the circle are painted in the front half of the circle so that the back edge of the line is at a tangent with the exact center of the circle.

Shot Put, Discus, and Hammer Throw—Circle Measurements

WA Shot Put: Inside diameter 2.135m +/- 5mm
Discus: Inside diameter 2.50m +/- 5mm
Hammer: Inside diameter 2.135m +/- 5mm

- USATF** Same as World Athletics.
- NCAA** Same as World Athletics.
- NFHS** Shot Put: Inside diameter 7' (2.134m)
Discus: Inside diameter 8' 2 1/2" (2.50m)

Note: When a white line borders any circle, the lines are not a part of the circle measurement.

Shot Put Stop Board

- WA** The board shall be white and made of wood or other suitable material. It shall be made in the form of an arc so that the inner edge coincides with the inner edge of the circle. The board shall be placed and securely anchored to the ground midway between the sector lines. It shall measure at least 0.112m wide and, when firmly in position, 0.10m +/- 0.002m high in relation to the level inside the circle.
- USATF** Same as World Athletics.
- NCAA** Same as World Athletics with the following exceptions: The board width shall measure 112mm (+/- 2) in the middle, 300 mm (+/- 2) at its ends, and 100mm (+/- 2) in height.
- NFHS** The board shall be constructed of concrete, fiberglass, metal, wood or other hard-surfaced material. It shall be made in the form of an arc so that the inner edge coincides with the inner edge of the circle and can be firmly secured in this position. The board shall measure 1.22m long along the inside, 10cm in height and 11.4cm in width. The inside edge of the line is the limit of the throwing circle.

Shot Put, Discus and Hammer Throw—Landing Areas

- WA** The landing sector shall consist of cinders, grass or other suitable material on which the implement makes an imprint. The overall downward inclination of the landing sector shall not exceed 1:1000 in the throwing direction. The landing sector shall be marked with white lines 5cm wide at an angle of 34.92 degrees such that the lines, if extended, would pass through the center of the circle.
- USATF** Same as World Athletics.
- NCAA** Same as World Athletics.
- NFHS** Effective with the 2006–2007 track season, all shot put and discus will utilize one standard sector of 34.92 degrees.

Note: Radial lines 5cm wide marking the sector are not a part of the sector (Fig. A-1).

Shot Put, Discus and Hammer Throw—Safety Cages

WA Shot Put: No cage required.

Discus and Hammer: Required. Detailed options can be found and explained in the World Athletics handbook as well as in the *World Athletics Track and Field Facilities Manual*.

USATF Same as World Athletics.

NCAA Same as World Athletics. See NCAA rules book for drawings.

NHFS Same as World Athletics (hammer not conducted). See NHFS rules book for drawings.

Note: Chapter 5 includes drawings for discus cages used at the collegiate and high school levels. Consult the governing body rules manual for current requirements.

Javelin Throw—Runway

WA The minimum length of the runway shall be 30m.

The maximum length of the runway is 36.5m.

It shall be marked by two parallel white lines 5cm wide and 4m apart (lines are not part of the 4m measurement).

The throw shall be made from behind an arc of a circle drawn with a radius of 8m. The arc shall consist of a strip painted or made of wood or metal 7cm wide.

The arc shall be painted white and be flush with the ground.

Lines shall be drawn from the extremities of the arc at right angles to the parallel lines marking the runway. These lines shall be white, 75cm long and 7cm wide.

The maximum lateral inclination of the runway shall be 1:100 and 1:1000 downward in the throwing direction.

USATF Same as World Athletics.

NCAA Same as World Athletics with the following exceptions:

For new construction, the minimum runway length shall be 33.5m. It is recommended that the entire runway be constructed of an artificial surface. When an artificial surface is used, the surface should extend 1m past the foul line for safety.

NFHS Recommended length of runway 36.5m(120 feet). Lateral inclination should be no more than 2:100 and 1:1000 from the throwing area to the landing area. No minimum runway length.

Javelin Throw—Implement Landing Area

WA The landing sector shall consist of natural grass or other suitable material on which the javelin makes an imprint. The landing sector is marked by 5cm wide white lines which form the inner edge of the sector and are not included in the sector (Fig. A-2). The sector is formed by laying out lines originating from the center point of the runway through the cross section of the throwing arc and the lines of the runway extending out approximately 100 meters running a line straight through the center of the runway. The distance between the sector lines at this point is approximately 50m. The distance can then be adjusted for the level of competition. This forms a sector of approximately 29 degrees.

USATF Same as World Athletics.

NCAA Same as World Athletics, except the maximum overall downward inclination permitted in the throwing direction for all landing sectors shall not exceed 1:1000, or one-tenth of one percent (0.1%).

NFHS Not defined.

Notes:

1. *Diagrams for field event specifications can be found in Chapter 5.*
2. *ASBA makes every effort to keep up to date with the various rules changes from each association. However, these rules are subject to change without notice to ASBA, and there are no assurances that the current rules are included in this volume. To make a specific rules determination, you are advised to contact the appropriate governing body for its current rules book.*
3. *The rules interpretations in these appendices have been summarized; for verbatim interpretations, please consult each governing body's rules manual.*

Appendix D—Calculation and Slope Notations

The following values are recommended for calculations for a proper balance of accuracy and workable measurements. Some common measurements used in track design and construction are as follows:

20cm = 0.656' or 7.87"	Distance from the start of lane 1 to the measure line (flush curb rule).
30cm = 0.984' or 11.81"	Distance from the start of lane 1 to the measure line (raised curb rule).
0.3048m = 1'	Meter equivalent of 1 foot
1.0668m = 3.5' or 42"	Width of standard lane (except WA)
1.2192m = 4.003' or 48"	Width of standard lane (WA)
1m = 3.28084'	Feet equivalent of 1 meter
pi = 3.1416	Calculation used for curve distances
400m = 1312.336'	Measure line distance in lane 1

The results of calculations should be expressed to the nearest 0.0001m (metric) or to the nearest 0.001' (imperial). These may be rounded to the nearest 0.001m or 0.01' for construction and marking. No minus tolerances are permissible for any race distance.

Note: Distances are rounded up to ensure that the distance is not less than 400m.

Slope Notation

Running tracks must be constructed in a single plane but are sloped (preferably to the inside) to facilitate drainage. Slopes will typically be noted as straight percentages. On occasion it may be helpful to convert slope percentages to inches per feet or to a ratio. Below are some typical slope conversions used in the design and construction of tracks:

Percentage	Inches per Feet	Ratio	Sample Use
0.1%	1" in 83.33'	1:1000	Maximum downward longitudinal slope measured over the full length of the track oval in running direction
0.4%	1" in 20.833'	1:250	Maximum downward slope in the direction of the cross bar on high jump for all but NFHS
1%	1" in 8.33'	1:100	Maximum cross-slope on track oval for all but NFHS (slope to the inside is preferred for NCAA and required for USATF)
2%*	1" in 4.17'	2:100	Maximum cross-slope on track oval for NFHS (slope to the inside is preferred)

**ASBA recommended slope for tracks without governing body limitations to facilitate drainage.*

Appendix E—Conversions

Metric Conversion

The procedure for converting an existing 440-yard track to a 400-meter track is fairly simple. If at all possible, it is best to perform the conversion on just one curve. This method is less invasive to the facility and, in most cases, less costly.

Fig. A-3 points out where the work should be performed with respect to the existing track. The contractor should make cuts where there is good ground stabilization between the existing and new materials. It is best to keep cuts on lane lines. All necessary excavation, drainage, base, asphalt and surfacing preparations are presented in Chapters 2 and 3.

Appendix F—400 Meter Track Verification Method

The following procedure can be used to determine if an existing track can be accurately striped to 400 meters. There are rare circumstances when this procedure is not suitable for determining the location of radius points (such as instances where the track is of double-bend configuration). Radius points used for the actual construction or striping of the track should be set by a Certified Track Builder, the track striping contractor or a land surveyor or professional engineer experienced in the design, configuration and layout of track and field facilities.

1. Estimate the location of the PC (juncture of the curve and straightaway) and measure from the inside edge of lane 1 from that PC across the infield to the inside edge of lane 1 at the PC on the opposite straightaway.
2. Divide the distance across the infield by two and install a temporary rod or pin (screwdriver) at this measurement.
3. Measure from the inside edge of lane 1 at the top (apex or mid-point) of the curve toward the temporary rod. Adjust the placement of the rod along the tape to match the measurement determined in step 1. Repeat these procedures until all three measurements match exactly (both PCs to pin and apex to pin).
4. Repeat steps 1 and 2 at the opposite end of the track. These are your radius points.
5. Once both radius points have been installed, measure the distance between them. This measurement will be used to calculate the size of the existing track. The measurements determined in steps 1, 2 and 3 are used for locating the radius points only and will not be used to calculate the size of the track. The following formula can now be used to determine if the existing track can be striped to 400-meter standards.

$400\text{m} - 2 \times \text{distance between the radius points} = 2 \times \text{arc length}$

$\div \text{by } \pi = \text{radius to measure line}$

The track side edge of the first 2”-wide painted line goes 20cm (0.66’) inside the measure line, (30cm [0.98’] for raised curb in place of painted line).

This is the radius to the start of lane 1. Check to see if this dimension fits on the track around both curves. It is also important to check that the outside line fits on the track surface. To do so, multiply the number of lanes by the required lane width. Add this number to the start of lane 1, as calculated above. Check to be sure that this dimension fits on the track around both curves. Lastly, check to be sure that the inside and outside lines will fit on the track straightaways and chutes using a string line or transit.

If it appears that the inside and outside land lines will not fit on the existing surface, contact a Certified Track Builder, qualified design professional or experienced track contractor to determine a method for modifying the track.

Appendix G—Suggested Running Track Marking Color Coding by Governing Body

STARTING LINES

EVENTS	NFHS	NCAA	USATF	WA
100m Dash/Hurdles	White	White	White	White
110m Hurdles	White	White	White	White
200m Dash	White	White	White	White
300m Hurdles	White	N/A	N/A	N/A
400m Dash/Hurdles/Relay	White	White	White	White
800m Run	Green	Green	White/Green Insert	White/Green Insert
4x100m Relay	White	White	White	White
4x200m Relay	Red*	White/Red Insert	N/A	N/A
4x400m Relay	Blue	White/Blue Insert	White/Blue Insert	White/Blue Insert
4x800m Relay (using 1-turn start)	Green	White/Green Insert	White/Green Insert	White/Green Insert
Distance Events Waterfall Starts	White	White	White	White

FINISH LINES

All Events	White	White	White	White
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BREAK LINE

All Events	Green	Green	Green	Green
------------	-------	-------	-------	-------

HURDLE MARKS

100m	Yellow	Yellow	Yellow	Yellow
110m	Blue	Blue	Blue	Blue
300m	Red	N/A	Red	N/A
400m	NA	Green	Green	Green
Steeplechase Barriers	NA	Black	Blue	Blue

RELAY EXCHANGE ZONES

Events	NFHS	NCAA	USATF	WA
4x100m	Yellow	Yellow	Yellow	Yellow
4x200m	Red	Red	N/A	N/A
4x400m	Blue	Blue	Blue	Blue
4x800m	Green	Blue	Blue	Blue

Note: The USATF and World Athletics acceleration marks for the 4x100m are red.

**For red tracks, use white.*

Appendix H—Checklists

The following checklists are useful in designing, preparing and maintaining your track facility. They are included on separate pages below for ease of reproduction.

1. Annual Maintenance Checklist
2. Bidding Checklist
3. Asphalt Mix Design Submittal Checklist

Annual Track Maintenance Checklist

Date Inspected: _____

Facility Name: _____

Drainage:

Check drains and remove leaves, debris, etc. Look for blockages that require professional assistance.

Repairs Needed? _____yes _____no

Track Surface:

Walk around entire track and look for areas of wear, delamination, bubbling, peeling, damage

Repairs Needed? _____yes _____no

Check for water to collect in low areas and not drain off after the rest of the track has dried.

Repairs Needed? _____yes _____no

Check and make sure sprinklers are not over saturating the track. Check surface for water marks and mineral content discoloration from sprinklers.

Repairs Needed? _____yes _____no

Check surface for rock, gravel, grass clippings, dirt or other debris, remove as necessary.

Repairs Needed? _____yes _____no

Check lane lines and markings, make sure they are clear and there is no fading, blurring or other damage.

Repairs Needed? _____yes _____no

Field Events:

Long jump Repairs Needed? _____yes _____no

Triple jump Repairs Needed? _____yes _____no

High jump Repairs Needed? _____yes _____no

Steeplechase Repairs Needed? _____yes _____no

Pole Vault Repairs Needed? _____yes _____no

Javelin Repairs Needed? _____yes _____no

Hammer throw Repairs Needed? _____yes _____no

Shot put Repairs Needed? _____yes _____no

Discus Repairs Needed? _____yes _____no

Hurdles Repairs Needed? _____yes _____no

Track Curbing Repairs Needed? _____yes _____no

Other Repairs Needed? _____yes _____no

Any Specific Concerns?

Bidding Checklist

Checklist will vary depending on bid type (public or private) and specifics, but a general sample bid checklist may include:

1. On time, secure delivery
2. All required paperwork complete
 - Bond/bid security
 - Noncollusion agreement
 - All addenda acknowledged
 - Unit prices and price breakouts completed and reasonable (not unbalanced and math checked)
 - Listing of subcontractors and major suppliers
 - Properly licensed and certified
 - Insurance certificate (proof of insurance)
 - Prebid conference attendance
 - Wage rate acknowledgement
3. All prequalification info complete, experience and references acceptable
4. CPM schedule acknowledgment and compliance certification received
5. Post bid interview to assure:
 - Schedule detail and analysis
 - Interpretation of plans and specs
 - Submitted review (major items)
 - Specified product compliance (major items)
 - Bidder has visited the site/understands conditions (access, soils, staging)
 - Understanding of the contractor's plan for construction
 - Superintendent/PM/project staffing
 - Equipment availability/familiarity
 - Financial stability/adequacy of resources
 - Sustainability (how waste will be handled/recycled)
 - Subcontract experience
 - Labor affiliations
 - Insurance experience modifier
 - Availability of major components or critical long-lead-time items
 - Detail of the bid takeoff (quantities)
 - Any local sourcing
 - Any value engineering or alternate methods of construction
 - Survey and control plan
 - QC and testing payment
 - Cost data (historical data bank)
 - Weather restrictions
 - Normal work hours as well as overtime and weekend work

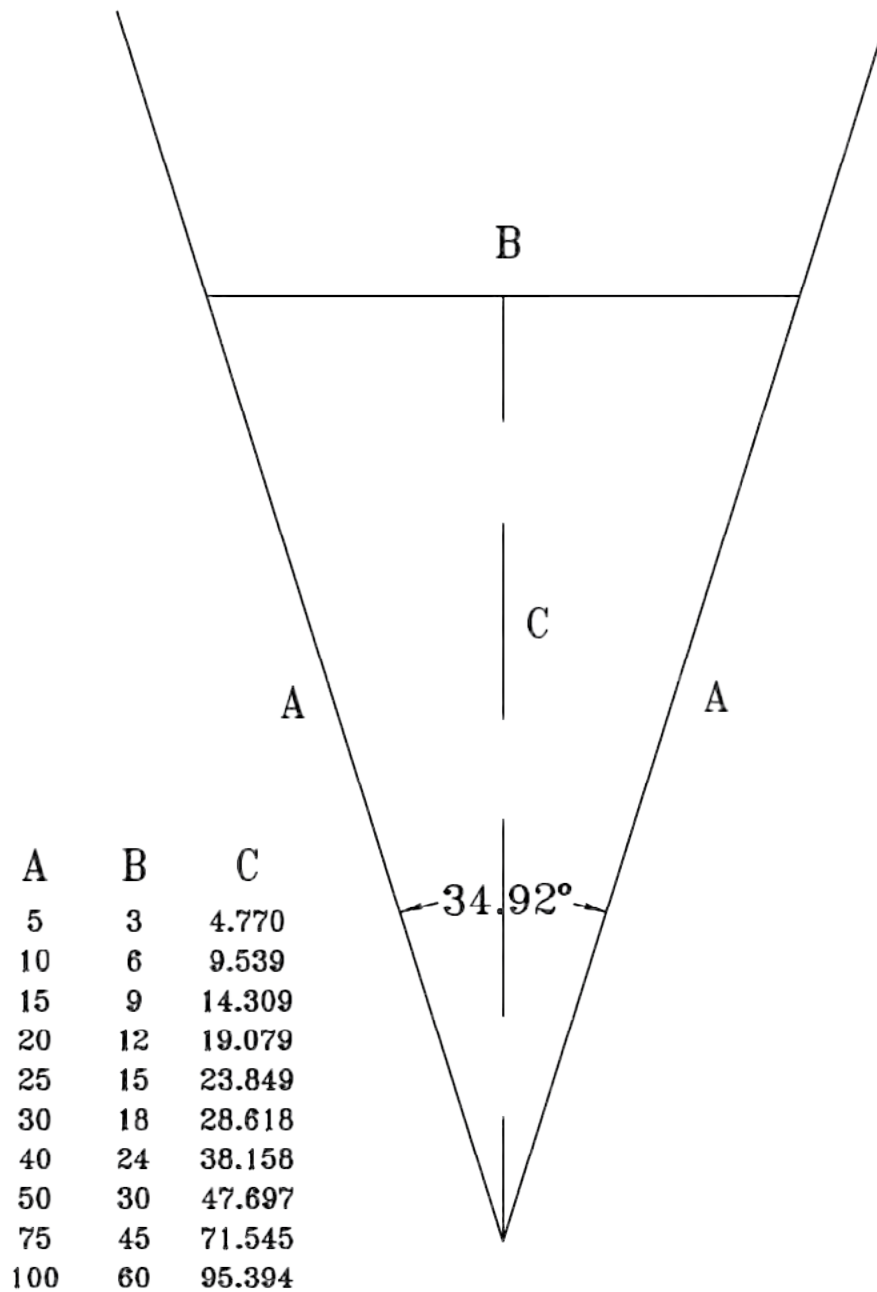


Mix Design Submittal Checklist			
Project:		Date:	
Supplier:		Mix Design:	Surface / Leveling / Base
Included	Missing	N/A	Required Information
			Contractor to select mix design method: (design shall be less than 24 months old)
			50-Blow Marshall
			50-Gyratation Superpave
			Hveem, Low Volume
			Other, Engineers Approval Req'd Before Bidding
			Proper Authorizing Signature for Mix Design
			All Aggregate Types, Gradations & % Crush
			FAA >= 40%, Maximum of 20% Natural Sand
			Plot (0.45 Power Graph) of Final Aggregate Blend
			Bulk (Dry) Specific Gravity of All Aggregates and Final Blend (Gsb), Include All Worksheets
			Optimum Binder Content (Pb)
			Mix Voids at Optimum (Va)
			VMA at Optimum
			Bulk Specific Gravity of Mix at Optimum (Gmb)
			Theoretical Maximum Specific Gravity at Optimum (Gmm)
			Dust to Total AC Ratio
			All Design Data and Associated Design Curves
			Recent Quality Control Production Charts
			Other Information per Specifications
Comments:			

ESTABLISHING THE SECTOR FOR THE SHOT PUT, DISCUS AND HAMMER

Not to Scale

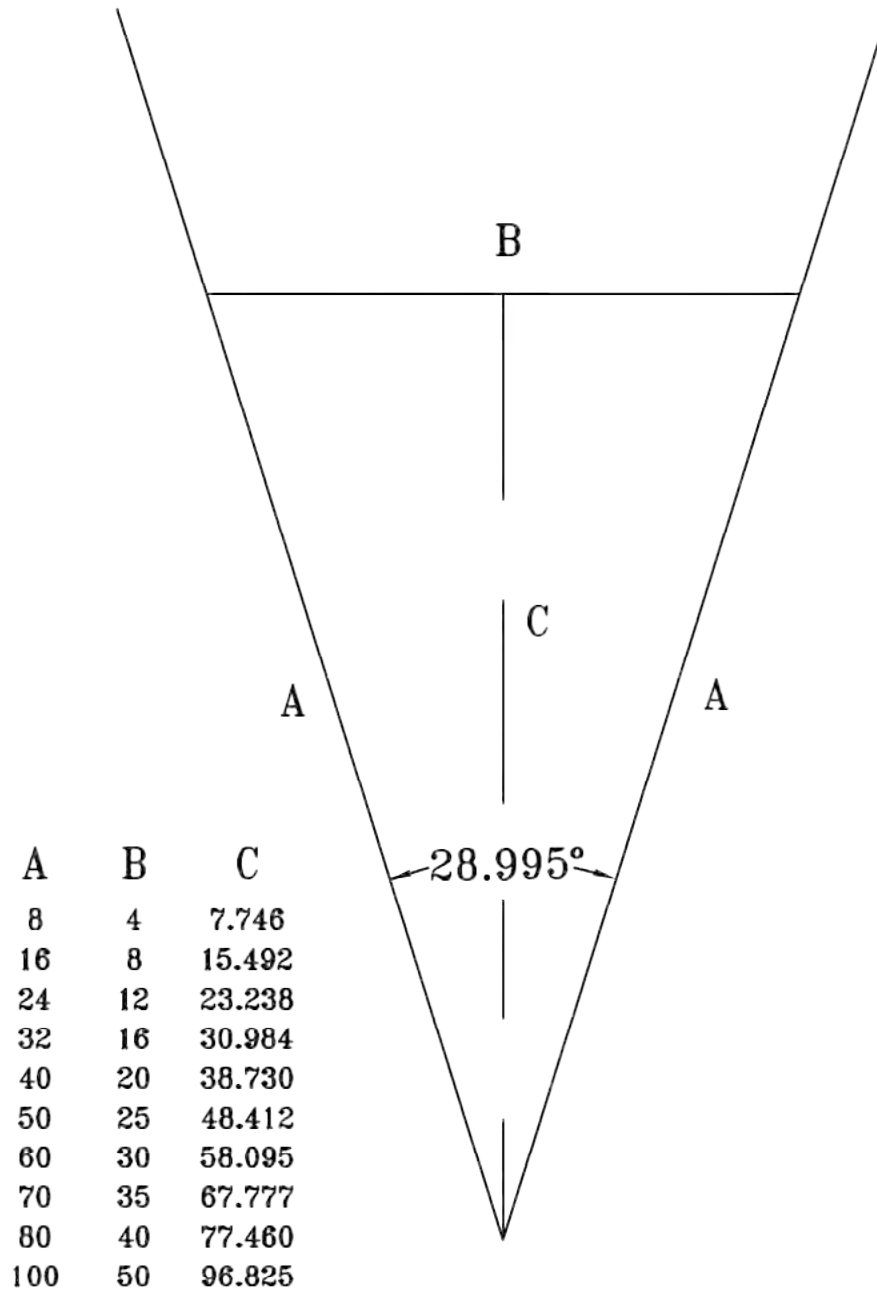
FIGURE A-1



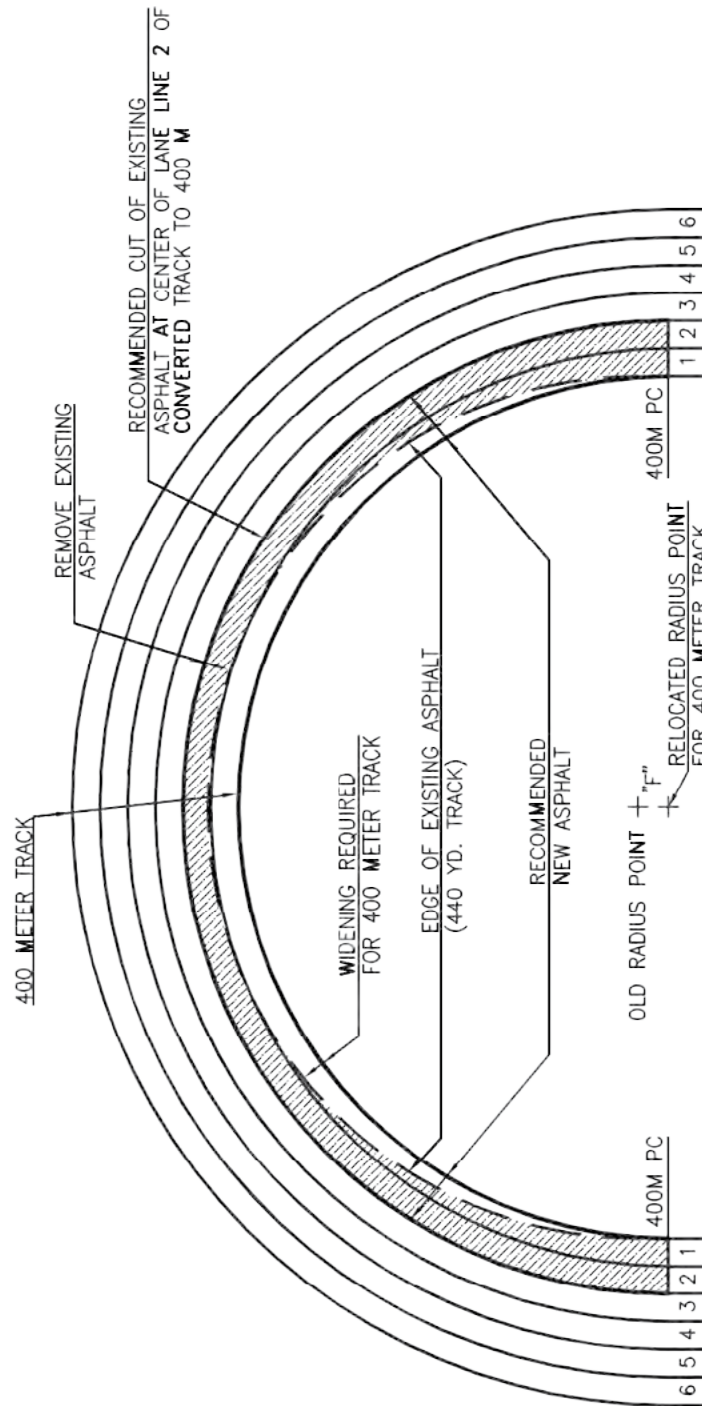
ESTABLISHING THE SECTOR FOR THE JAVELIN

Not to Scale

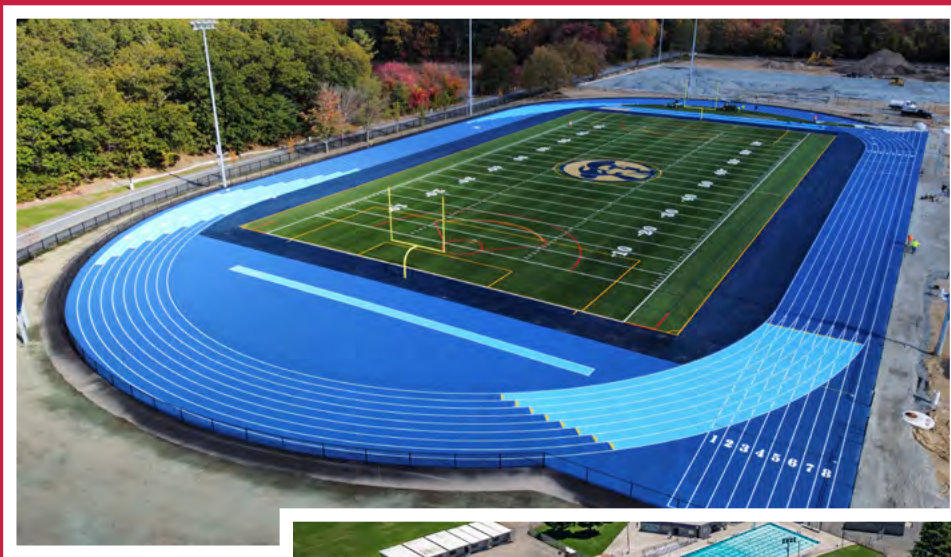
FIGURE A-2



METRIC CONVERSION
FIGURE A-3



Glossary of Terms



Glossary of Terms

This glossary contains terms from the body of the manual that could use more explanation, as well as common terms within the industry.

Abrasion Resistance:

The ability of a surface to withstand wearing away by rubbing or scraping. In synthetic surfaces, it's expressed in terms of grams lost.

Acrylic Cushion Systems:

Surfacing designation, generally referring to an acrylic and finely ground rubber mixture, squeegee or spray applied.

Aggregate:

Particles of construction materials, usually referring to crushed or natural sand, stone, or rock products. Mixed stone may include stone of various sizes, shapes and/or types.

Aliphatic:

A top-coating for synthetic surfaces with a high UV resistance.

ASTM International:

Formerly the American Society for Testing and Materials. Global organization that oversees the standards for consumer products and materials.

Back Straightaway:

Straightaway opposite the home straightaway between PC#2 and PC#3 (visitor's side).

Base:

Material placed over the subgrade and compacted prior to placing the pavement, also sometimes known as the aggregate base.

Base Mat:

Surfacing designation generally referring to a premanufactured or machine-applied (paved) mixture of rubber granules and polyurethane binders.

Bituminous Asphalt Mixture:

Mixture of asphalt (hot or cold) and aggregate.

Box Alley Start:

A curved start line for runners sharing lanes #1 and #2, #3 and #4, #5 and #6, or #7 and #8, or for runners sharing lanes #1 through #4, and #5 through #8. This start is sometimes used for the 800m race.

Break Line:

Also called cut line. The marking at the end of a curve (PC#2 or PC#4) denoting the location where runners may depart their respective lanes and move to the inside lane in specific races.

Broken Back Track:

See “Double-Bend Track.”

Calibration:

Mathematical formulation utilized to determine various distances designated by governing rules.

Certification:

The generation of a document stating that the responsible party has set or verified the calculations and markings and determined the course to be in compliance with all of the rules of the appropriate governing body, including specified slopes, and that course measurements for each lane are equal to or greater than the specified length.

Checking:

Also known as “heat checking;” a pattern of hairline cracks in asphalt, which often occurs when asphalt is compacted in thin lifts or at inappropriate temperatures. Cracking occurs when the surface has already cooled and hardened before it is rolled. The pattern of cracking follows the direction of rolling.

Chute:

Extension of the straightaway beyond the transition. Chutes and transitions are used as staging areas and to run the 100m and 110m events on both the home and visitor straightaways. They are sometimes used for other events as well. Chutes are identified by the same number as the nearest point of curve; i.e., Chute #1 is nearest to PC#1.

Clear Zone:

Sometimes called “safety zone;” an area free of obstructions, such as retaining walls, fences, light poles, bleachers, etc., or severe grade changes. WA recommends a clear zone 1m in width on the inside and outside of the running surface. NCAA and NFHS recommends a 1m obstacle-free zone on the inside and outside of the track “whenever possible.” Common exceptions include finish line poles and raised inside curbs.

Coefficient of Friction:

Wet spike resistance

Cohesion:

The ability to stick together tightly. Cohesion of a soil is the shearing strength not due to friction.

Compressibility:

The property of materials that enable them to decrease in volume when subject to pressure from all sides.

Crotch:

The intersection of the outside lane line of the curve and the inside lane line of the chute. Also the terminating point of the transition and beginning of the chute.

CTB:

The designation of a Certified Track Builder. CTBs are certified through an intensive exam and continuing education program hosted by ASBA.

Curing (pavement):

The development of the mechanical properties of the asphalt binder. This occurs after the emulsion has broken and the emulsion particles coalesce and bond to the aggregate. Due to the properties of various products used in track construction, cure times can be affected by atmospheric conditions, product type, temperature, etc. Curing time can take several weeks in some instances.

Cut Line:

See “Break Line.”

D-Zone:

The area of the infield that is made up of all or part of the semicircle inside the curve of the running track. The D-Zone may also be called the D-Area.

Daylight:

A termination area that is lower in elevation than adjacent areas, allowing water to flow away from the facility.

Design Professional:

An architect, landscape architect or civil engineer.

Double-Bend Track:

A track with a compound radius curve (each usually two small curves and a large curve formed by three different radii and, together, forming one complex curve).

Durometer:

A measure of the hardness of a substance. Sport surfacing materials are measured on a Shore A scale. The higher the rating, the harder the substance. Often used to refer to the hardness of a rubber granule.

EDM:

Electronic Distance Measuring device.

Elasticity:

Ability of a material to return to its original shape after having been deformed by a load for a short period of time.

Elongation:

The capacity for being stretched or lengthened before breaking. Expressed in a percentage of its original size.

EPDM:

Ethylene propylene diene monomer (EPDM) rubber, virgin rubber specifically made to be granulated and available in black and colors.

Equal Quadrant Track:

A track consisting of two straightaways of 100m each and two curves of 100m each.

Exchange Zones:

Areas (zones) in each lane where runners must pass the baton during relay races. These zones are 30m in length for the 4x100m and 4x200m. Relays longer than the 4x200m relay have 20m exchange zones with an acceleration mark 10m ahead of the runner's entrance of the exchange zone. Exchange zones are most commonly denoted by large triangles or check marks. Zones are color-coded for specific races (e.g., yellow for the 4x100m relay). Some relay zones are denoted by a change in track surface color.

Force Reduction:

A quantitative measure of the impact forces on an athletic surface. Expressed as a percentage, it is often used as an indicator of the performance, safety, comfort or suitability of the surface.

Friction:

See "Coefficient of Friction."

Frost Susceptibility:

The degree to which soil is affected by frost. Where freezing temperatures are prolonged and the frost line penetrates deep into the soil, ice layers form in the soil, causing it to heave at the surface in an amount equal to the thickness of the ice layers.

Full Pour:

Surface designation generally referring to the nonporous, poured polyurethane installations.

Geotextile Membranes:

A manufactured fabric, either woven or nonwoven, used in various civil engineering and construction applications to separate materials, filter materials and/or provide added stability.

Home Straightaway:

The straightaway between PC#4 and PC#1, on which the primary finish line(s) for all events is located (home side).

Implement Landing Area:

Landing area defined by governing bodies for discus, shotput, javelin, hammer and other implements.

Interceptor Drain:

A drain that collects water from the adjacent areas and disburses it to other locations.

International Standard Track:

A track design recommended by World Athletics; a nonequal quadrant track with straightaways of 84.39m and measure line radii of 36.8m.

Kilogram (kg):

A common measurement of weight of the products used for building synthetic surfaces. 1kg = 2.205lbs.

Lane Width:

The area from the edge of the stripe on the runner's left through and including the full width of the stripe on the runner's right.

Latex-Bound Track System:

Track surface made of rubber particles bound together by one or more latex products.

Measure Line:

A theoretical line that measures the length of the race course. A painted inside lane line is located 20cm towards the infield. A raised curb is located 30cm towards the infield.

Modified Vertical Deformation:

A measure of surface compression resulting from an applied load.

NAIA:

National Association of Intercollegiate Athletics. A governing body for collegiate competition; currently follows NCAA rules.

NCAA:

National Collegiate Athletic Association. A governing body for collegiate competition.

NFHS:

National Federation of State High School Associations. Governing body for high school competition.

NIAAA:

National Interscholastic Athletic Administrators Association. A governing body for high school competition; currently follows NFHS rules.

Nonequal Quadrant Track:

A track with two equal curves and two equal straightaways that may be either shorter or longer than the curves.

NR:

Abbreviation for natural rubber.

Oxidation:

Structural and cosmetic deterioration due to chemical reaction with oxygen, normally associated with long exposure to the sun.

Pavement:

Asphalt or concrete that is placed over the aggregate base, specialized fill or subgrade, onto which the synthetic surfacing is to be applied.

PC:

The PC (Point of Curve) is the exact location of the transition between the straightaway and the curve or the curve and the straightaway. PC#1 is nearest the finish line, which is typically at or near the counterclockwise end of the home straightaway. PC#2 is directly across the field from PC#1 at the clockwise end of the back straightaway (180° from PC#1). PC#3 is at the opposite (counterclockwise) end of the back straightaway. PC#4 is directly across the field from PC#3, at the clockwise end of the home straightaway (180° from PC#3).

Permeability:

The ease with which water will flow or pass through the pores of a particular stratum. Texture, gradation, degree of compaction and primary structure strongly influence a given stratum's relative permeability.

Planarity:

A term referring to how level a surface is, or how level it is within a geometric plane.

Plasticity:

The ability of a material to be deformed without cracking or crumbling, and then to maintain that deformed shape after the deforming force has been released.

Polyurethane:

Synthetic rubber polymers with high solvent/heat resistance and low permeability.

Polyurethane-Bound Systems:

Track surface made of rubber particles bound together by one or more polyurethane products.

Ponding:

When water accumulates or becomes trapped on a track surface and cannot run off or otherwise escape.

Post-Tensioned Concrete:

A type of concrete in which steel cables are positioned inside the concrete as it's placed and then pulled tight to reinforce the concrete before the weight load is applied.

Premanufactured Rubber Mat:

Sheet good, normally of vulcanized rubber or polyurethane-bound rubber. Usually adhered to the asphalt/concrete pavement with adhesive.

Pulverization:

The crushing of asphalt to provide particle type materials that can be mixed with aggregate to form a track base.

Radius Point(s):

A point used to locate a curve of a track. Because of their importance in layout and marking a track, radius points often are permanently marked on the site by radius monuments. A standard track has two radius points, one at each end. A double-bend track, with its complex curves, will have six radius points.

Raised Curb:

A curb approximately 5cm x 5cm (2" x 2") that is secured to or placed on the track surface to denote the inside edge of lane #1. Where no raised curb exists, a painted line denotes the inside edge of lane #1. A raised curb is placed 30cm from the measure line towards the infield of the facility.

Resilient:

Having the ability to bounce or spring back into shape after being stretched, bent or compressed.

Rubberized Asphalt Track:

Track surface made of rubber particles, asphalt binders and, sometimes, stone or sand aggregate.

Runner's Edge:

The edge of the lane line to the runner's left where the left-hand lane line is out of bounds as it pertains to that lane. Typically used to define which edge of a lane line a measurement is made to, such as RLL (Radius to Lane Line).

Sandwich:

Track surface of two or more distinct strata, usually with a black base and a colored top.

SBR:

Styrene-butadiene rubber (SBR) granules, black in color and produced from rejected new or recycled tires or scrap rubber.

Shearing:

Of a soil, the result of friction loss between the particles and cohesion.

Sieve Analysis:

Describes particle percentages that will pass through various sized screens (sieves).

Specialized Fill Materials:

Sand, lime and similar materials that may be used to enhance the stability of the subgrade prior to placement of the aggregate base.

Staggers:

Markings designed to equalize the distance that all contestants run, regardless of which lane they are in. Staggers relate to the number of curves in a race, not the total length; i.e., a one-turn stagger means the runner will round one curve of the track before either finishing the race or cutting to lane one. Staggers are used not only for start lines, but also for the placement of hurdles and exchange zones, to ensure that the distance to those features is the same for all contestants.

Structural Spray:

Spray material containing pigmented binder and rubber particles.

Subbase:

See "Subgrade."

Subgrade:

Existing site materials and select fill that are used to level and stabilize. Sometimes referred to as subbase.

Swale:

Designed depressions to direct the flow of surface water.

Tensile Strength:

Measurement for the strength required to pull an object to its breaking point.

Total Station:

An electronic theodolite with a built-in EDM capable of being set up on site and used to layout points provided from an electronic file for track construction and track markings.

Transition:

The portion of track that starts at the point of curve. The transition area ends at the outside track radius. See Figure 2-12 on page 40.

USA Track and Field:

USA Track and Field. Governing body for open competition in the USA.

Viscosity:

A fluid's resistance to flow. Low-viscosity materials are free-flowing (water); high-viscosity materials exhibit slow mobility (syrup).

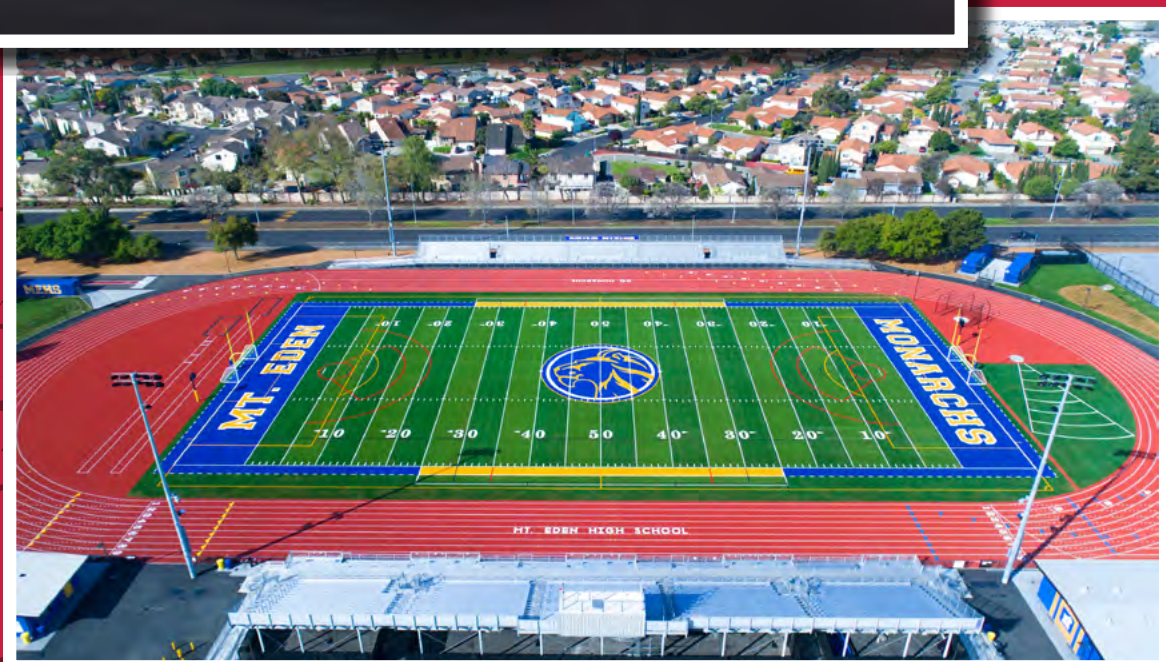
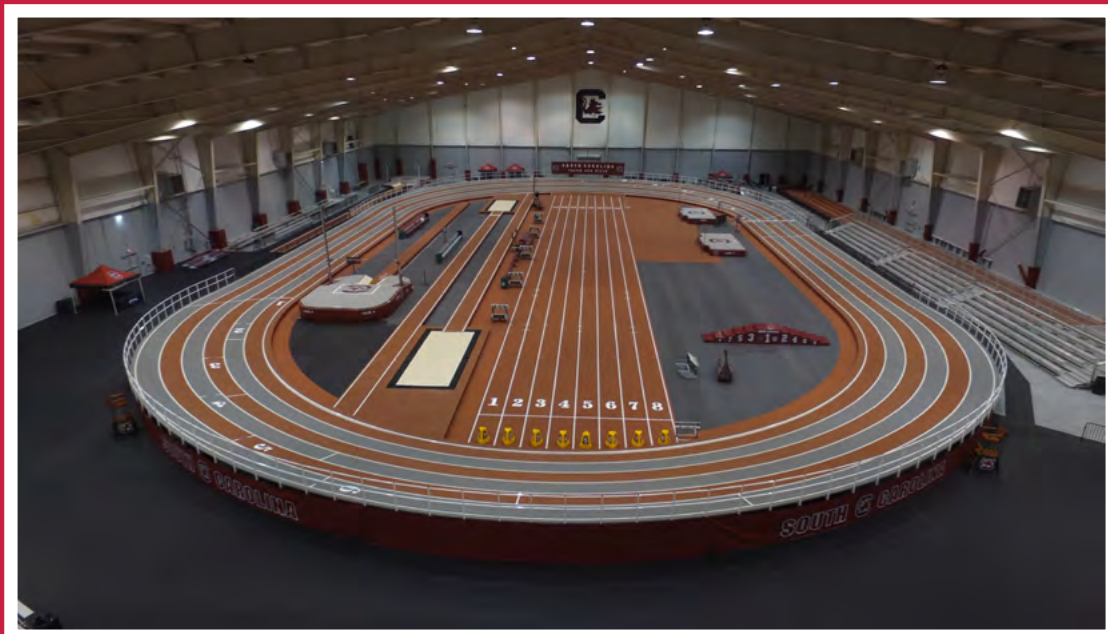
Waterfall Start:

A curved start line used for distance races of 800m or greater. Runners line up at their respective locations along the curved start, and at the commencement of the race, may run directly to lane #1.

World Athletics:

Governing body for international competition. Formerly known as IAAF.

Index



Index

A

- accessibility issues, 15, 97
- agreements, negotiated, 5–6
- aggregate base, 24, 28
- asphalt, 74–84
 - and drainage, 25, 27
 - track surfaces, 56, 65
 - installation, 74–84
 - repair of, 147–148
 - and indoor tracks, 154
- associations, trade, 159

B

- base, construction of, 28–29
- bid process, 2–7, 181
- binders, surface, 66
- black mat system, 57
- budgeting, 10–11

C

- calculation, 18–19, 155
- certification of track, 113, 117–136
- Certified Track Builder, ix–x, 4, 7, 71
- cleaning, 139–140, 146, 155
- clearances (safety zone), 98
- coatings, surface, 60–61, 66
- codes and permits, 21
- color coding, 178
- colored rubber particles, 67–68
- community-acquired MRSA (CA-MRSA)
 - infections, 145
- concrete pavement, 84–87
- configurations of tracks, 16–18
- construction timetable, 88
- consultants, 7
- contractor, choosing, 71–73
- control joints, 82, 86
- cooperative purchasing, 2–3
- crossing area, 140
- curbs and edging, 27–28, 74

D

- D-zone, 19
- design/bid, 3–5

- design/build, 5
- design professionals, 7–8
- dimensions
 - field events, 19
 - site requirements, 20
- discus/hammer, 91, 98, 171–173
 - See also* throwing events
- double-bend track, 18
- drainage of site, 24–27, 74
- drinking water, 97

E

- edging (curbs and perimeter), 27–29, 74
- environmental considerations, 9, 15
- equal quadrant track, 17
- expansion joints, 85
- extension of pavement, 27

F

- fencing, 96–97, 141, 149
- fertilizer, 142–143
- field events, 89–98
 - See also* specific events
- fill materials, 23, 29
- flush curb, 28
- footwear, impact on track, 139
- fountains, drinking, 97
- 400-meter tracks
 - formulas for, 18
 - verification method, 177
- freeze-thaw cycles, 21
- French drain, 26
- full pour surface (polyurethane), 58
- funding considerations, 8

G

- gates, 97
- geography, 21
- geometry, 152–153
- governing bodies, 160–161
- grading of site, 23–24
- graphics and symbols, 113–114

H

- hazards, 148–149
- health and hygiene, 144–145
- herbicides, 24
- high jump, 92
 - dimensions, 19
 - specifications, 166
 - See also* jumping and vaulting hurdles
 - placements (steeplechase), 92
 - specifications, 164–165

I

- impermeable (nonporous) surfaces, 24
- impermeable polyurethane systems, 58, 62
- indoor tracks, 151–155
- infections, MRSA, 145

J

- javelin, 90–91, 173–174
 - See also* throwing events
- joints
 - in asphalt pavement, 81–82, 85
 - in concrete pavement, 85
- jumping and vaulting, 98

L

- landing area, landing pits
 - See* field events or specific event
- landscaping, 8–10, 20–21, 73–74, 143
- latex primer, binders, and coating, 66
- latex surfacing systems, 55, 57
- LEED, 9–10
- lighting, 87, 94–95, 98, 148
- long jump, 16, 20, 89–91, 114, 141, 154, 168, 170
 - See also* jumping and vaulting

M

- maintenance, 8–9, 22, 27, 139–150
 - checklist, 180
 - curbs, 74
 - equipment, 140, 149
 - fencing, 96–97, 141
 - landing areas, 141
 - landscaping, 142
 - lighting, 95, 149, 150
- measure line, 17–19, 27, 110–113, 162, 175
- metric conversion, xii, 176
- MRSA infections, 145

N

- natural material surfaces, 65
- negotiated agreements, 5–6
- night events (lighting), 94–95, 150
- nonequal quadrant track, 17

P

- painting, 112–113
 - See also* marking a track pavement
 - repainting, 144
 - verification and, 118–119, 130
- perimeter. *See* edging
- permeable latex systems, 57
- permeable polyurethane systems, 58
- permeable surfaces, 24, 28, 56, 66
- permits, 21, 29, 73, 88
- planning, 1–11, Chapter 2, 151–152
- pole vault, 19, 89–90, 98, 154, 166–168
 - See also* jumping and vaulting
- polyurethane
 - primers, binder and coatings, 65–66
 - systems, 58–65
- post-tensioned concrete, 84–85
- precast channel drain systems, 25–26
- primers, surface, 65–66

R

- radius point(s), 17–18, 112, 177
- raised curb, 27
- reconstruction of surface, 147–148
- reinforcement of concrete base, 84
- repainting, 144
- repairs, 146–147
- Request for Proposal (RFP), 6–7
- resurfacing, 66, 82, 146–147
- rolling, of asphalt pavement, 81, 84
- roofing asphalt, 65
- rubber surfaces, 57–70
- runways
 - jumping events, 89–90. *See also* specific events
 - throwing events, 91–92. *See also* specific events

S

- safety
 - field events, 94–95
 - jumping events, 98
 - maintenance, 148–150

- oval clearance, 98
- spectator, 90, 93
- throwing events, 90, 98
- safety zone (clearances), 98, 148, 153
- sand, for landing pits, 91
- sandwich systems, 57–58
- SAR systems (asphalt), 65
- SBR (Styrene-Butadiene-Rubber) latex binder, 67
- shot put, 91, 141, 171–173
 - See also* throwing events
- site for track
 - grading of, 23–24
 - considerations, 15
- slope notation, 172
- soccer, impact on field sports, 89
- soil
 - conditions, 21–22
 - sterilization of, 22
 - testing of, 1, 14, 21
- sound systems, 94
- specifications
 - asphalt, 83–84
 - indoor tracks, 152–153
 - by governing body, 161–174
 - lighting, 94–95
 - and track surface, 63
- spectator viewing, 93
- spikes, 139
- stains, 140–141
- start and finish lines, 110, 152, 164
 - and verification, 124, 131, 135
- steeplechase, 92, 164–165
- storage shelters, 20, 92
- sun, impact of, 21, 98, 141
- surfaces and surfacing systems, 55–68
 - asphalt, 65
 - installation, 55
 - latex, 57
 - maintenance, 63–64
 - natural, 65
 - polyurethane, 58–65
 - primers, binders and coatings, 60, 66
 - rubber, 67–68
 - stains, 140–141
 - testing of, 68–70
- survey of site, 15, 74, 83
- swales, 26
- symbols and graphics, 113–114
- synthetic turf, 26–27

T

- testing
 - of soil, 1, 14, 21–22
 - of surfaces, 63, 67, 83–85
- throwing events, 91, 98
- timing systems, 93
- tolerance, plus and minus, 109
- track certification, 113
- track configurations, 16–18
- track markings, 109–114
- trees and vegetation, 21–22.
 - See also* environmental considerations, landscaping maintenance
- triple jump, 20, 89–91, 154, 170
 - See also* jumping and vaulting

U

- utilities, locating, 15

V

- verification, Chapter 7, 155
- view, of spectators, 93

W

- water
 - drinking, 97
 - effect on tracks, 143
- water jump, 90, 164–166
 - See also* steeplechase
- wear considerations, 142–143
 - See also* maintenance
- weeds and grass control, 142–143
- weight, of track surface, 69
- winterizing, 144



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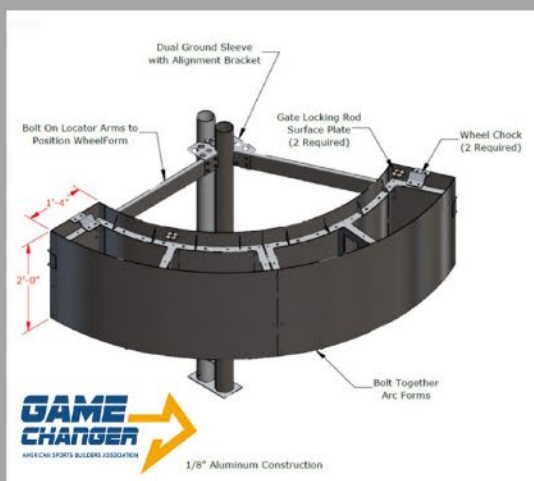


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